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(54) **METHOD OF OPERATING A COMBINATION
ULTRASONIC / BIPOLAR RF SURGICAL
DEVICE WITH A COMBINATION ENERGY
MODALITY END-EFFECTOR**

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(US); **Stephen M. Leuck**, Milford, OH
(US); **Nina Mastroianni**, Cincinnati,
OH (US); **John E. Brady**, Cincinnati,
OH (US); **Wei Guo**, Mason, OH (US);
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Kemper**, Cincinnati, OH (US)

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(60) Provisional application No. 62/955,292, filed on Dec. 30, 2019.

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(51) **Int. Cl.**

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A61B 17/32 (2006.01)

A61B 18/12 (2006.01)

A61B 18/14 (2006.01)

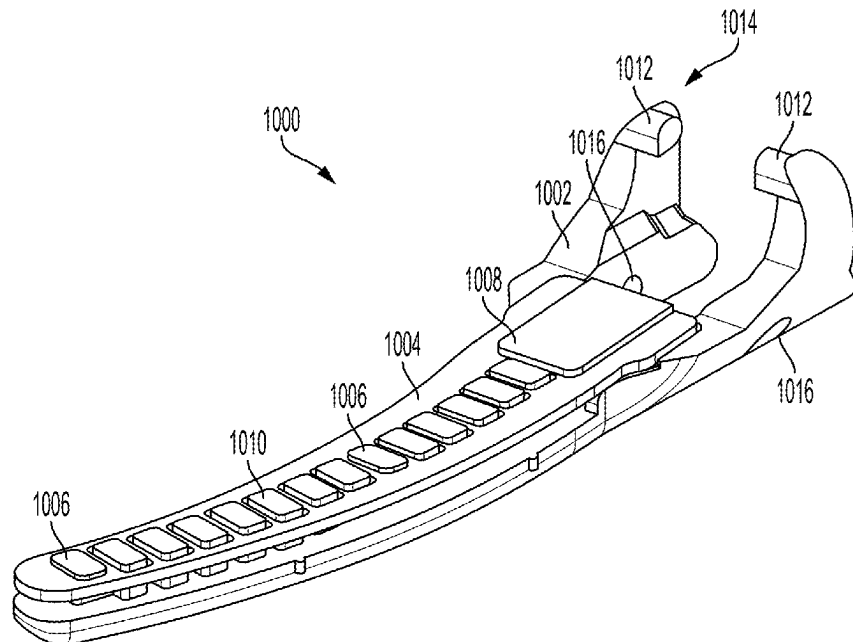
(52) **U.S. Cl.**

CPC **A61B 18/00** (2013.01); **A61B 18/1445**
(2013.01); **A61B 2017/320075** (2017.08);
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2017/320095 (2017.08); **A61B 2018/00077**
(2013.01); **A61B 2018/00083** (2013.01); **A61B**
2018/00601 (2013.01); **A61B 2018/0063**
(2013.01); **A61B 2018/00994** (2013.01); **A61B**
2018/126 (2013.01); **A61B 2018/1452**
(2013.01)

(57)

ABSTRACT

Disclosed is a surgical instrument with an end-effector adapted and configured to deliver a plurality of energy modalities to tissue at a distal end thereof. The energy modalities may be applied simultaneously, independently, or sequentially. A generator is electrically coupled to the surgical instrument and is configured to supply a plurality of energy modalities to the end-effector. In one aspect, the generator is configured to supply electrosurgical energy (e.g., monopolar or bipolar radio frequency (RF) energy) and ultrasonic energy to the end-effector to allow the end-effector to interact with the tissue. The energy modalities may be supplied to the end-effector by a single generator or multiple generators.



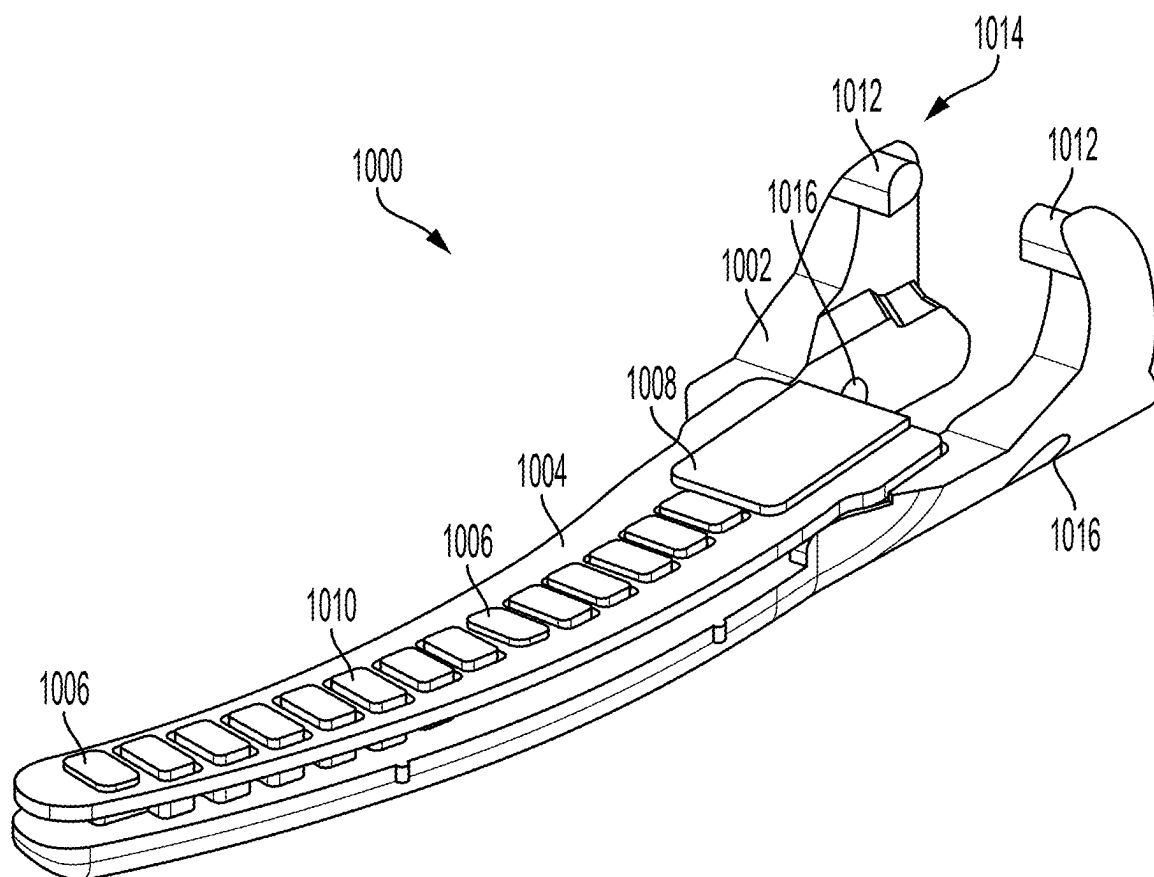


FIG. 1

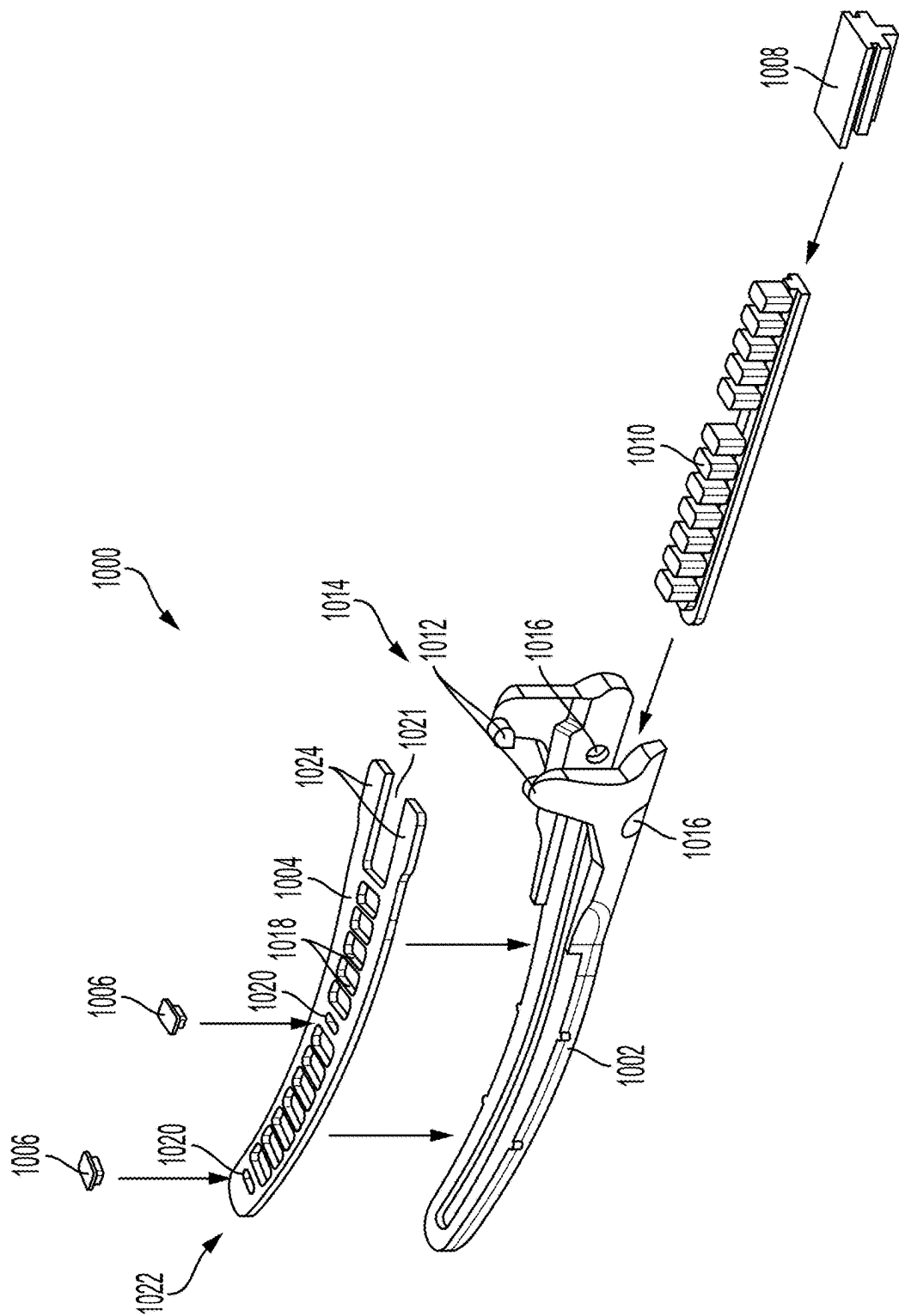


FIG. 2

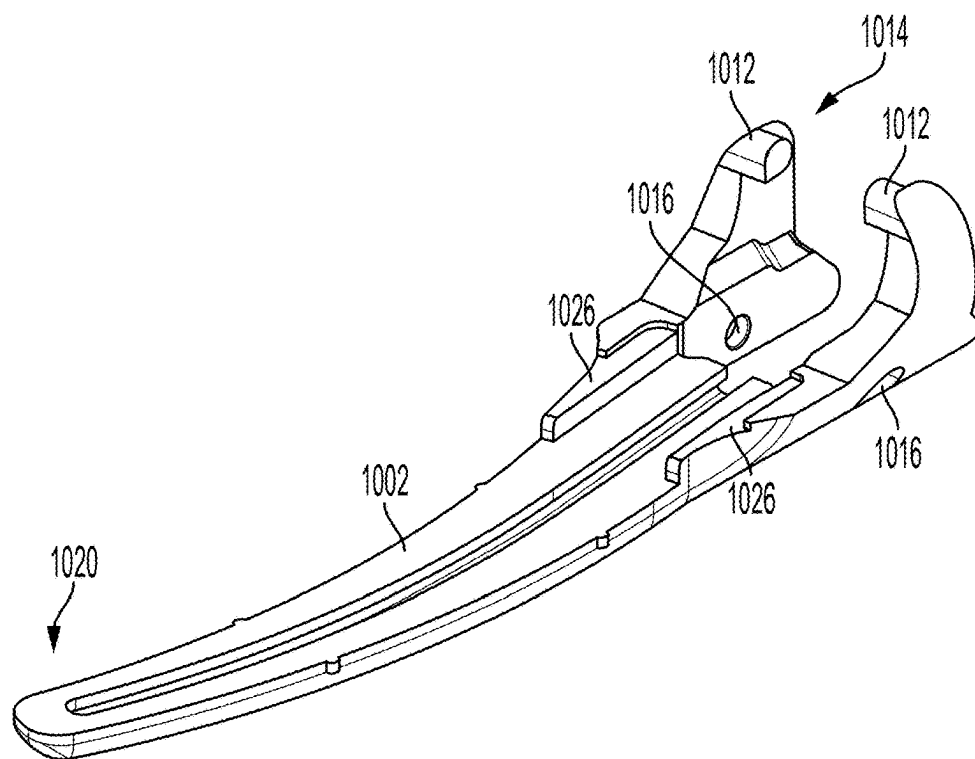


FIG. 3

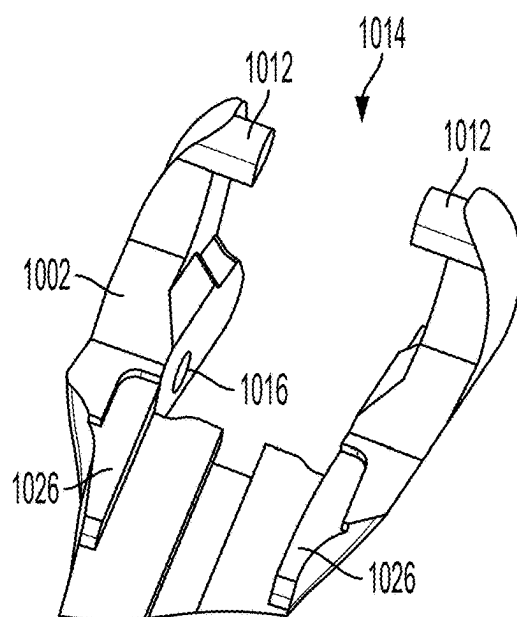


FIG. 4

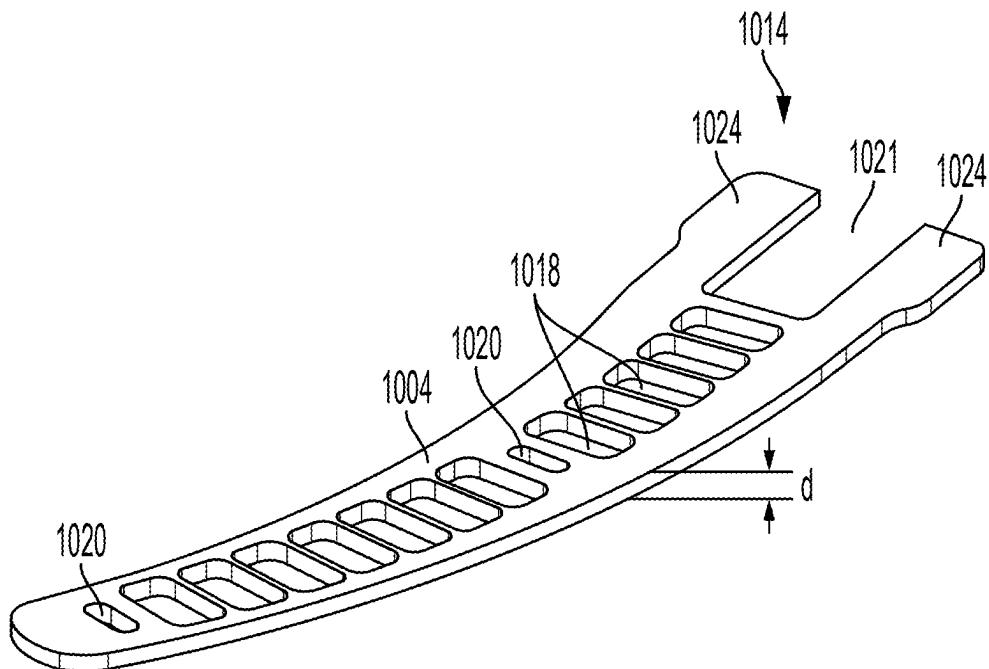


FIG. 5

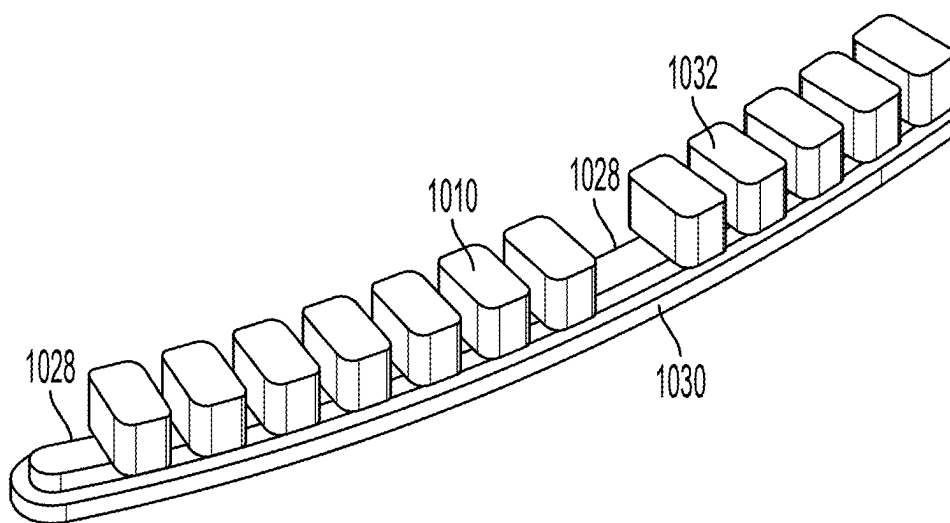


FIG. 6

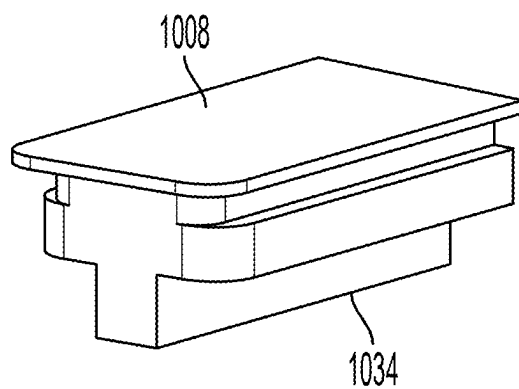


FIG. 7

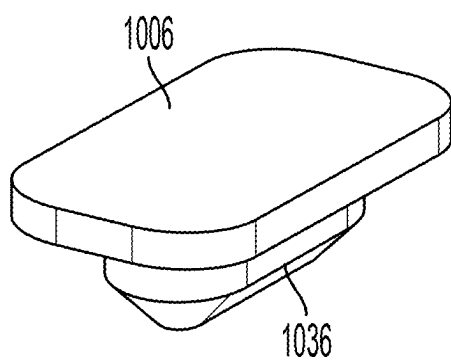


FIG. 8

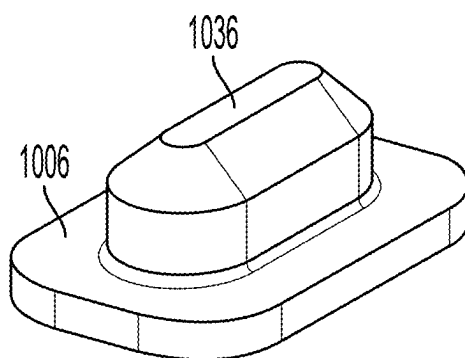


FIG. 9

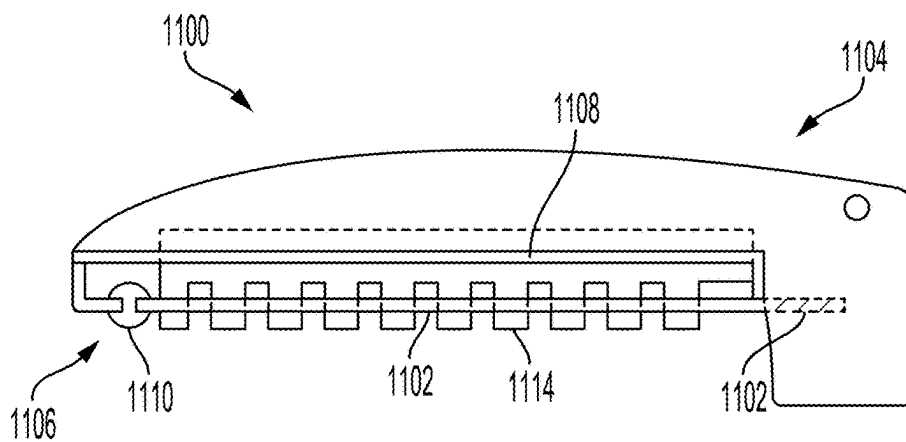


FIG. 10A

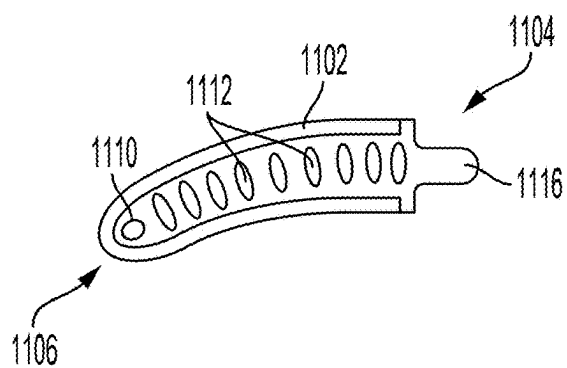


FIG. 10B

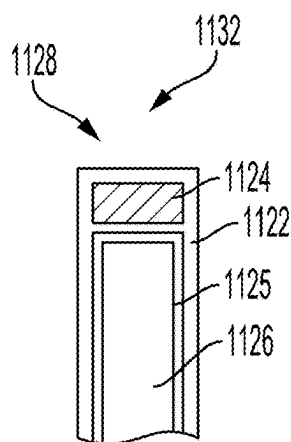


FIG. 11A

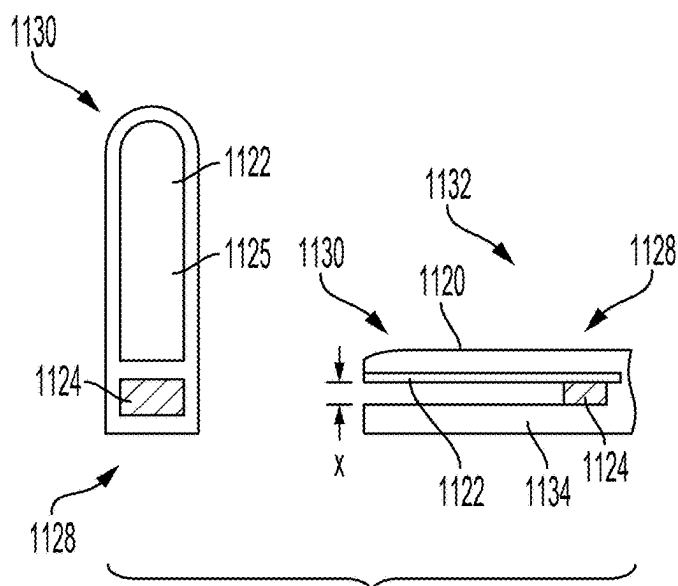


FIG. 11B

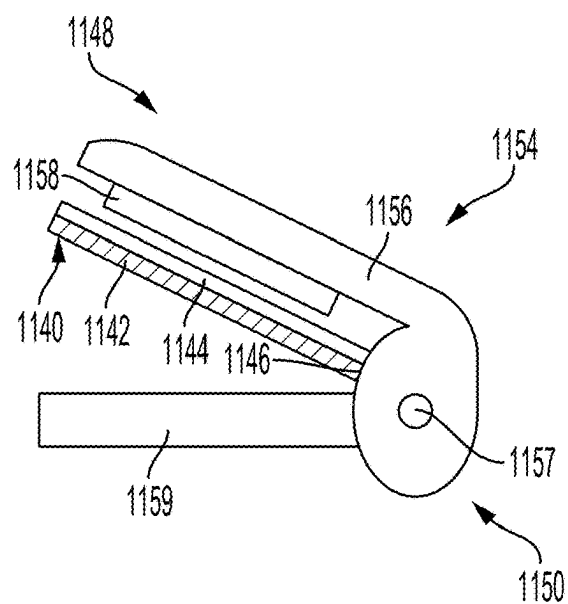
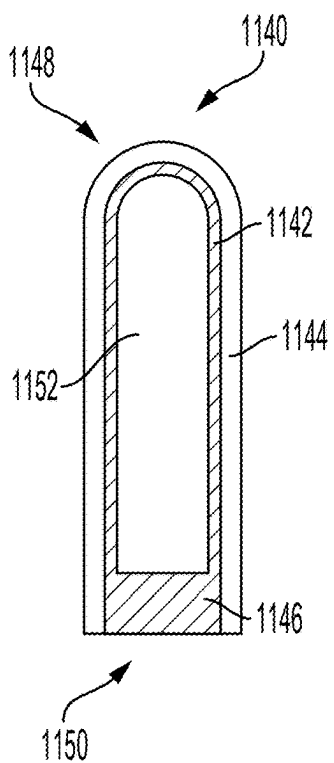


FIG. 11C

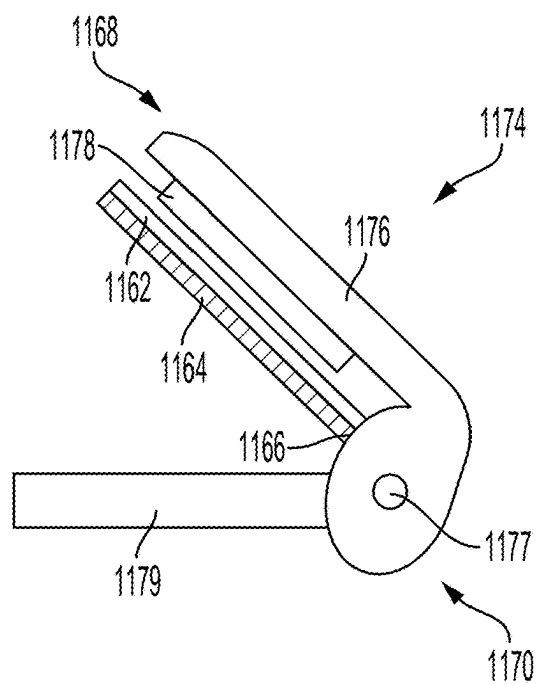
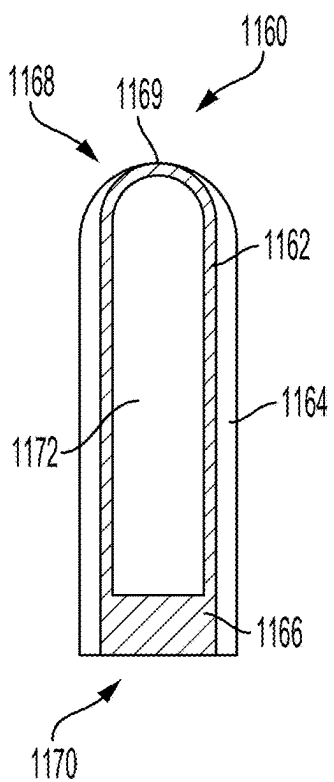


FIG. 11D

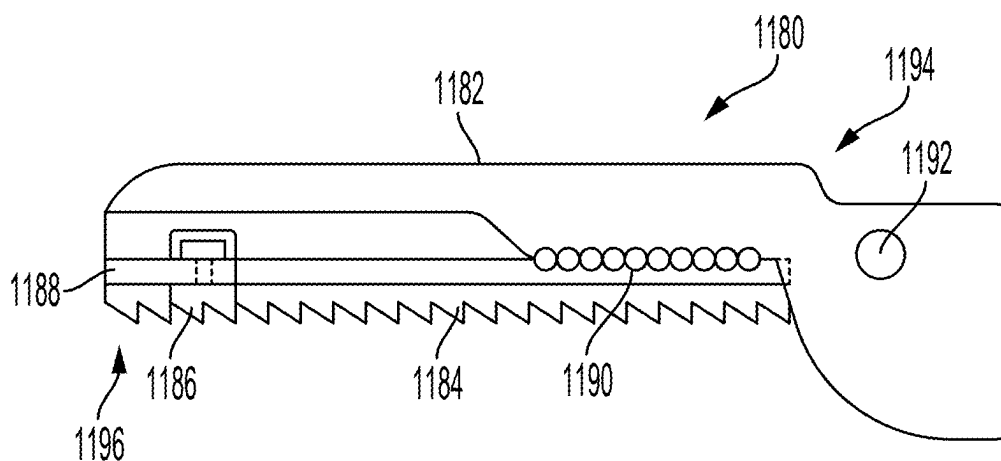


FIG. 12A

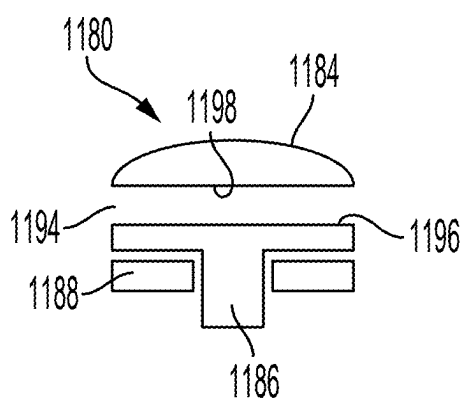


FIG. 12B

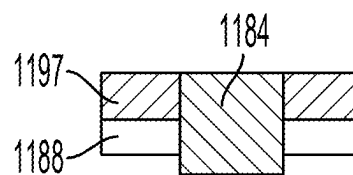


FIG. 12C

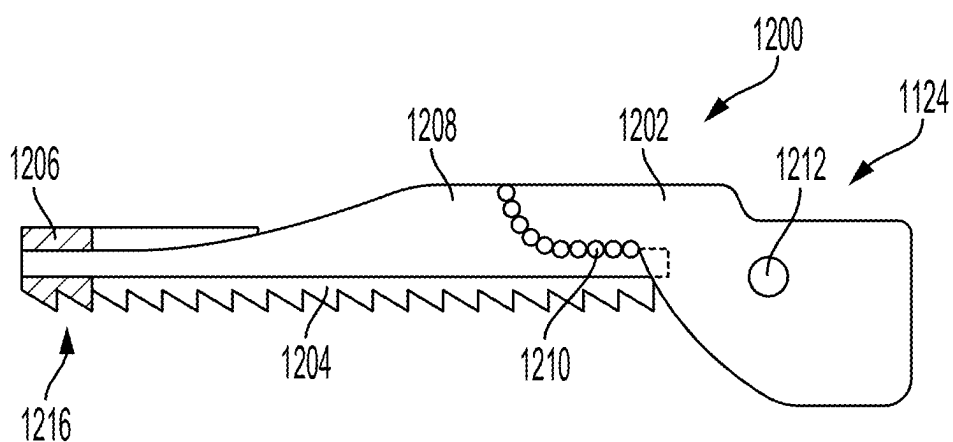


FIG. 12D

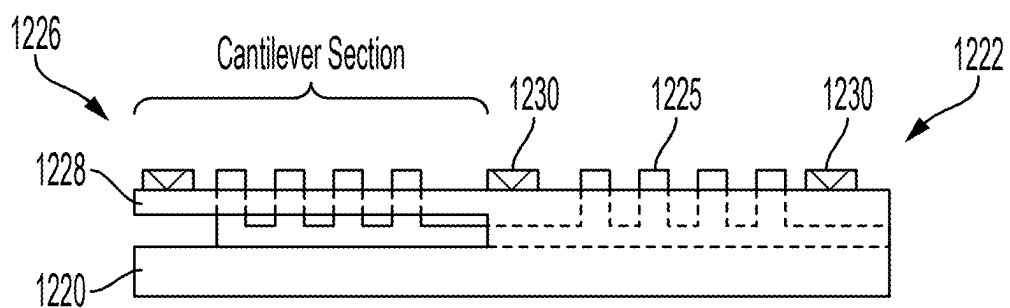


FIG. 13A

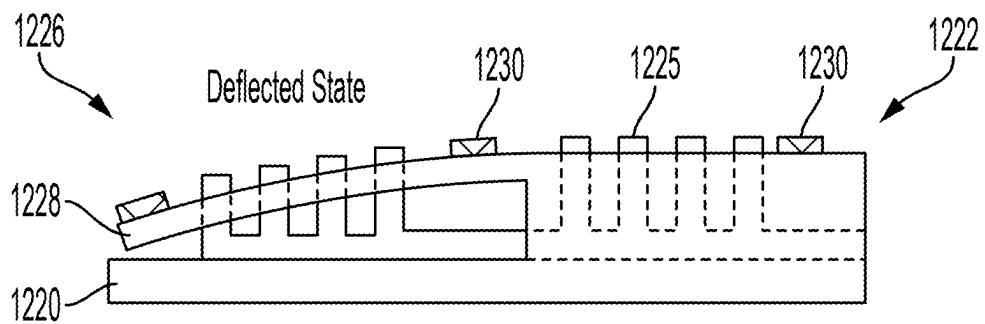


FIG. 13B

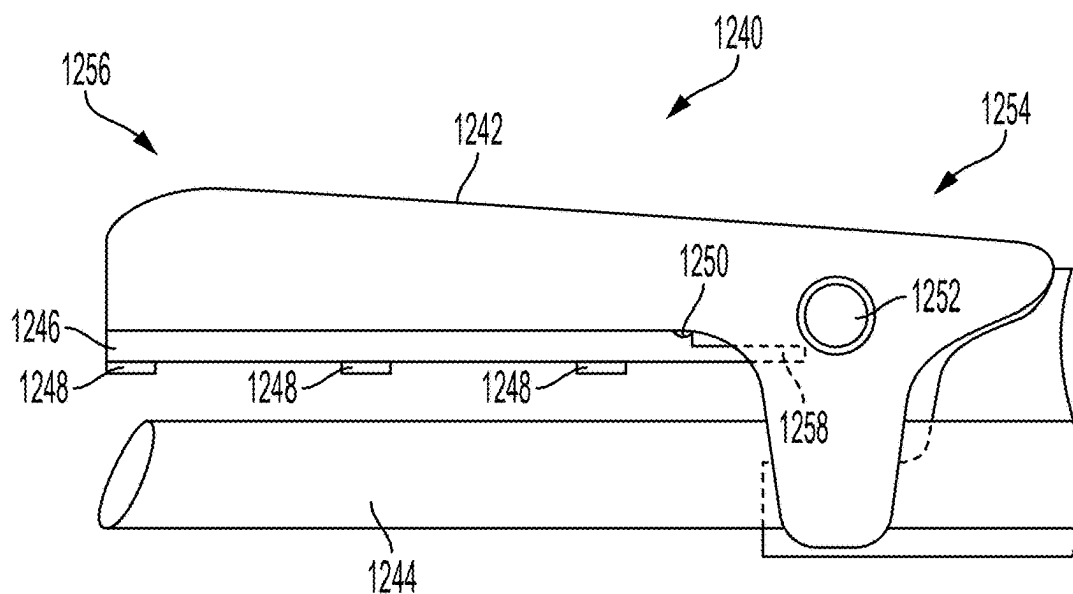


FIG. 14A

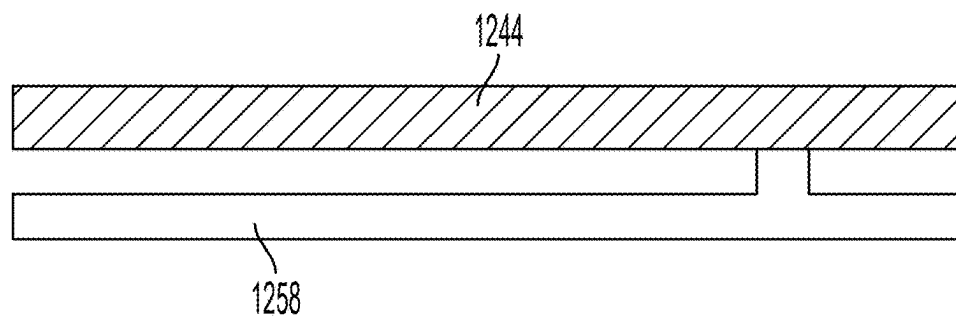


FIG. 14B

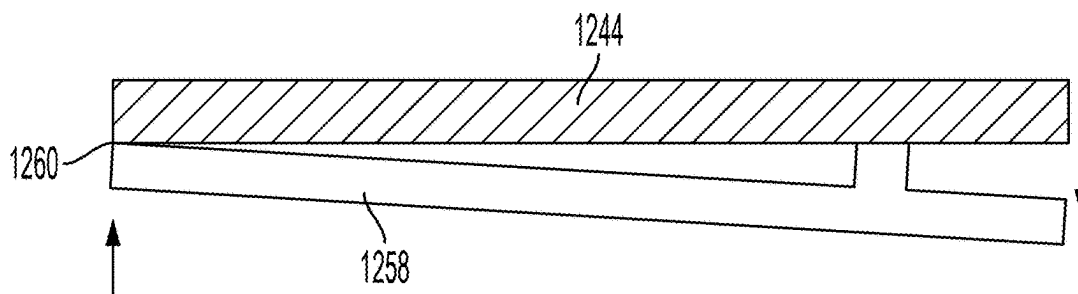


FIG. 14C

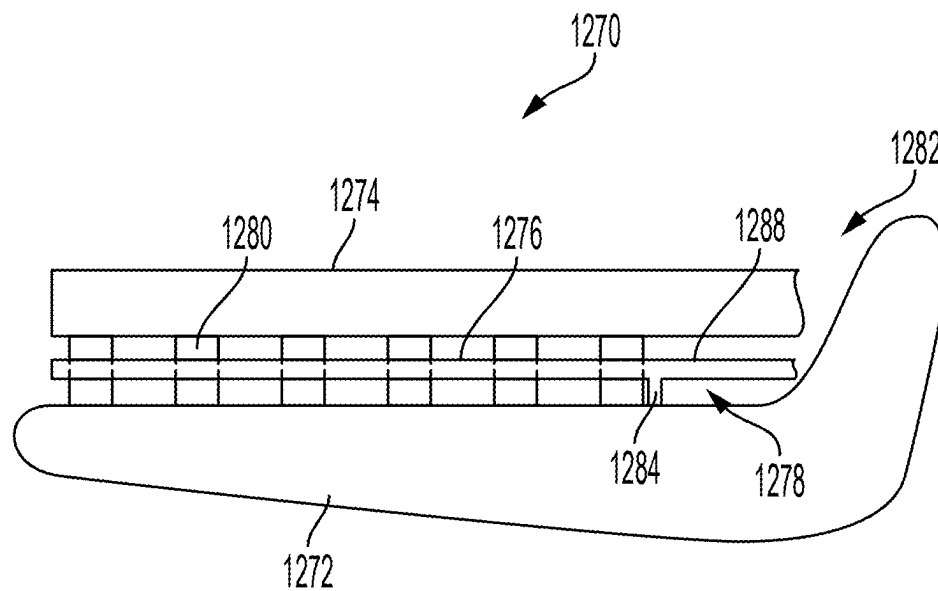


FIG. 15A

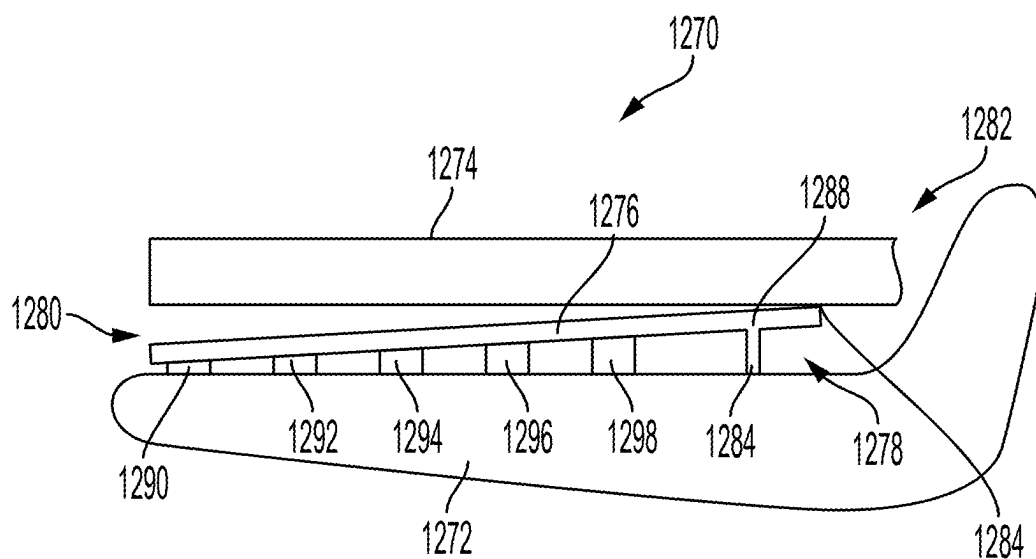


FIG. 15B

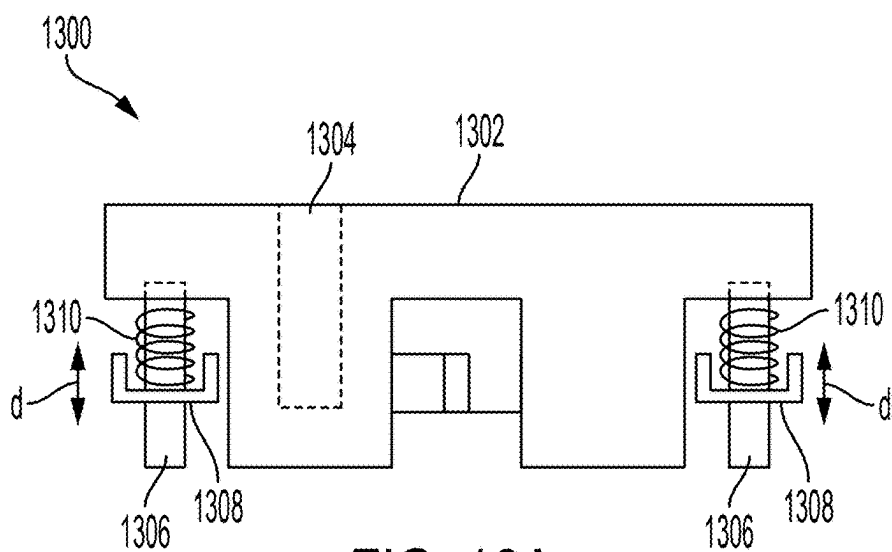


FIG. 16A

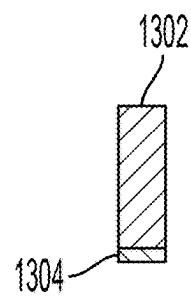


FIG. 16B

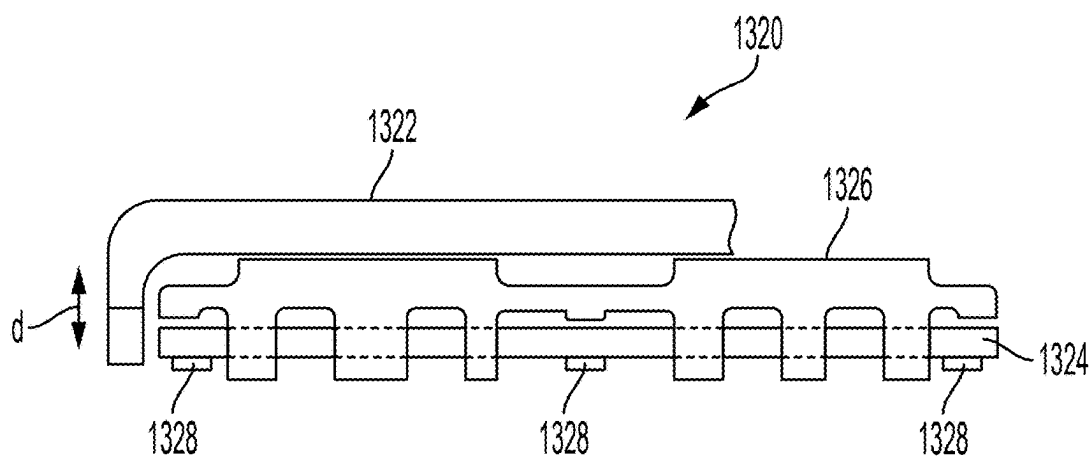


FIG. 17

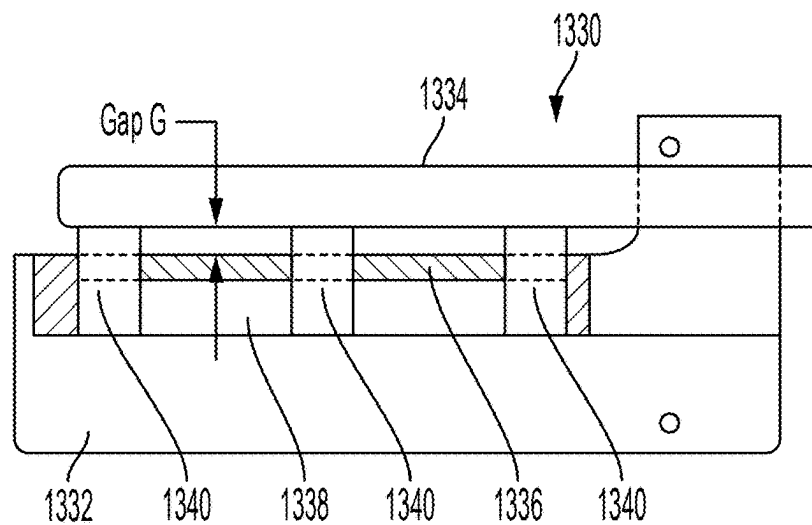


FIG. 18A

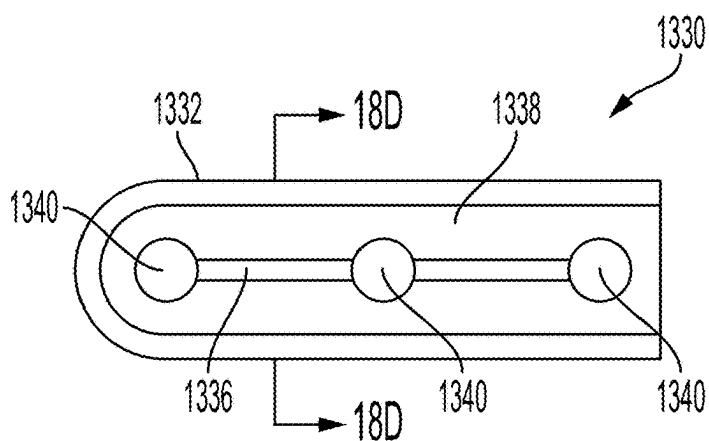


FIG. 18B

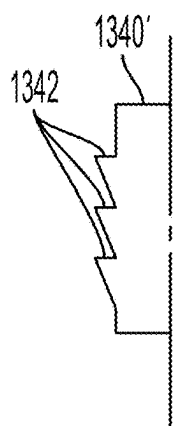


FIG. 18C



FIG. 18D

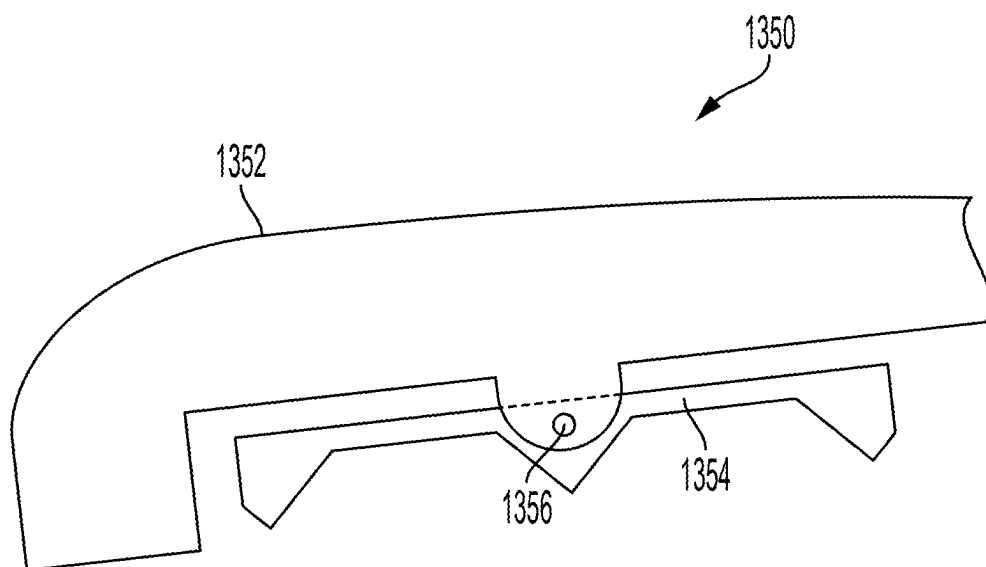


FIG. 19A

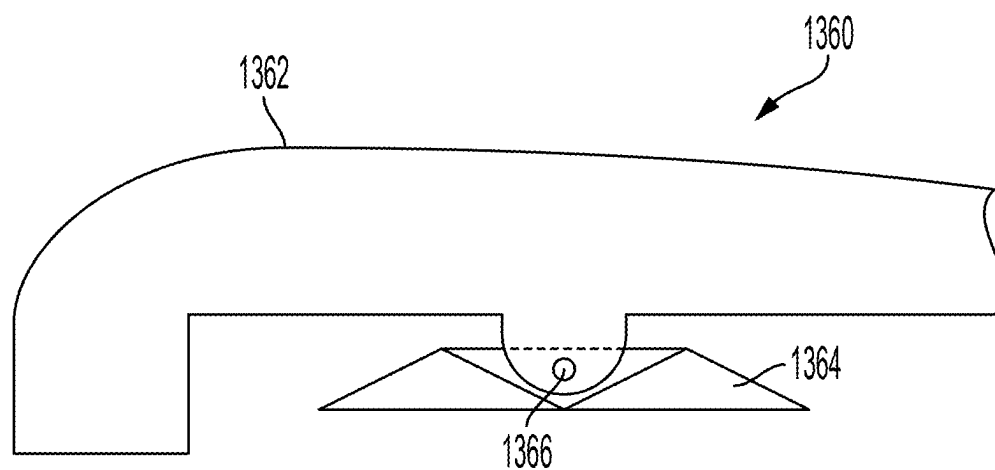


FIG. 19B

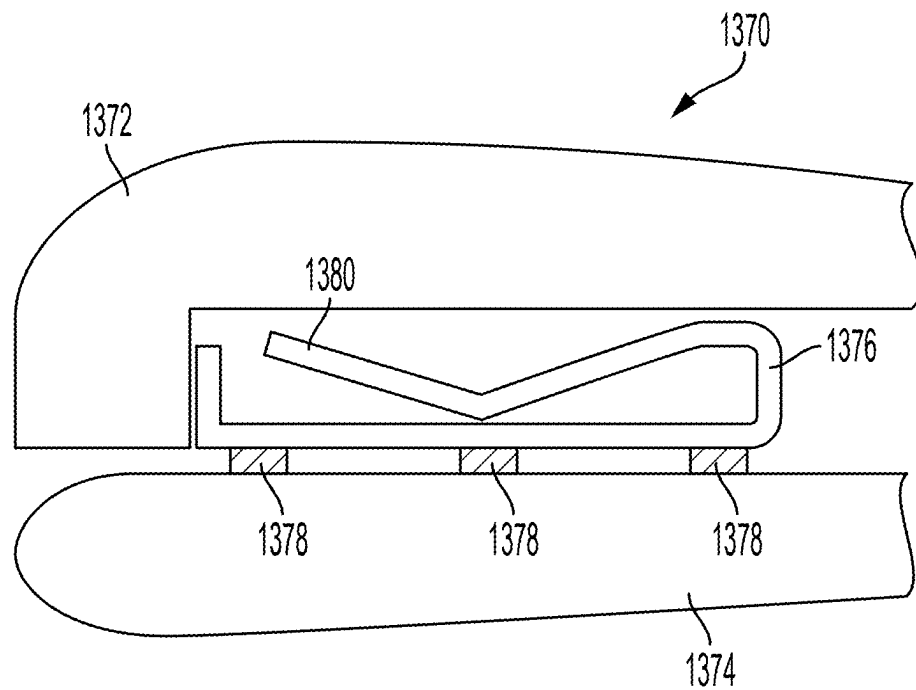


FIG. 20

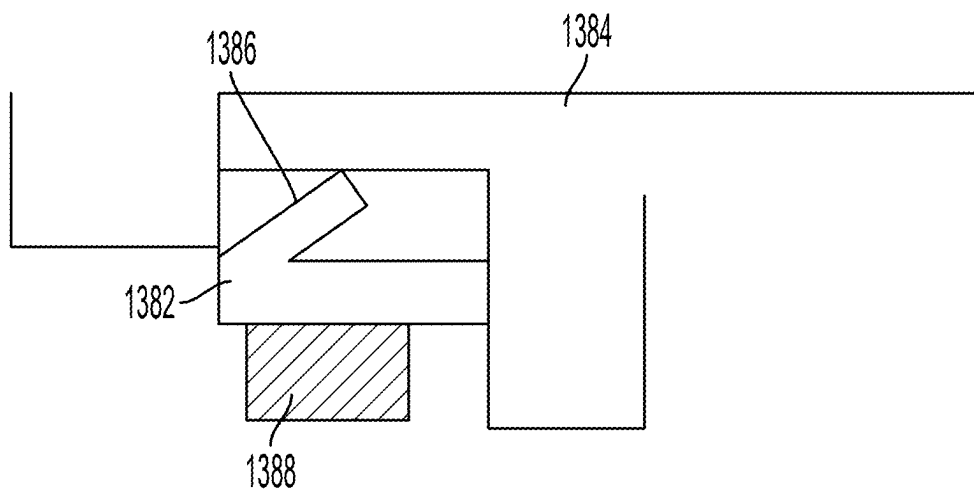


FIG. 21

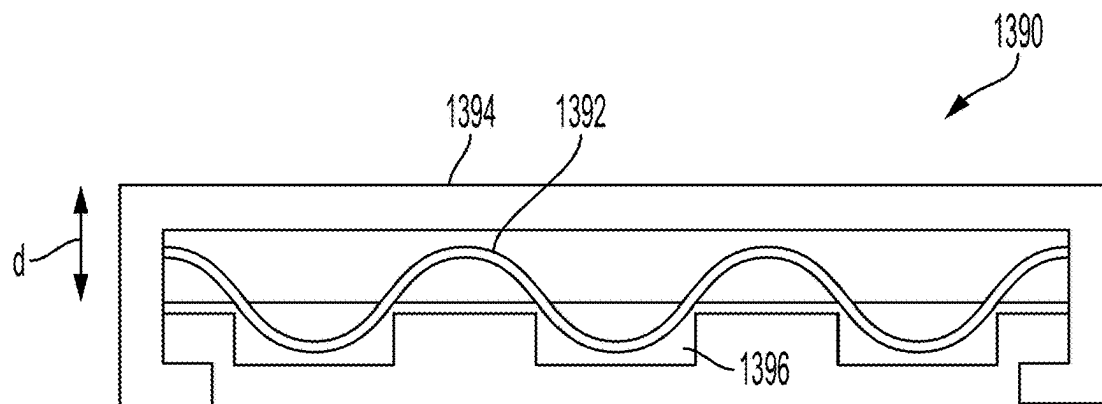


FIG. 22A

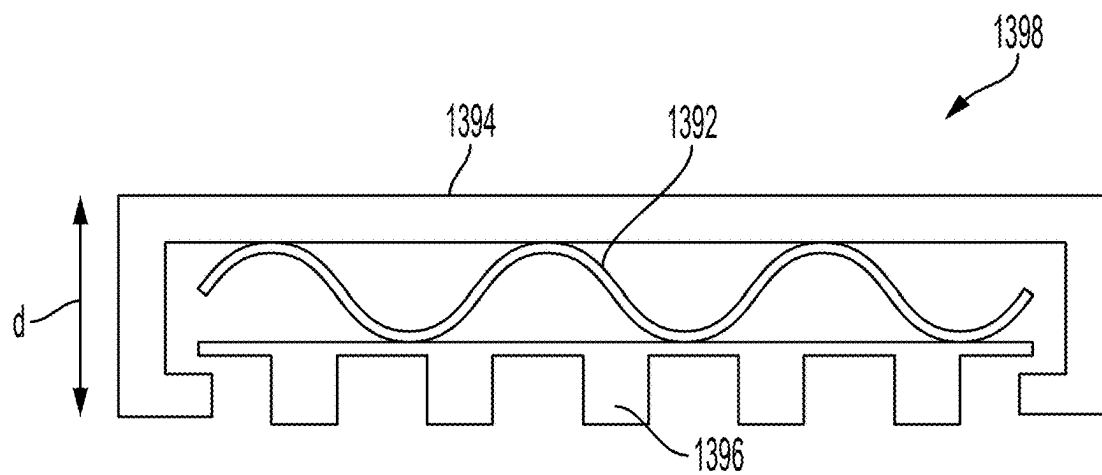


FIG. 22B

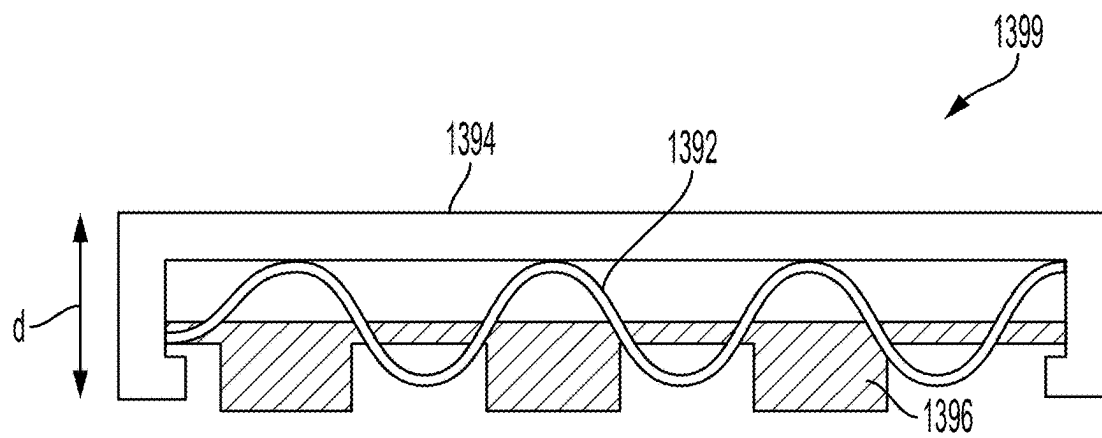


FIG. 22C

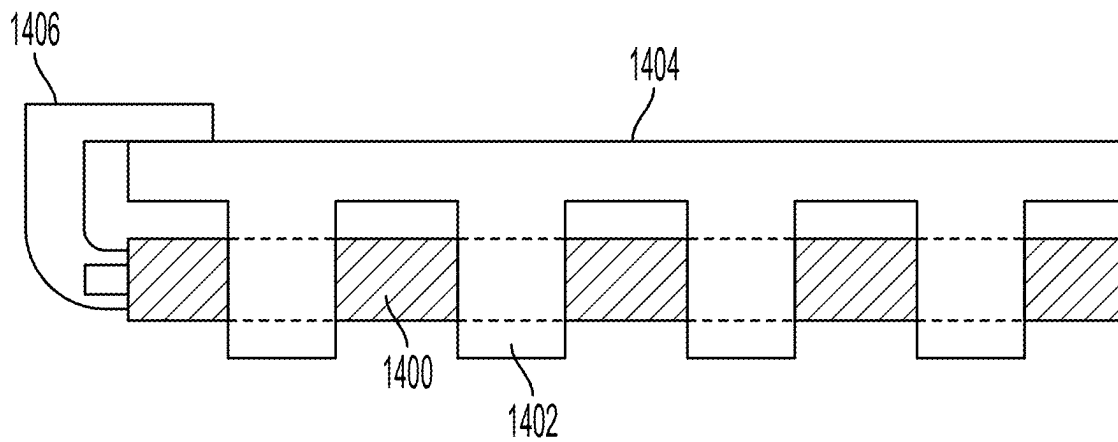


FIG. 23A

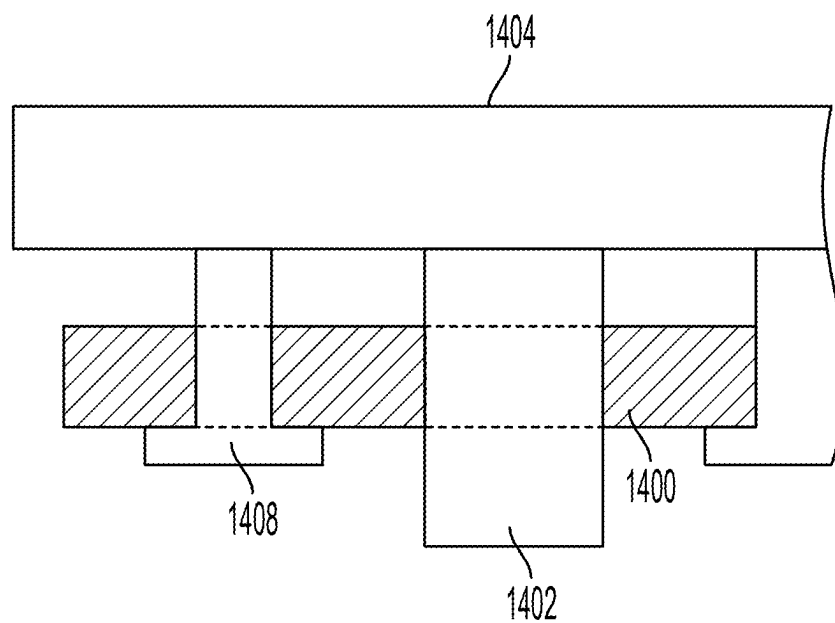


FIG. 23B

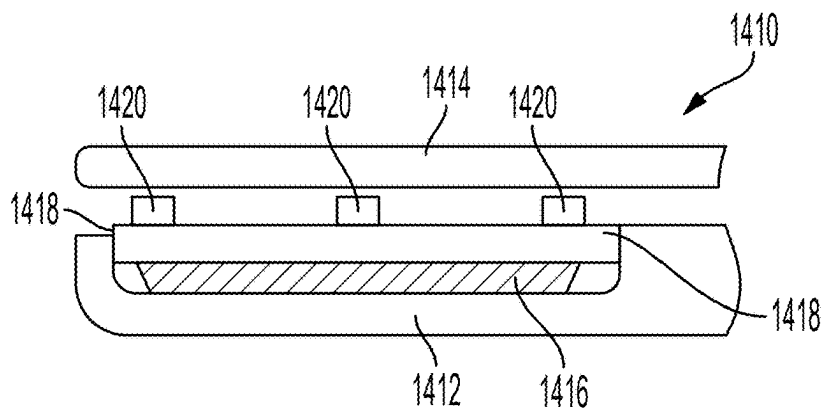


FIG. 24A

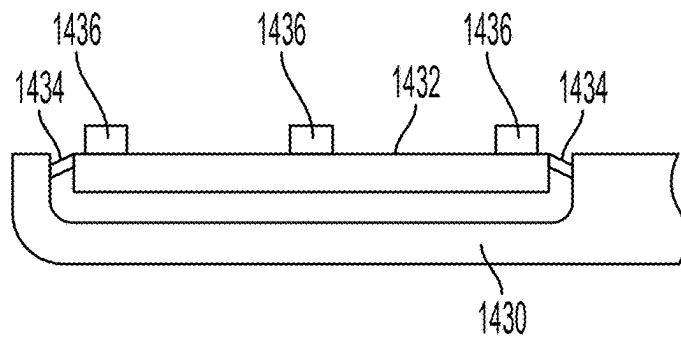


FIG. 24B

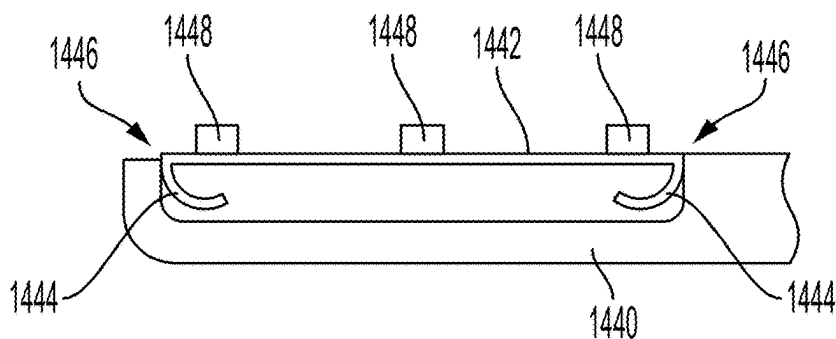


FIG. 24C

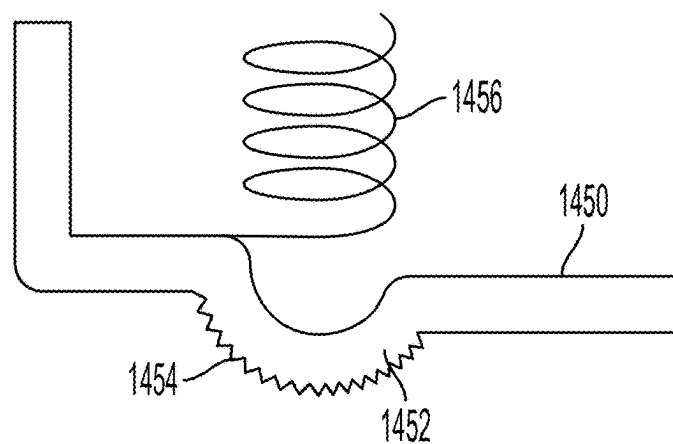


FIG. 25

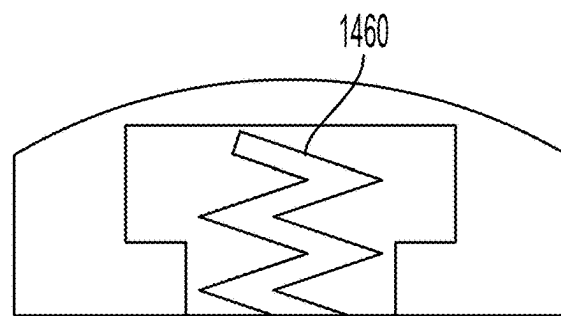


FIG. 26

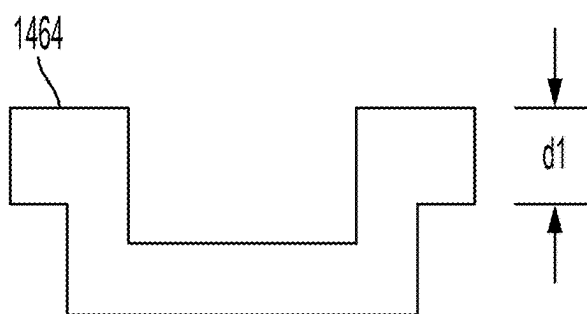


FIG. 27

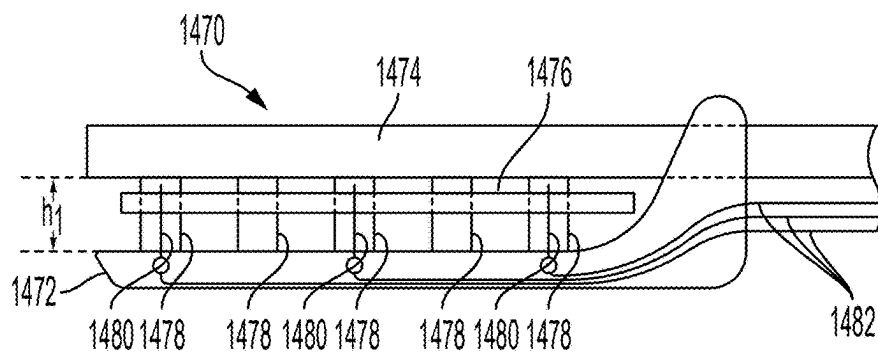


FIG. 28A

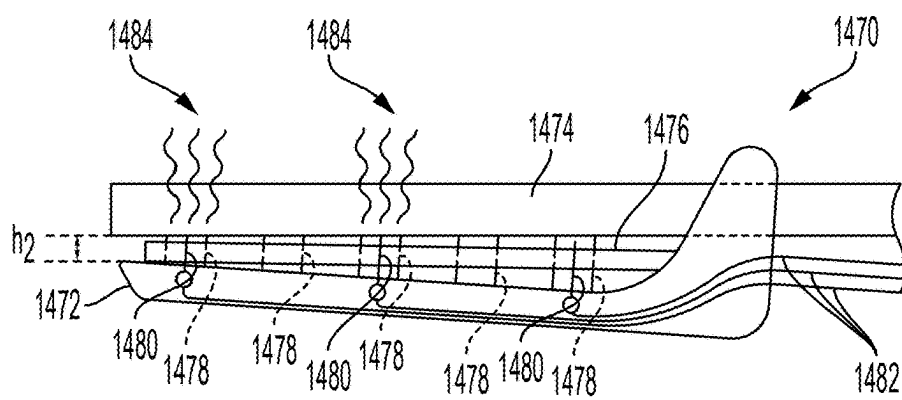


FIG. 28B

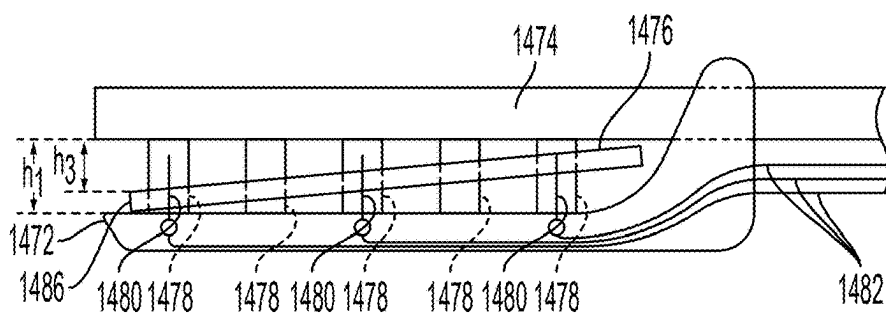


FIG. 28C

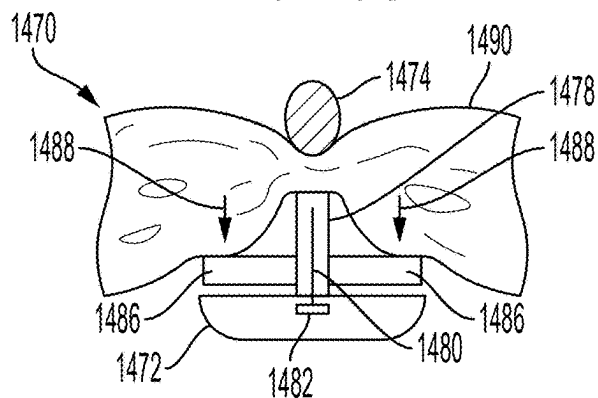


FIG. 28D

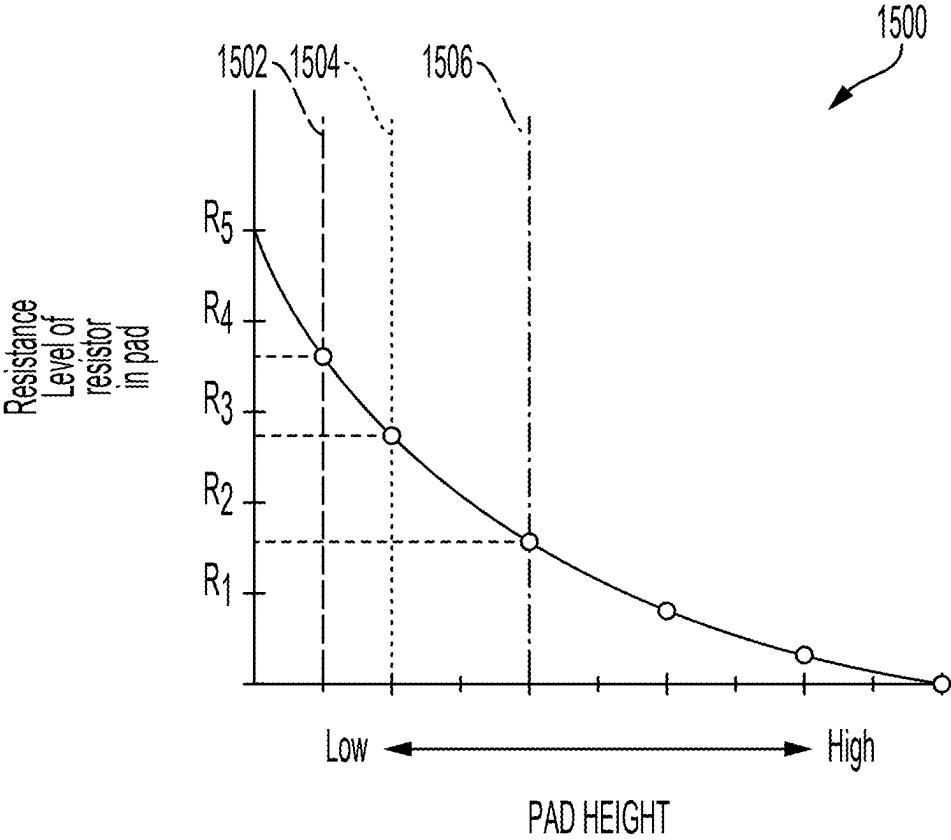


FIG. 29

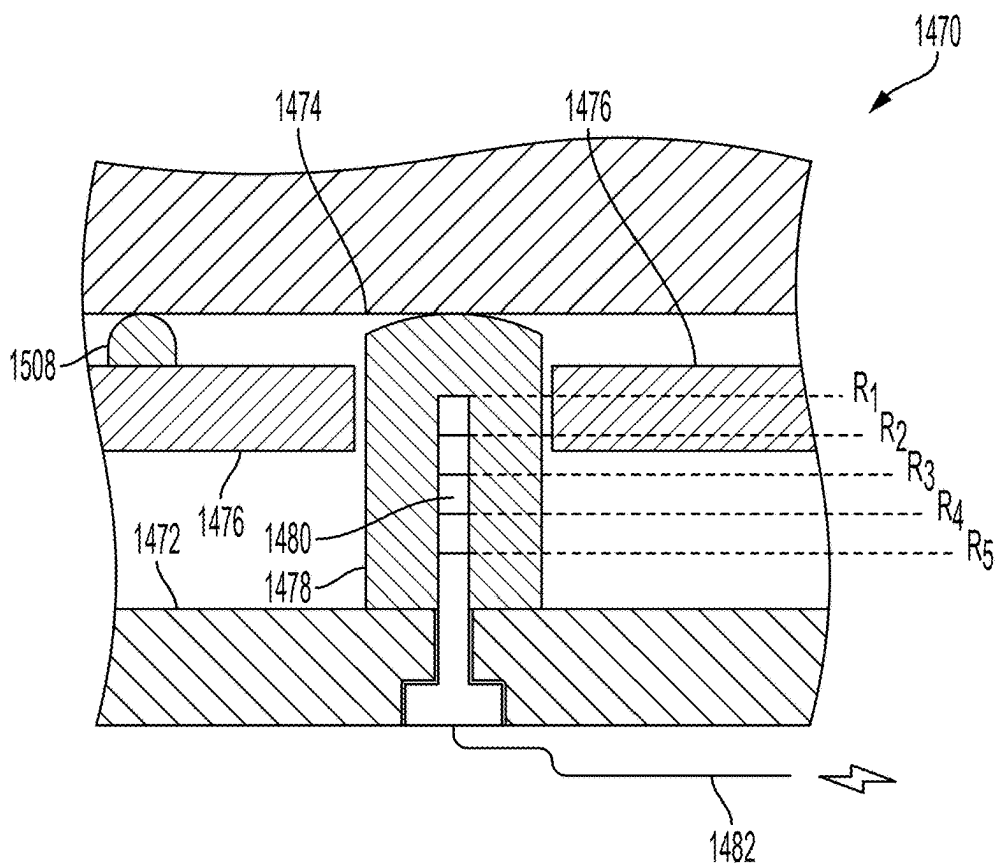


FIG. 30A

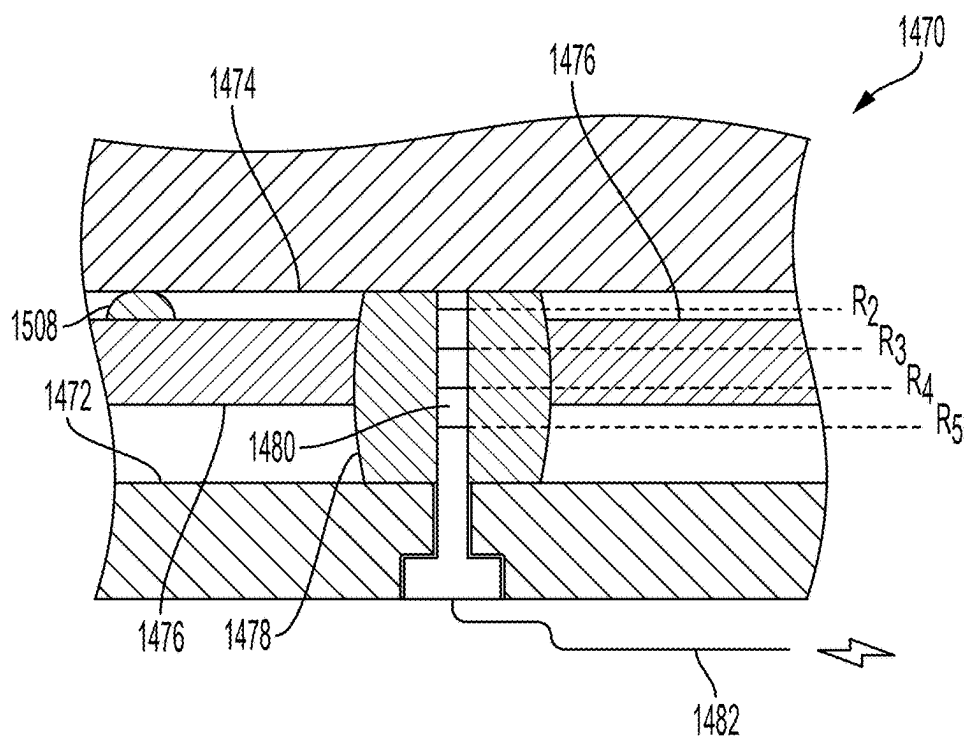


FIG. 30B

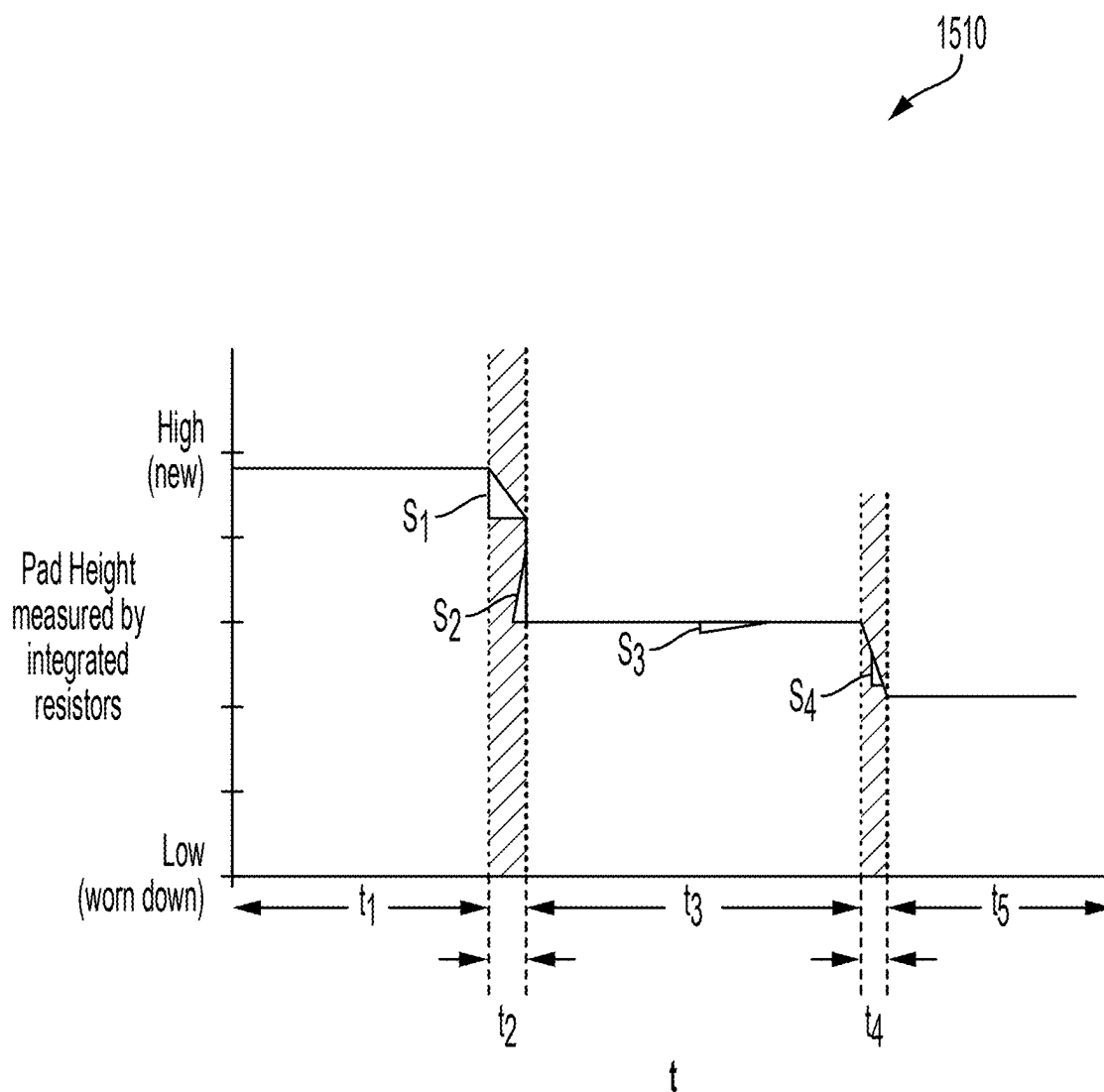


FIG. 31

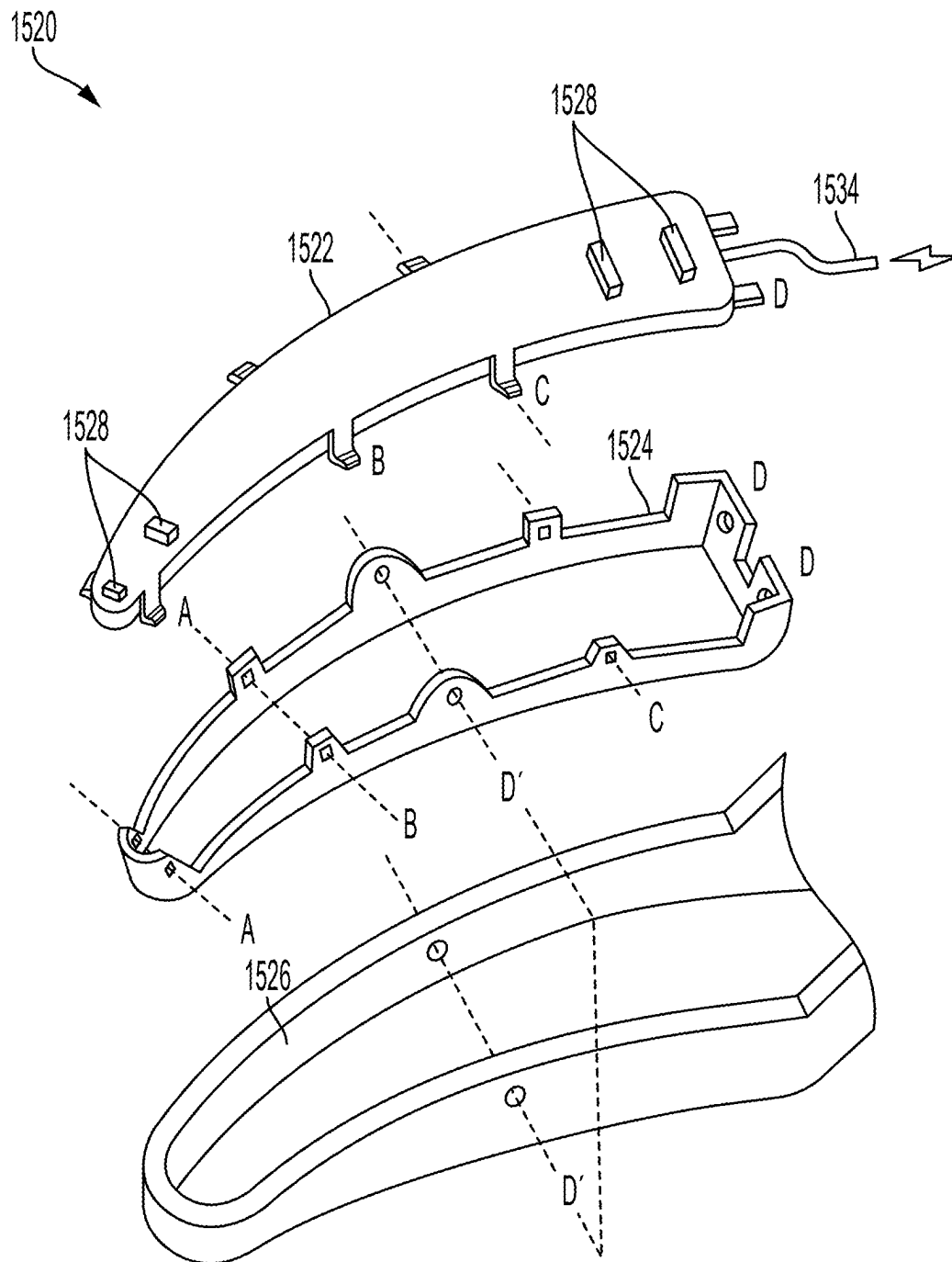


FIG. 32

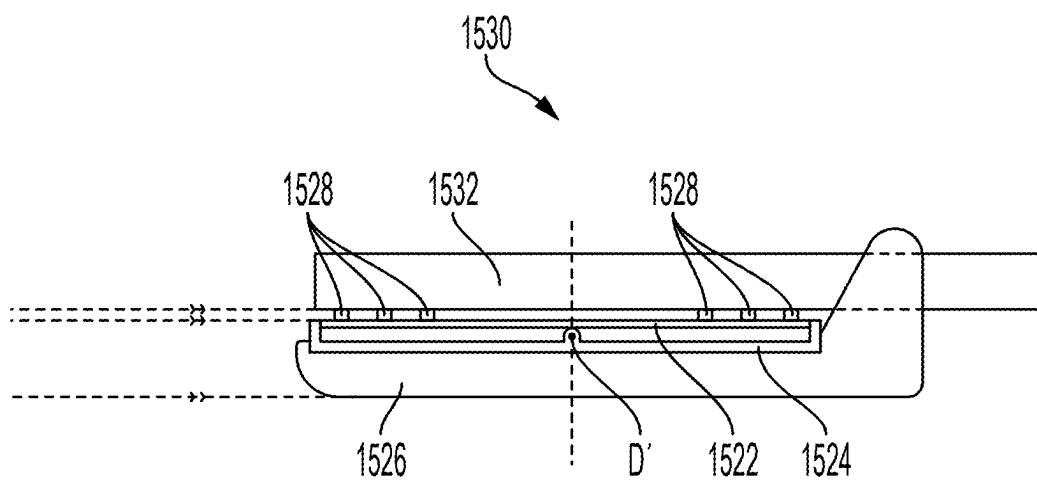


FIG. 33A

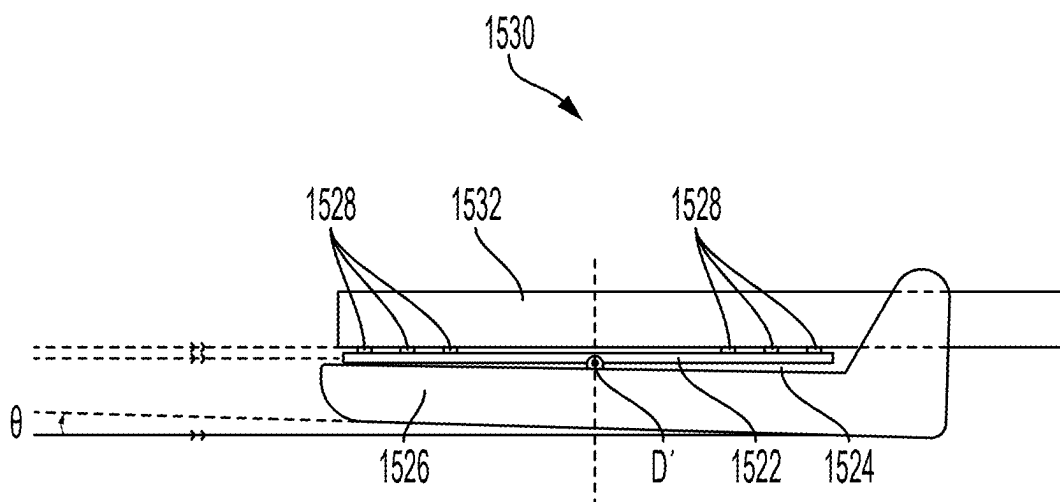


FIG. 33B

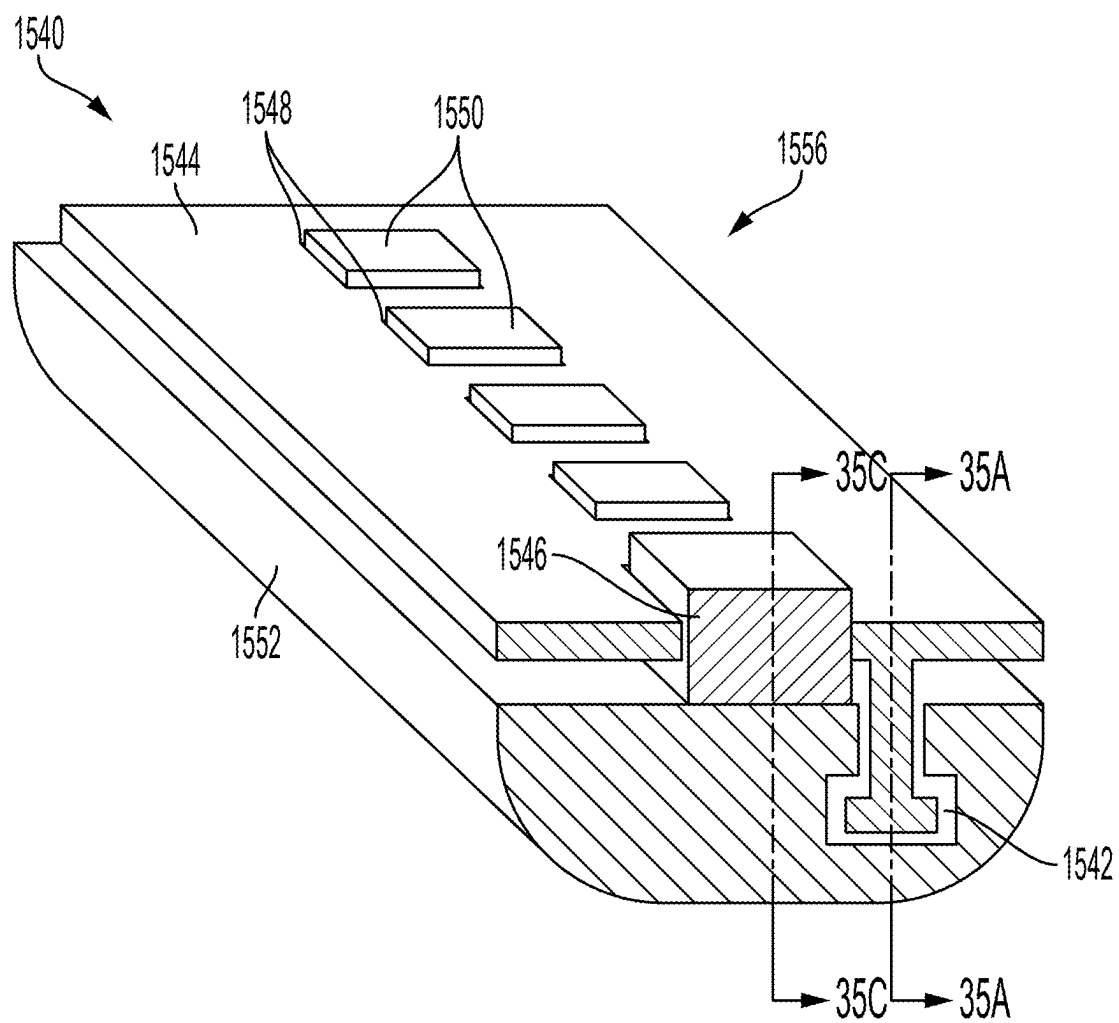


FIG. 34

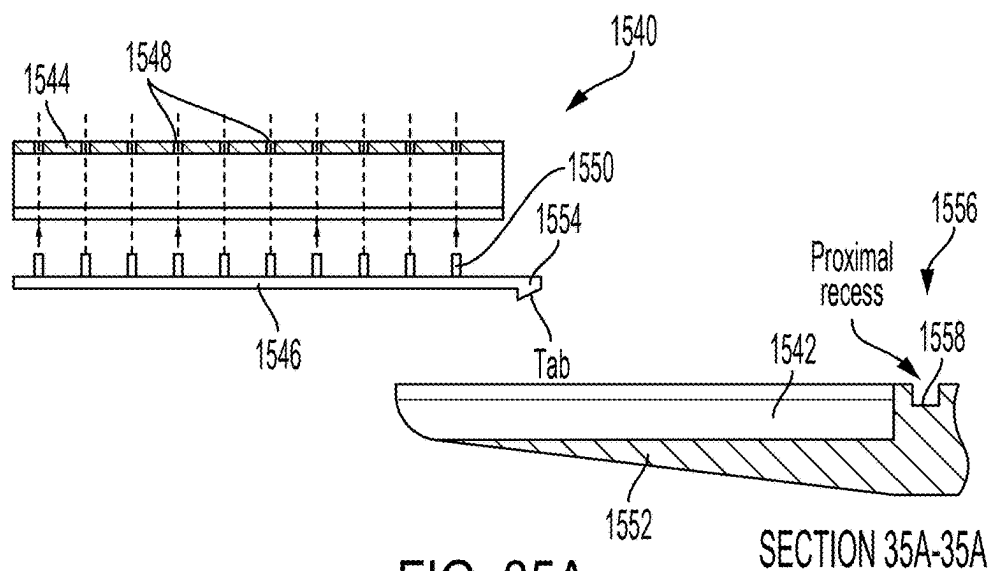


FIG. 35A

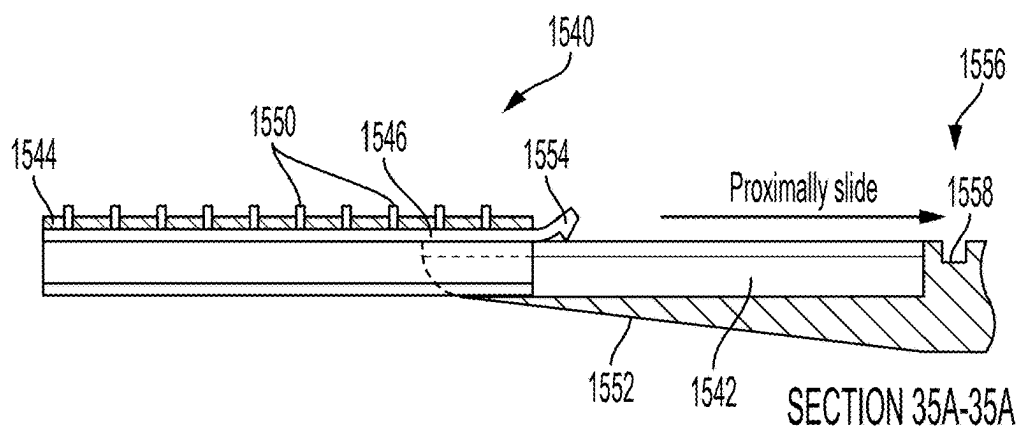


FIG. 35B

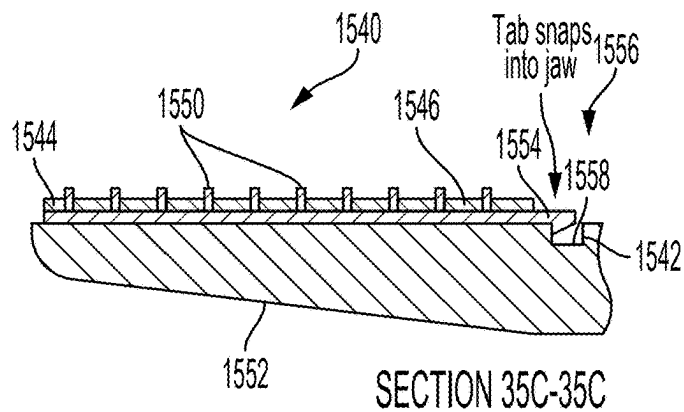


FIG. 35C

1570

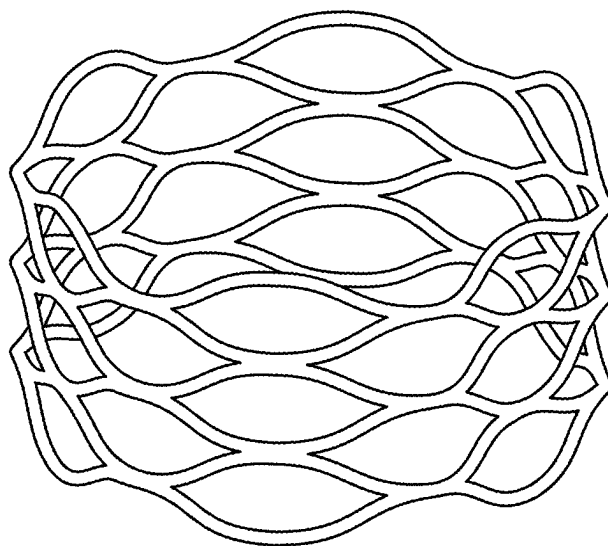


FIG. 36

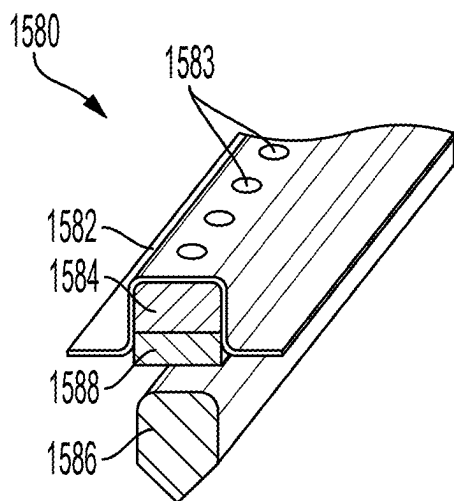


FIG. 37

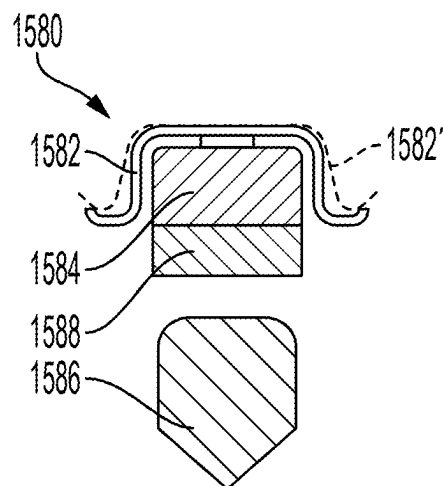


FIG. 38

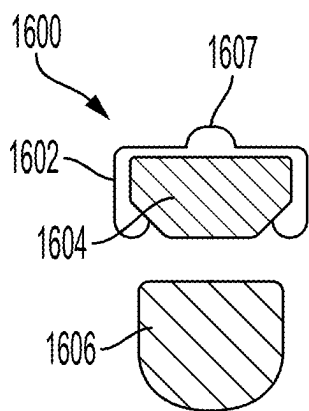


FIG. 39

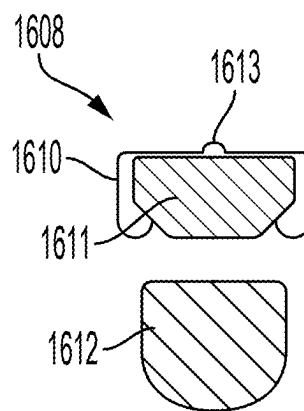


FIG. 40

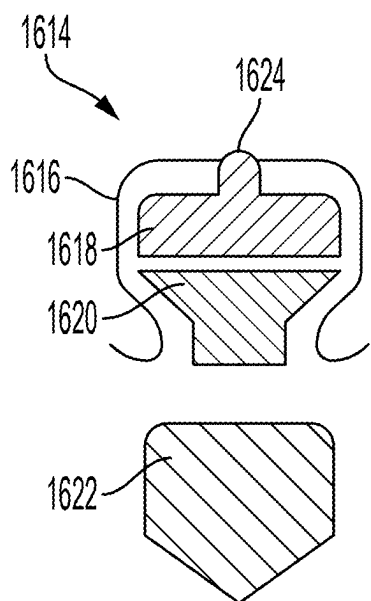


FIG. 41

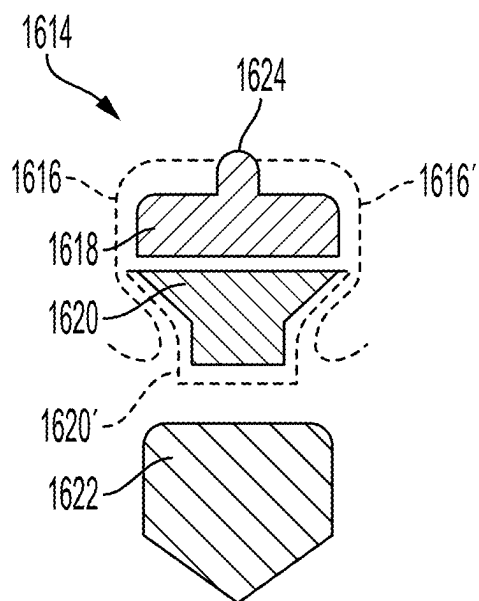


FIG. 42

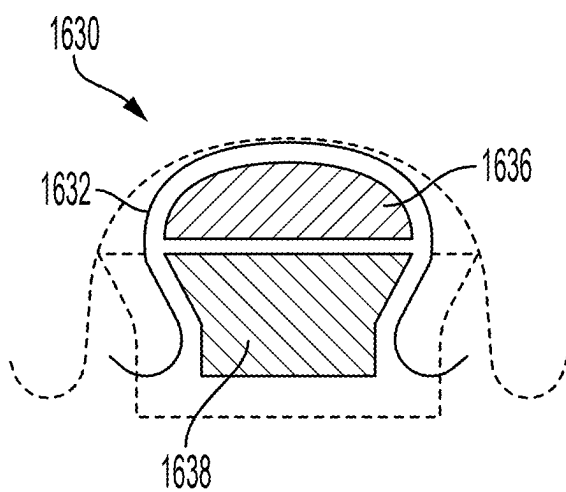


FIG. 43

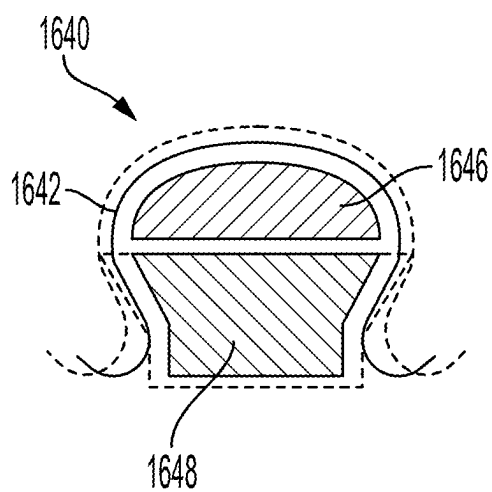


FIG. 44

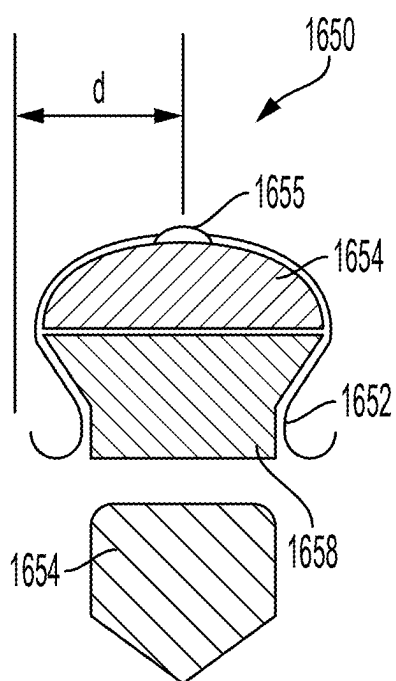


FIG. 45

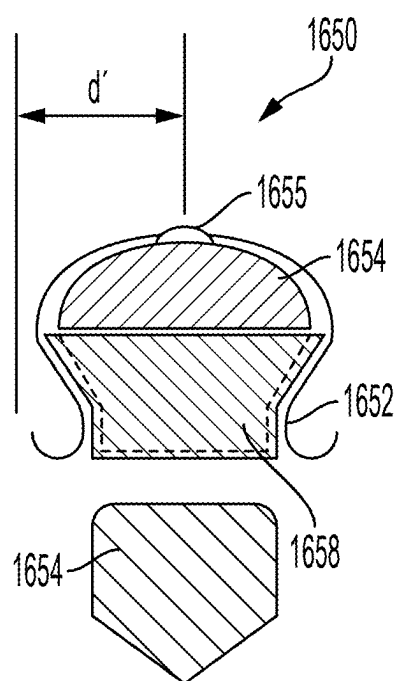


FIG. 46

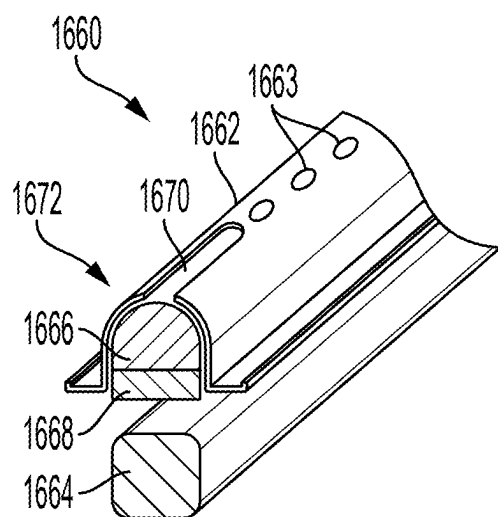


FIG. 47

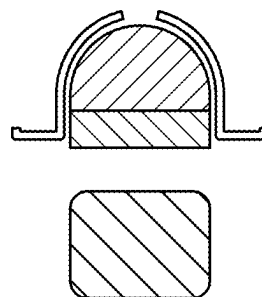


FIG. 48

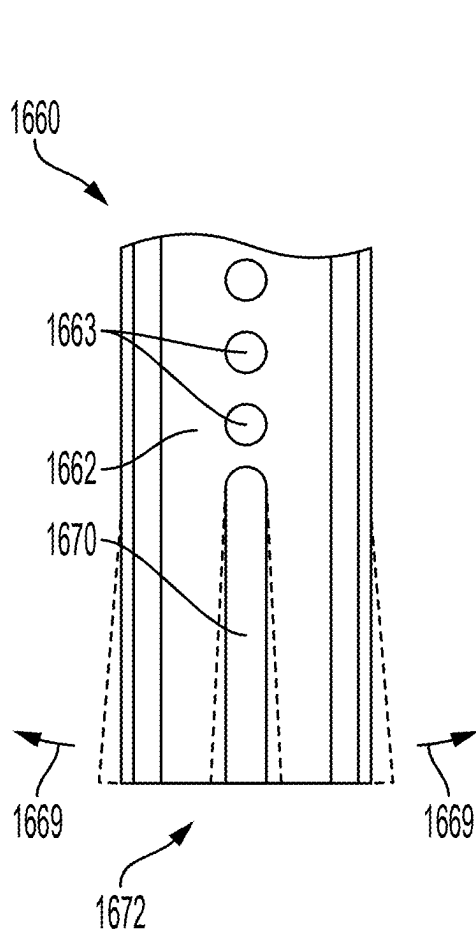


FIG. 49

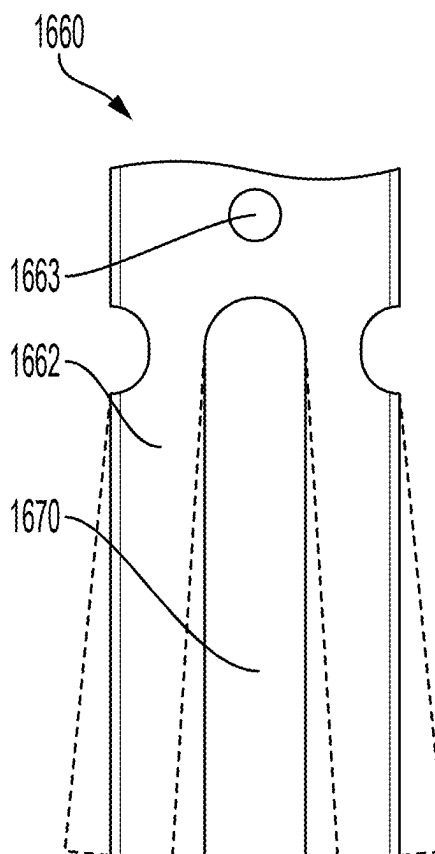


FIG. 50

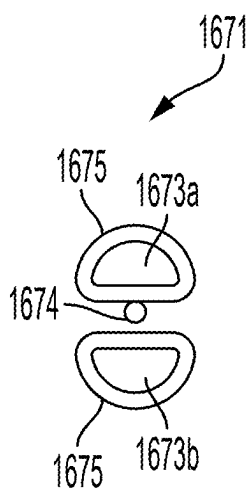


FIG. 51

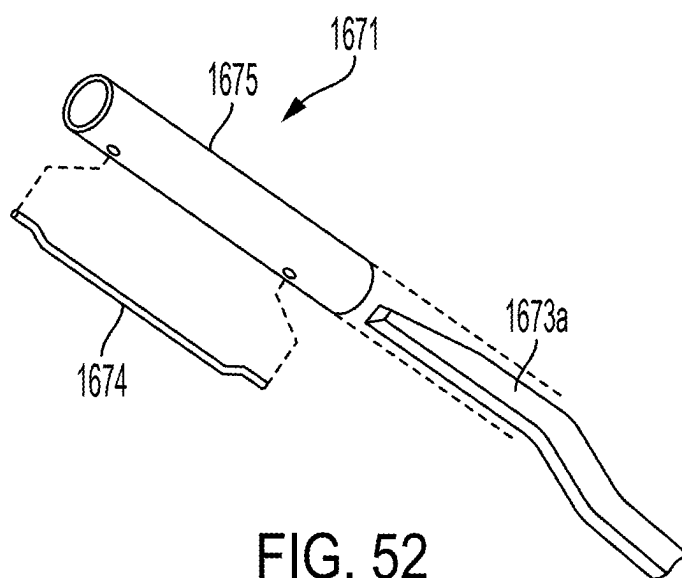


FIG. 52

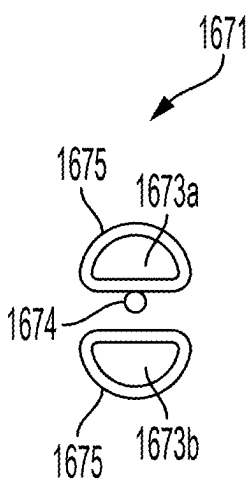


FIG. 53

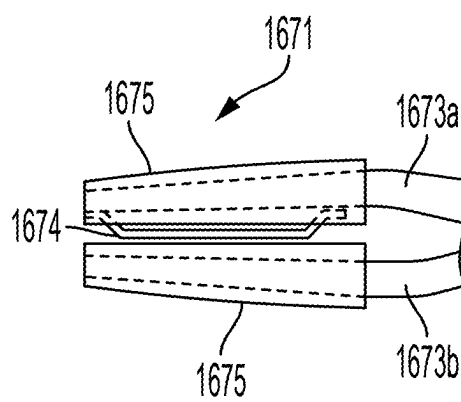


FIG. 54

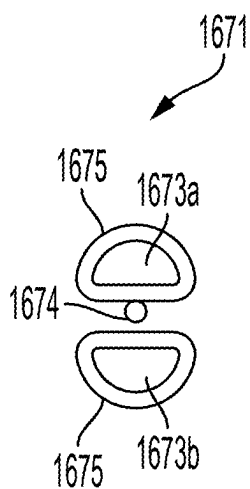


FIG. 55

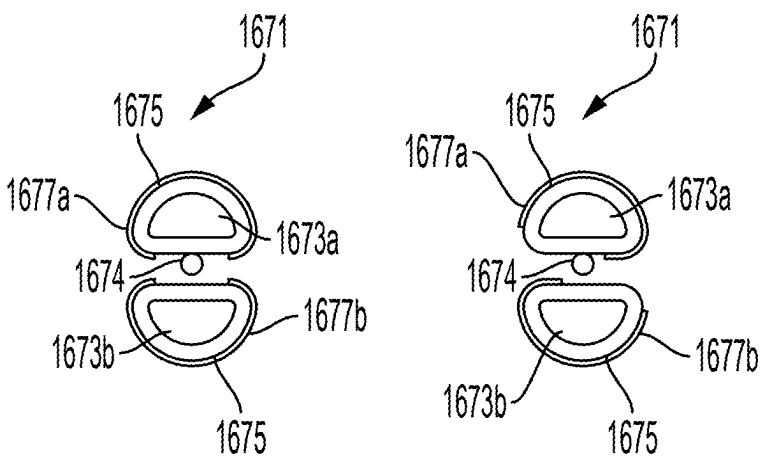


FIG. 56

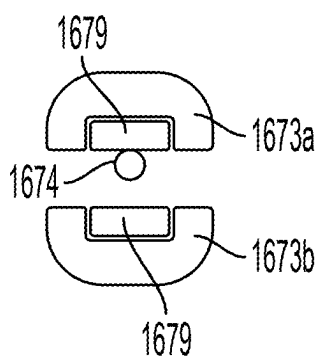


FIG. 57

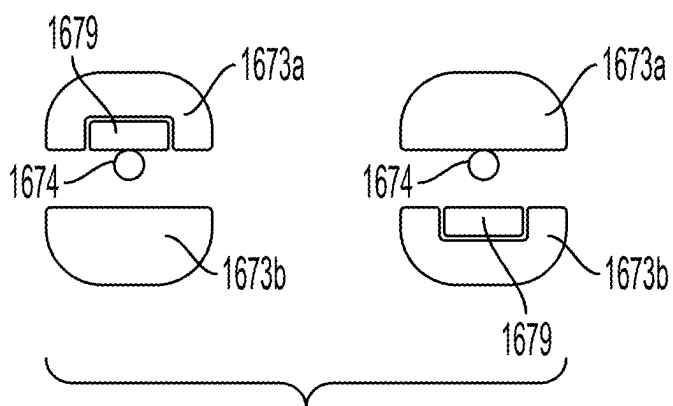


FIG. 58

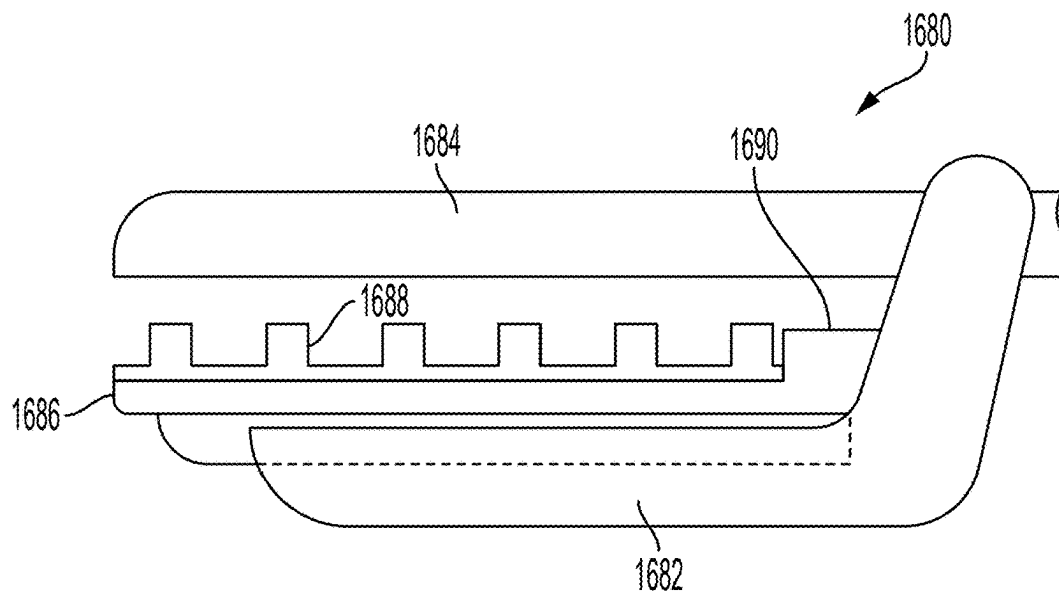


FIG. 59

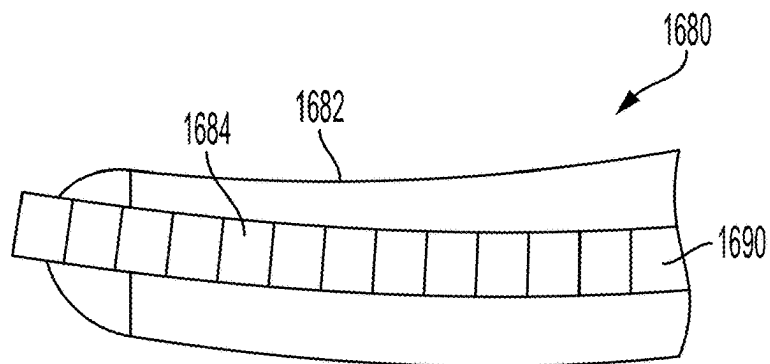


FIG. 60

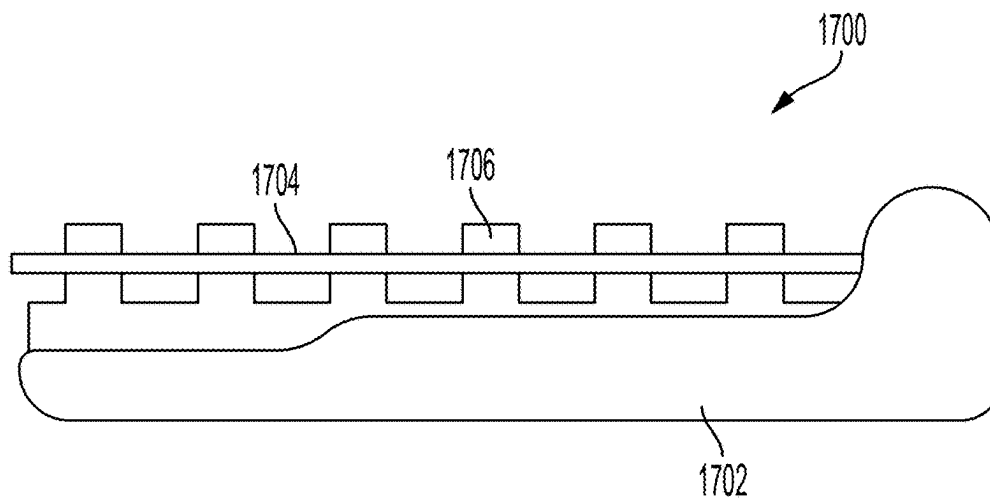


FIG. 61

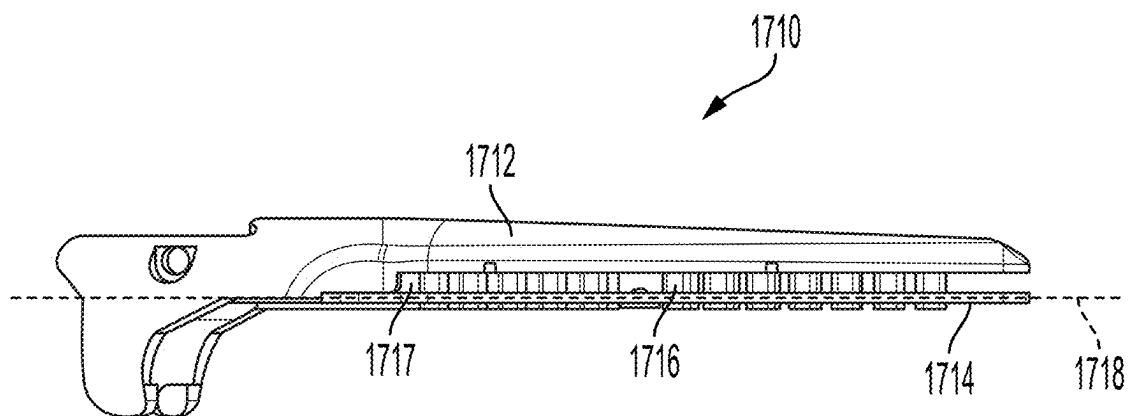


FIG. 62

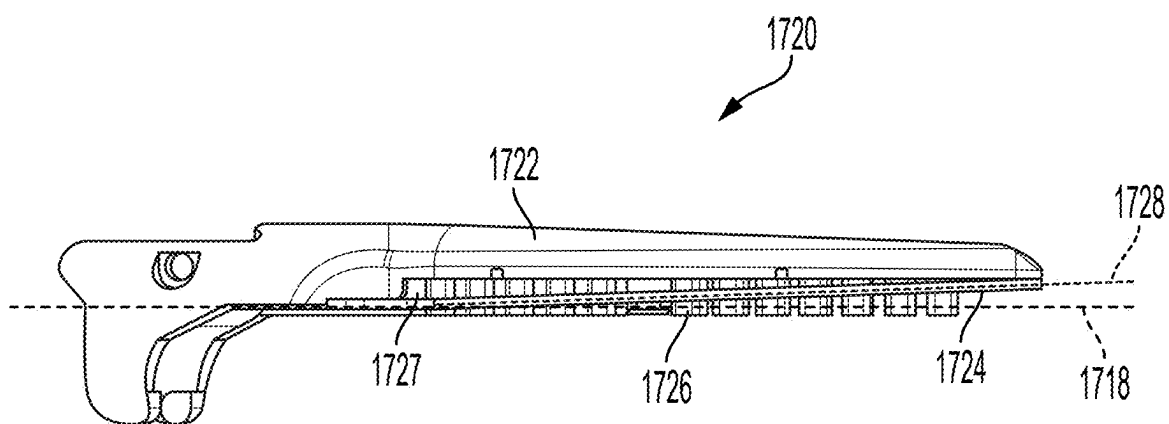


FIG. 63

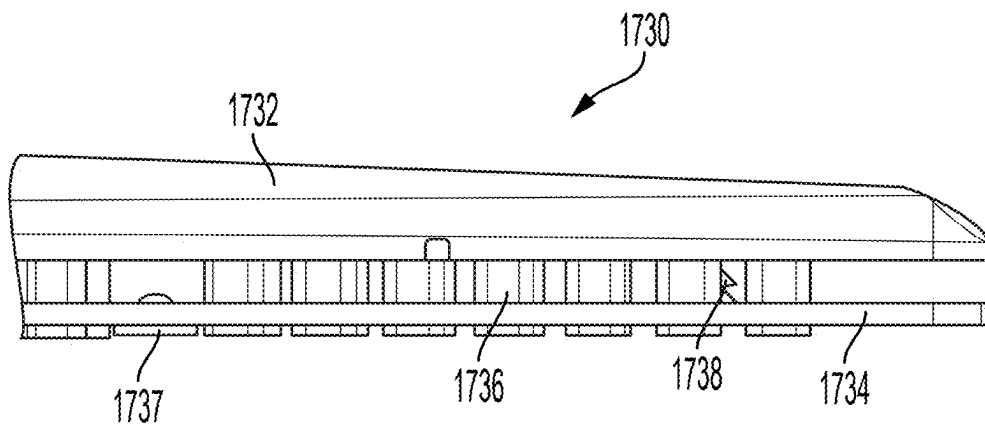


FIG. 64

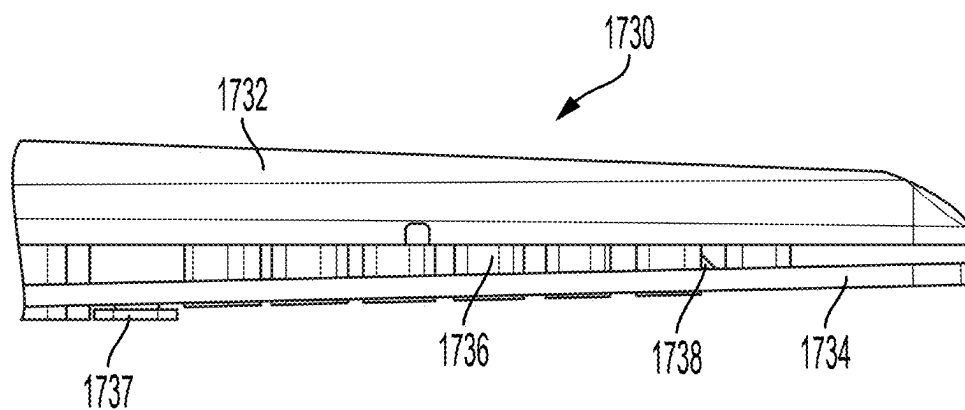


FIG. 65

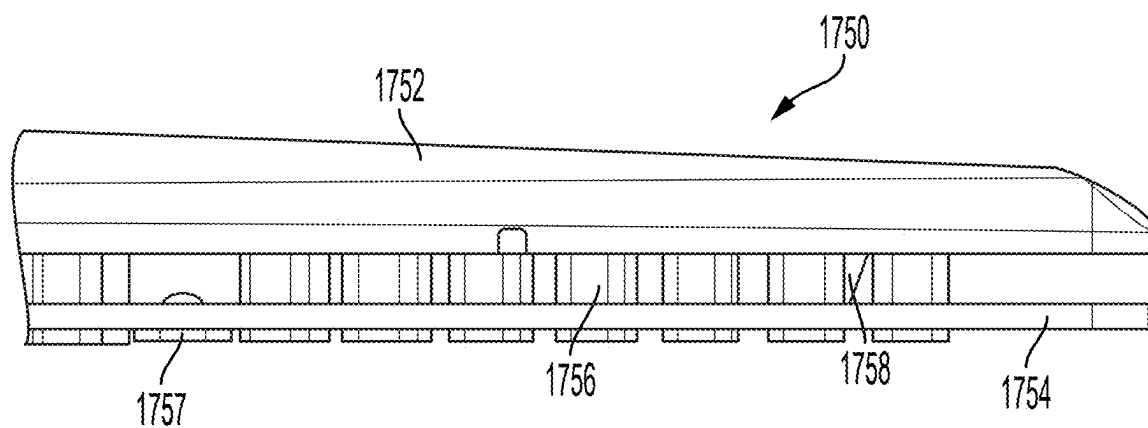


FIG. 66

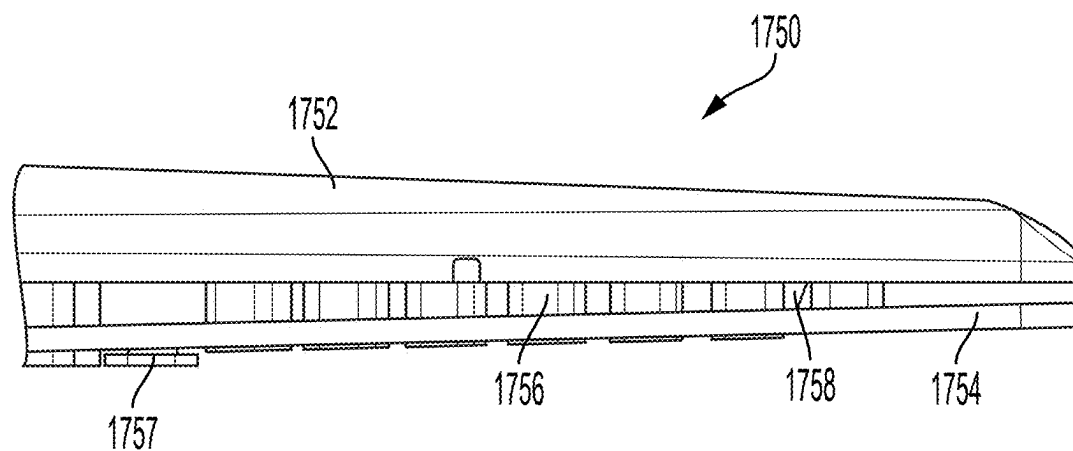


FIG. 67

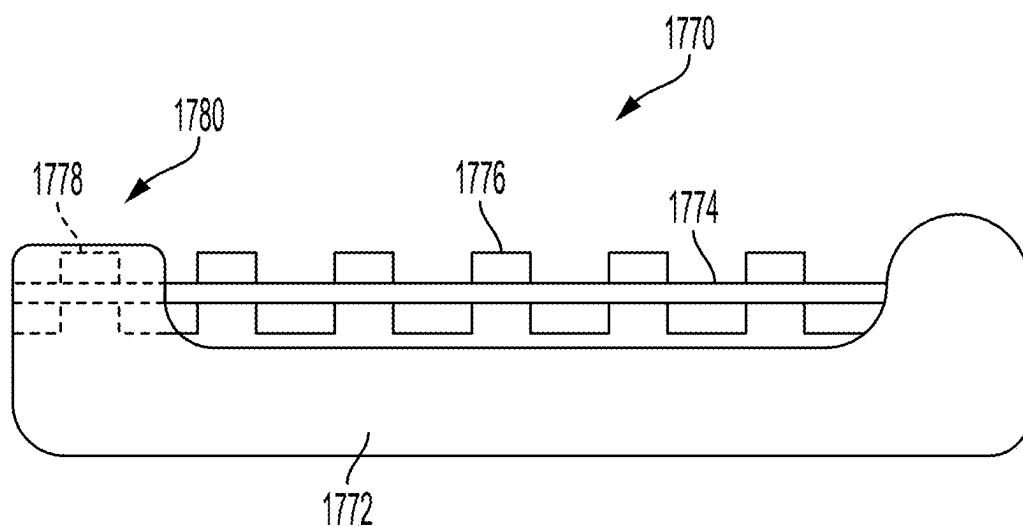


FIG. 68

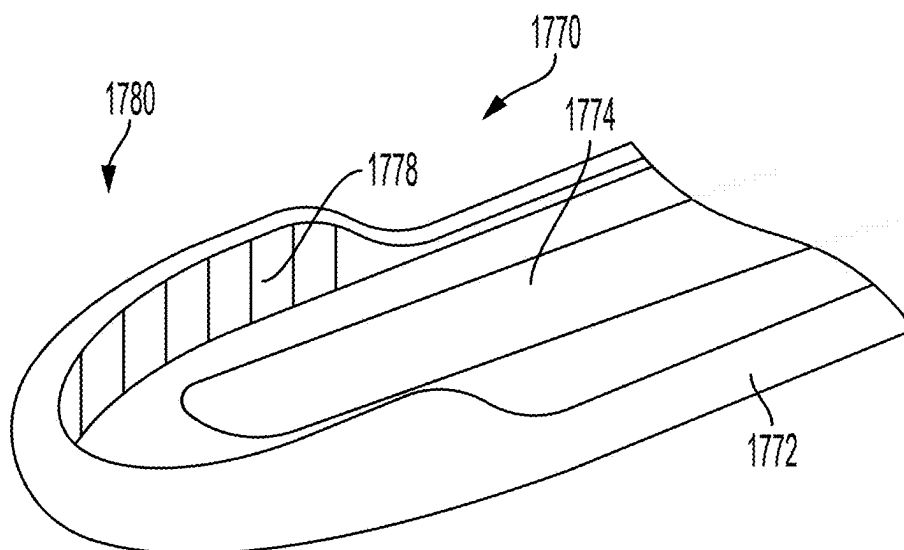


FIG. 69

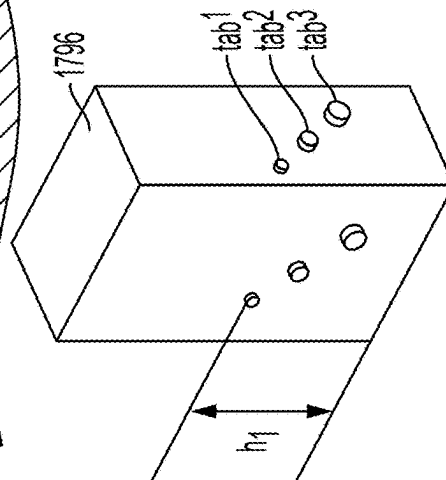
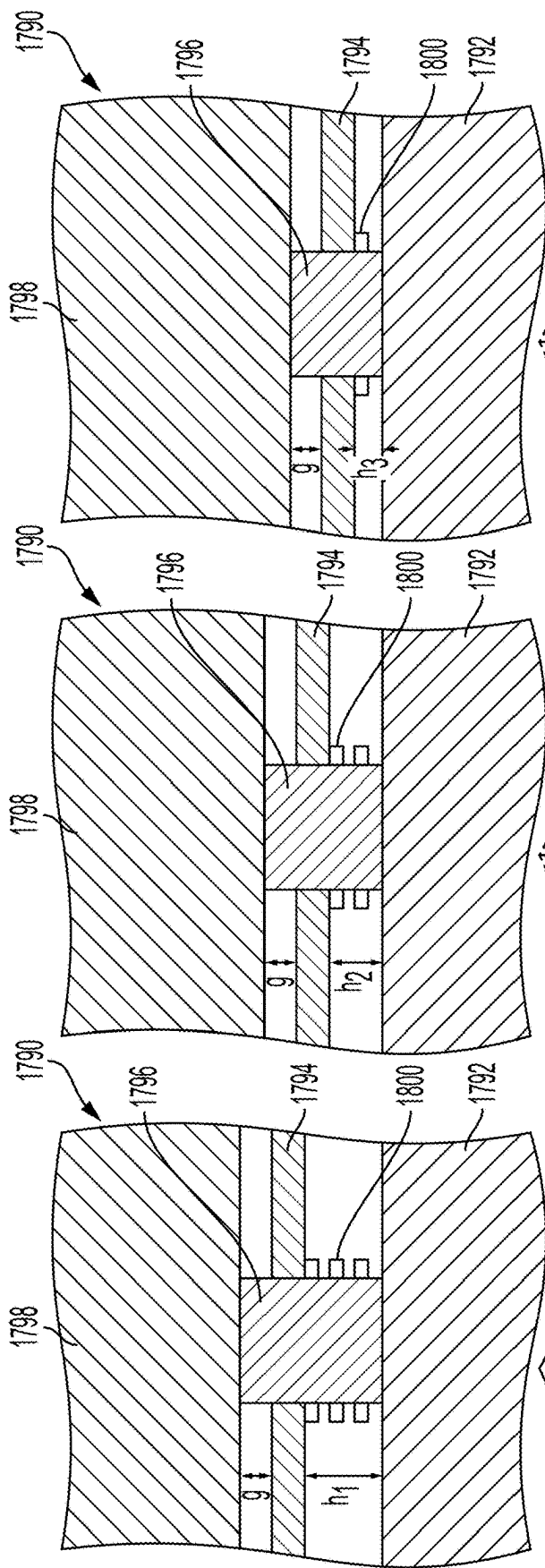


FIG. 70A

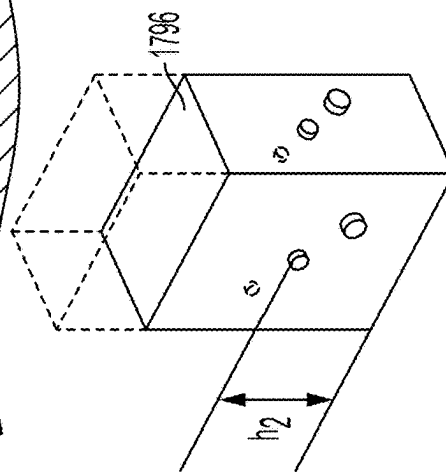


FIG. 70B

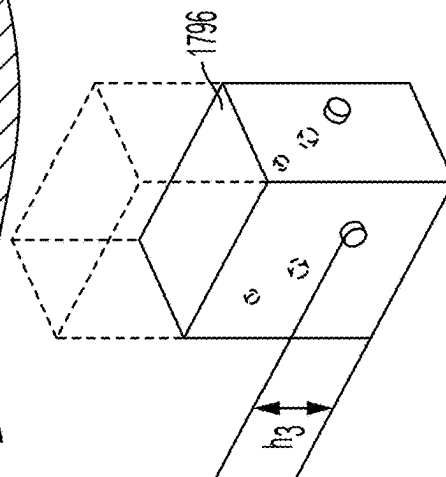


FIG. 70C

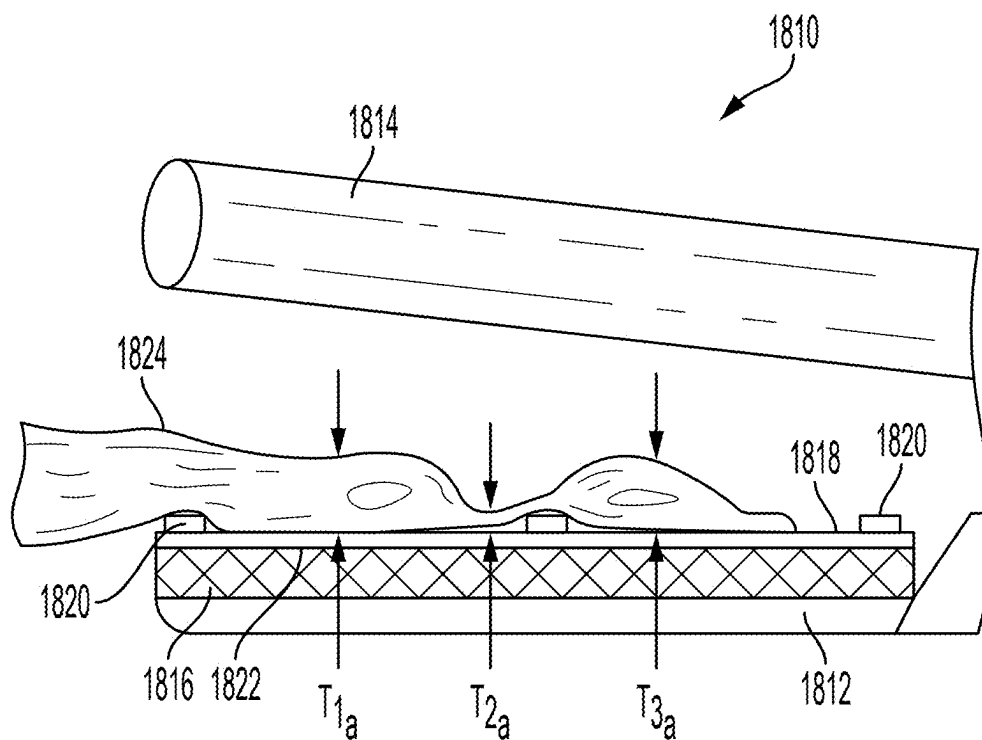


FIG. 71

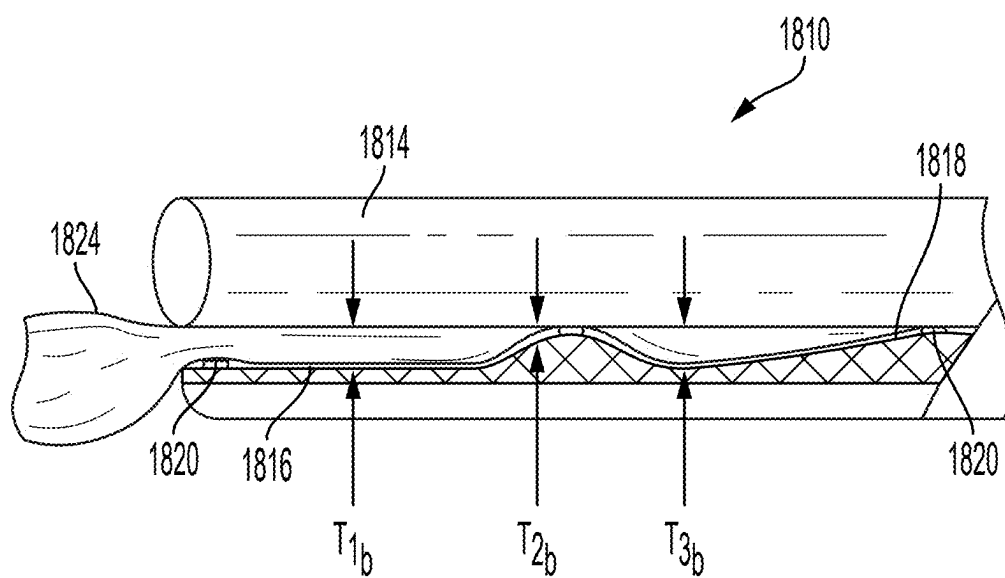


FIG. 72

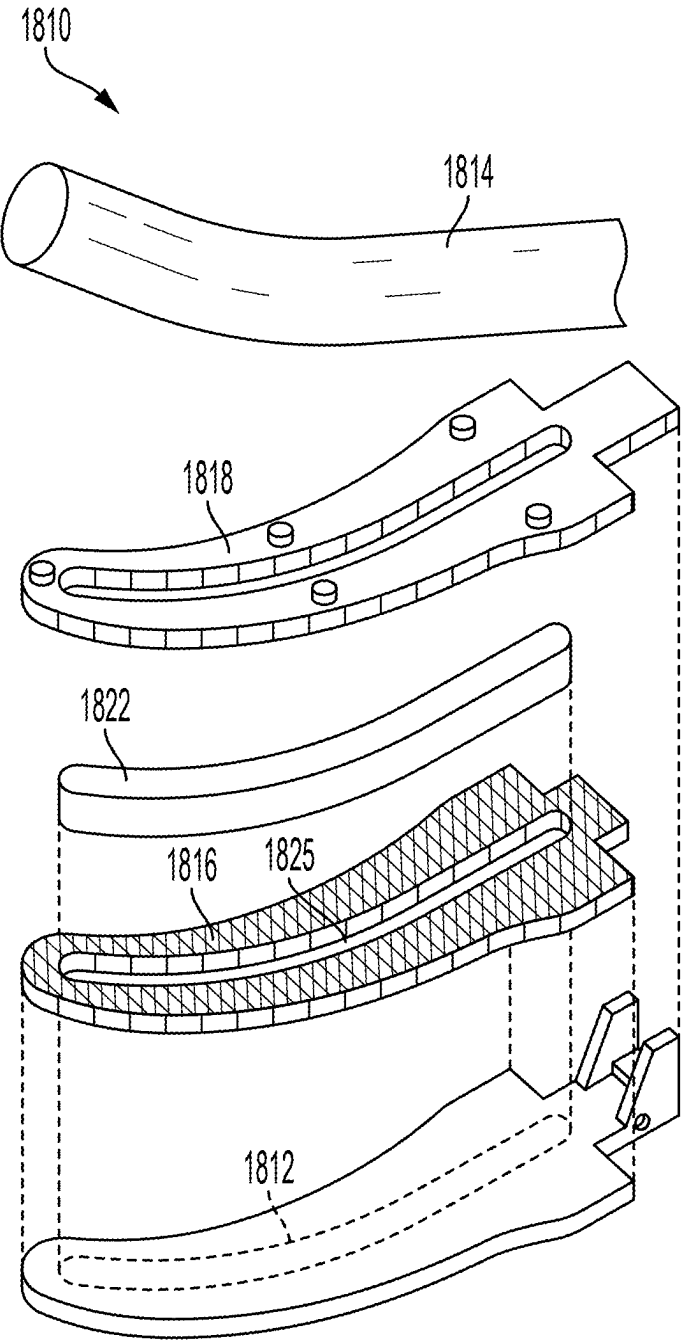


FIG. 73

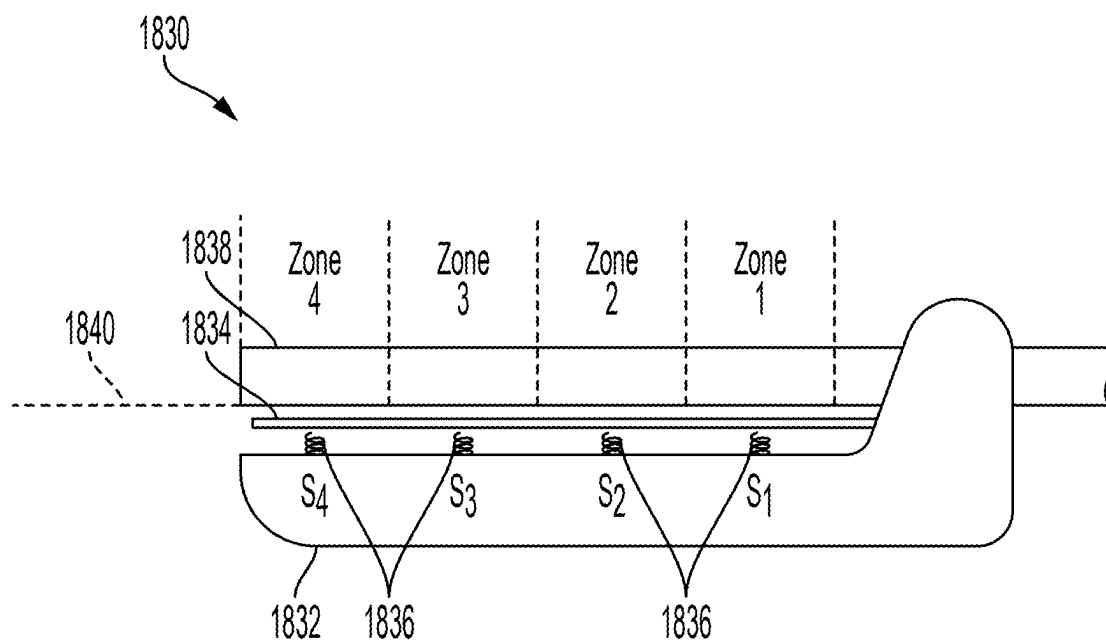


FIG. 74

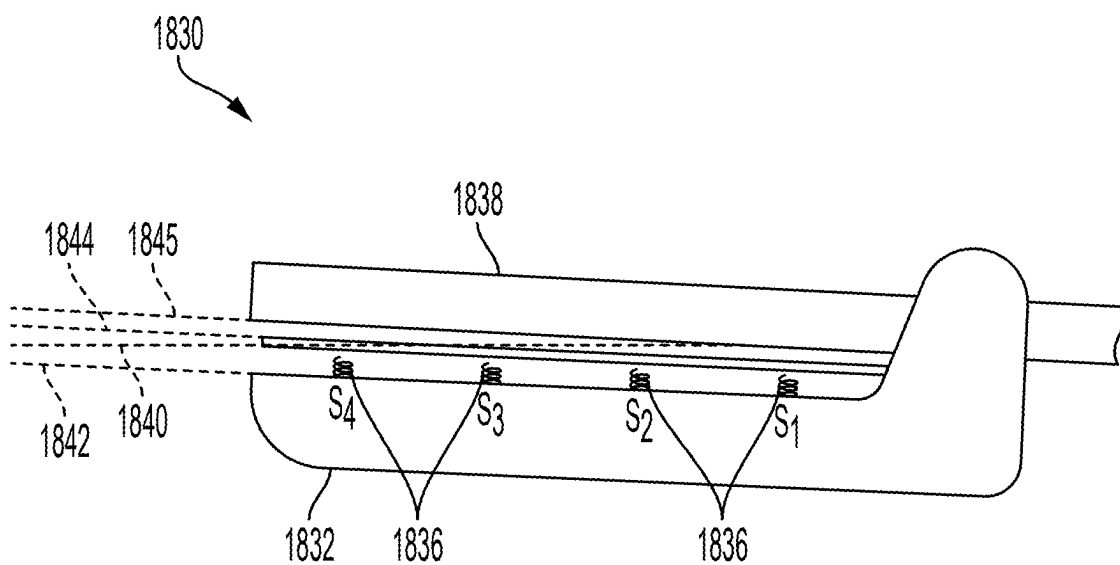


FIG. 75



FIG. 76

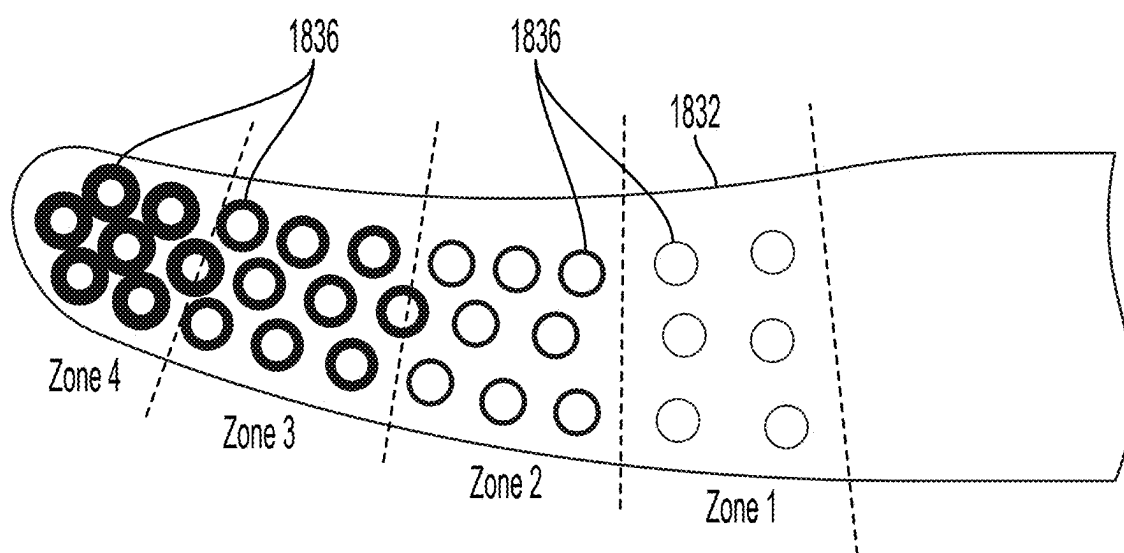


FIG. 77

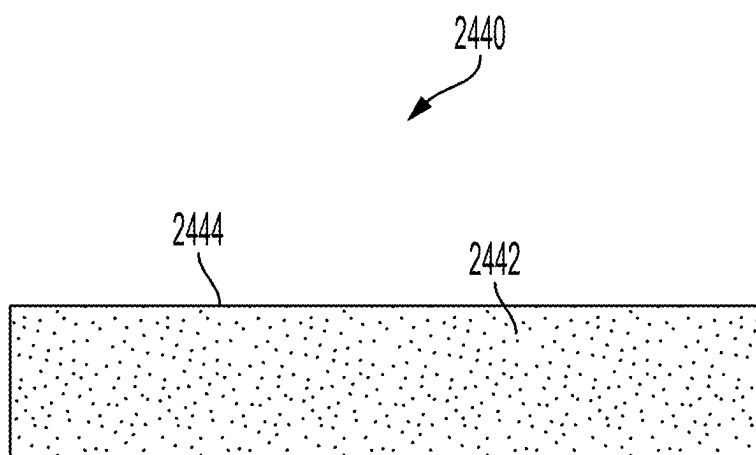


FIG. 78

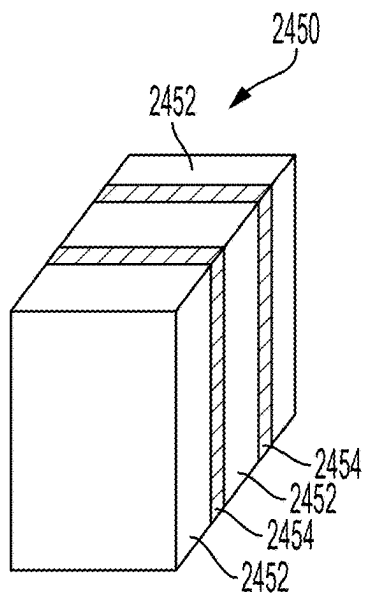


FIG. 79

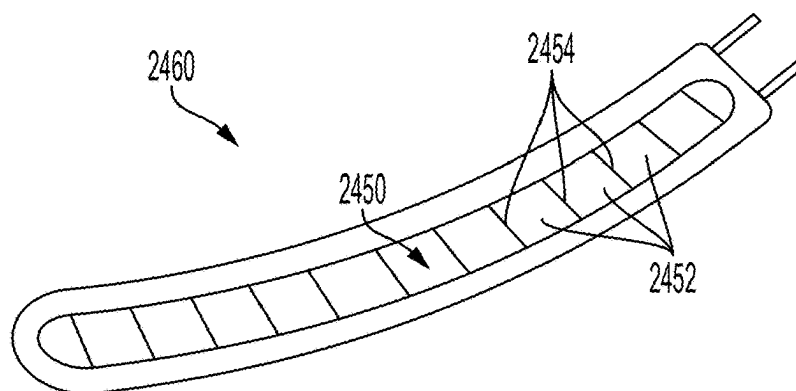


FIG. 80

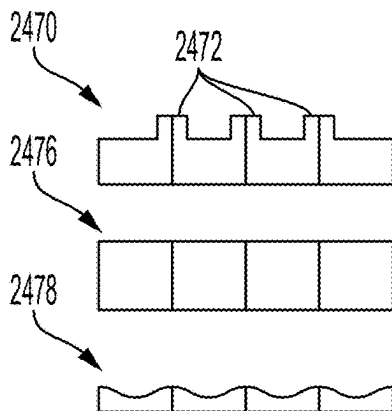


FIG. 81

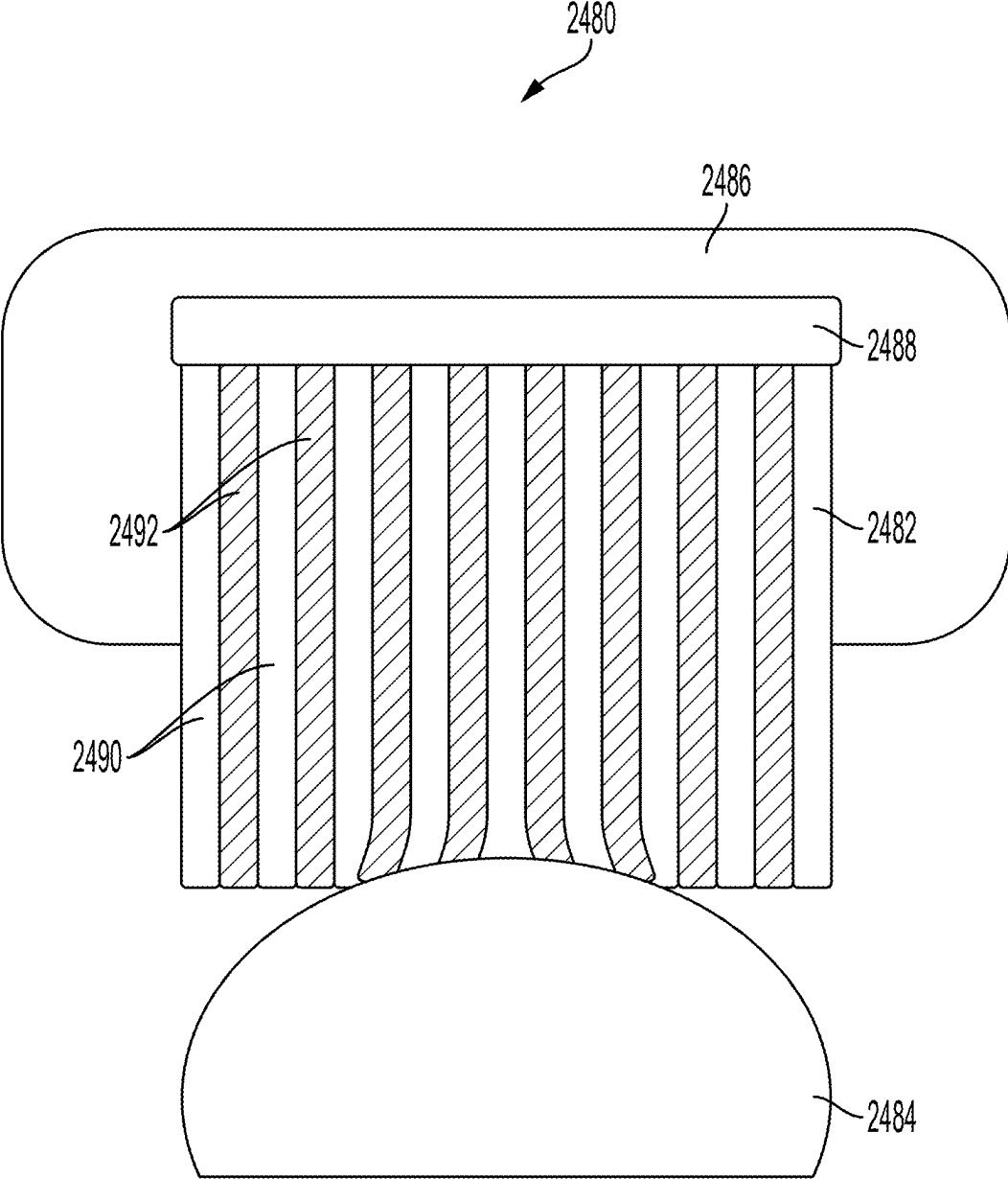


FIG. 82

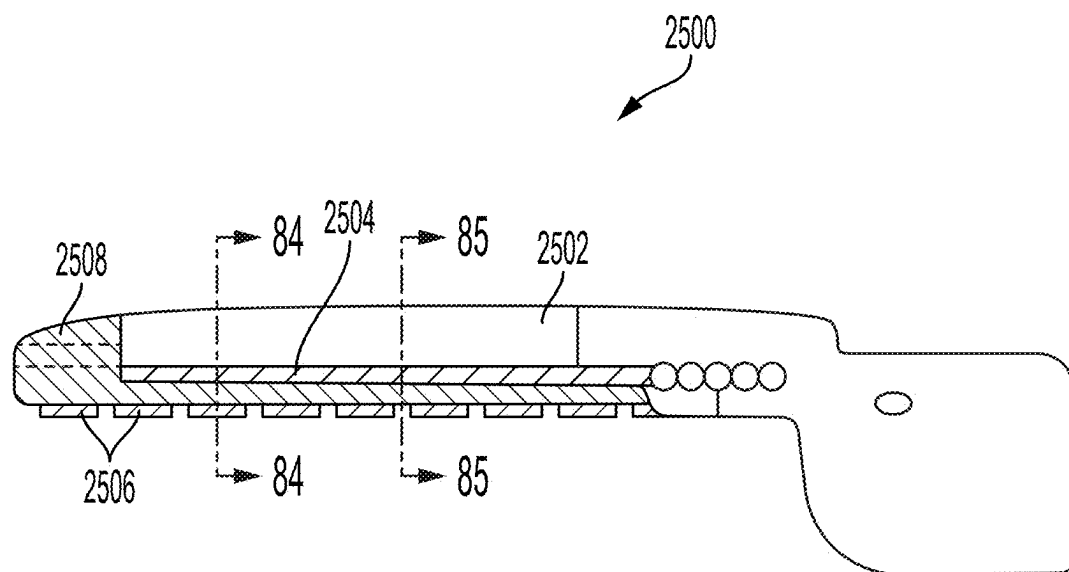


FIG. 83

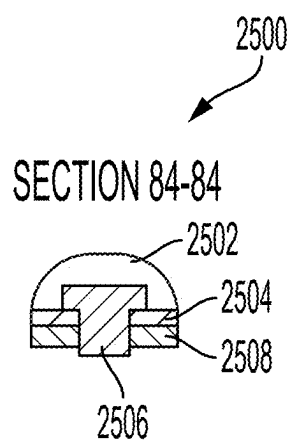


FIG. 84

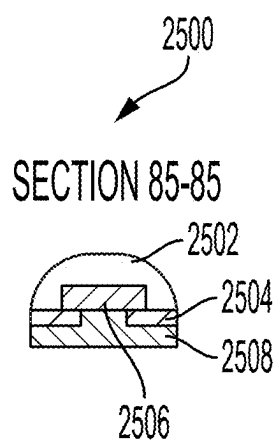


FIG. 85

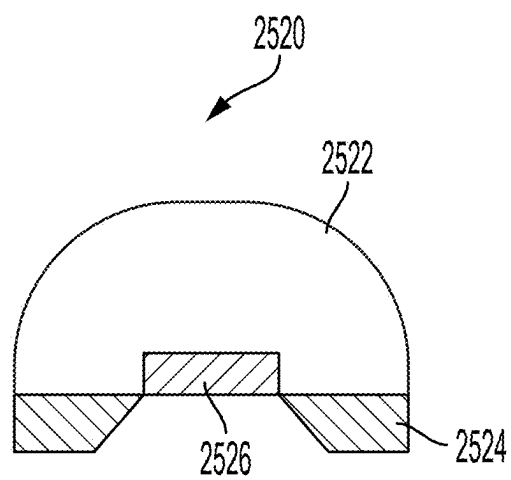


FIG. 86

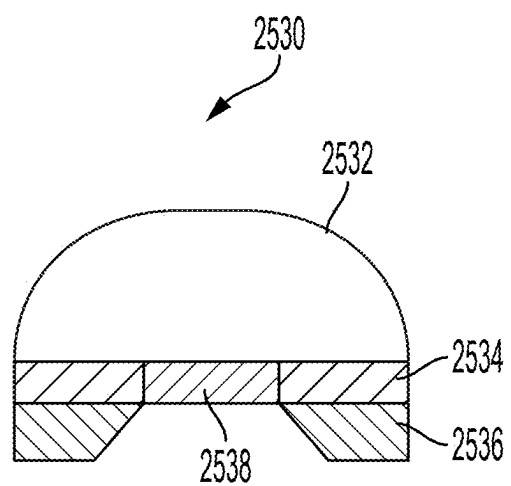


FIG. 87

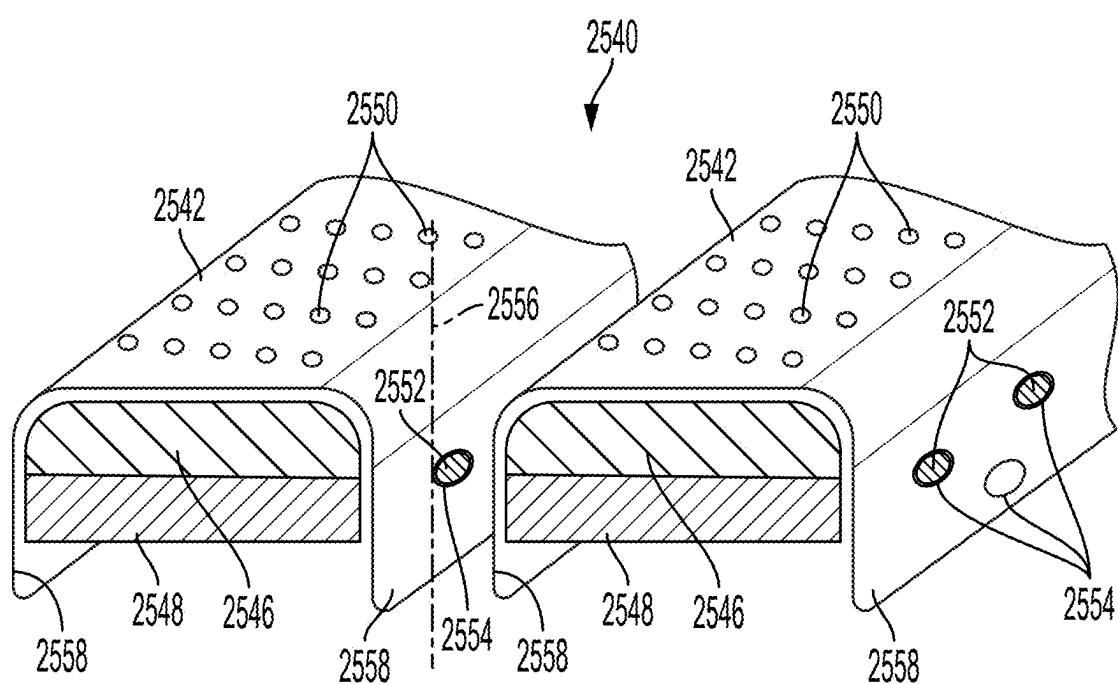


FIG. 88

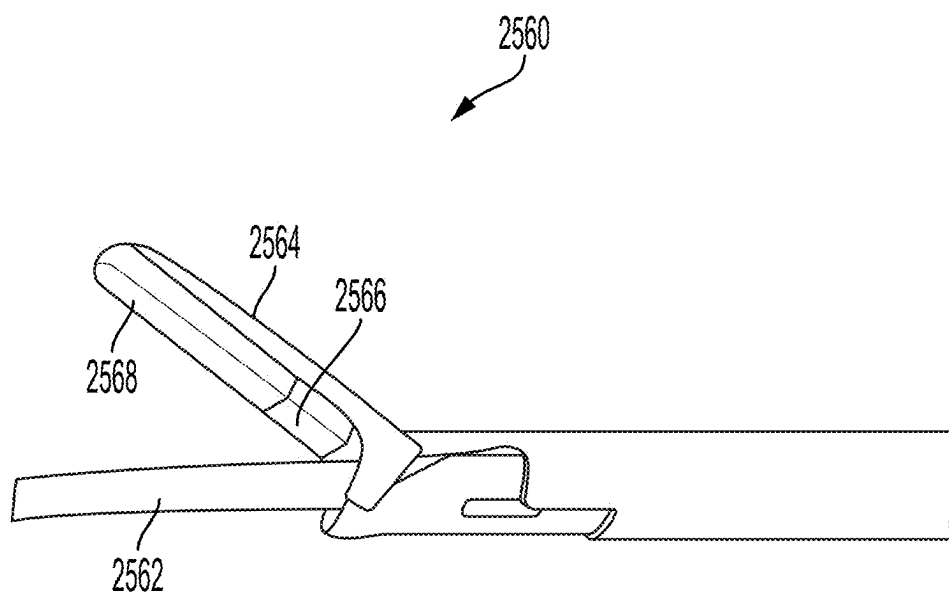


FIG. 89

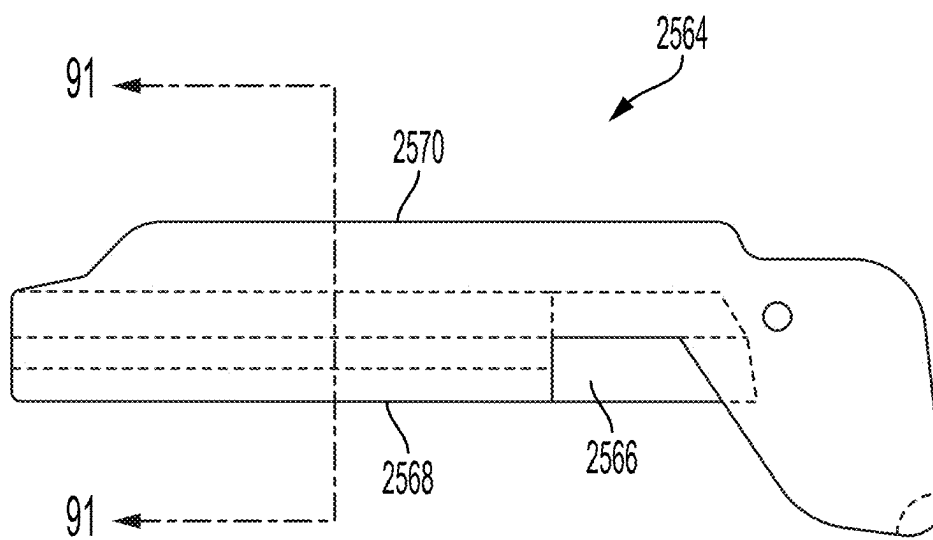


FIG. 90

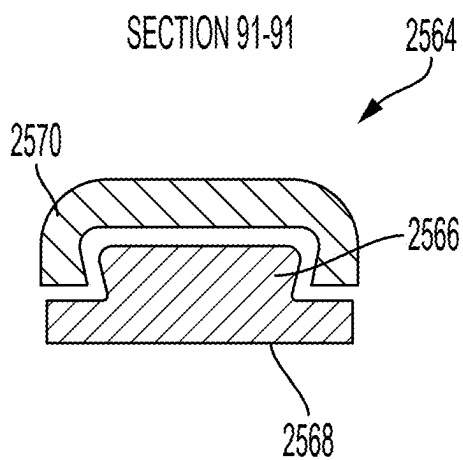


FIG. 91

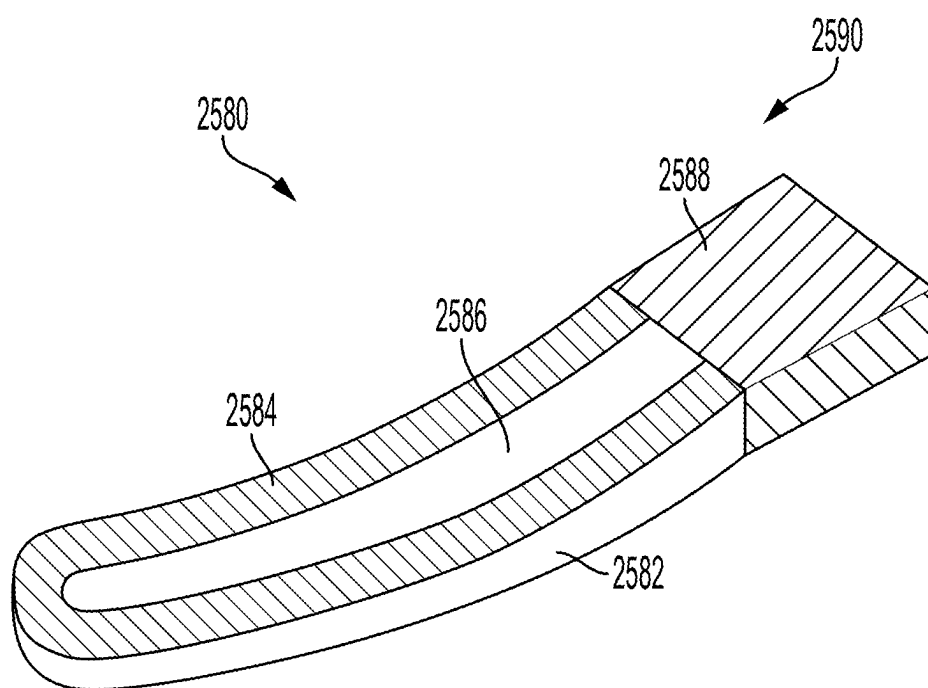


FIG. 92

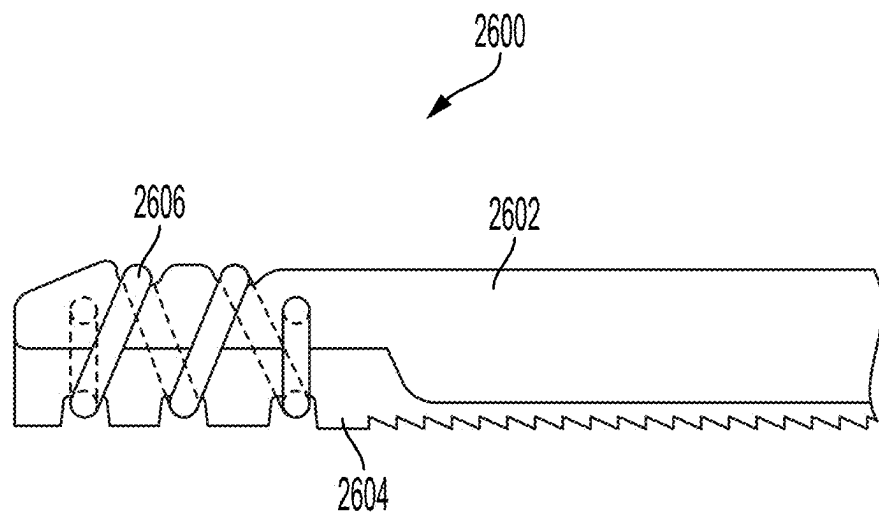


FIG. 93

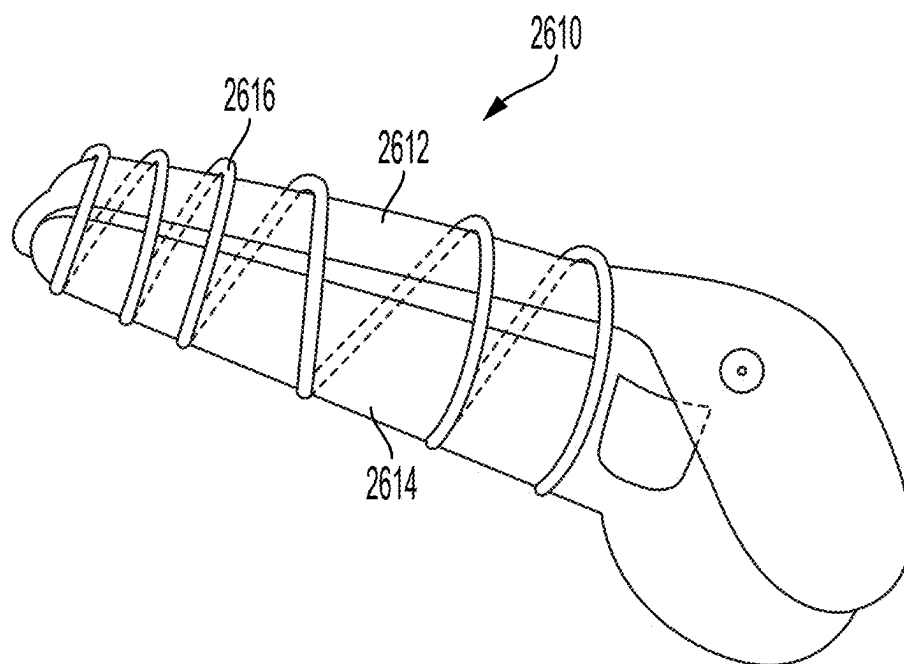


FIG. 94

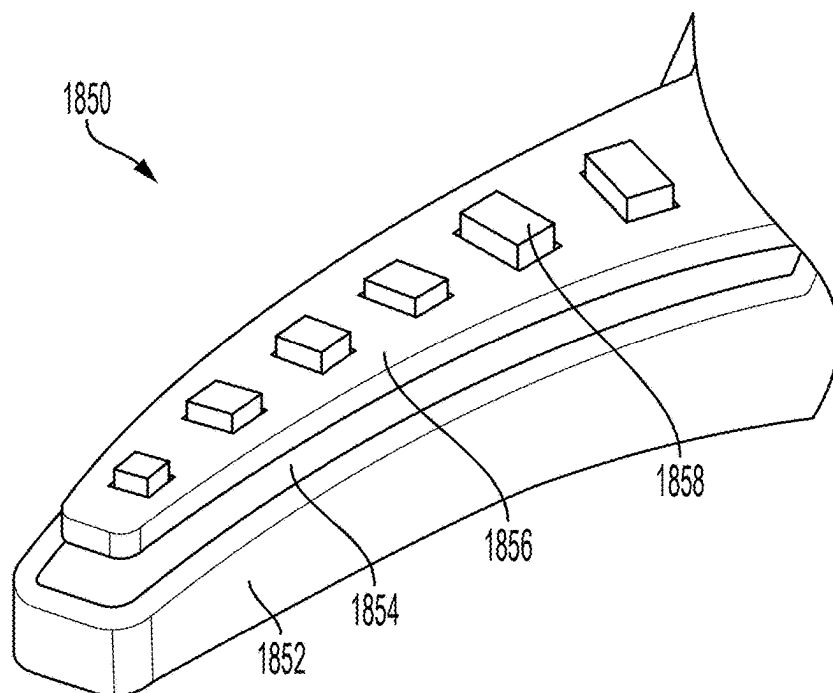


FIG. 95

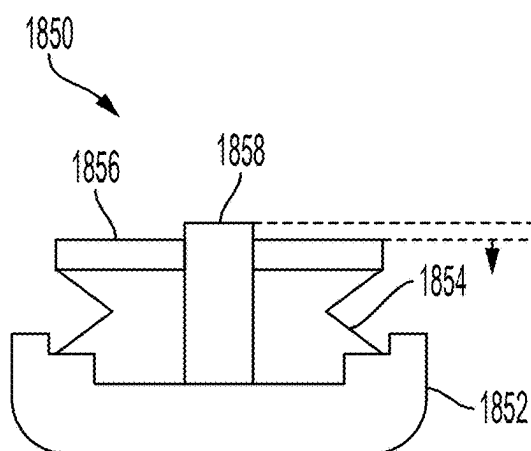


FIG. 96

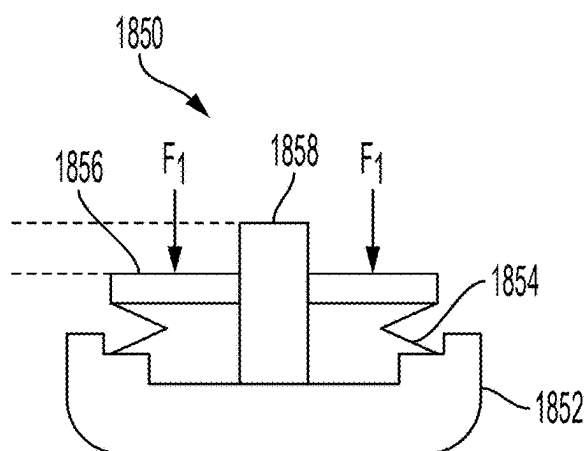


FIG. 97

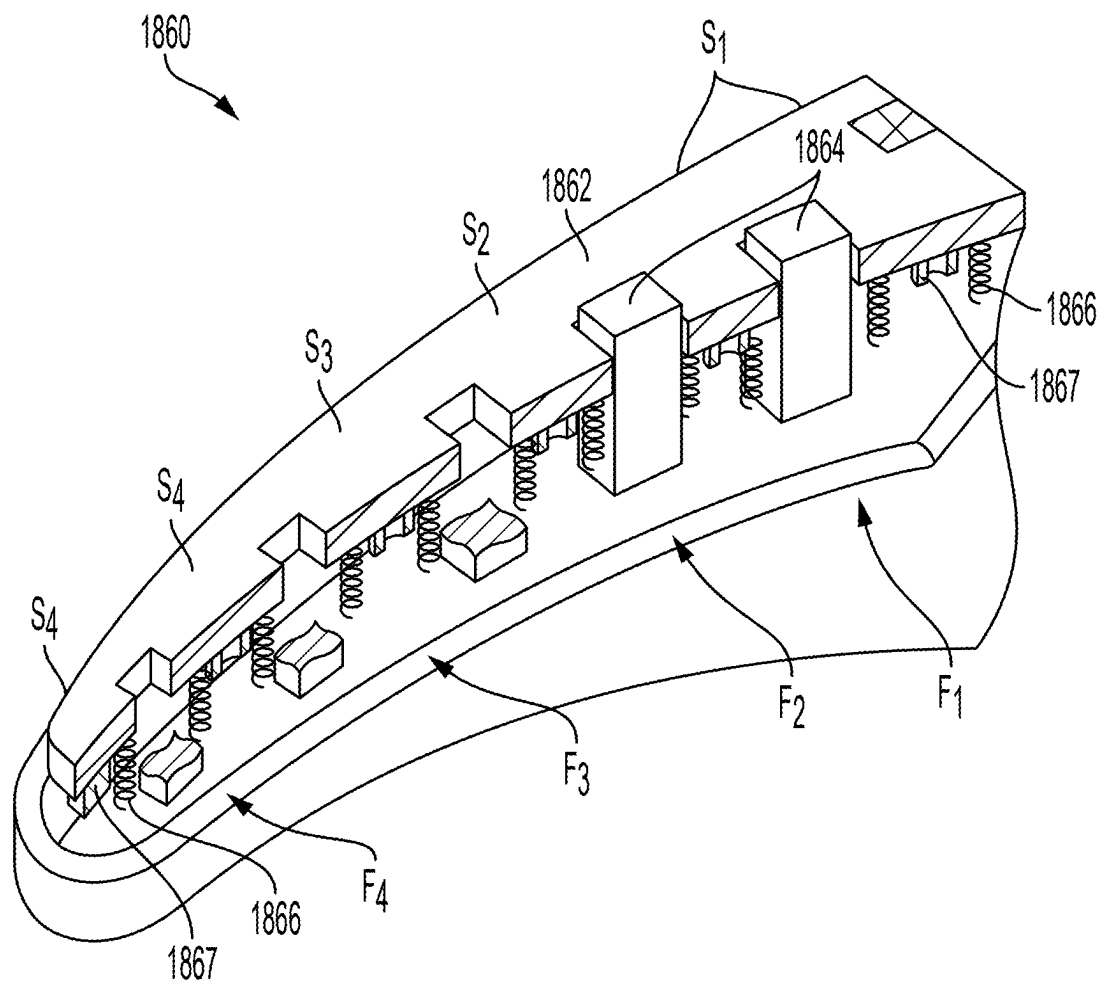


FIG. 98

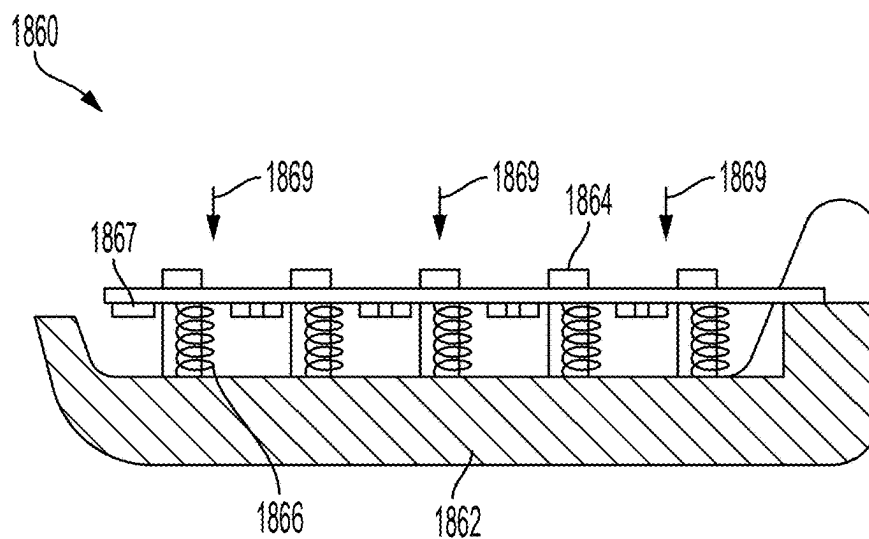


FIG. 99A

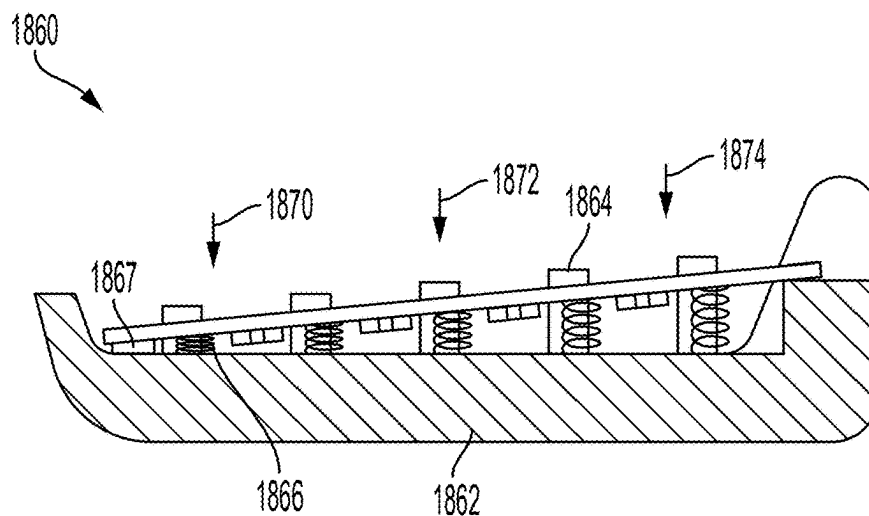


FIG. 99B

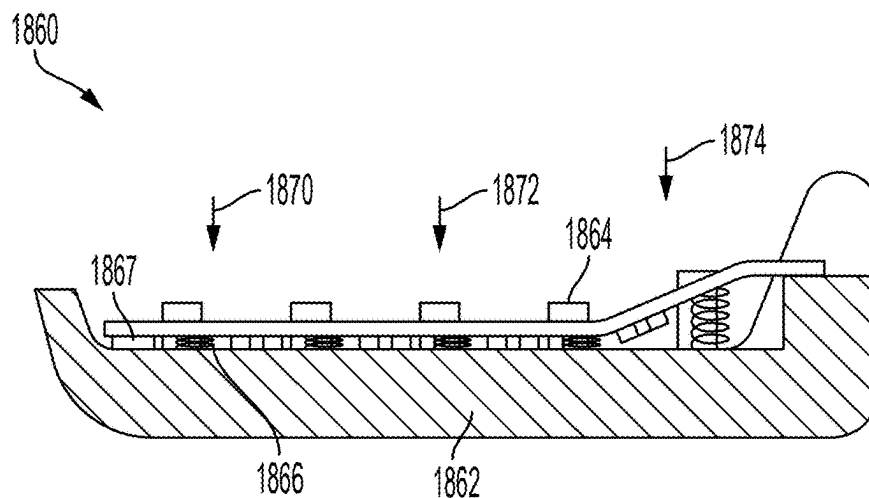


FIG. 99C

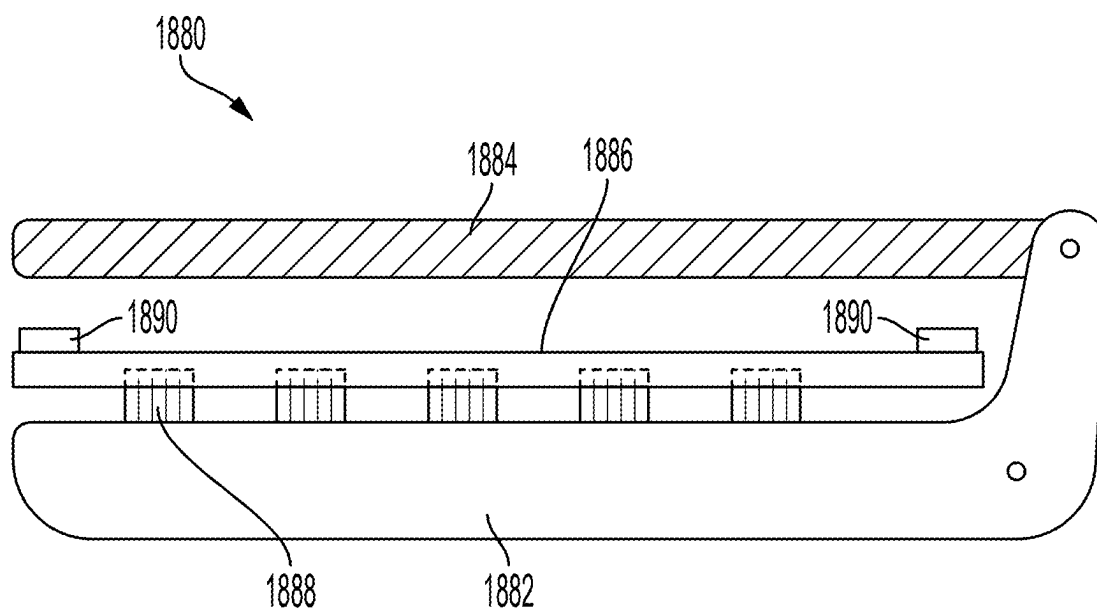


FIG. 100

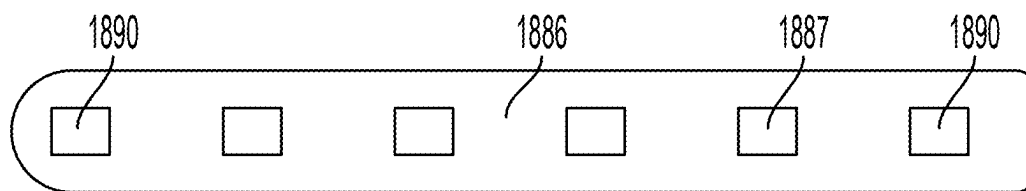


FIG. 101

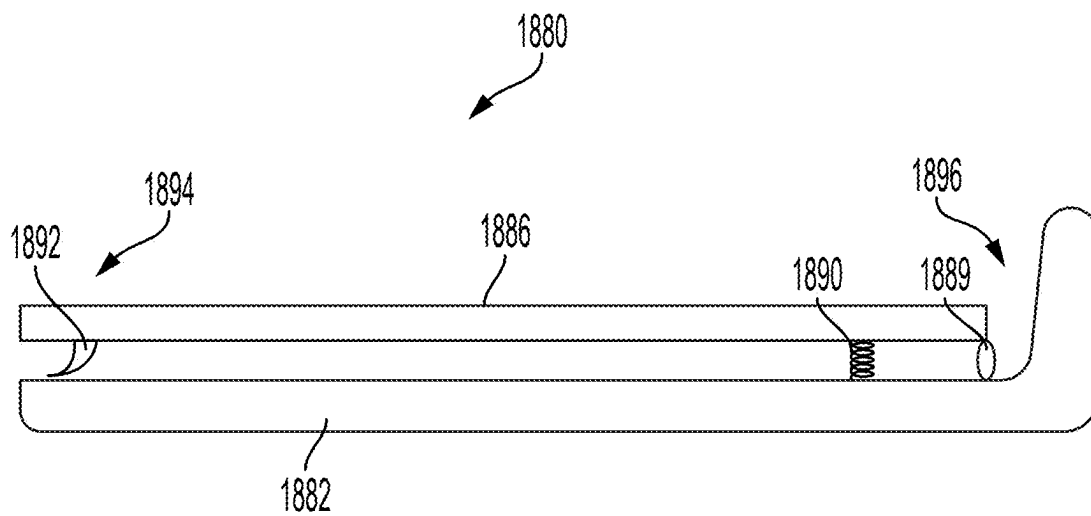


FIG. 102

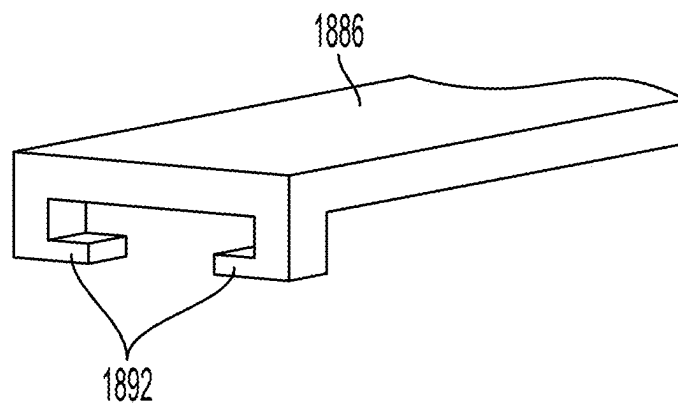


FIG. 103

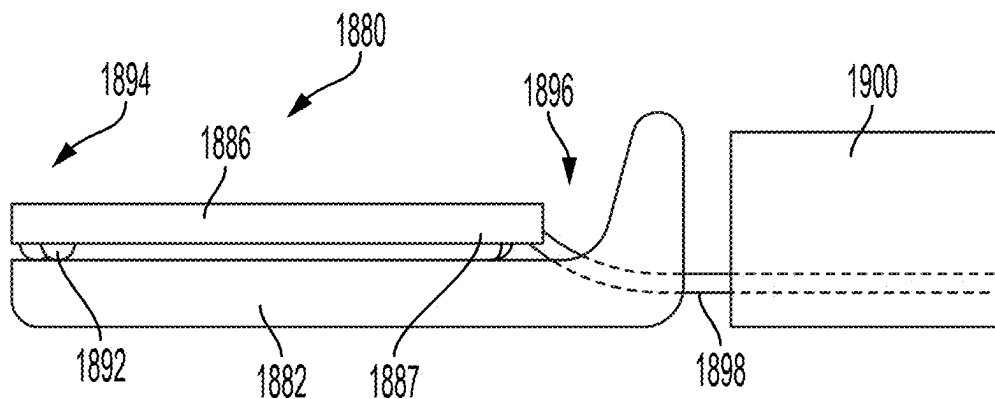


FIG. 104

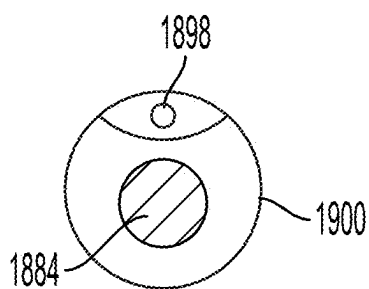


FIG. 105

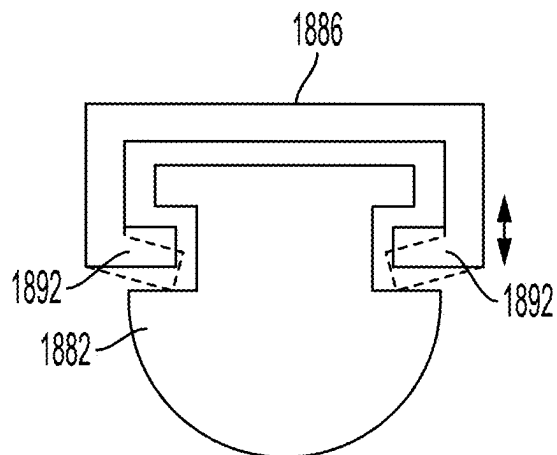


FIG. 106

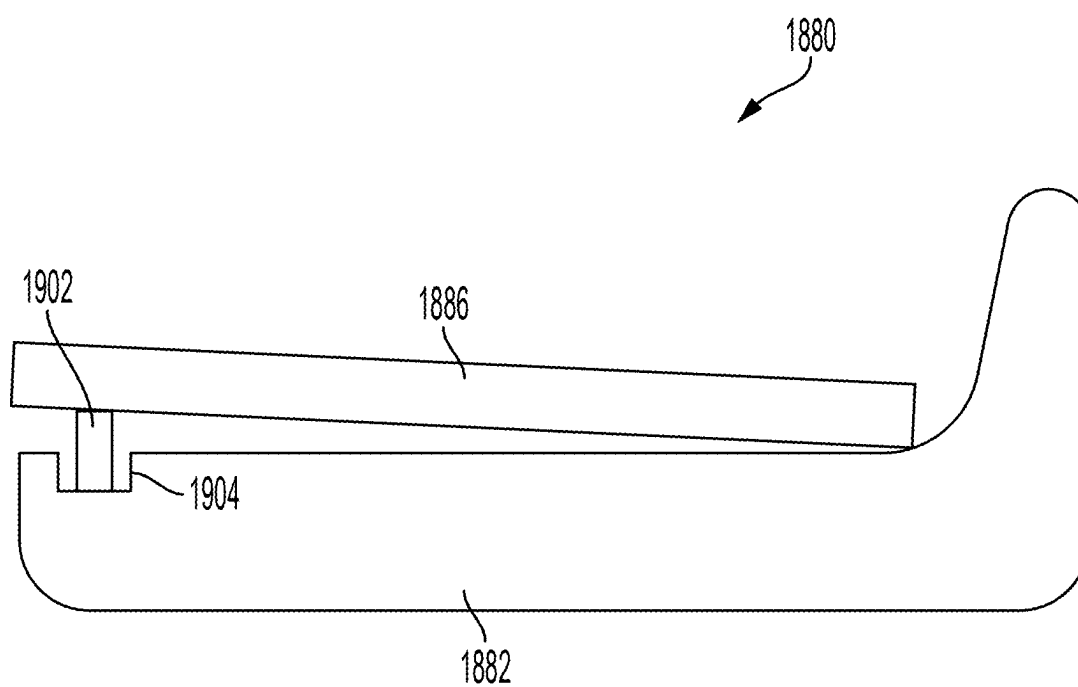


FIG. 107

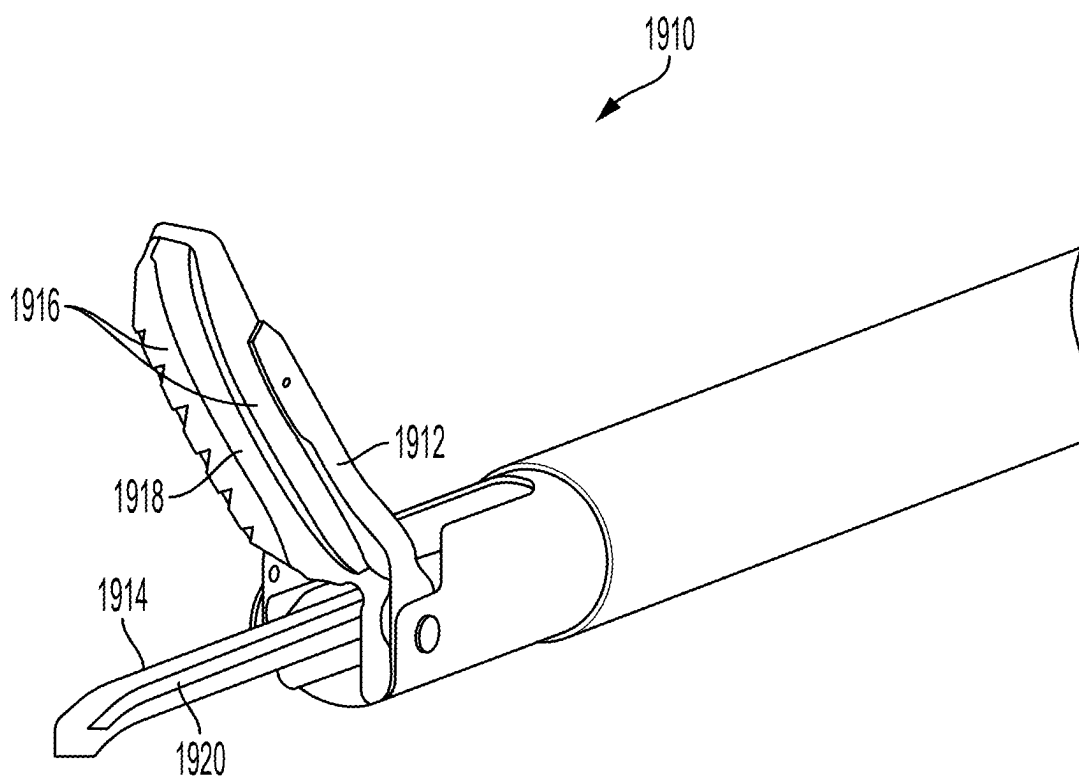
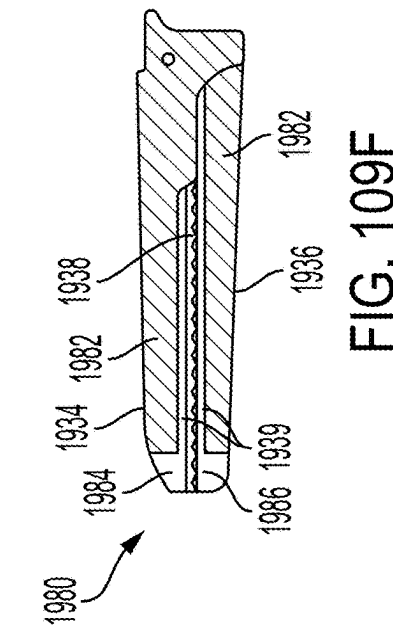
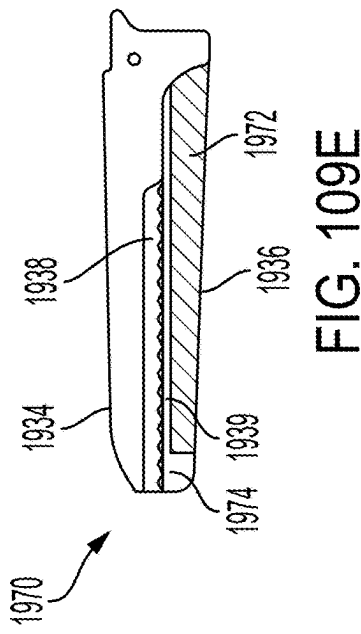
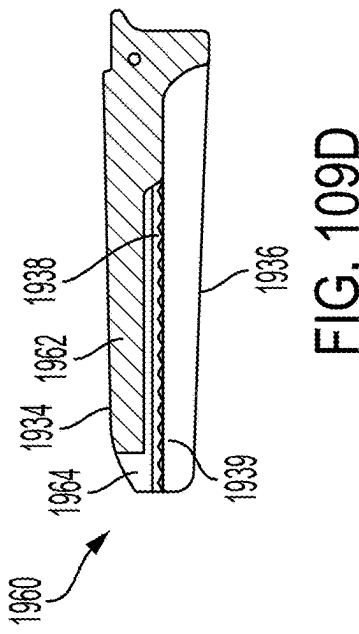
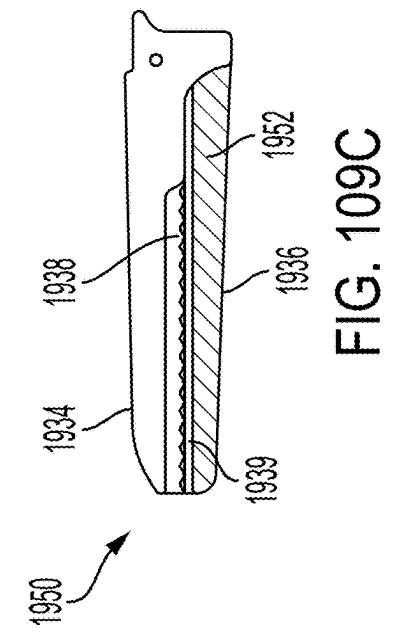
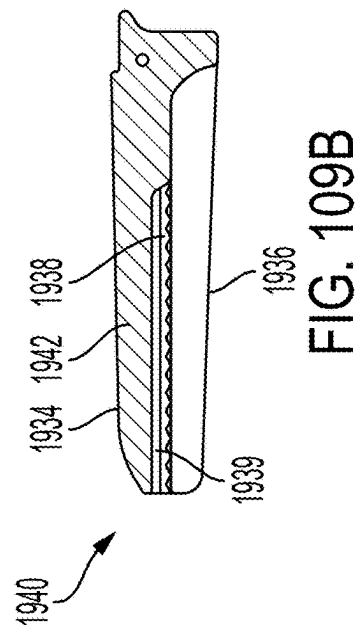
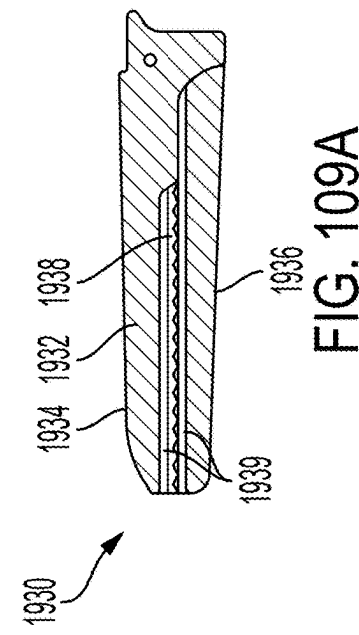


FIG. 108



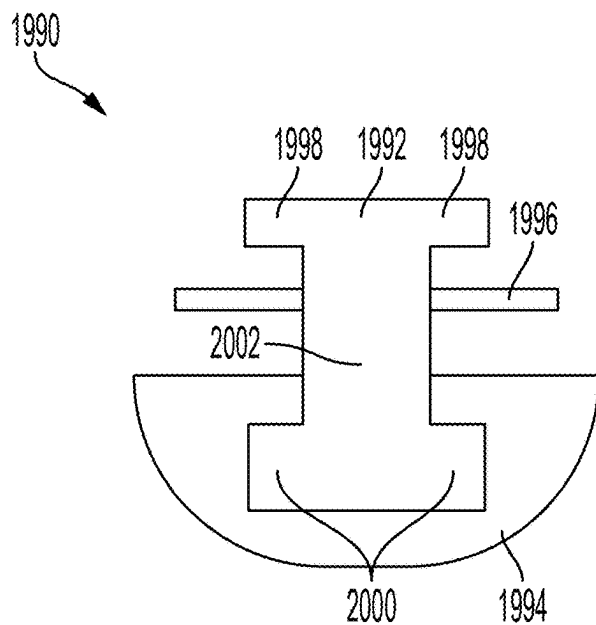


FIG. 110

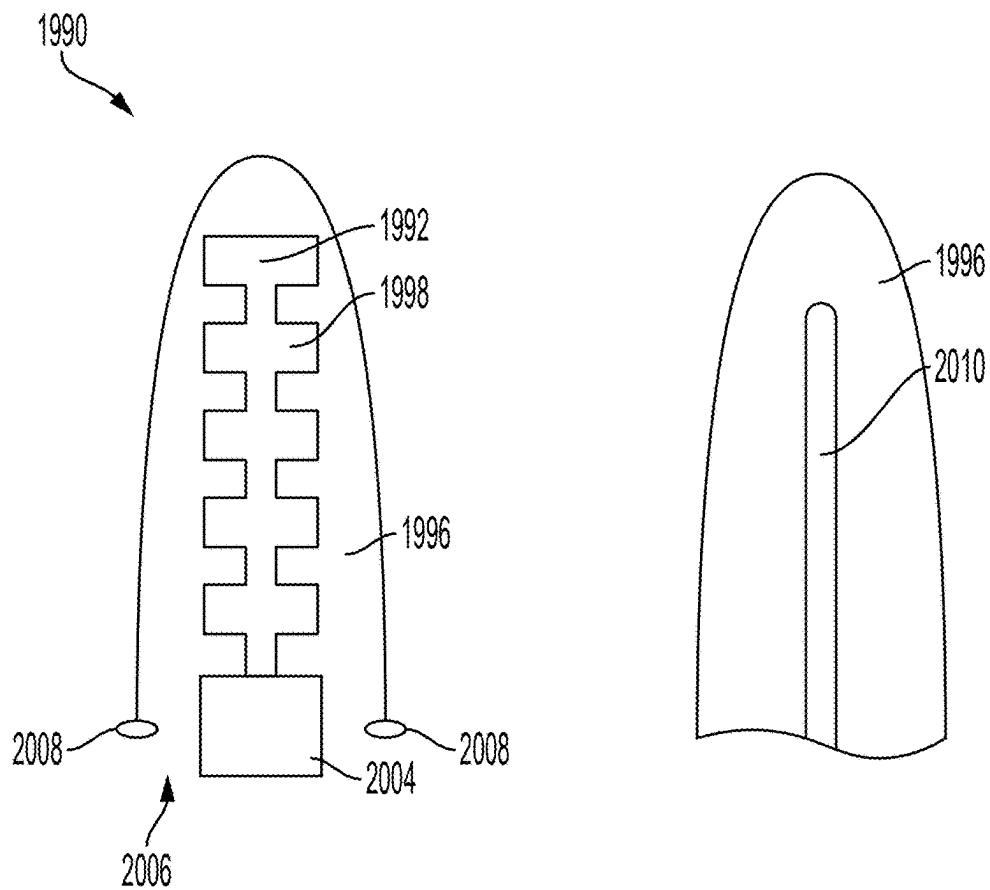


FIG. 111

FIG. 112

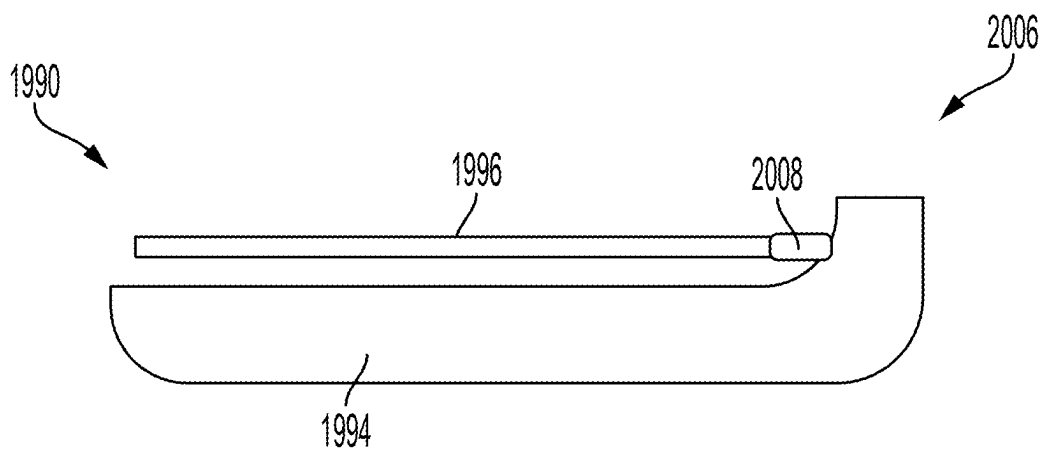


FIG. 113A

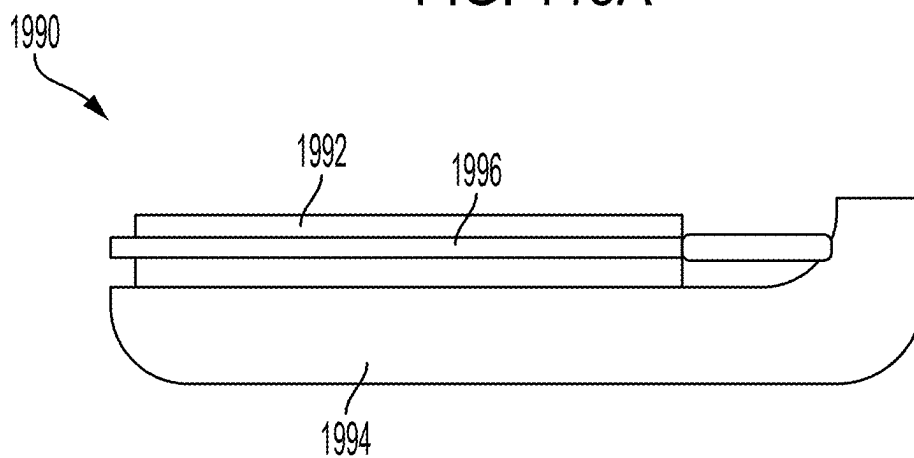


FIG. 113B

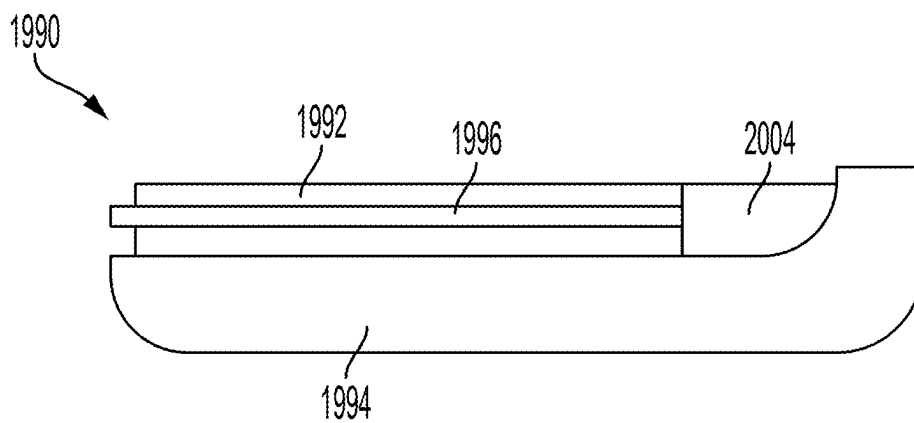


FIG. 113C

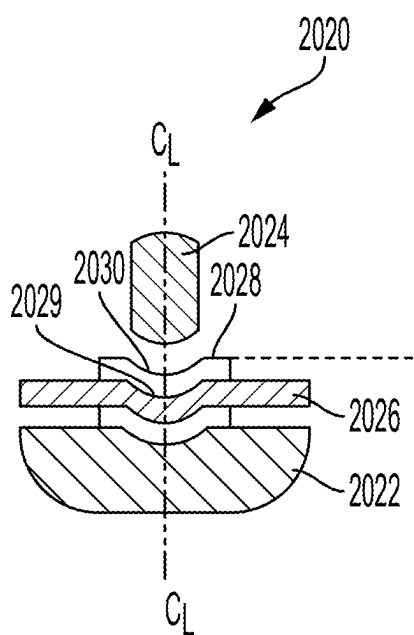


FIG. 114

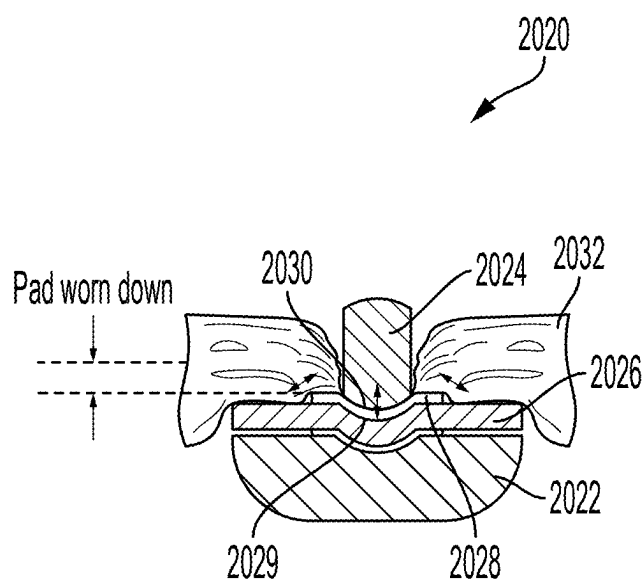


FIG. 115

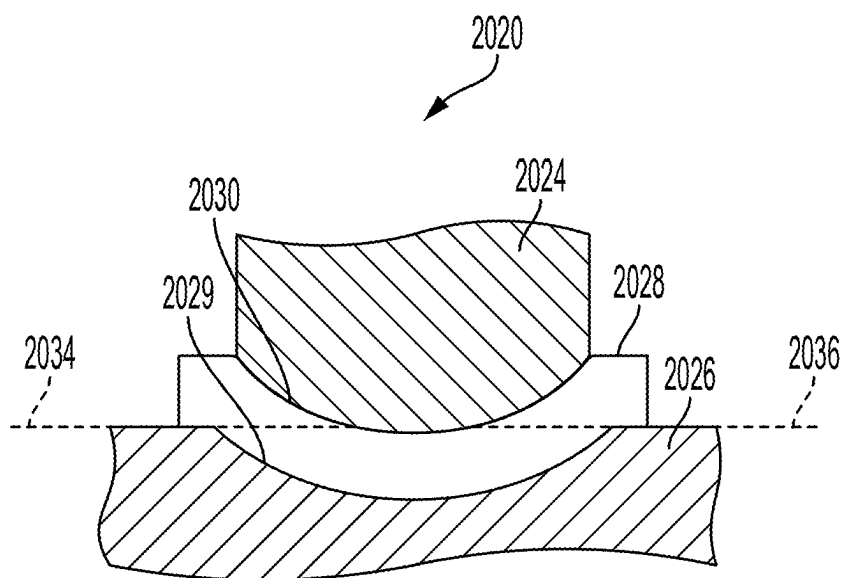


FIG. 116

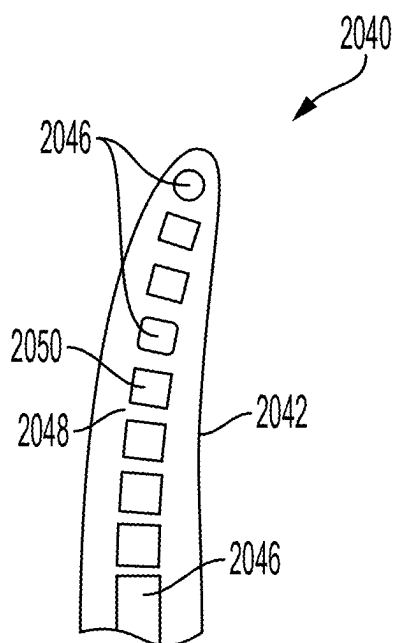


FIG. 117

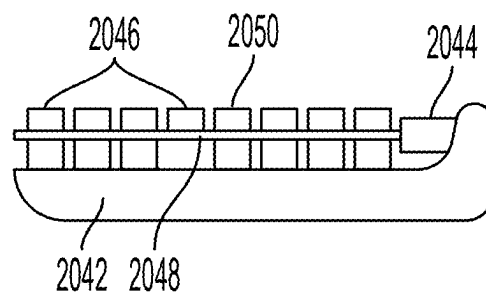


FIG. 118

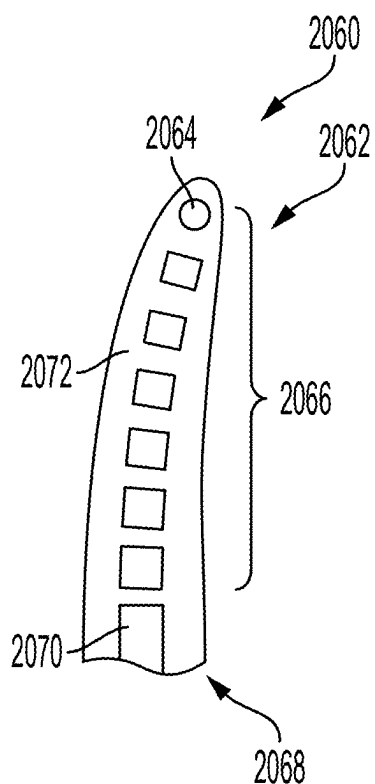


FIG. 119

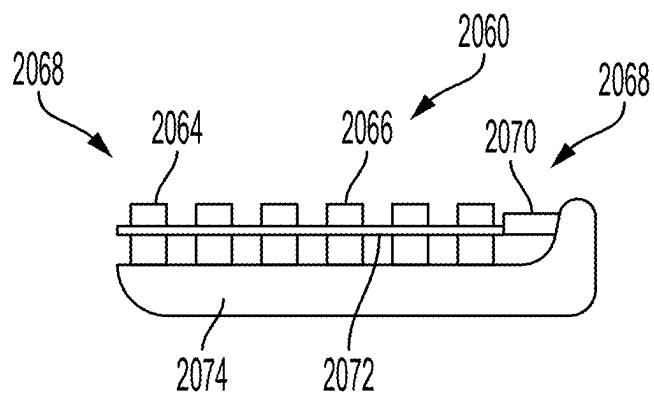


FIG. 120

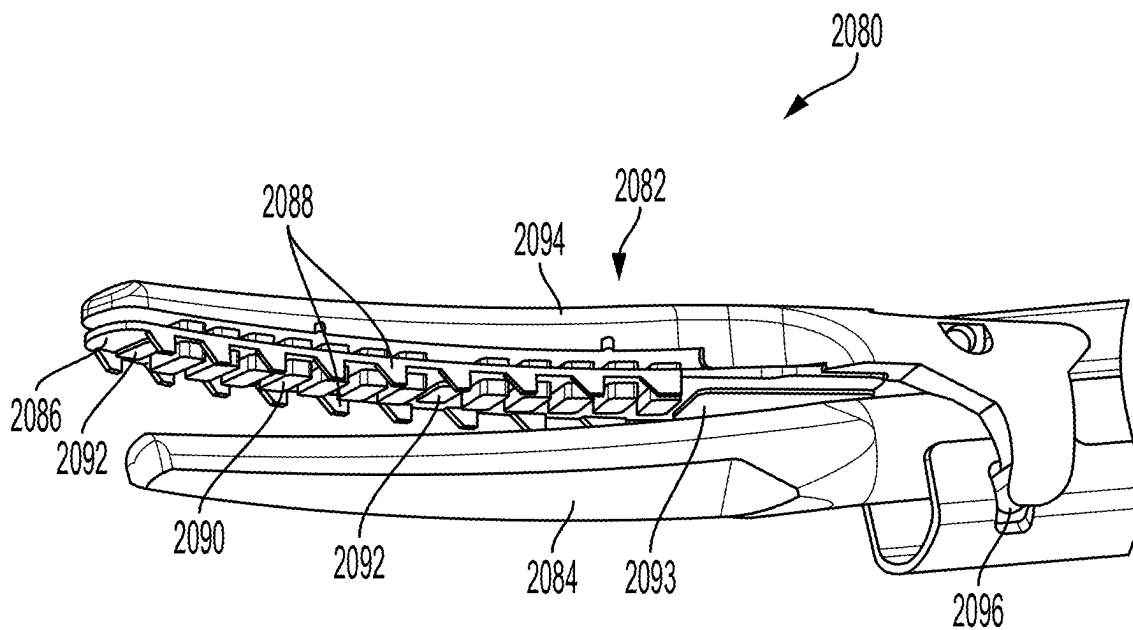


FIG. 121

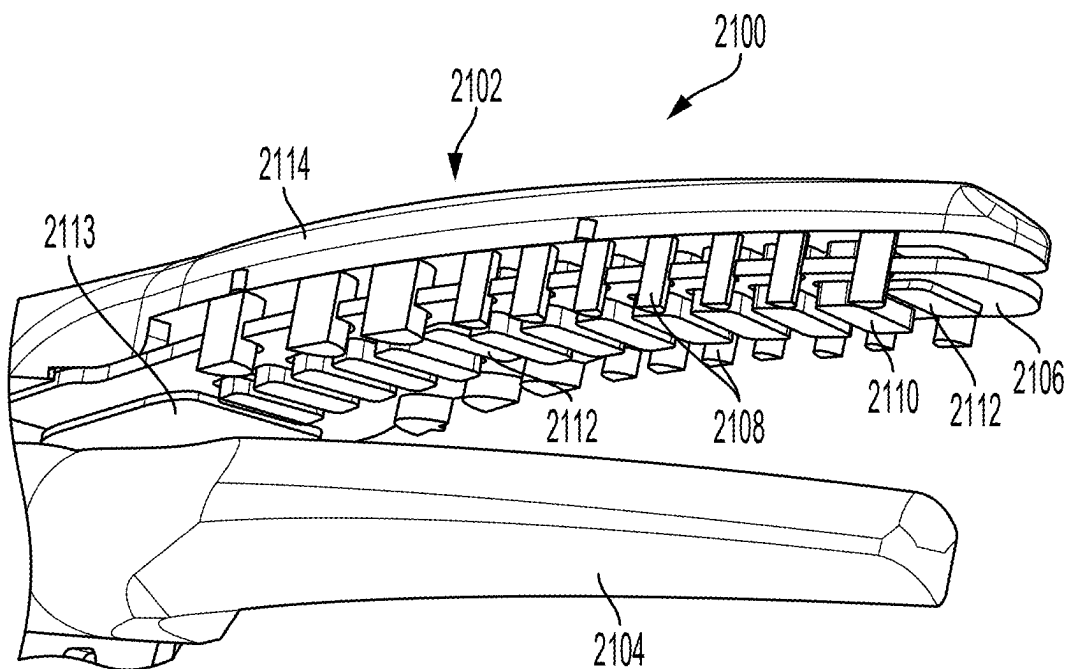


FIG. 122

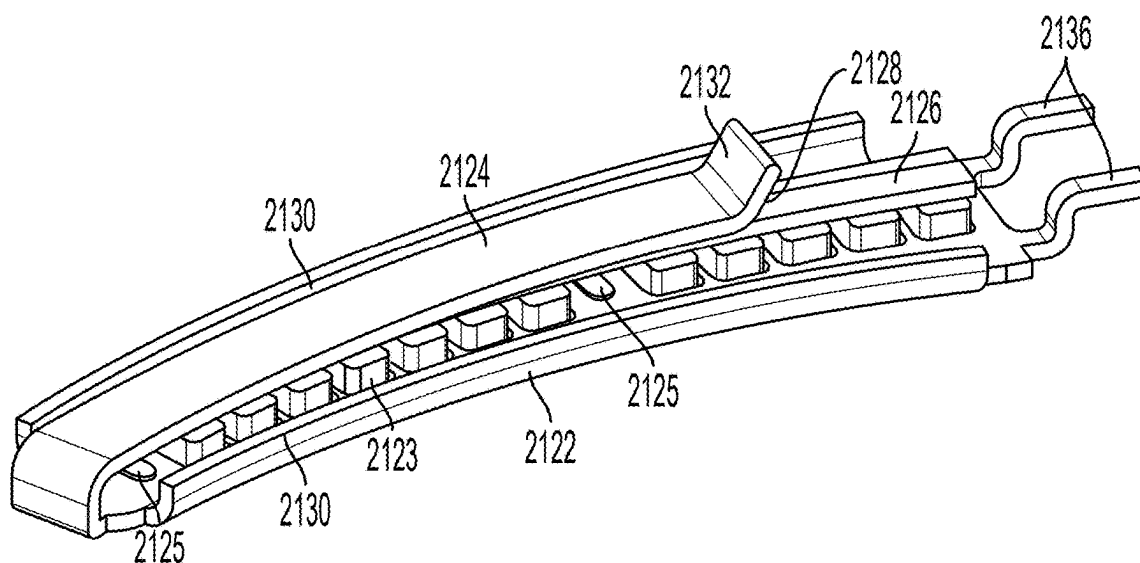


FIG. 123

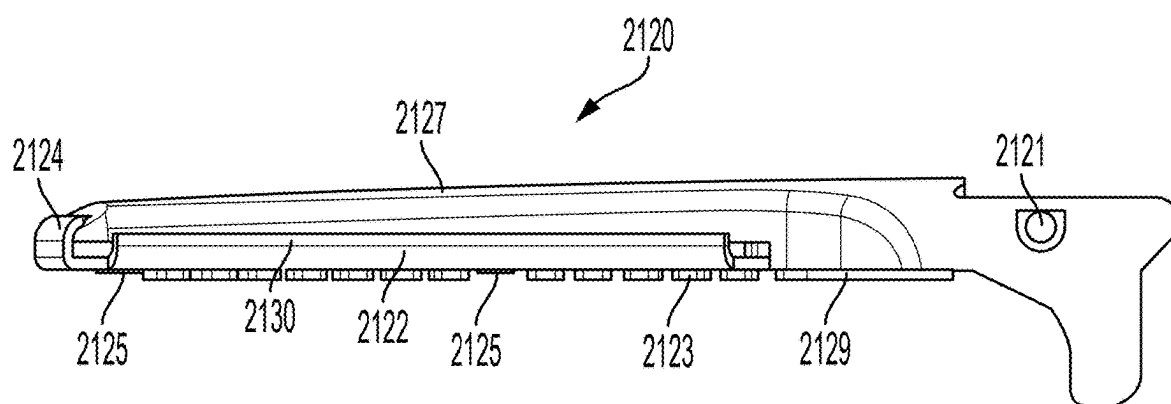


FIG. 124

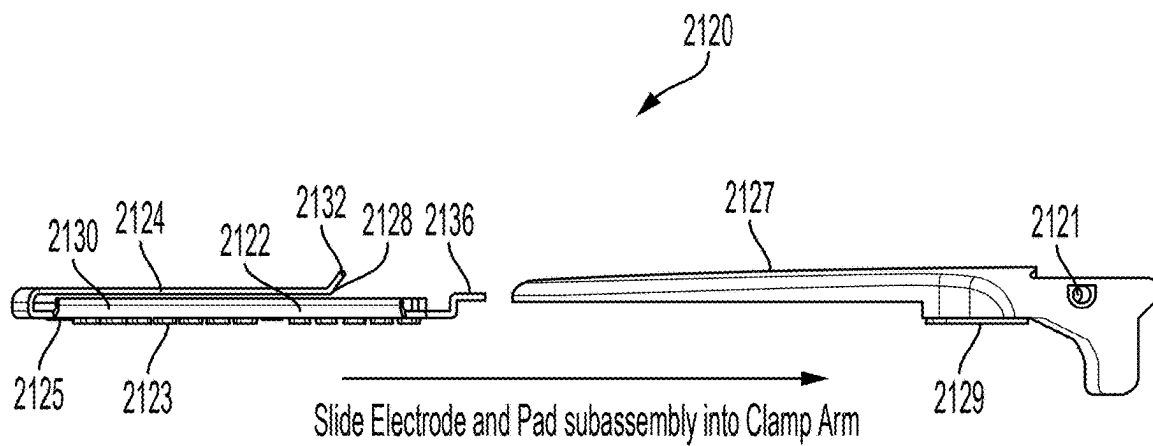


FIG. 125

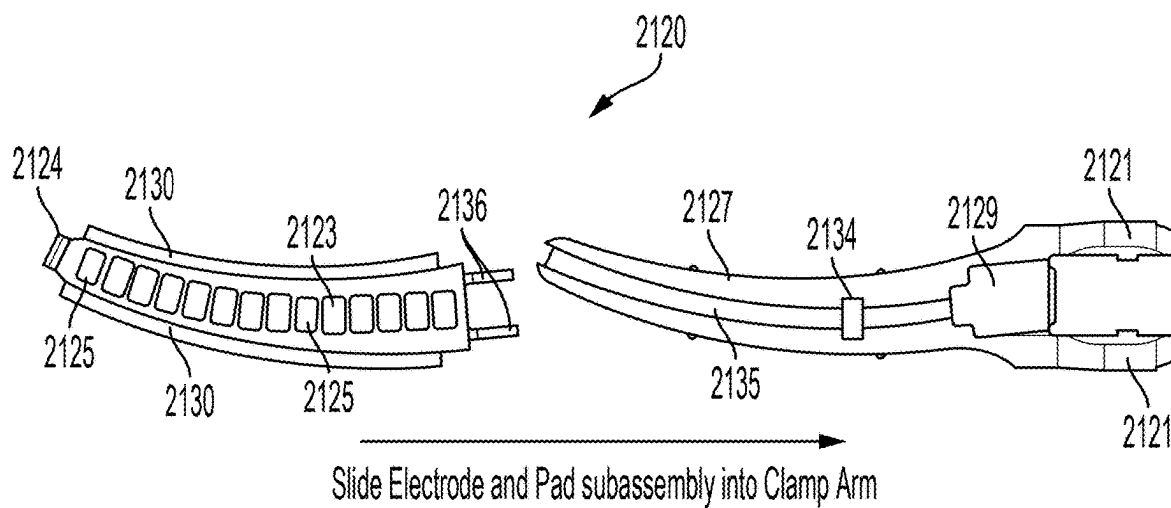


FIG. 126

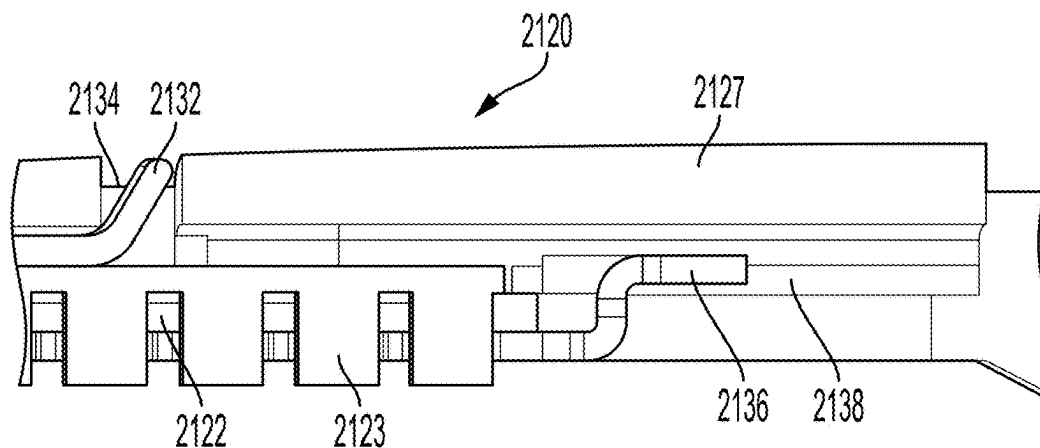


FIG. 127

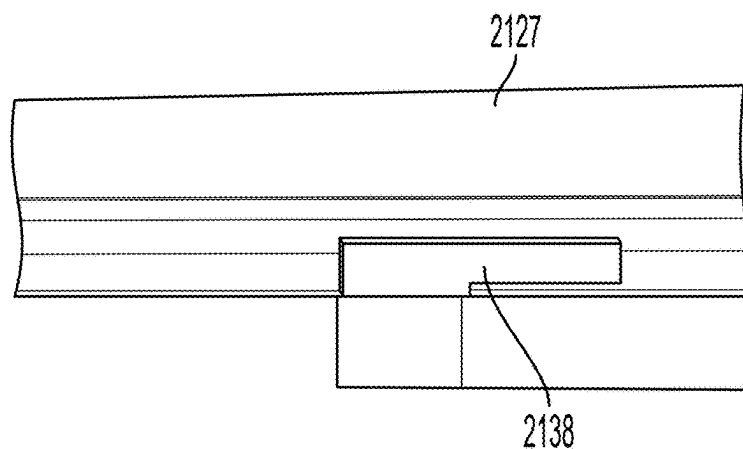


FIG. 128

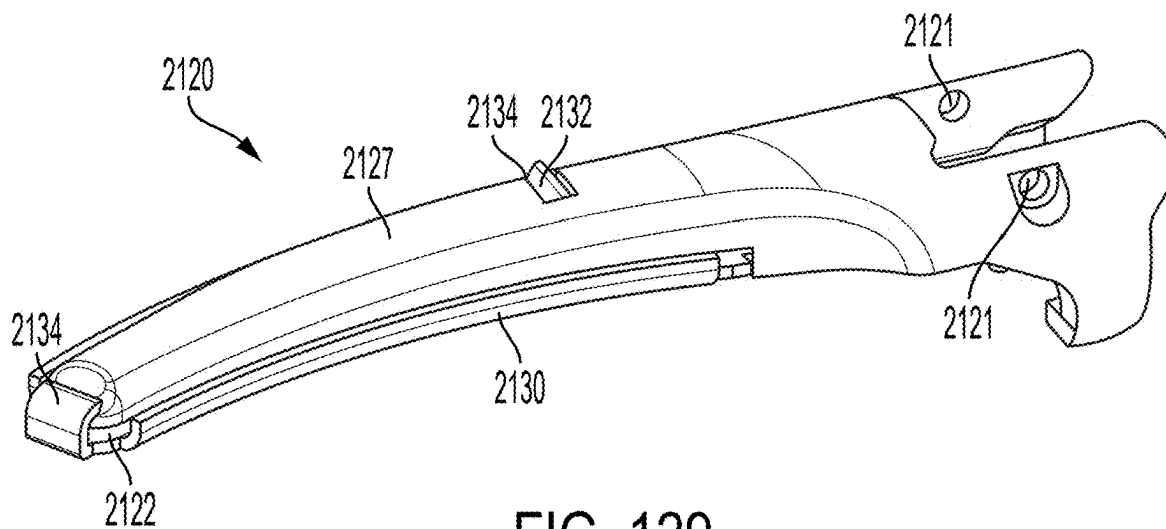


FIG. 129

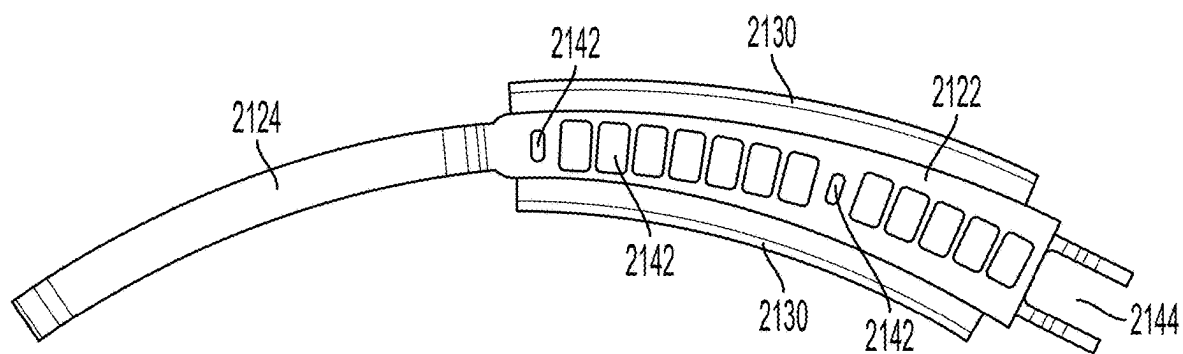


FIG. 130

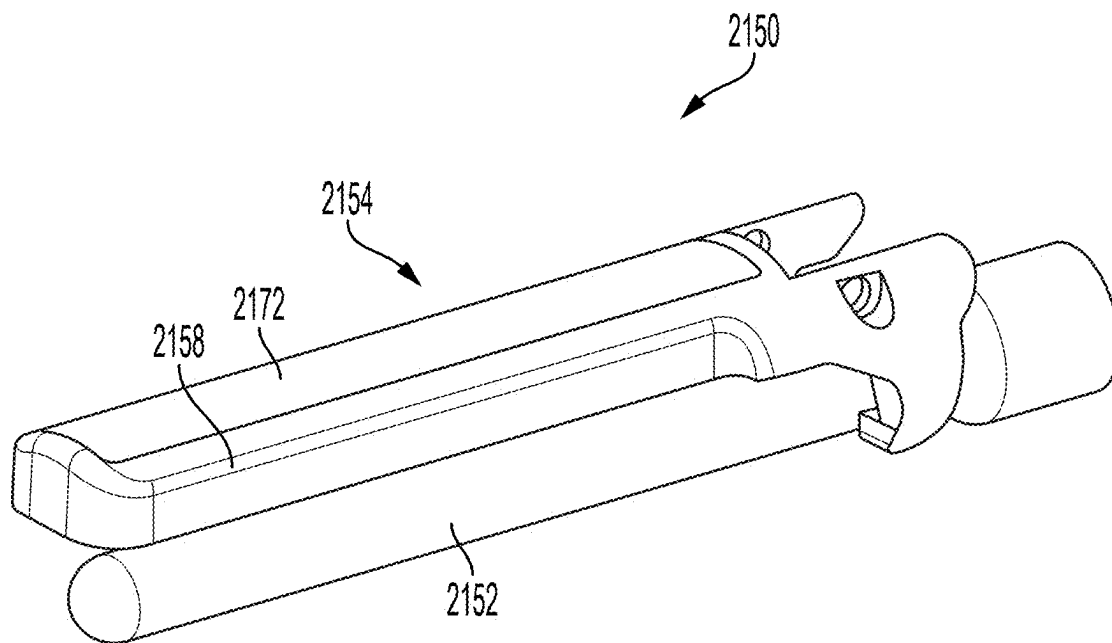


FIG. 131

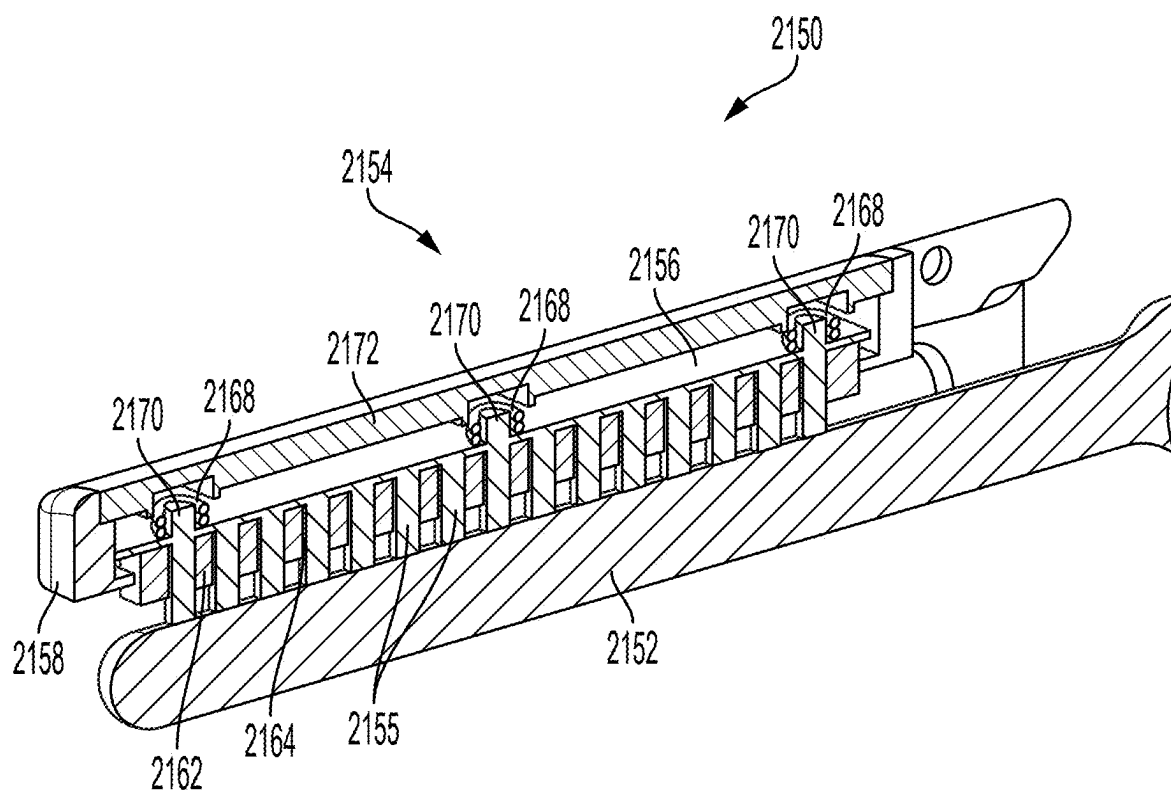


FIG. 132

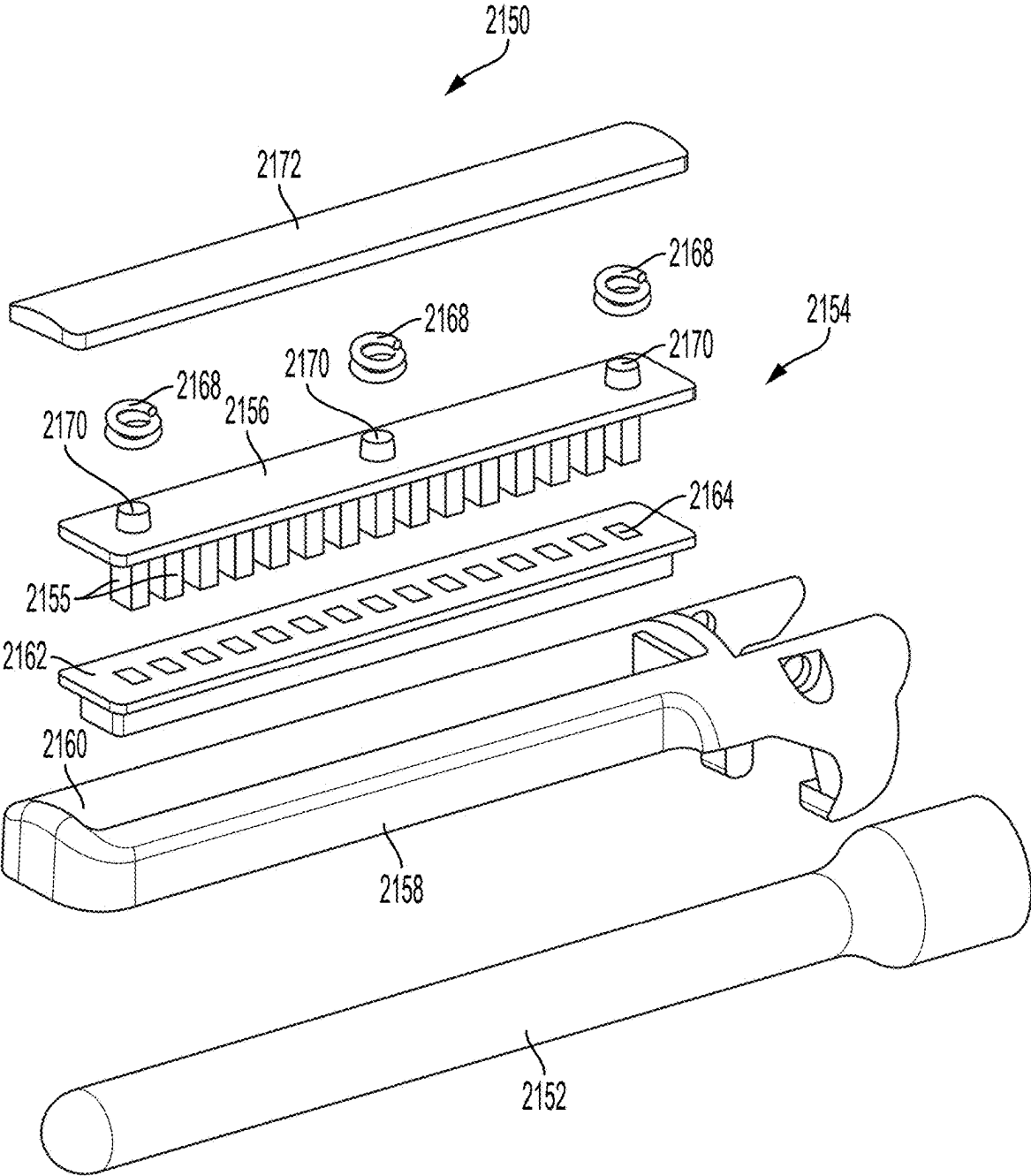


FIG. 133

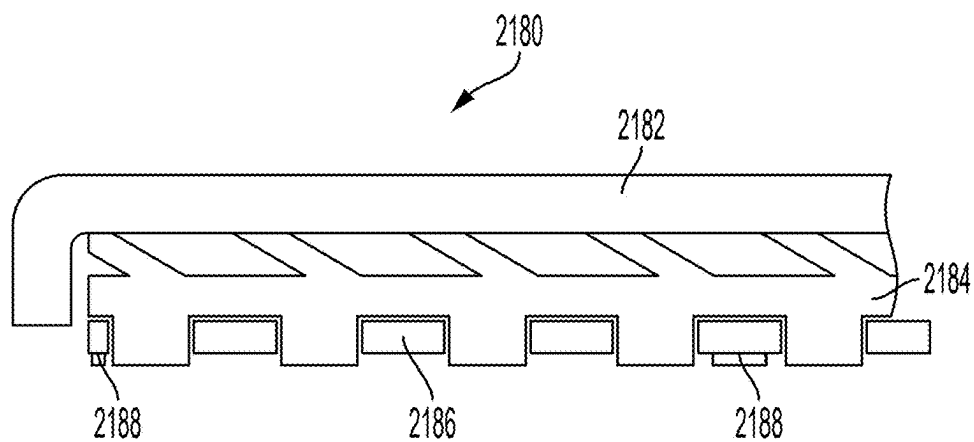


FIG. 134

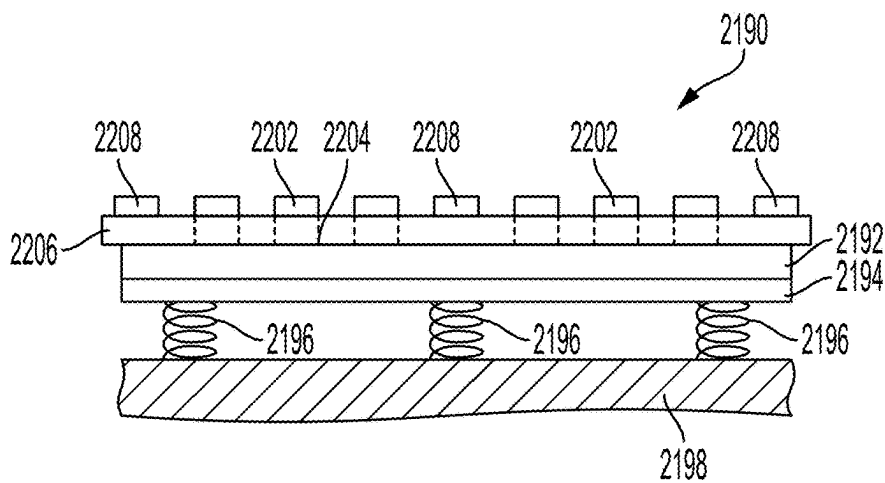


FIG. 135

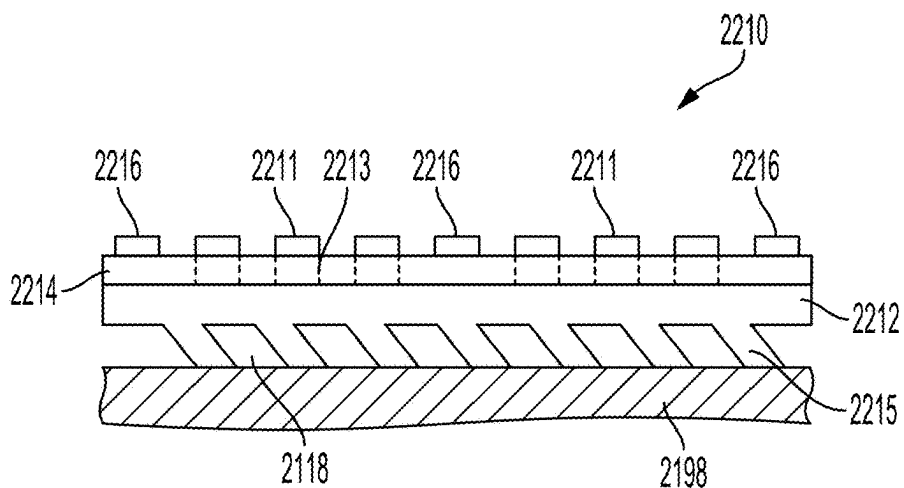


FIG. 136

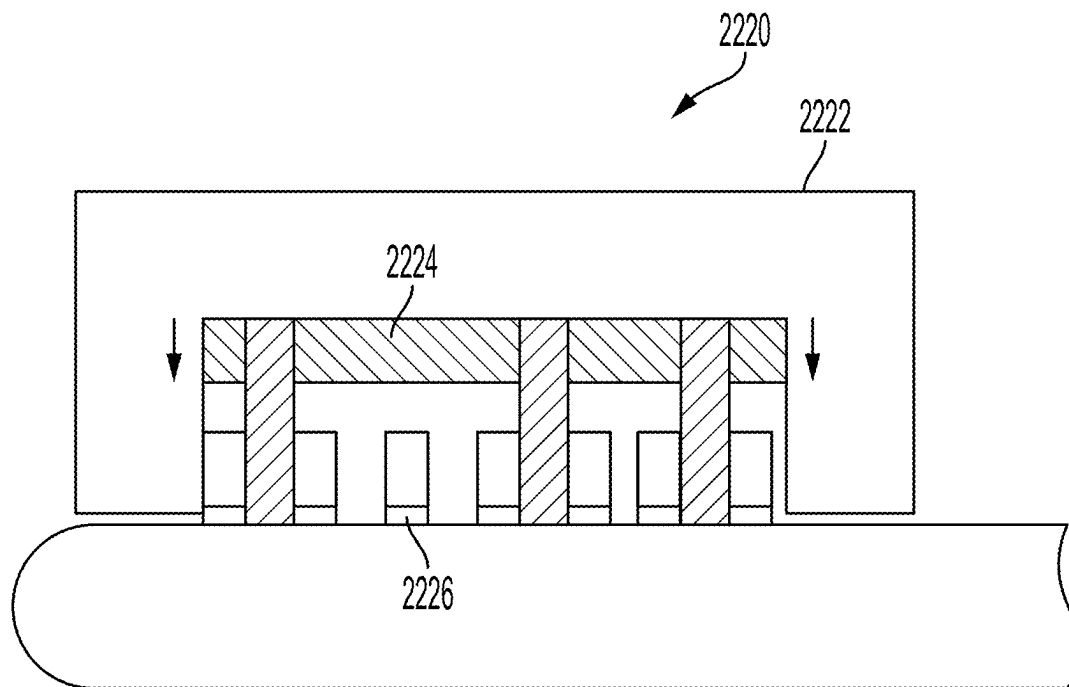


FIG. 137

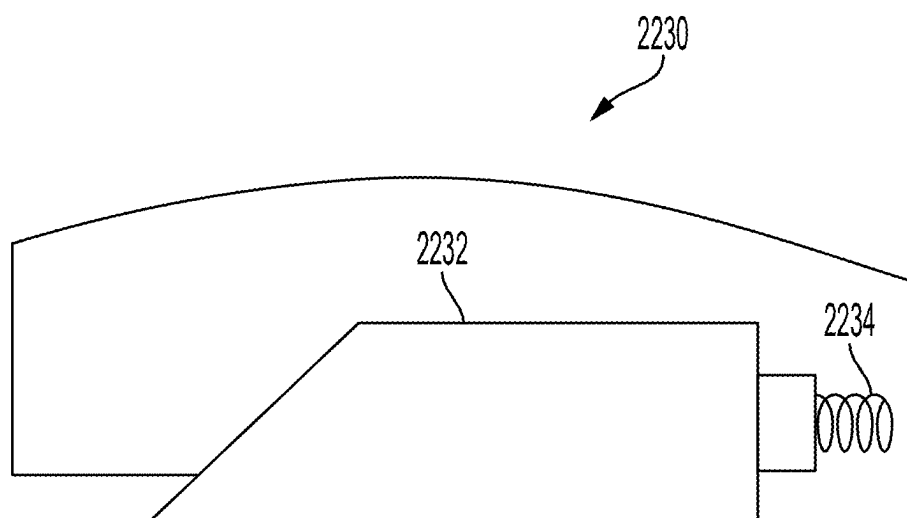


FIG. 138

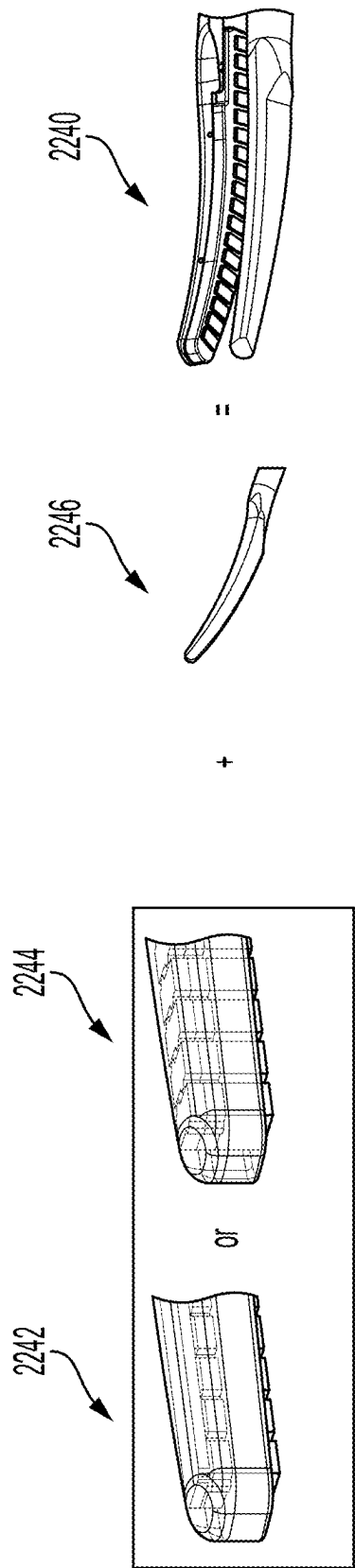


FIG. 139

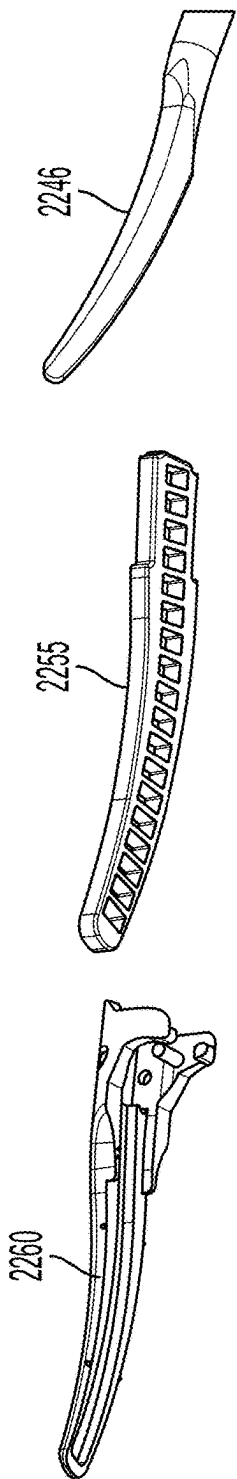


FIG. 140

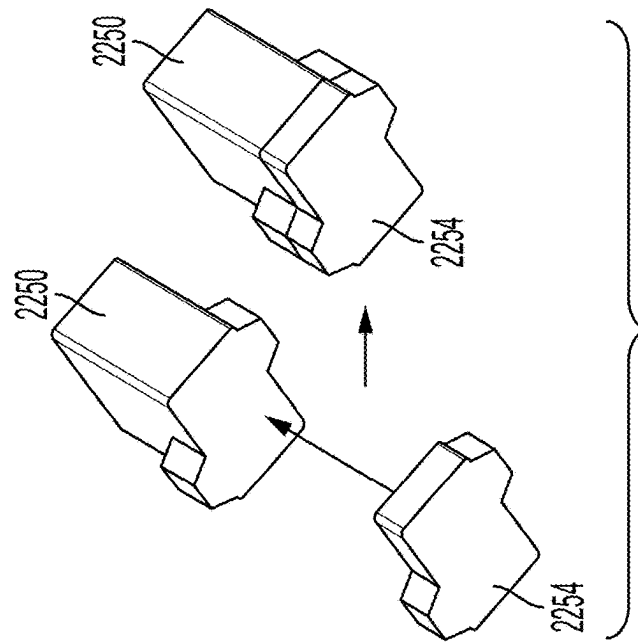


FIG. 142

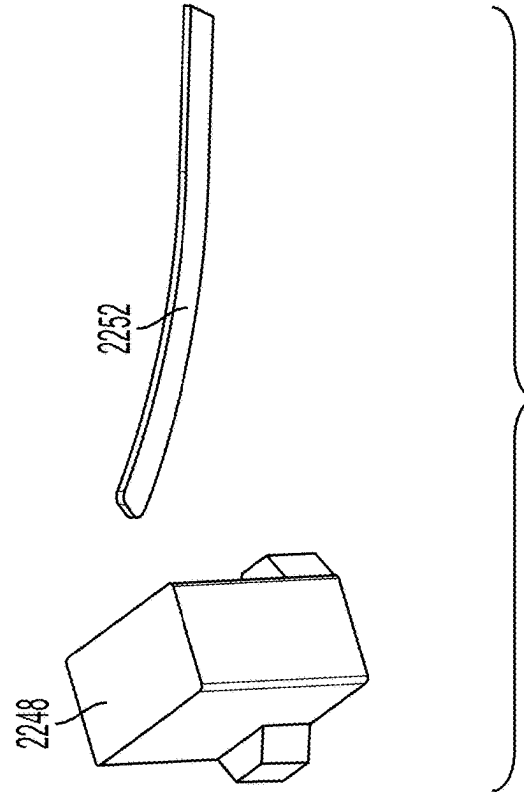


FIG. 141

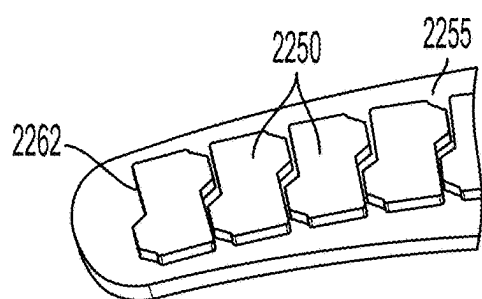


FIG. 143

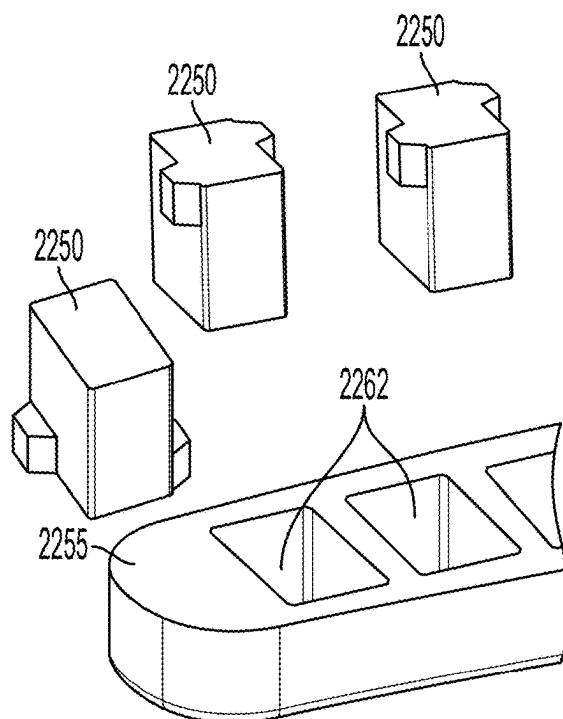


FIG. 144

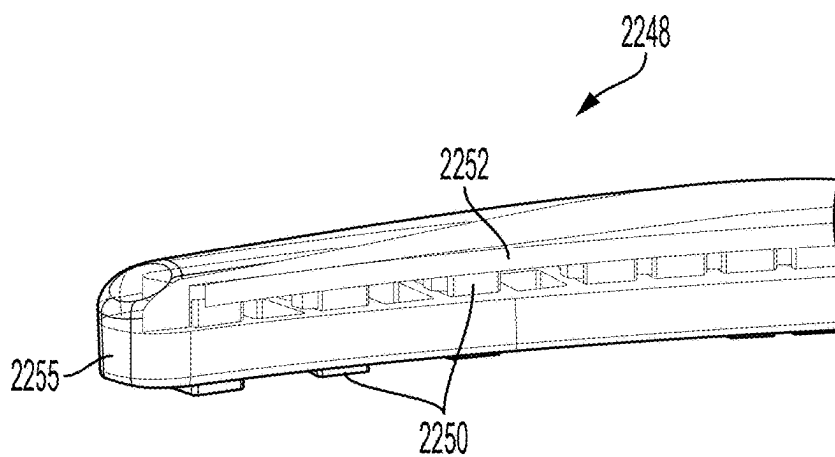


FIG. 145

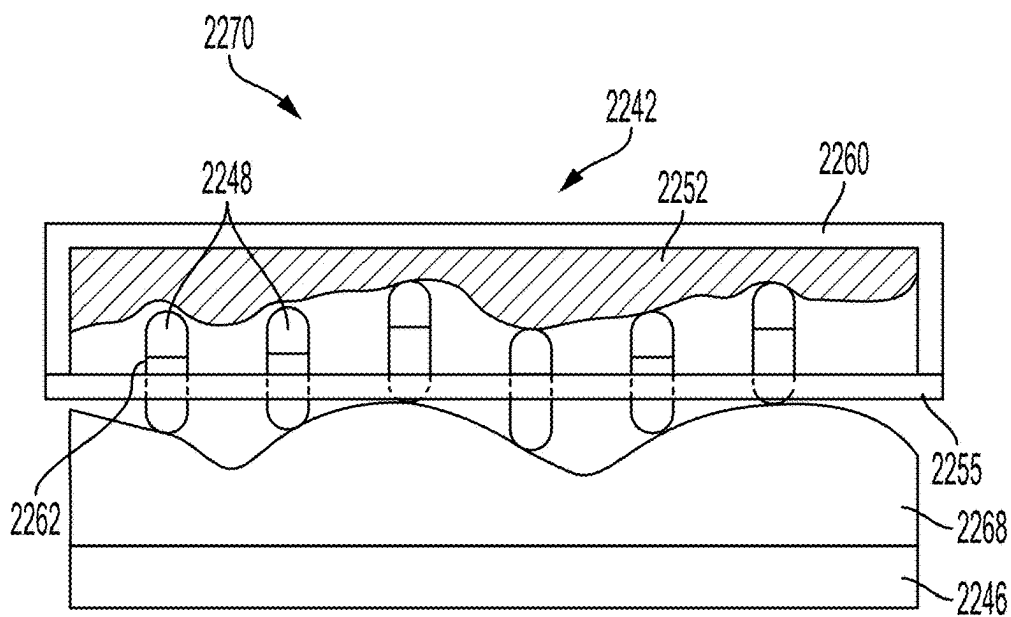


FIG. 146

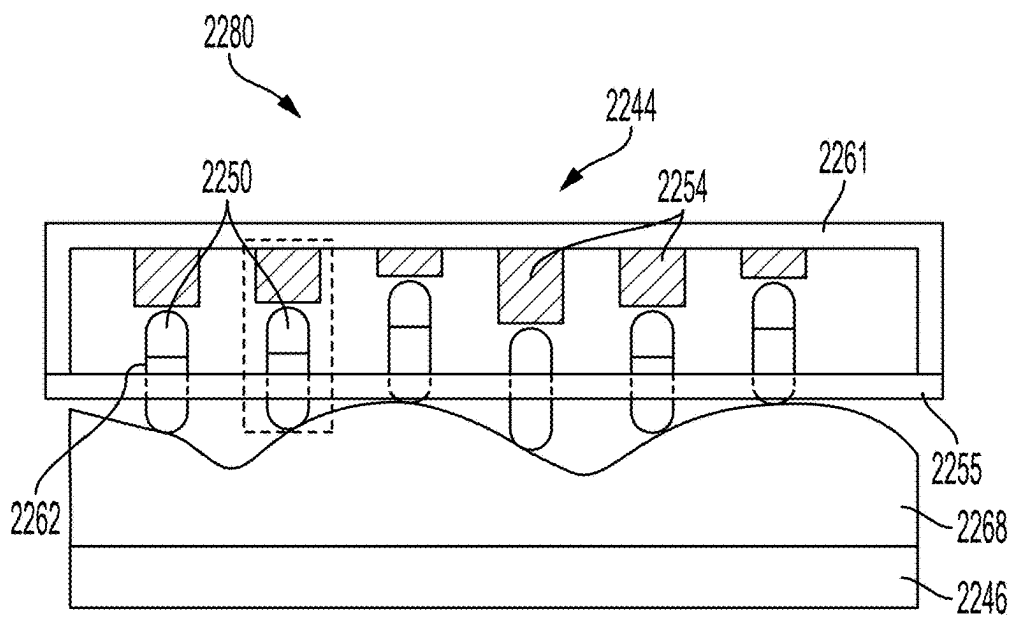


FIG. 147

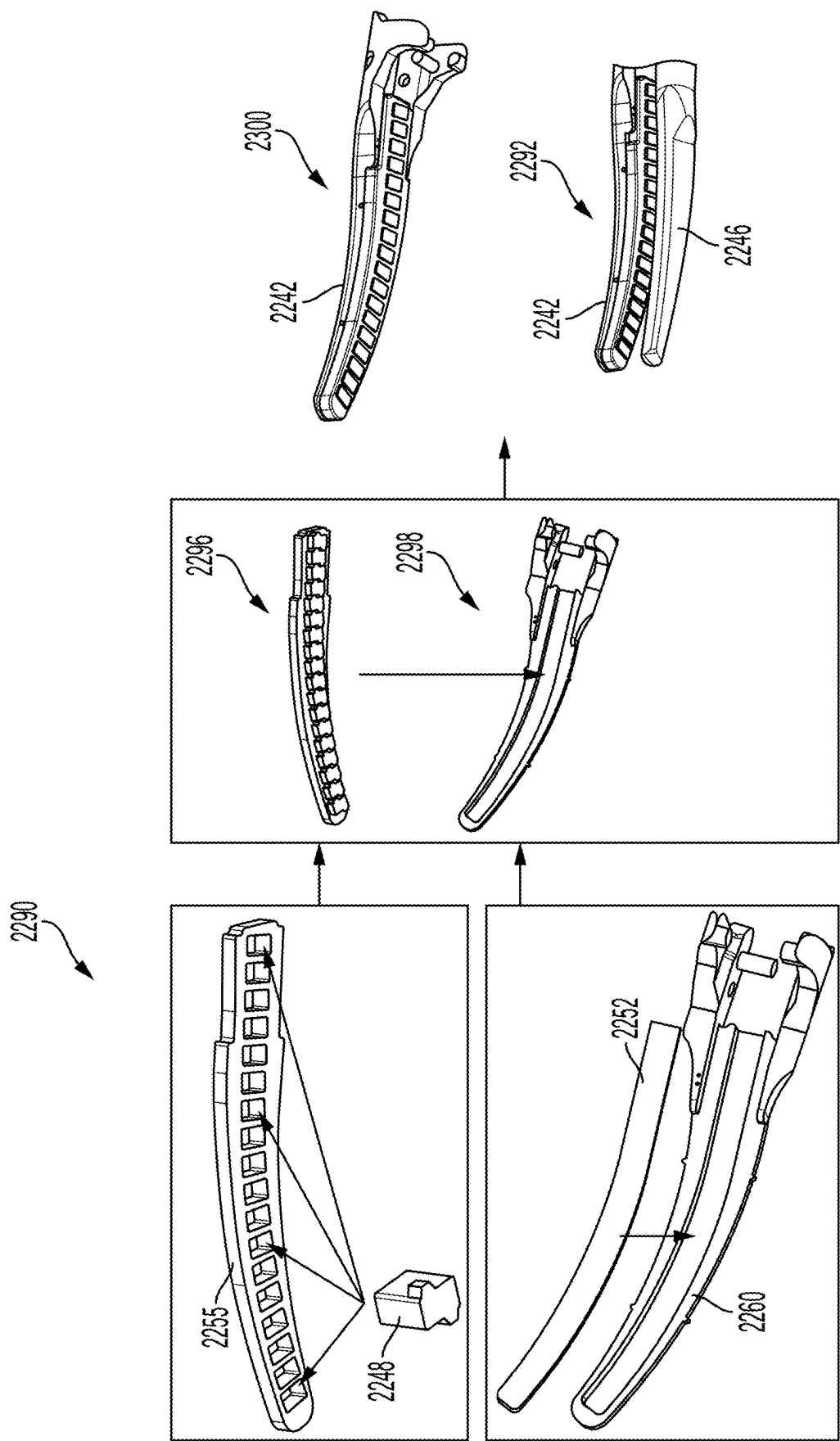


FIG. 148

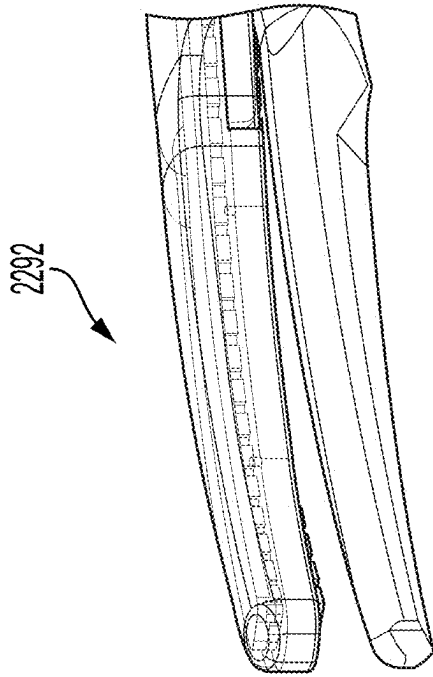


FIG. 149

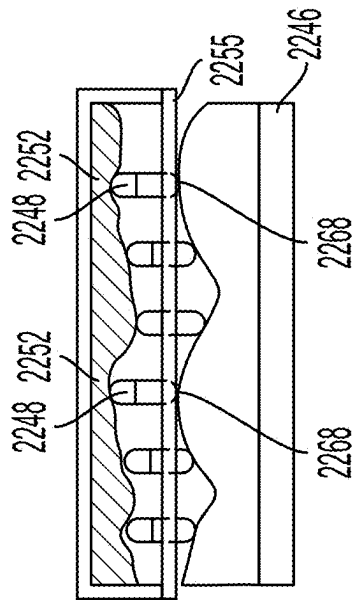


FIG. 150

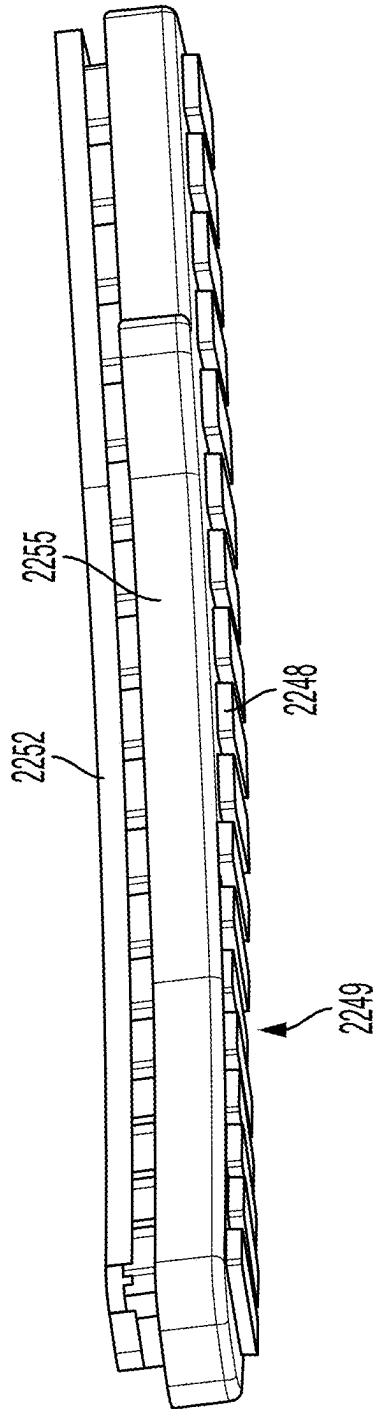


FIG. 151

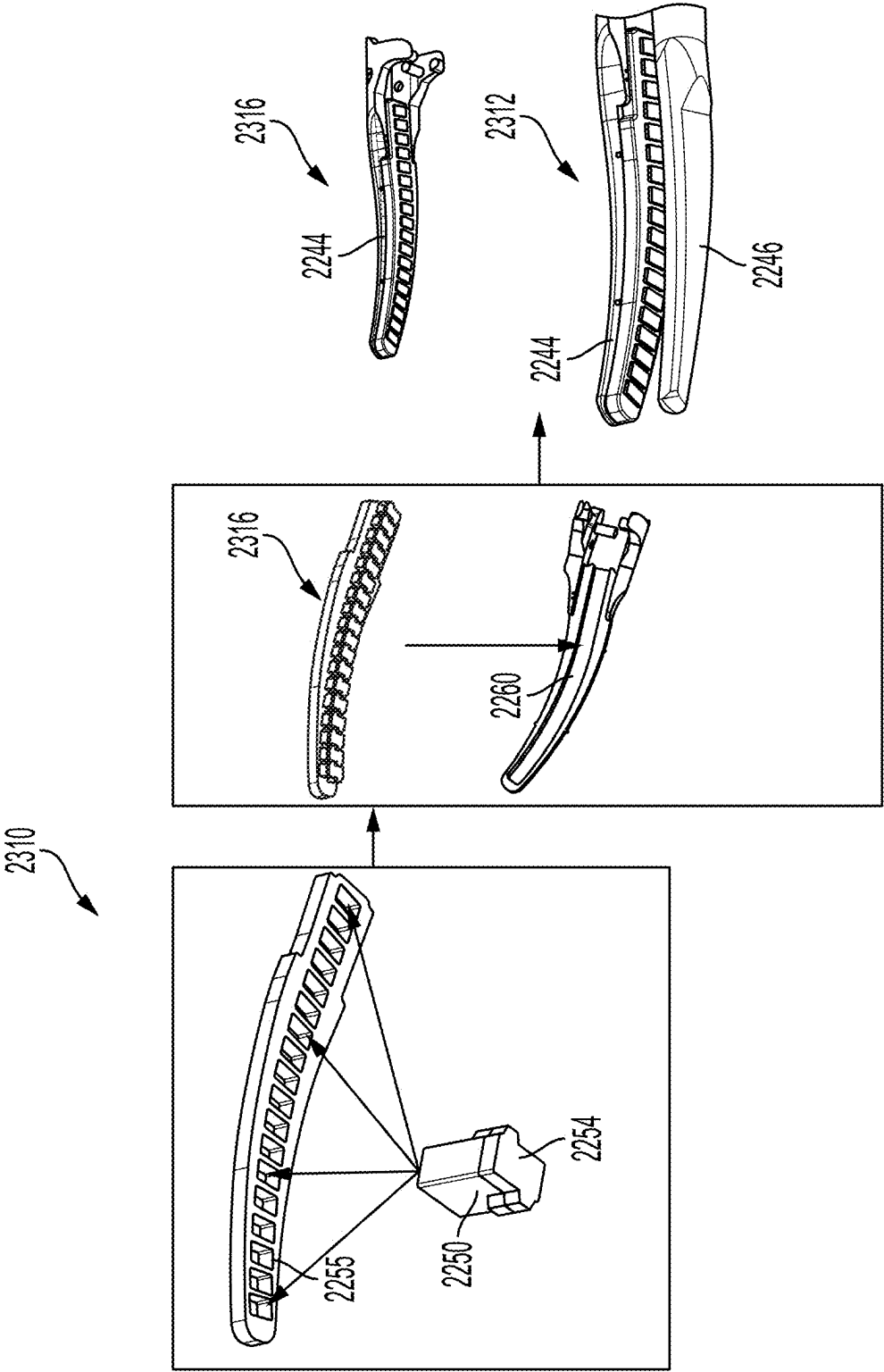


FIG. 152

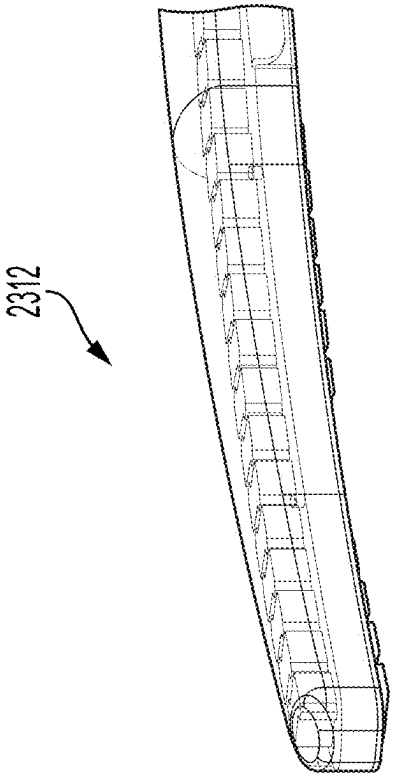


FIG. 153

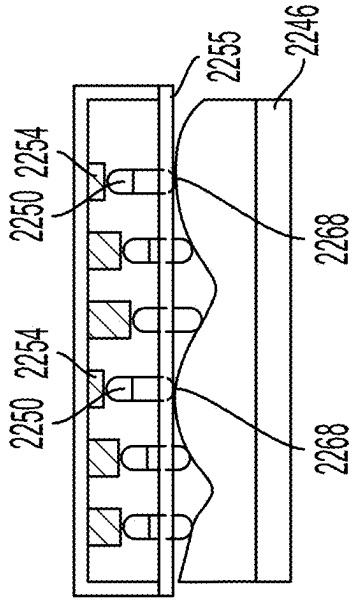


FIG. 154

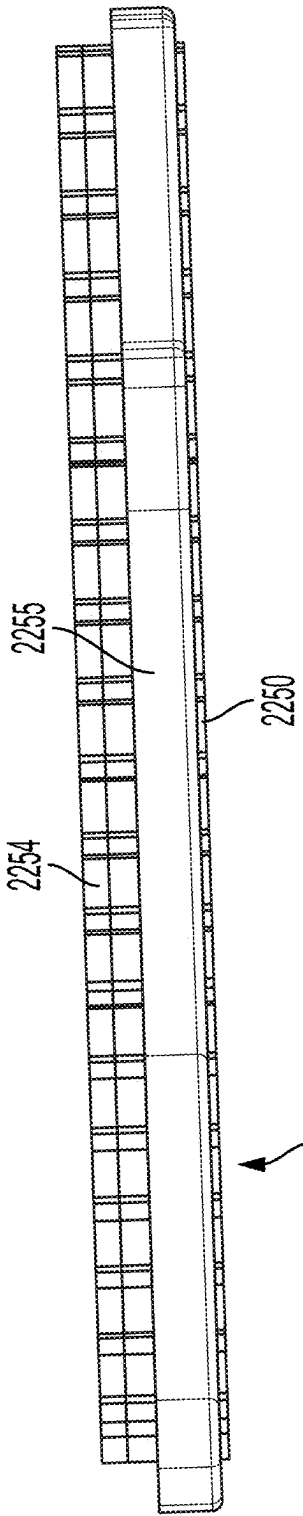


FIG. 155

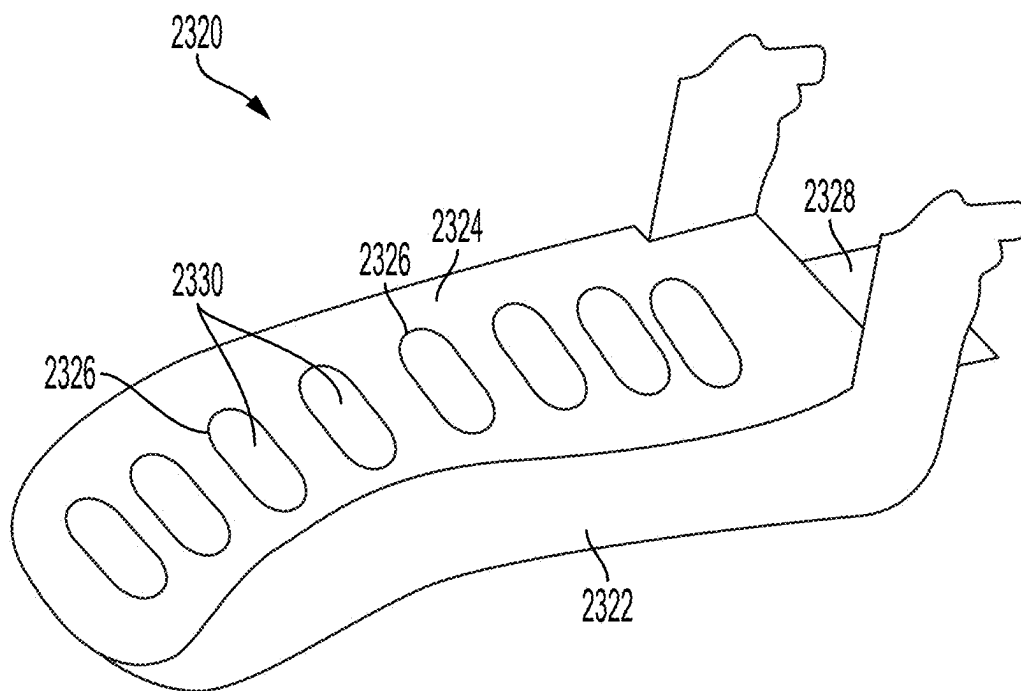


FIG. 156

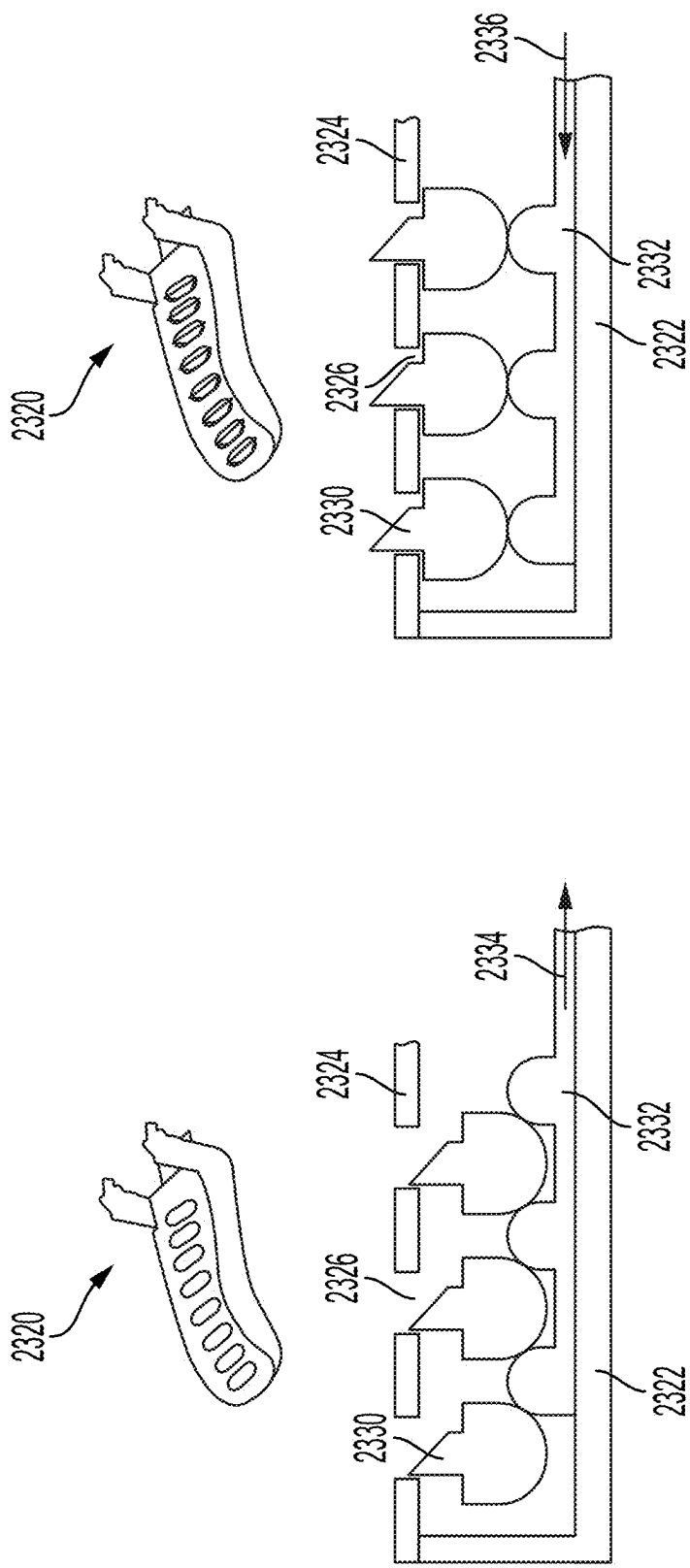


FIG. 158

FIG. 157

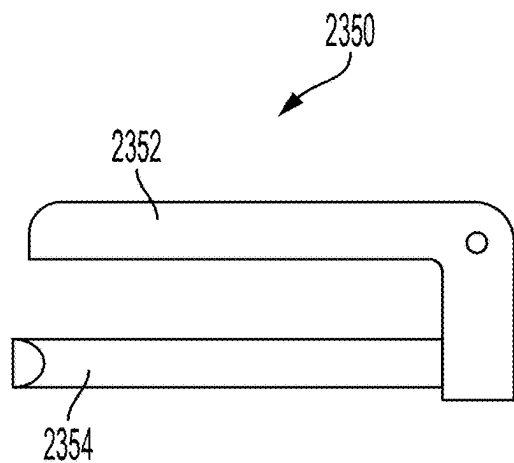


FIG. 160A

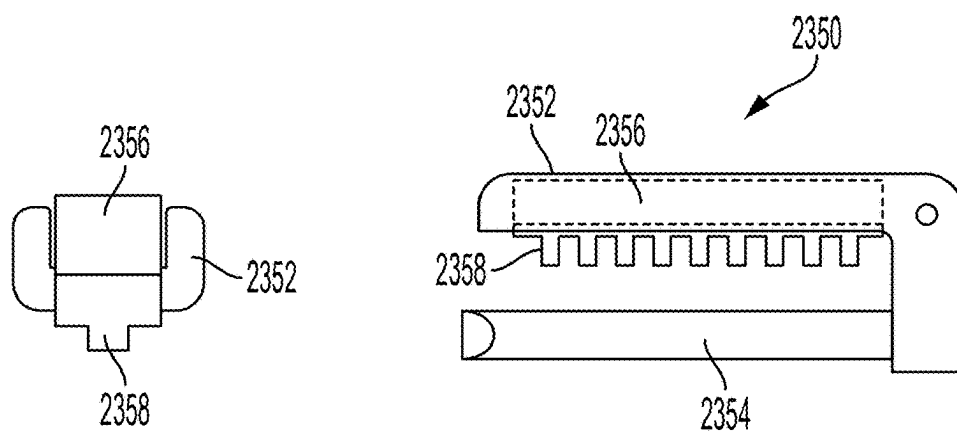


FIG. 160B

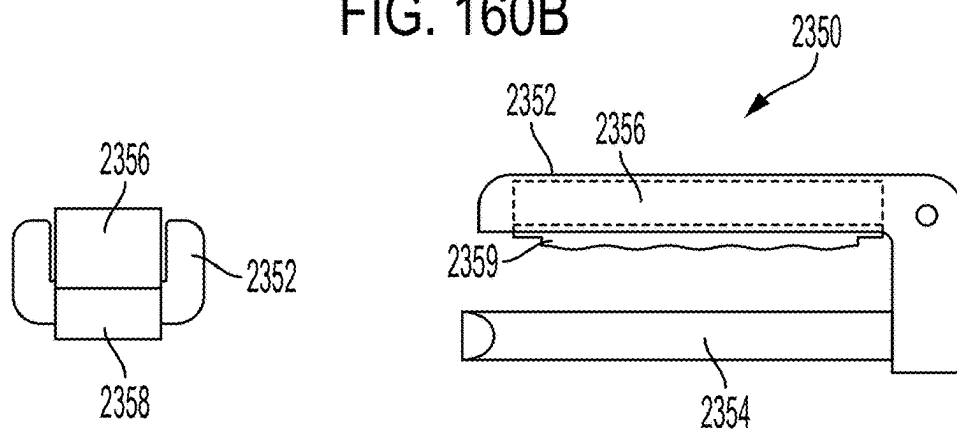


FIG. 160C

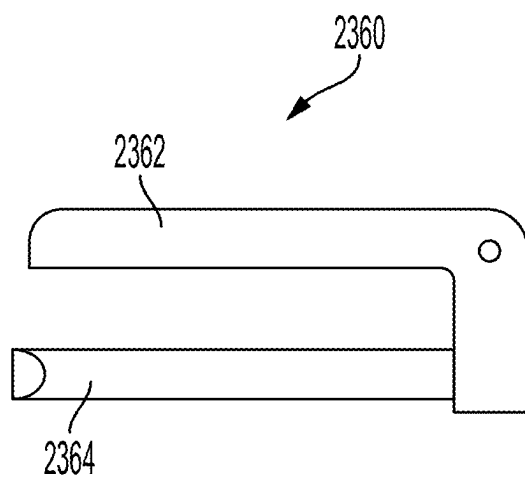


FIG. 161A

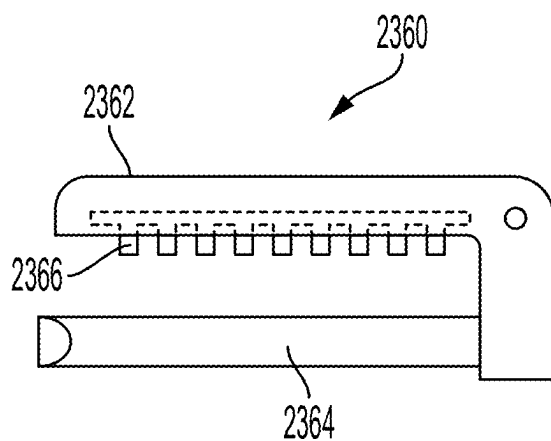


FIG. 161B

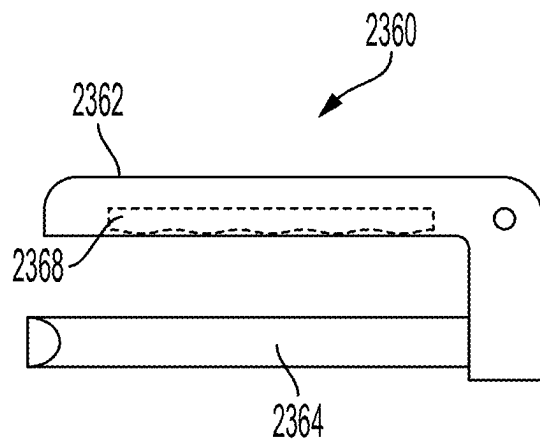


FIG. 161C

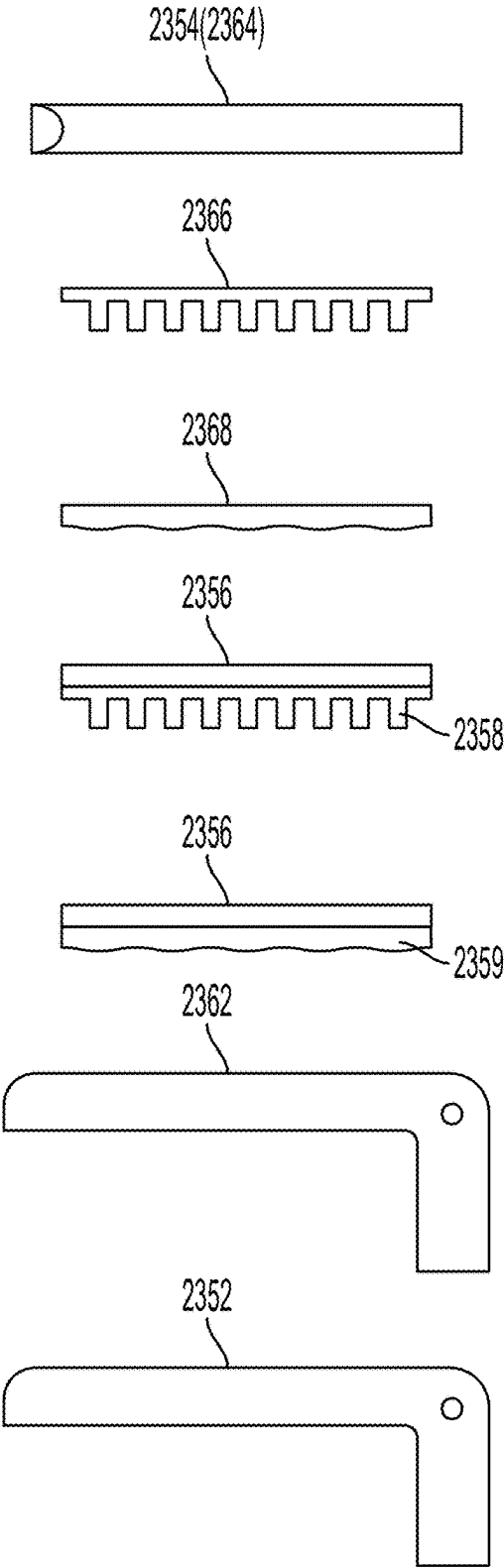


FIG. 162

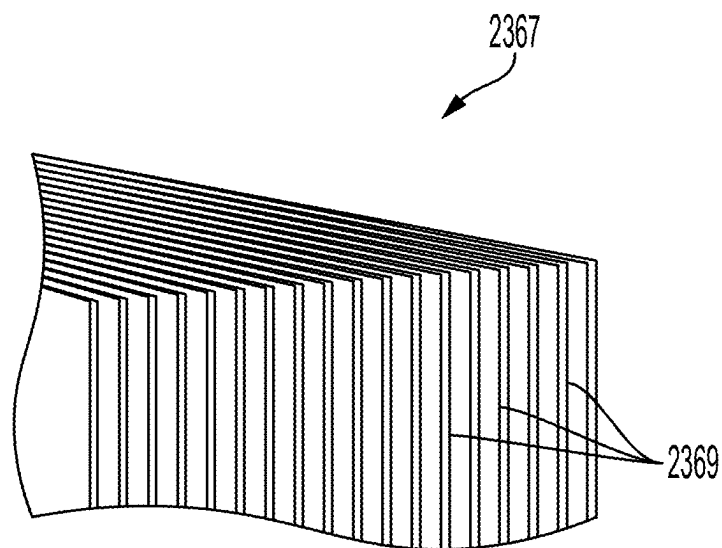


FIG. 163

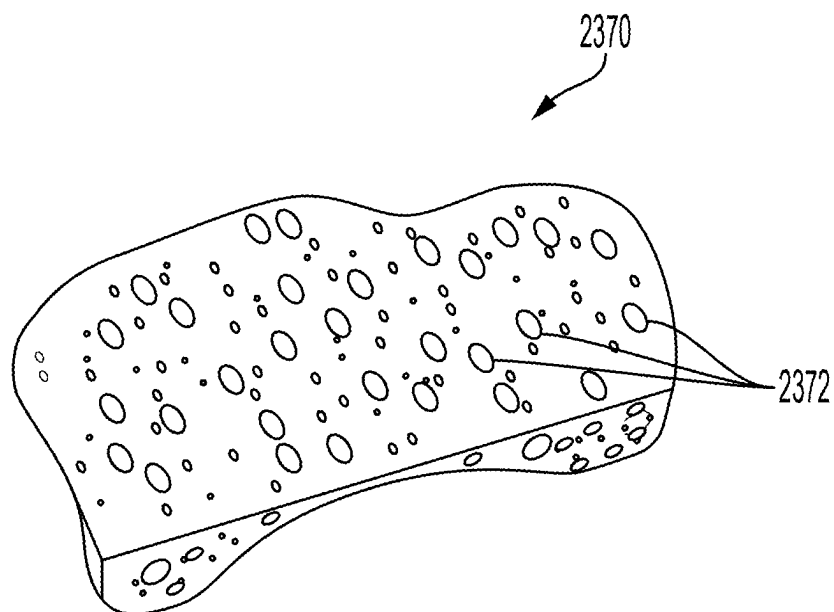


FIG. 164

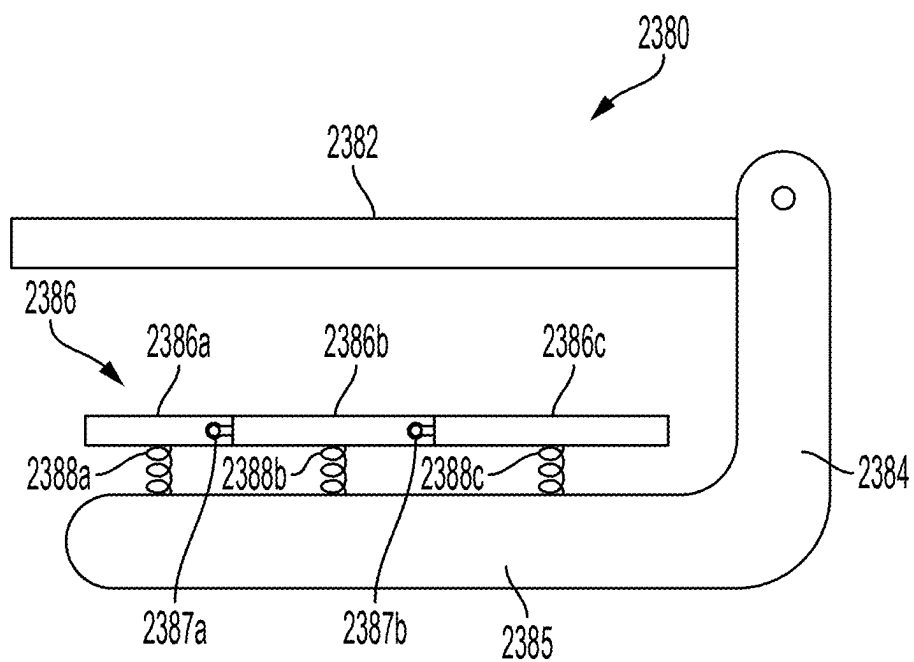


FIG. 165

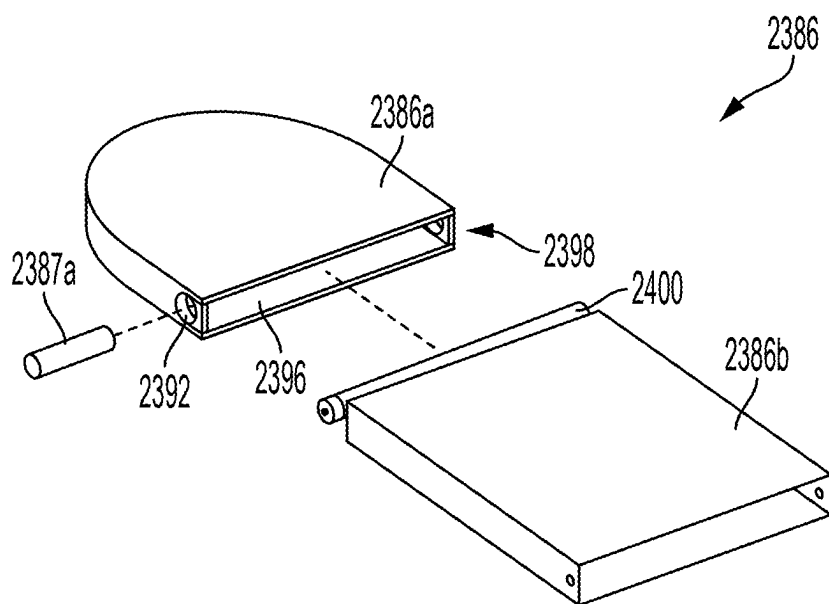


FIG. 166

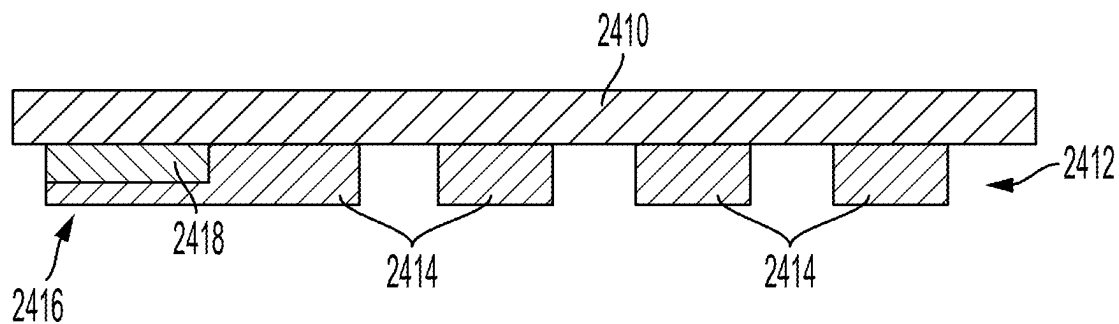


FIG. 167

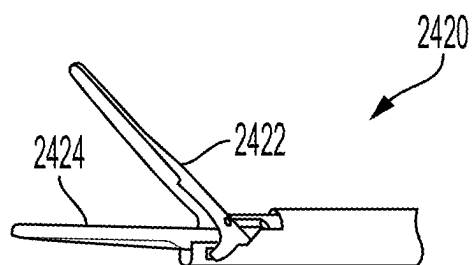


FIG. 168

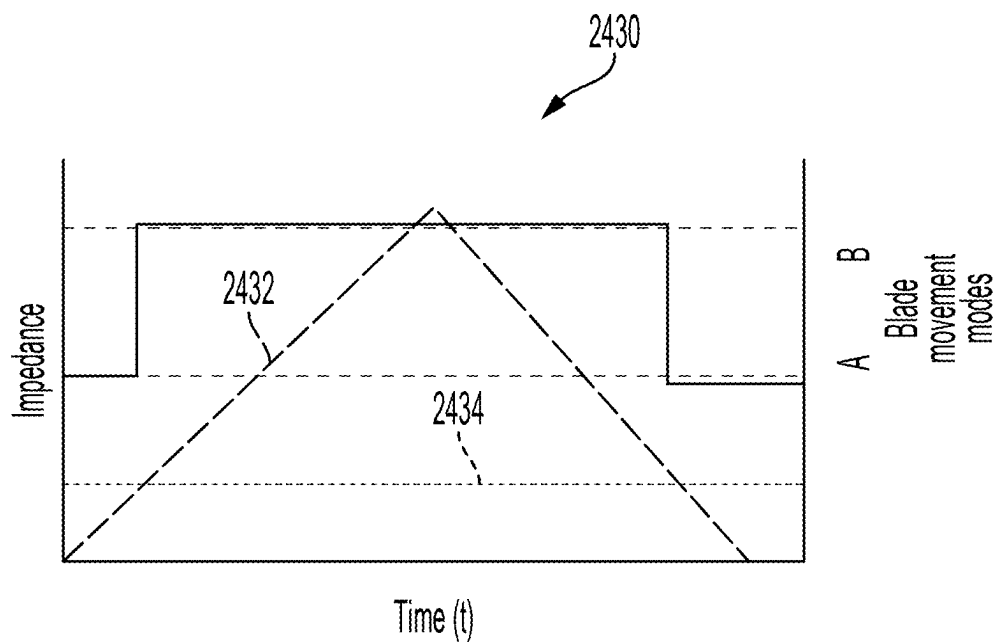


FIG. 169

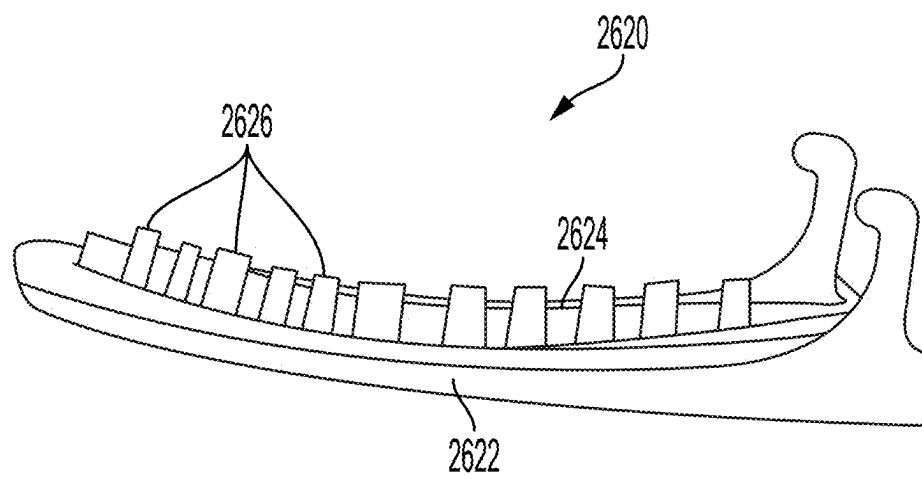


FIG. 170

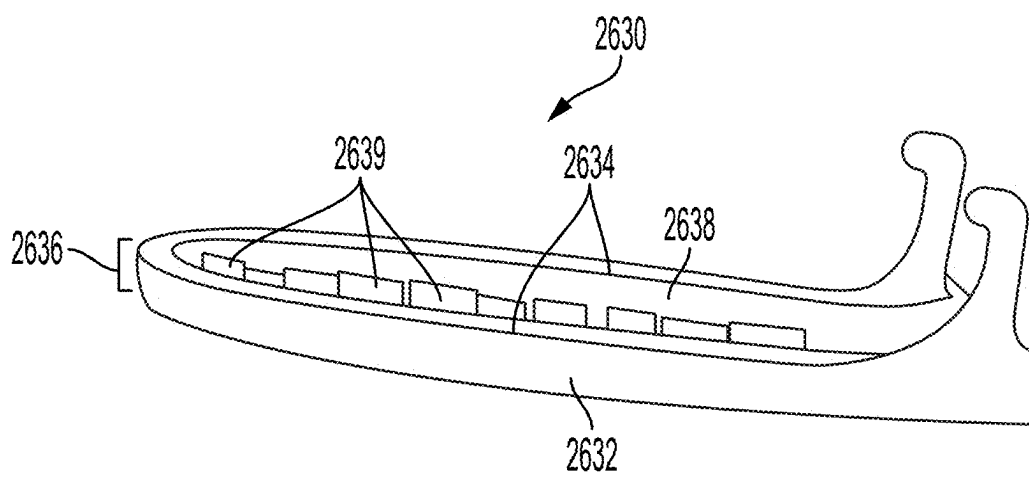


FIG. 171

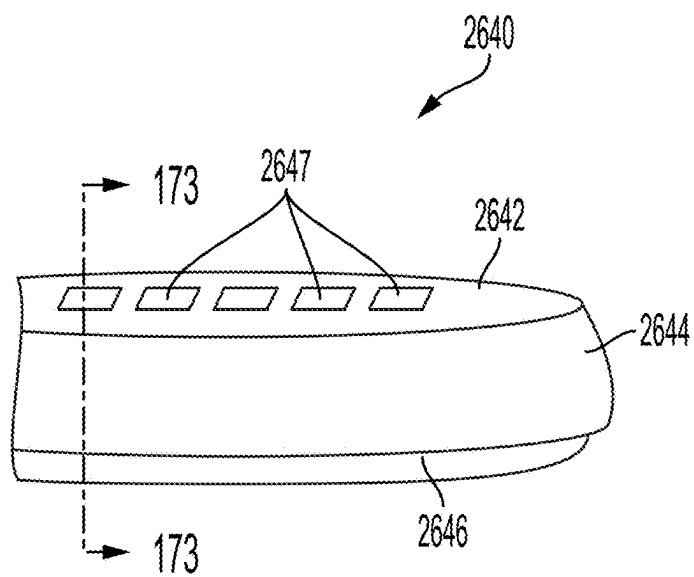


FIG. 172

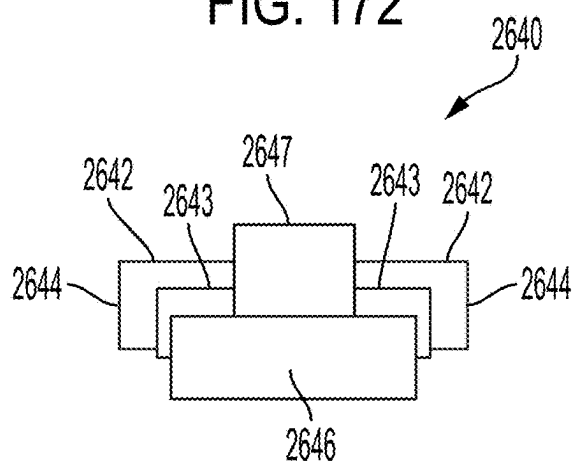


FIG. 173

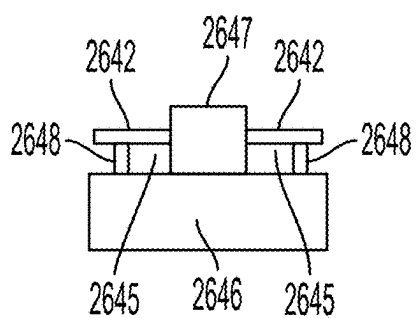


FIG. 174

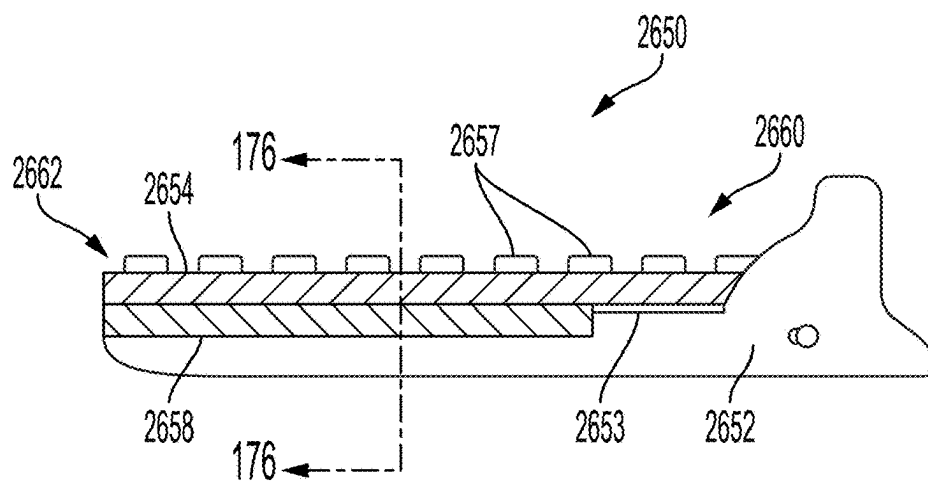


FIG. 175

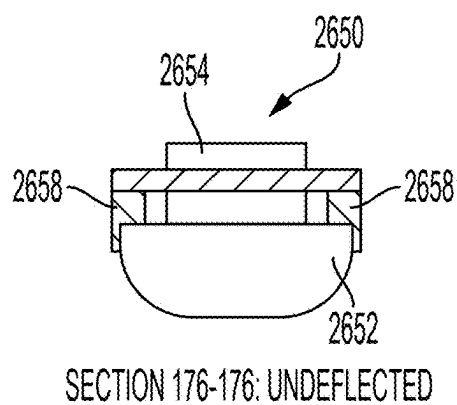


FIG. 176

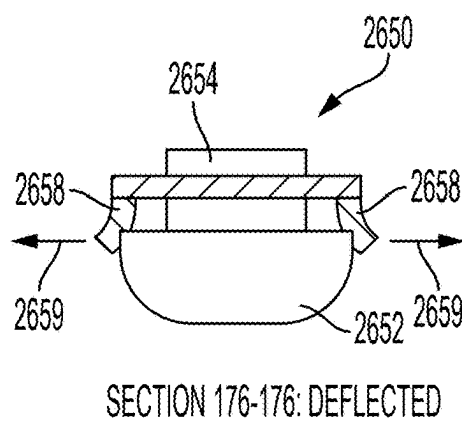


FIG. 177

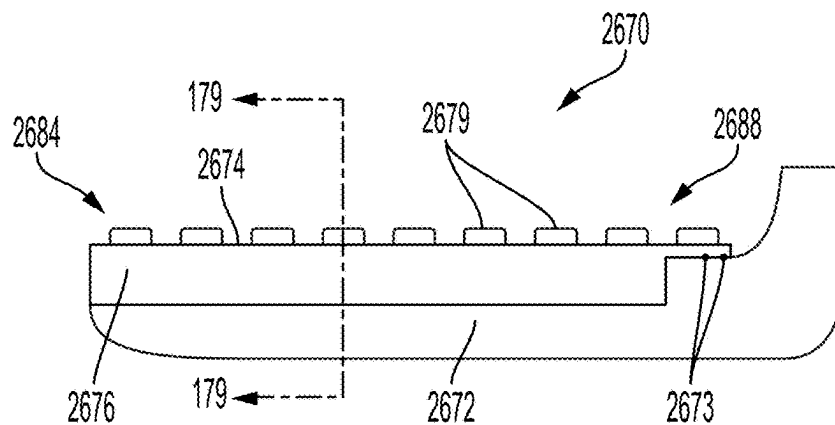


FIG. 178

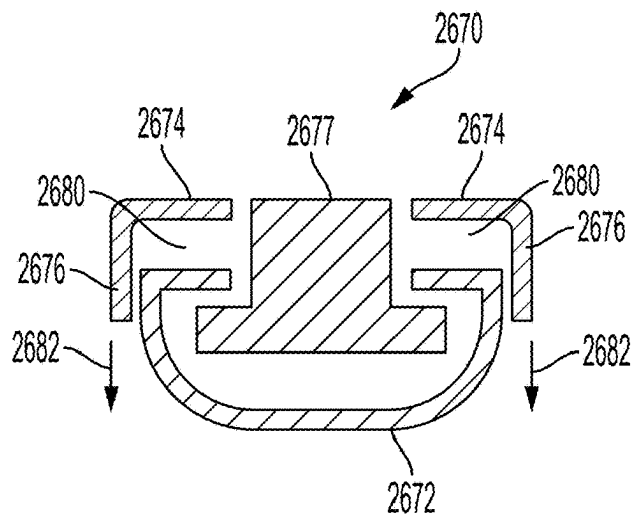


FIG. 179

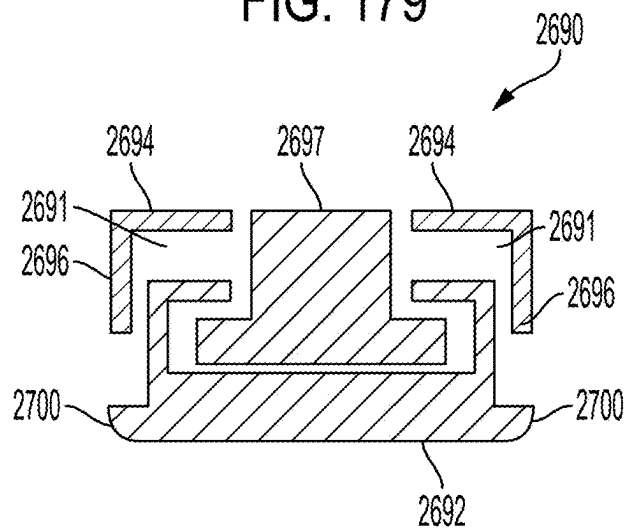


FIG. 180

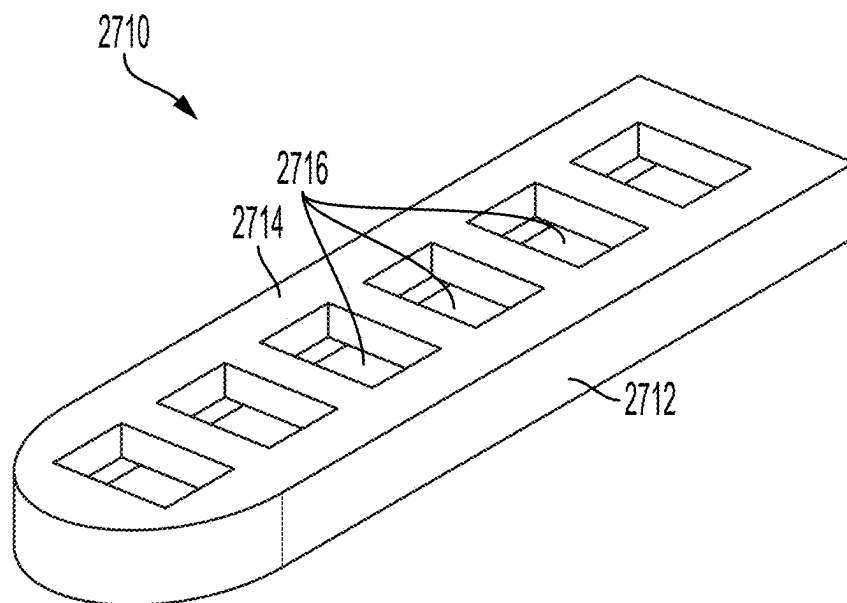


FIG. 181

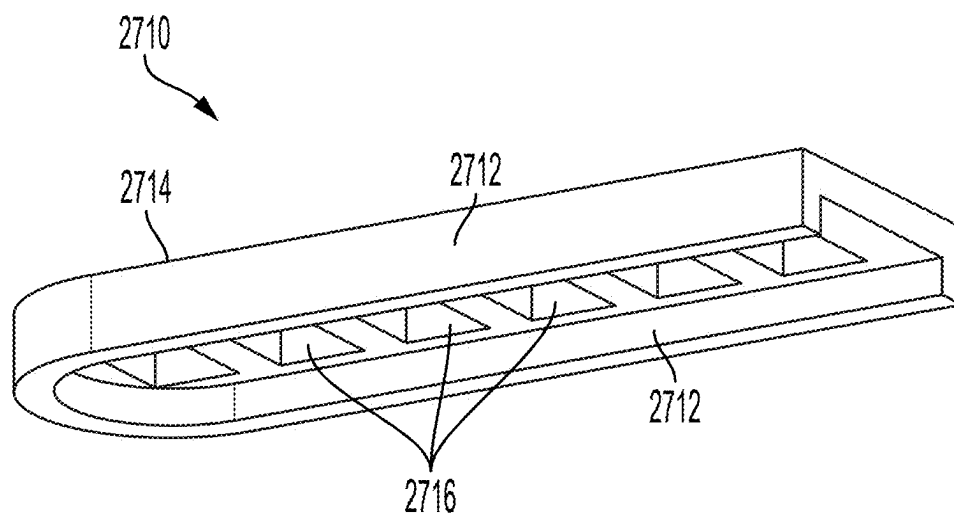


FIG. 182

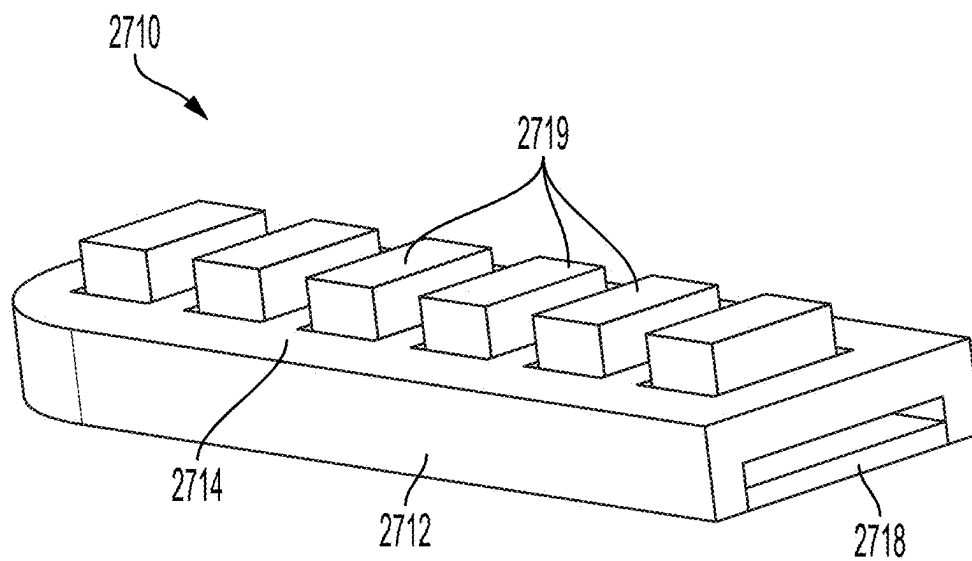


FIG. 183

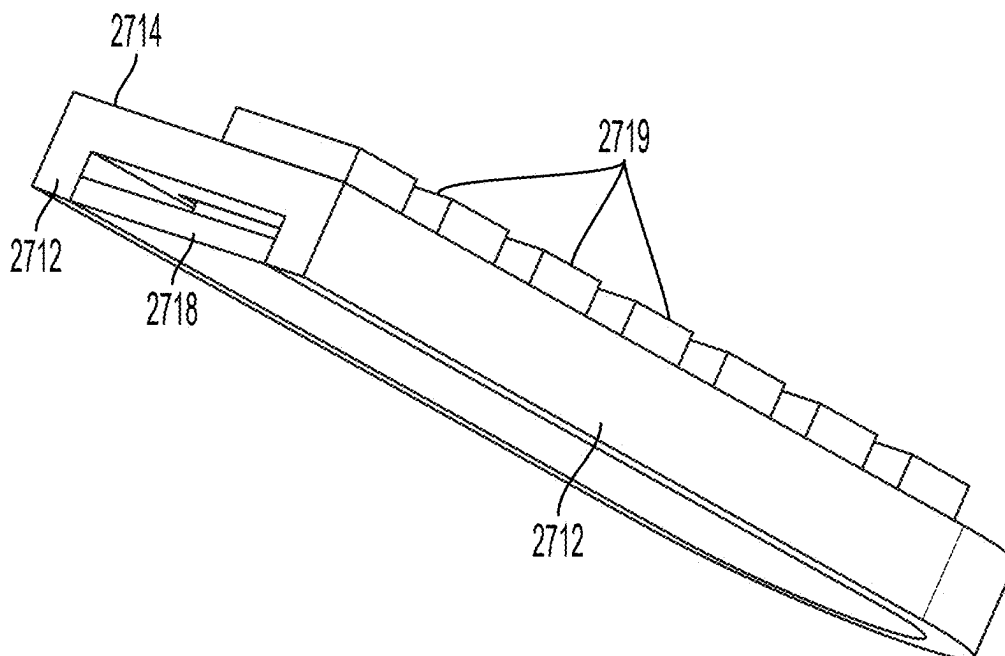
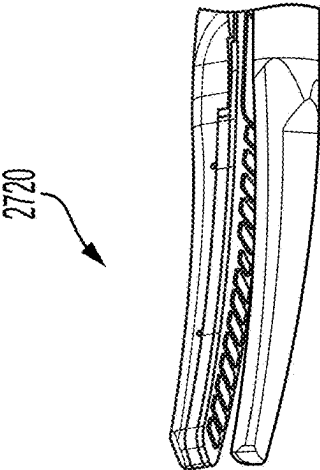
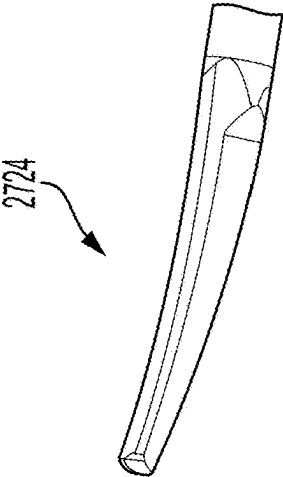


FIG. 184



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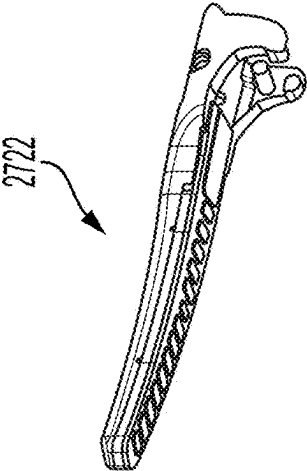


FIG. 185

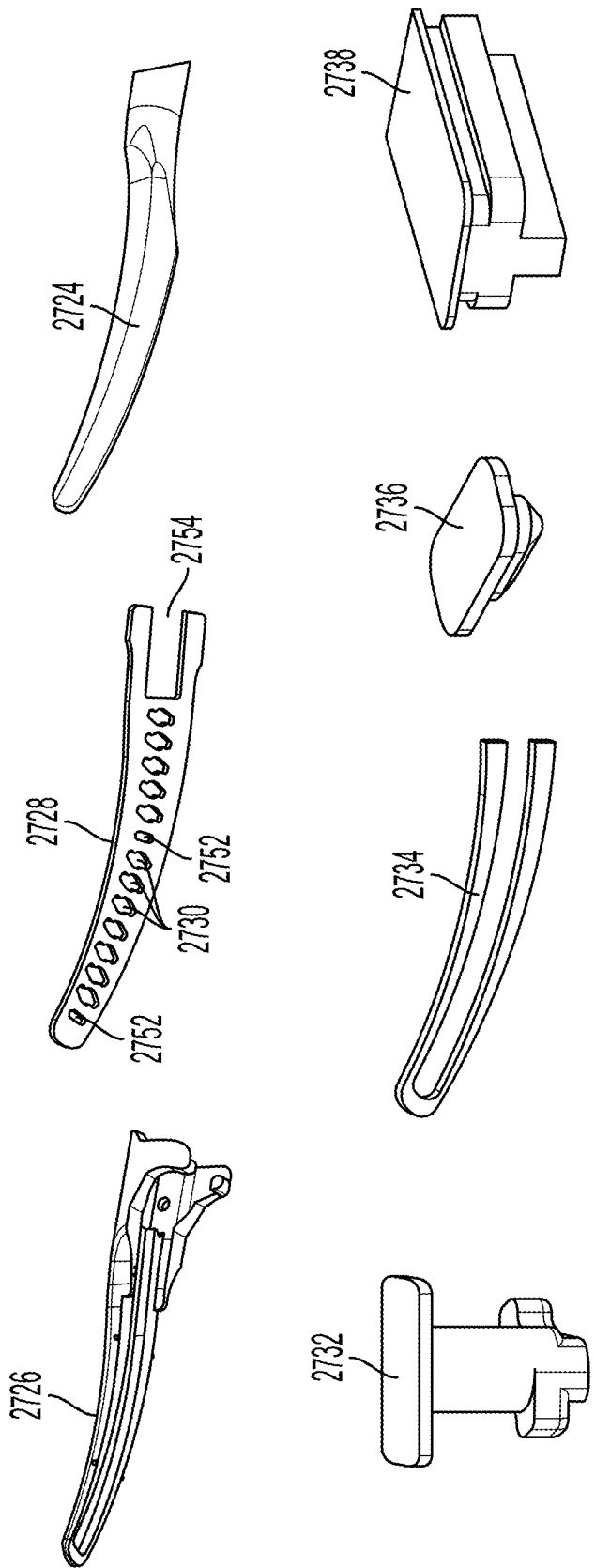


FIG. 186

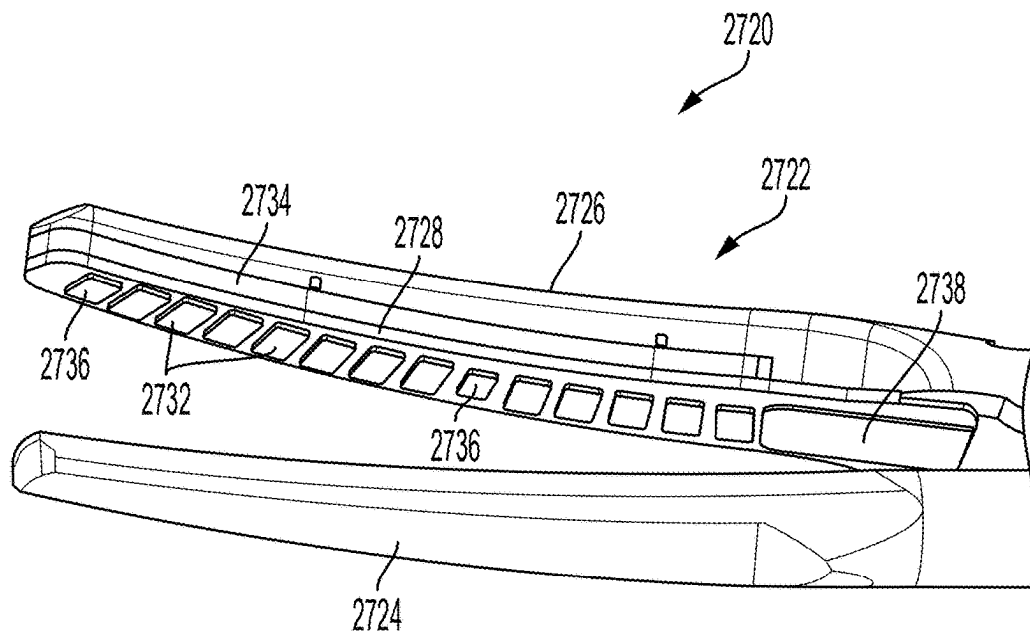


FIG. 187

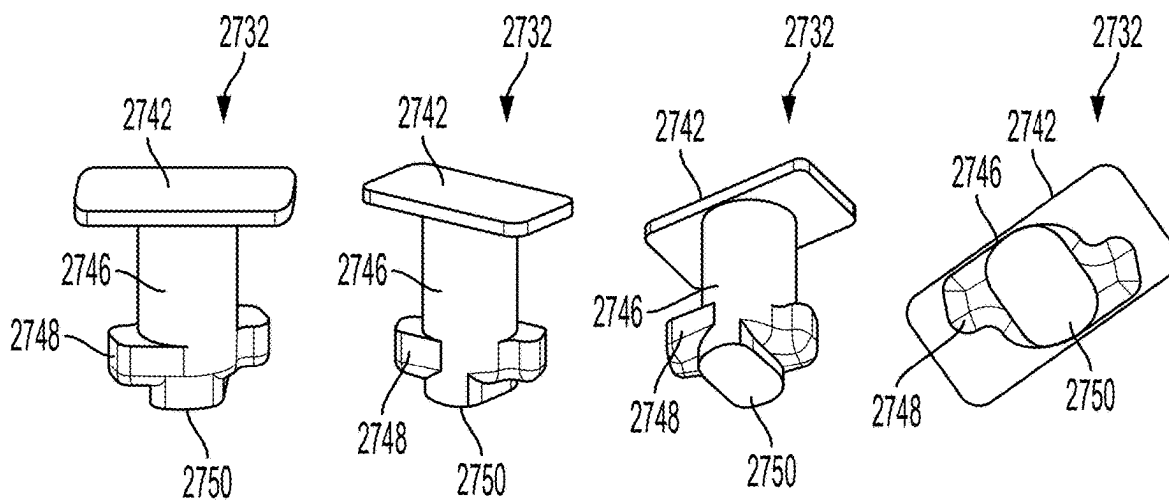


FIG. 188

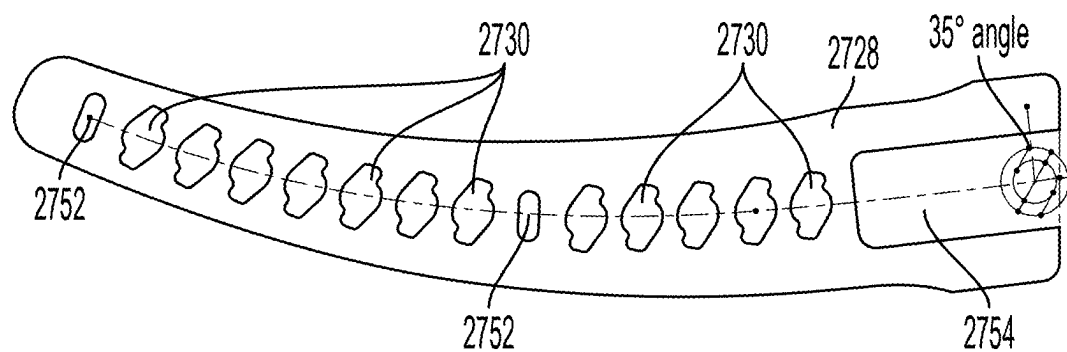


FIG. 189

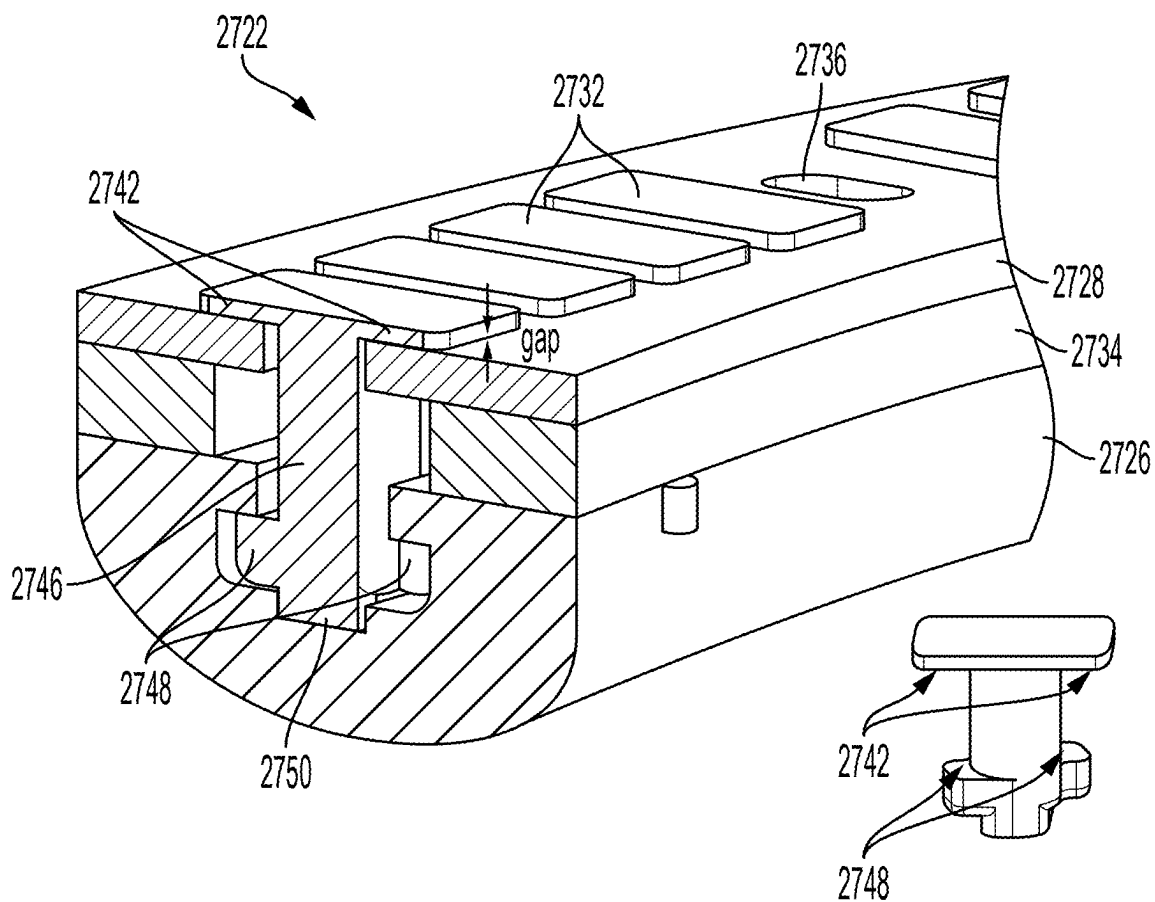


FIG. 190

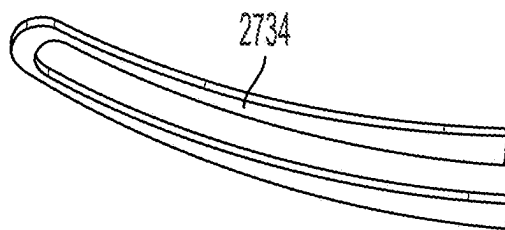


FIG. 191

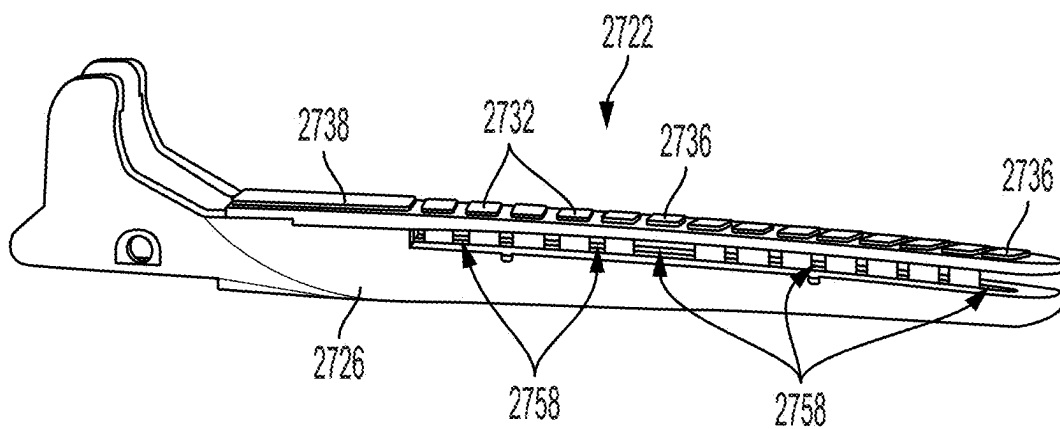


FIG. 192

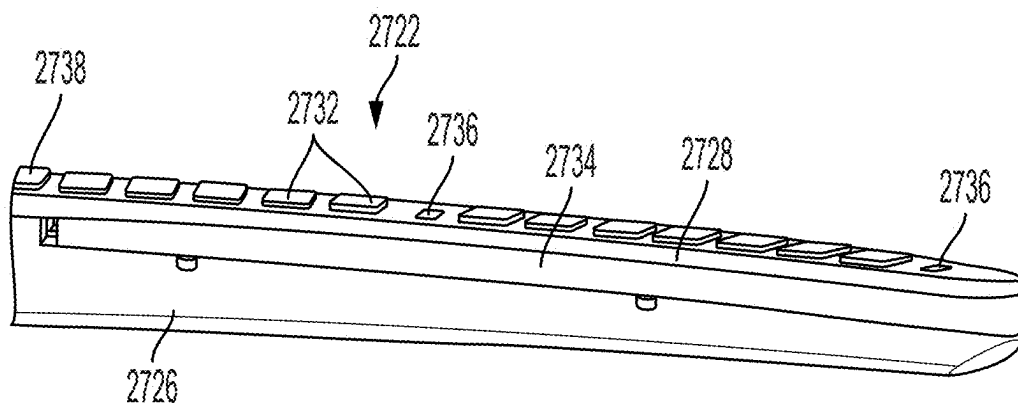


FIG. 193

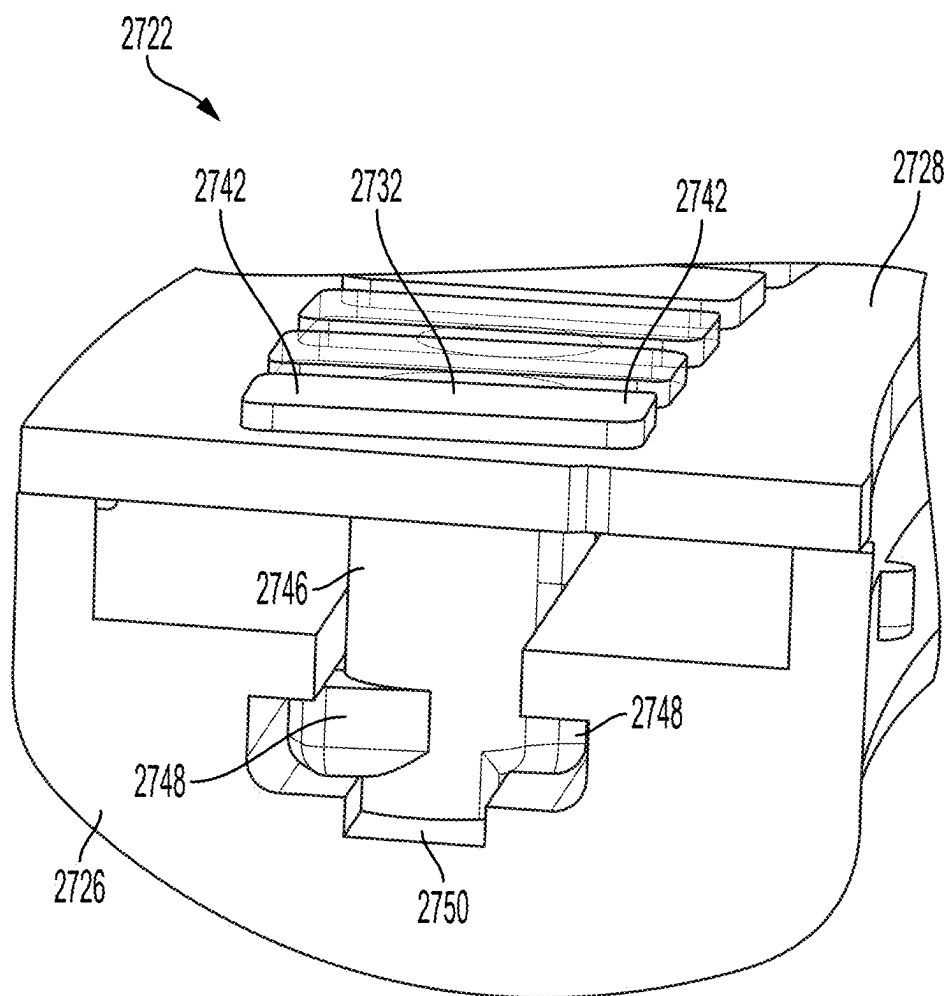


FIG. 194

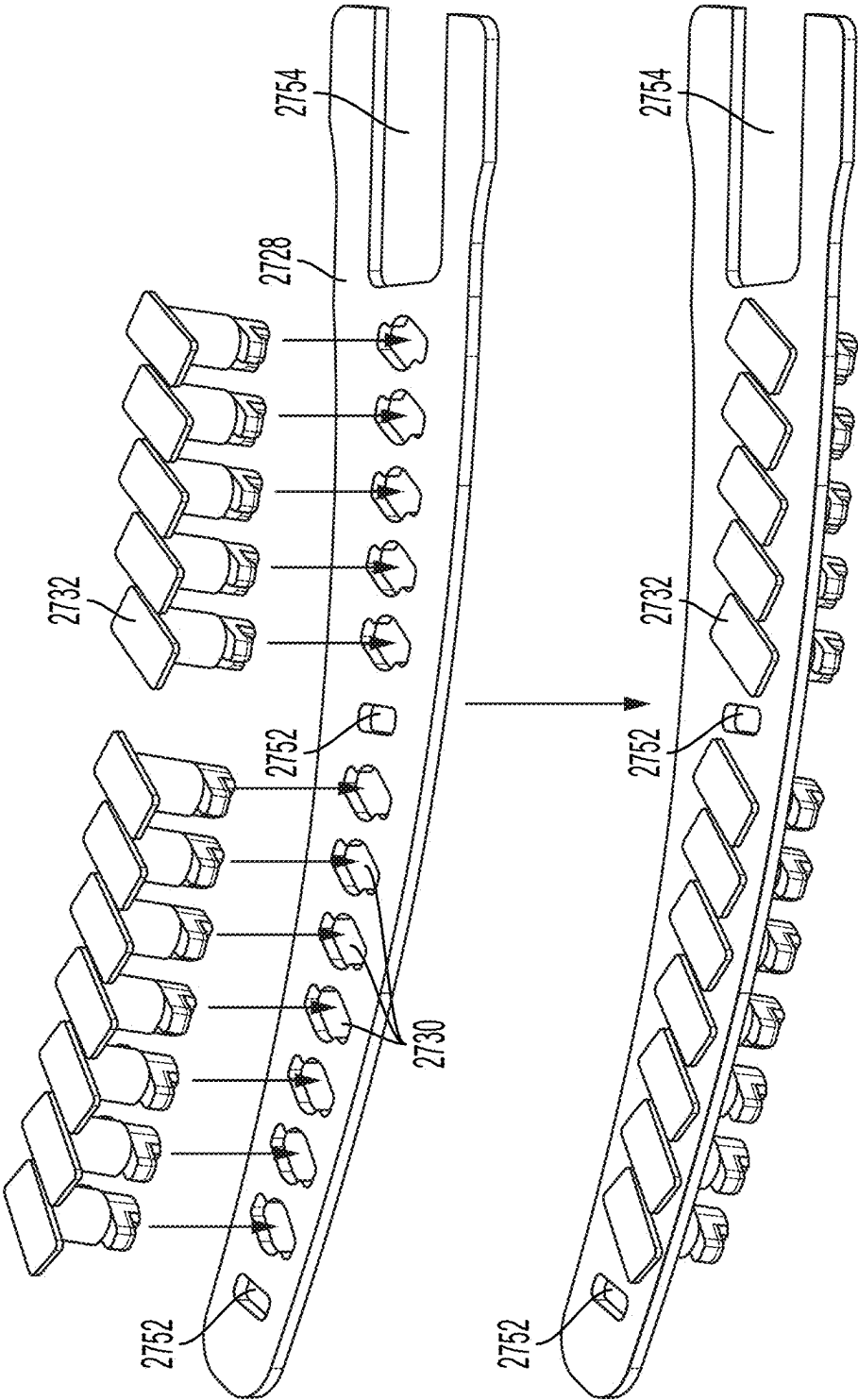


FIG. 195

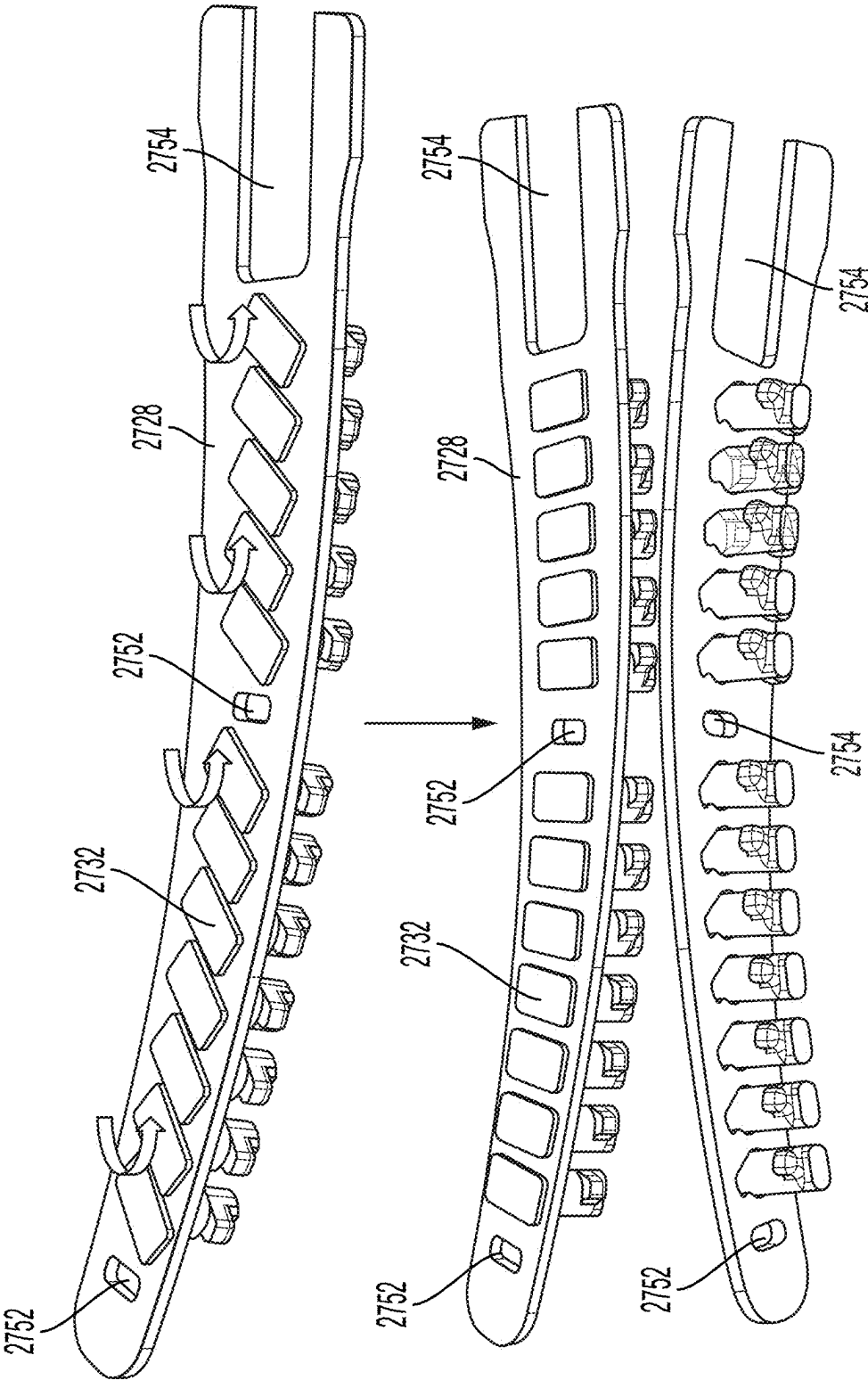


FIG. 196

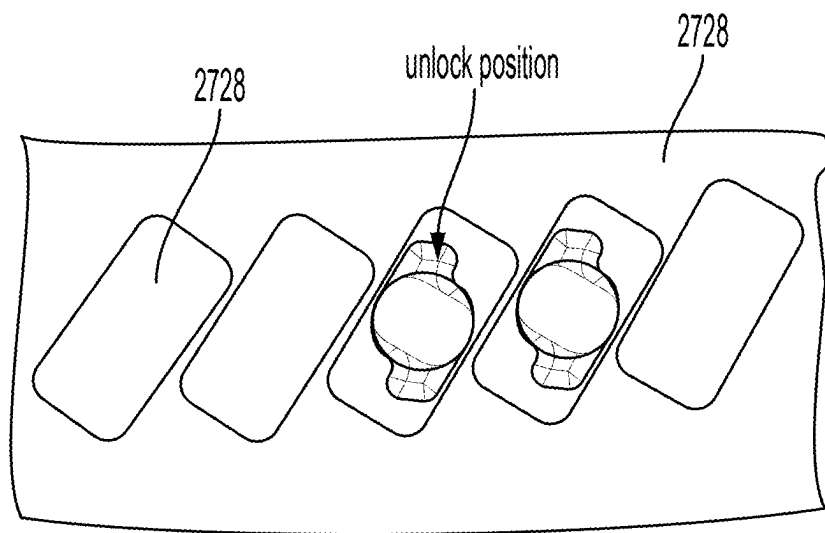


FIG. 197

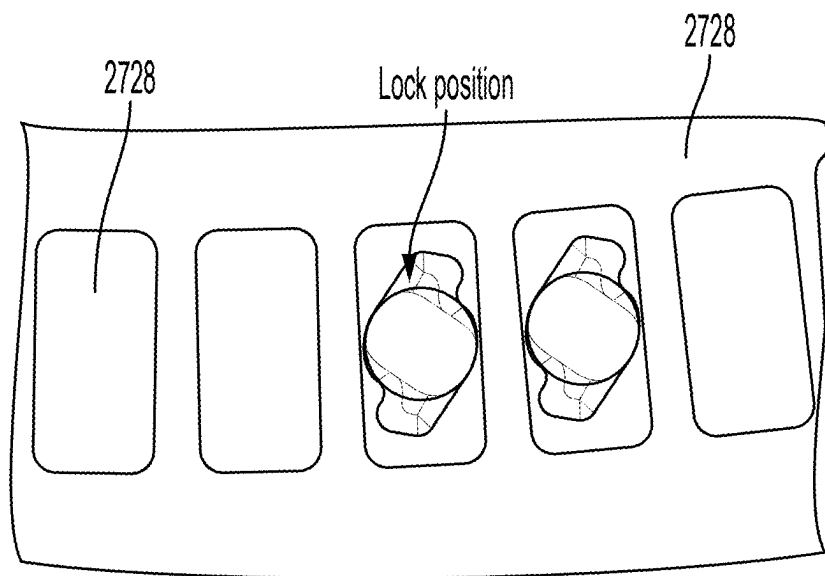


FIG. 198

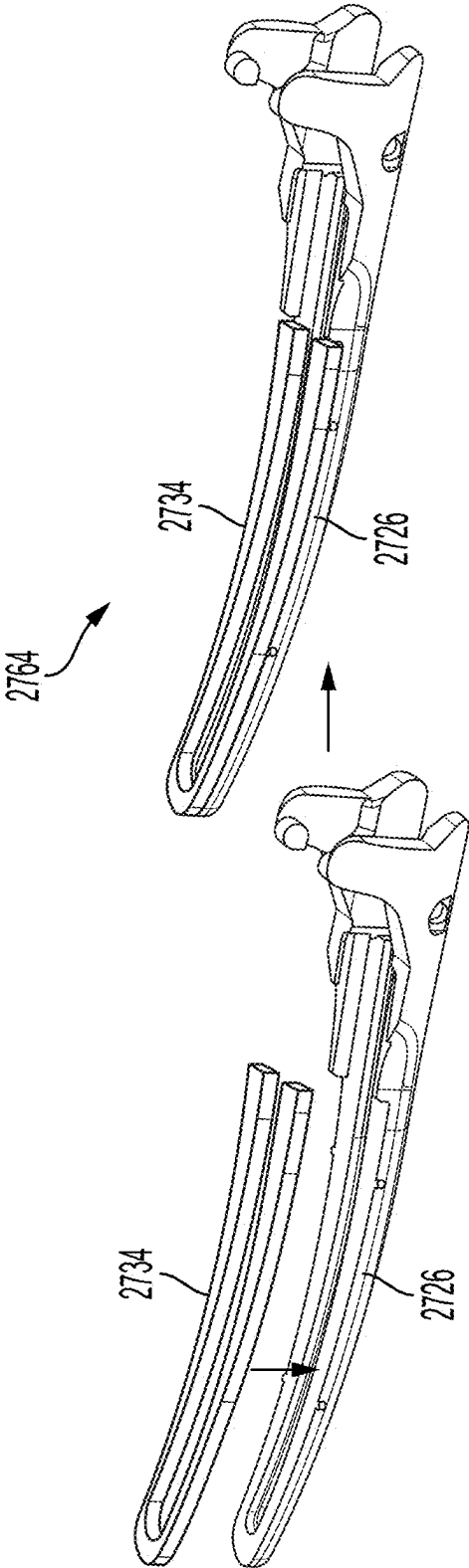


FIG. 199

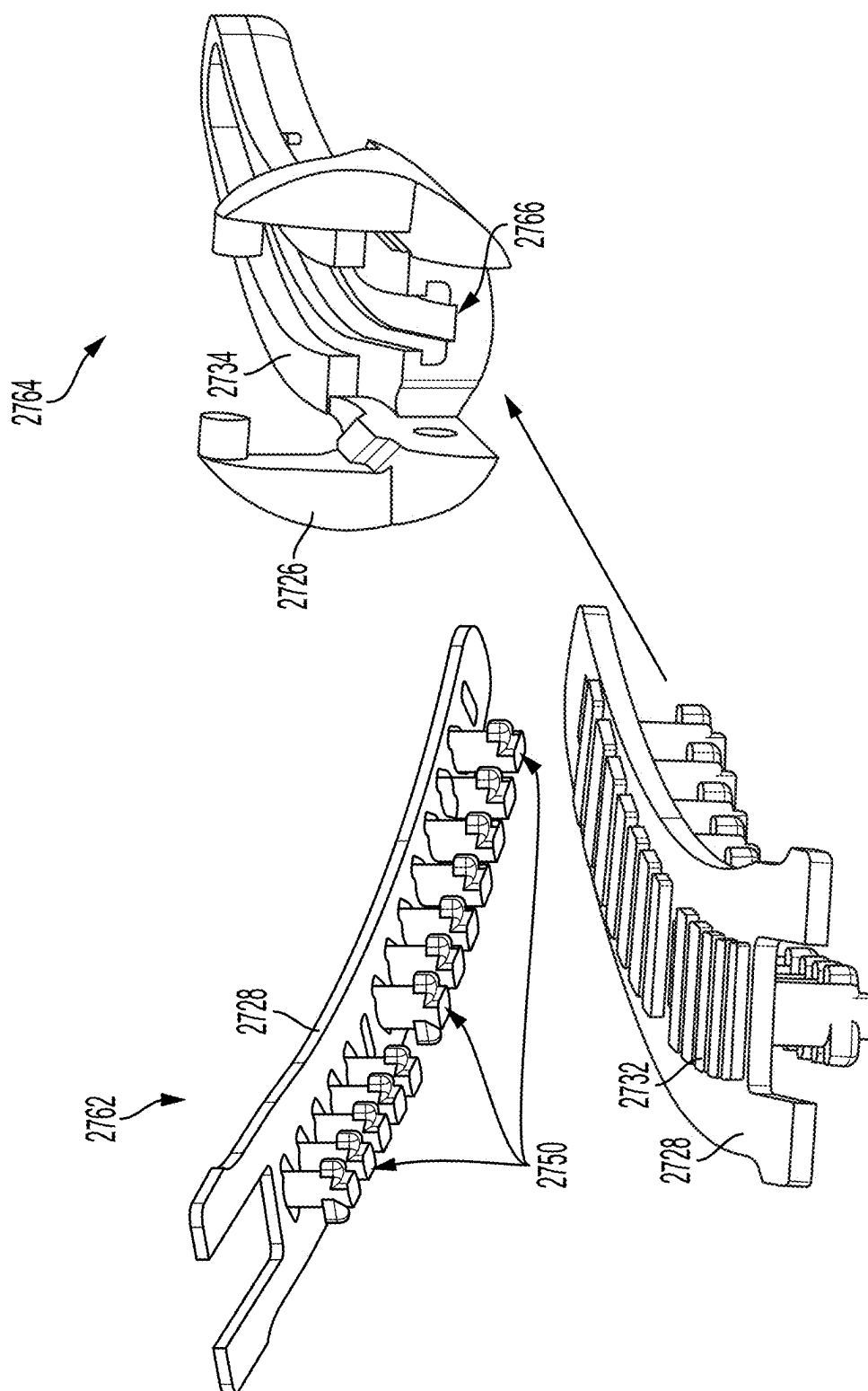


FIG. 200

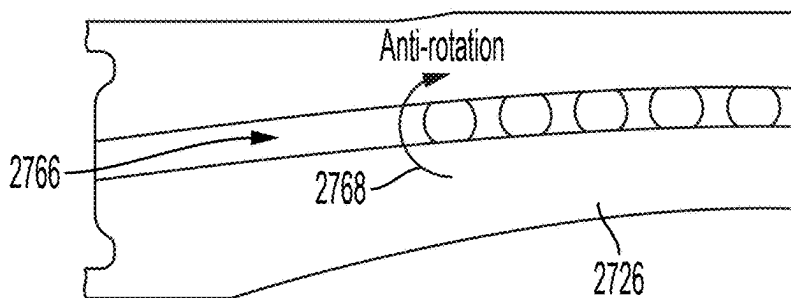


FIG. 201

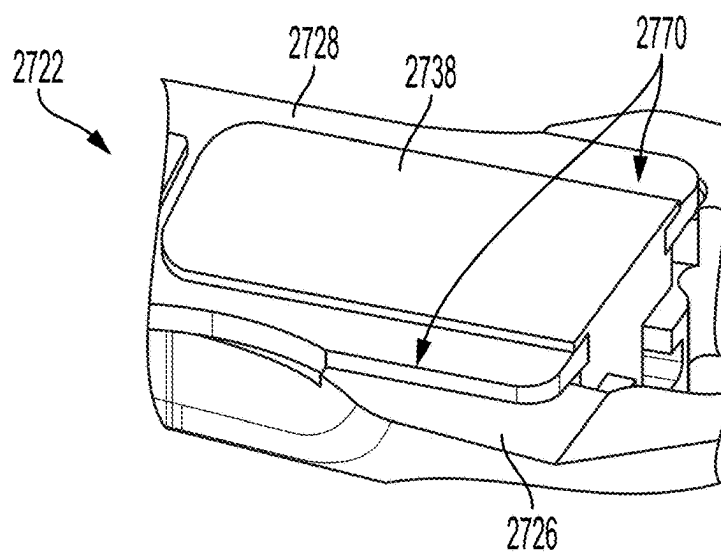


FIG. 202

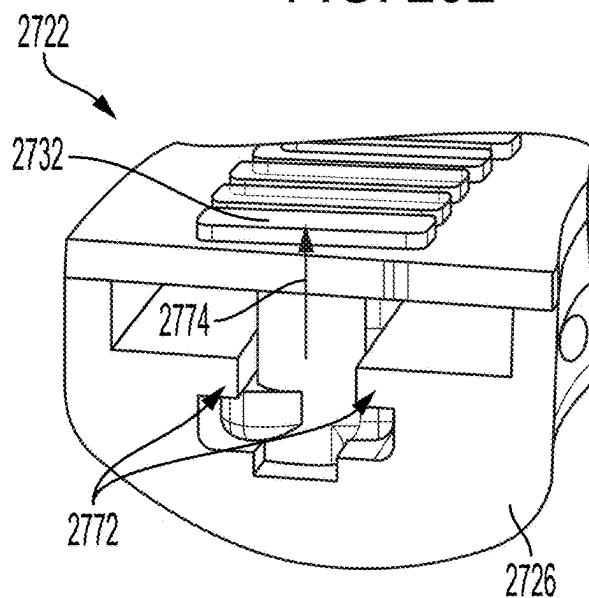


FIG. 203

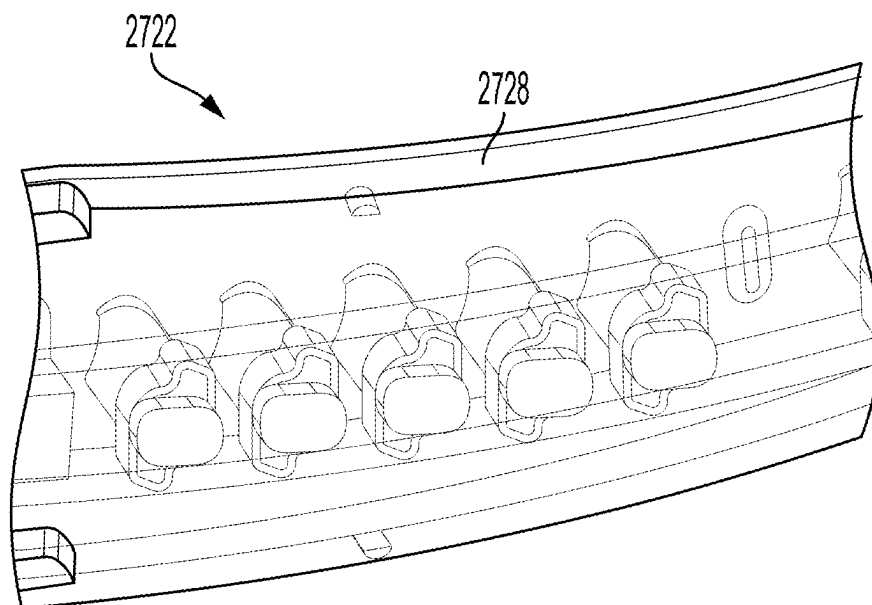


FIG. 204

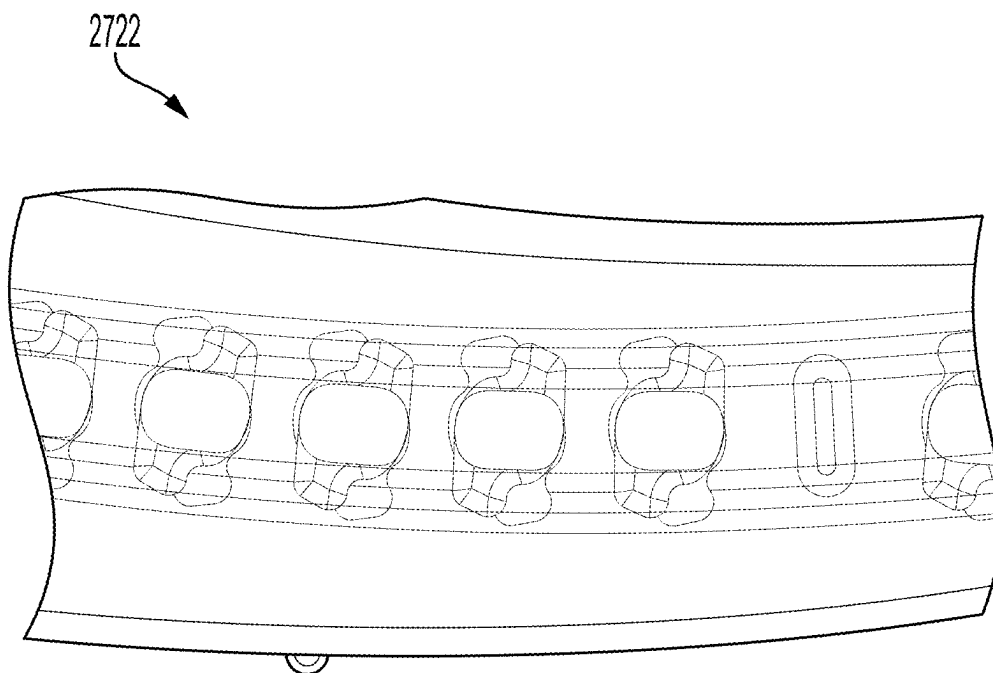


FIG. 205

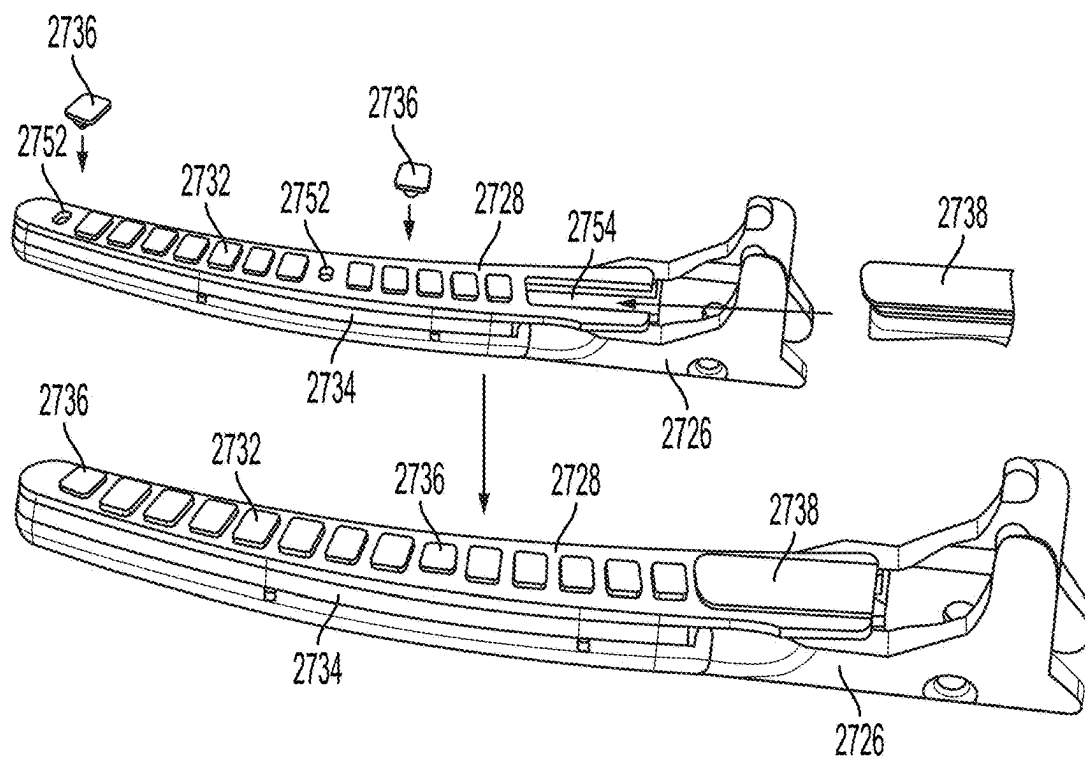


FIG. 206

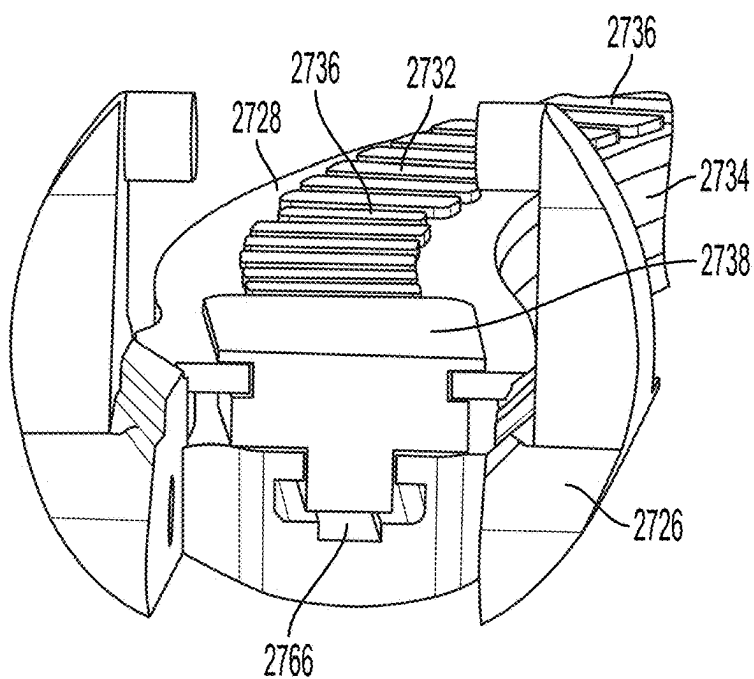


FIG. 207

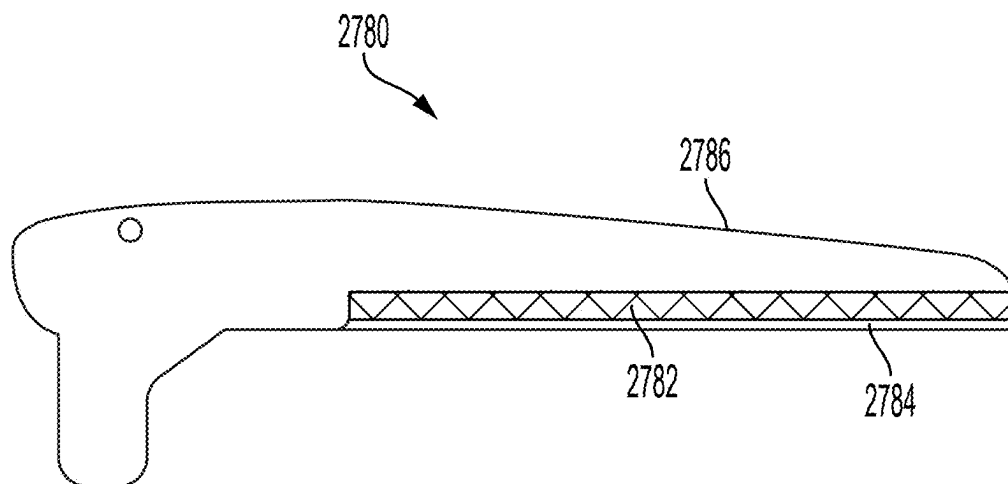


FIG. 208

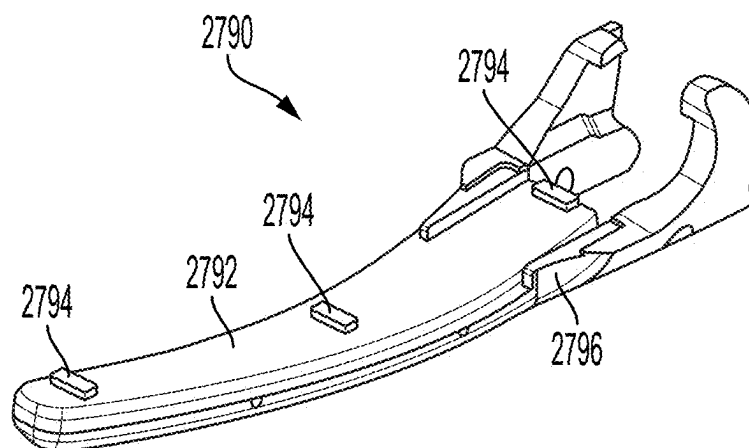


FIG. 209

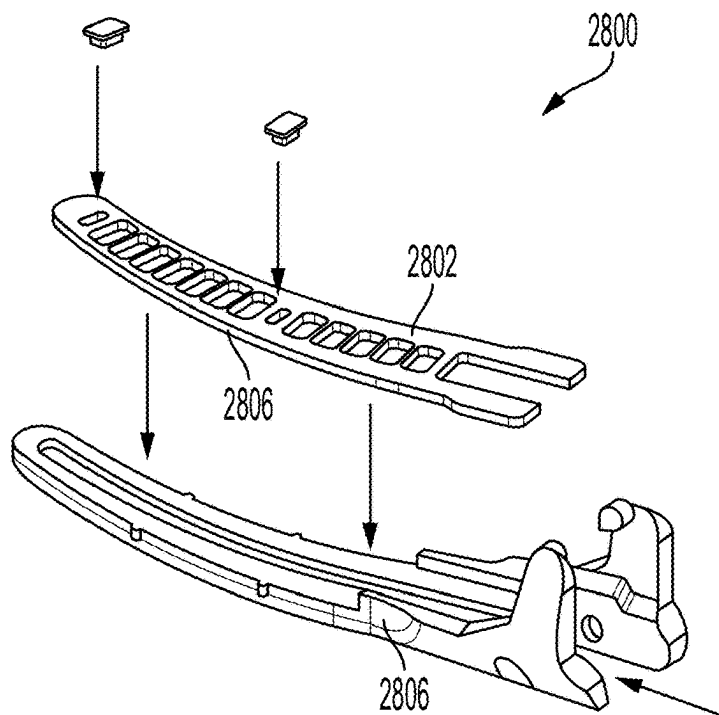


FIG. 210

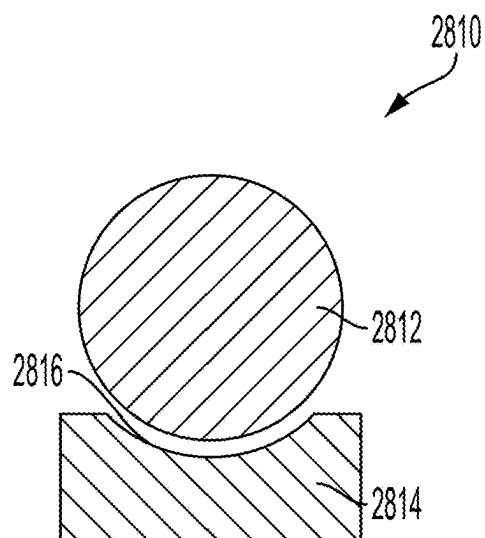


FIG. 211

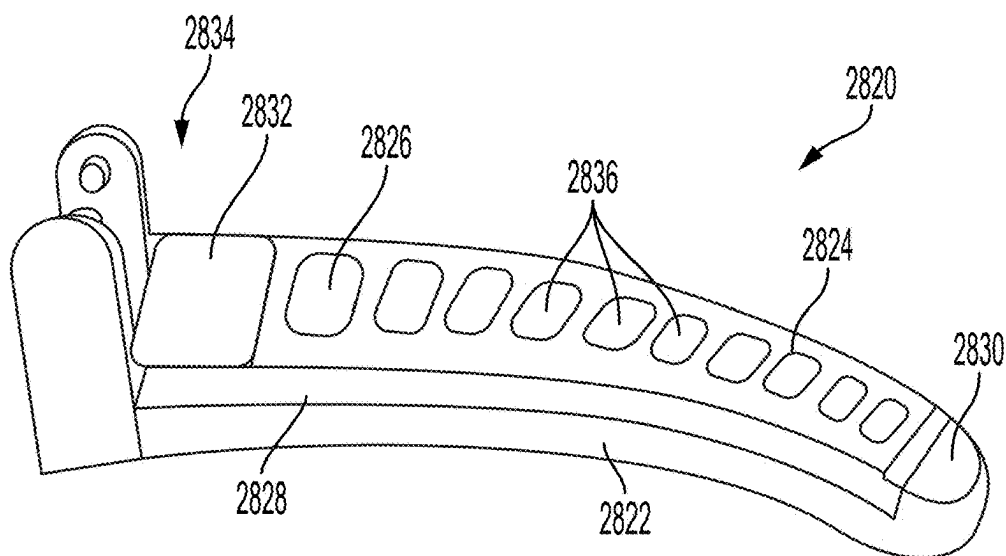


FIG. 212

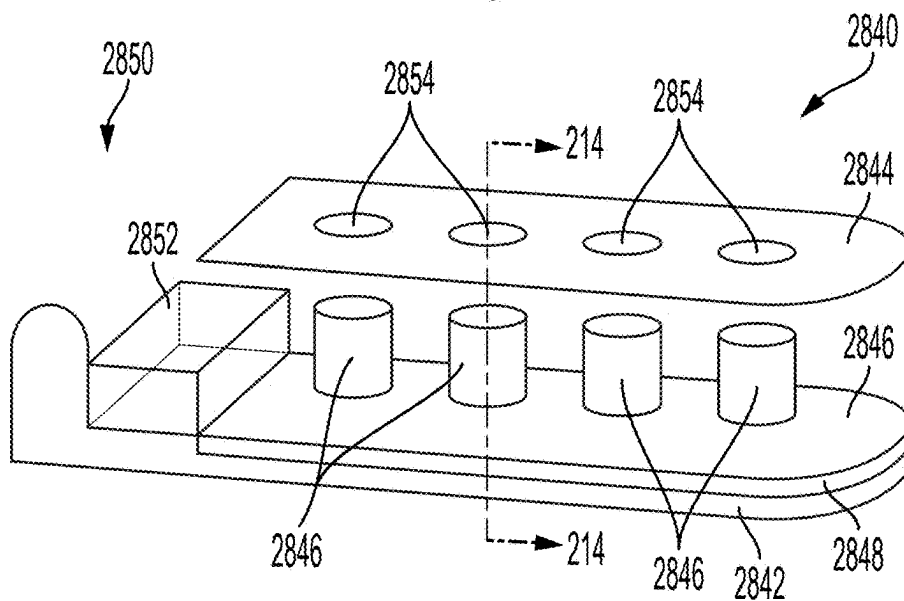


FIG. 213

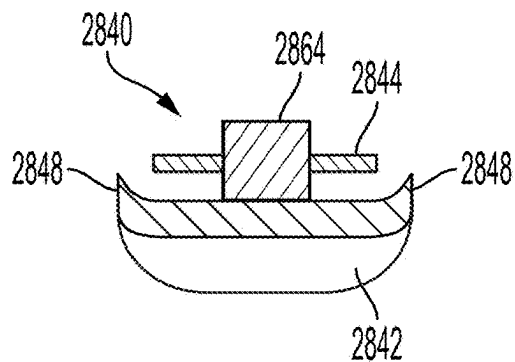


FIG. 214

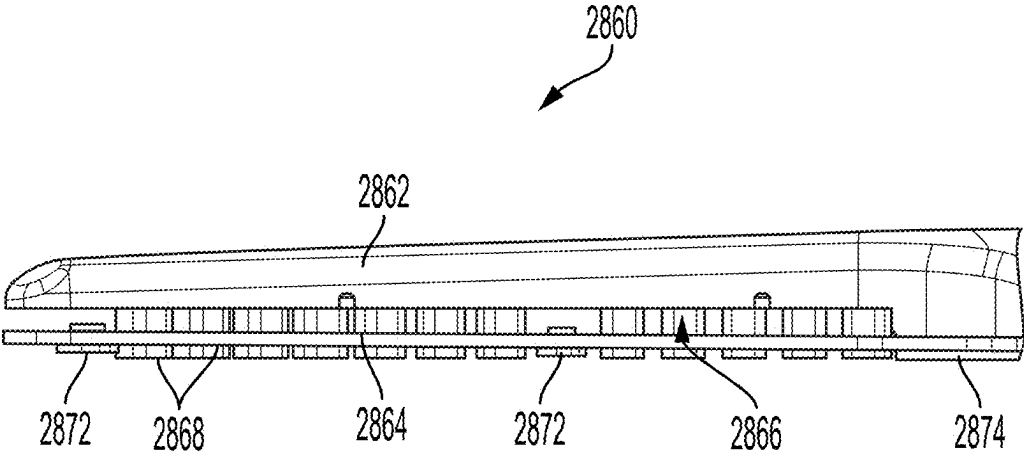


FIG. 215

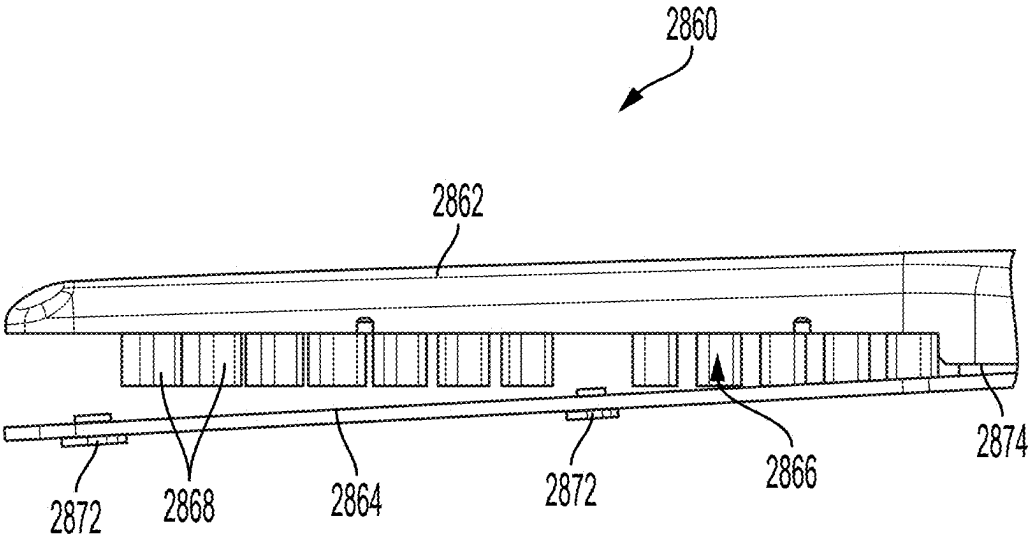


FIG. 216

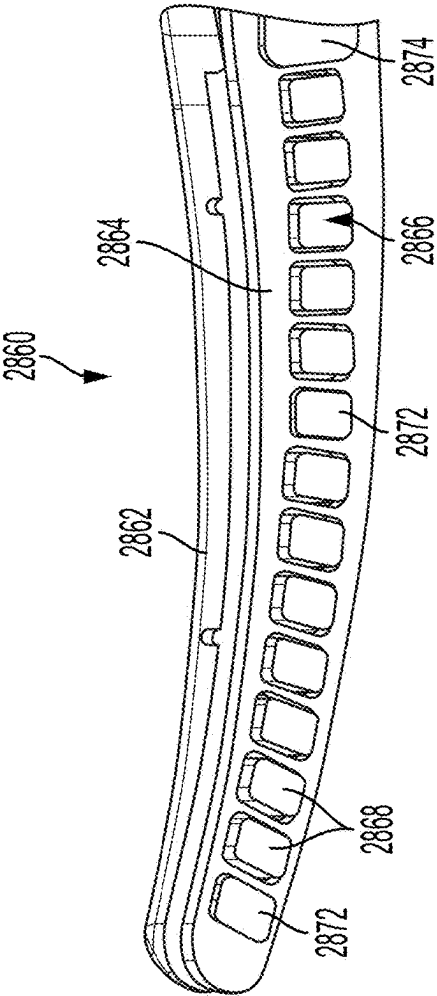


FIG. 217

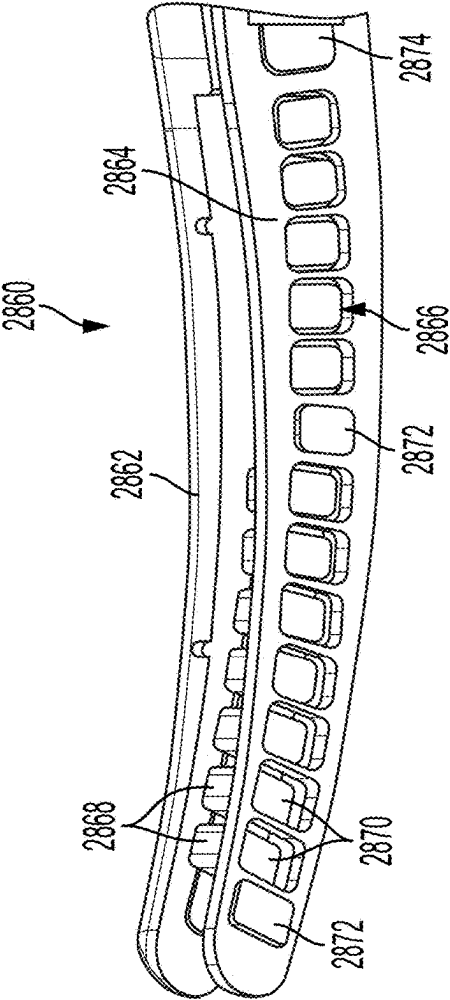


FIG. 218

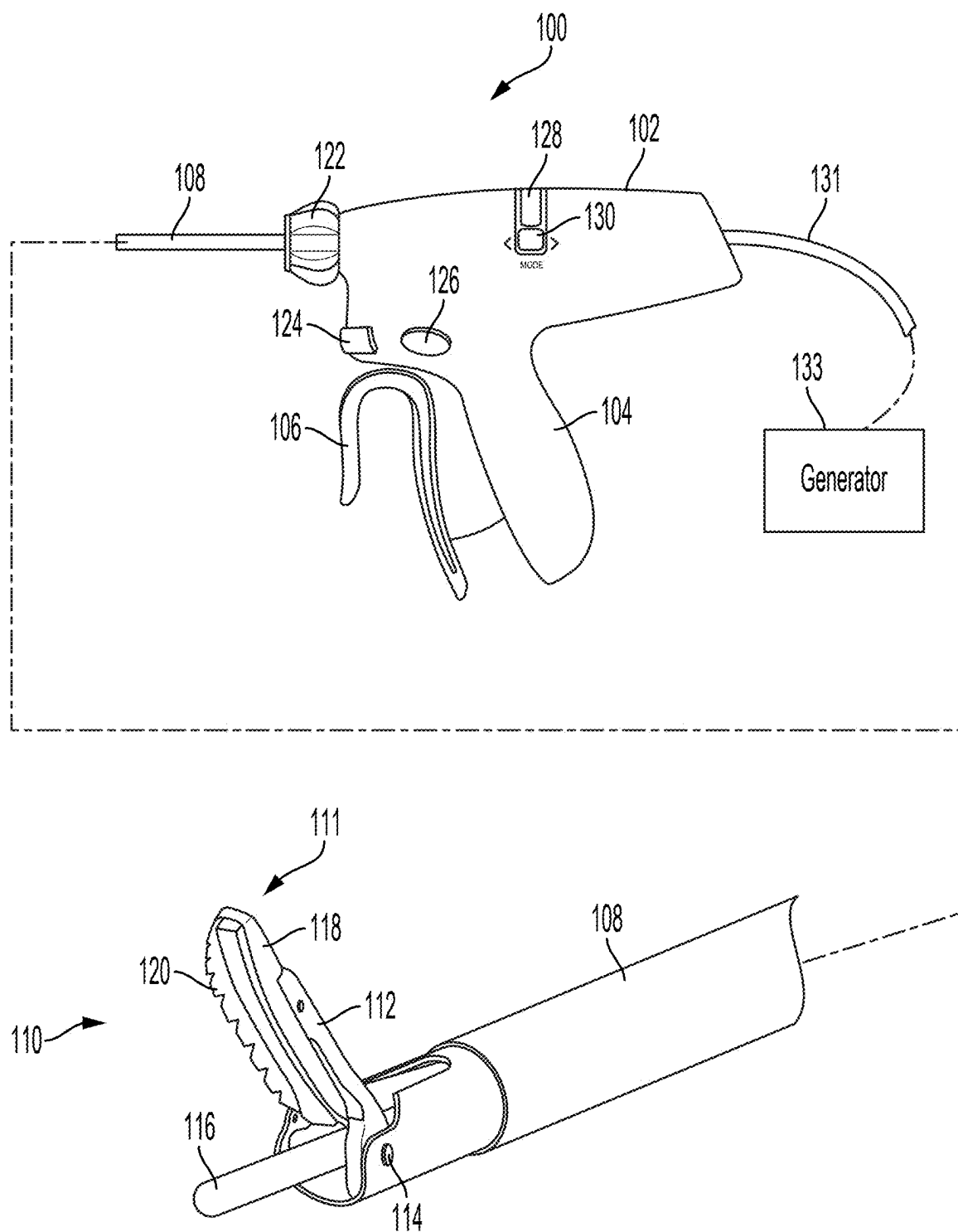


FIG. 219

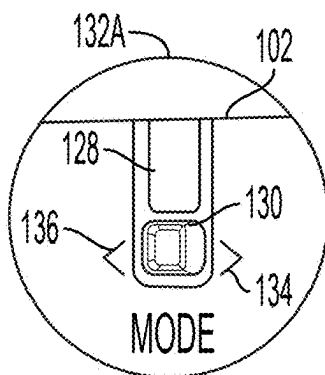


FIG. 220A

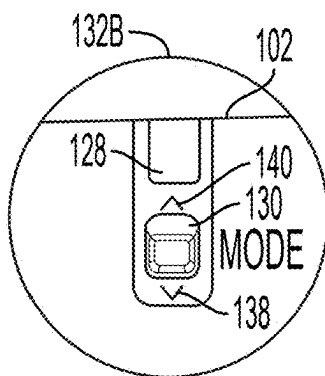


FIG. 220B

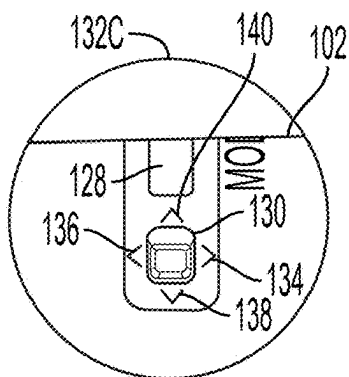


FIG. 220C

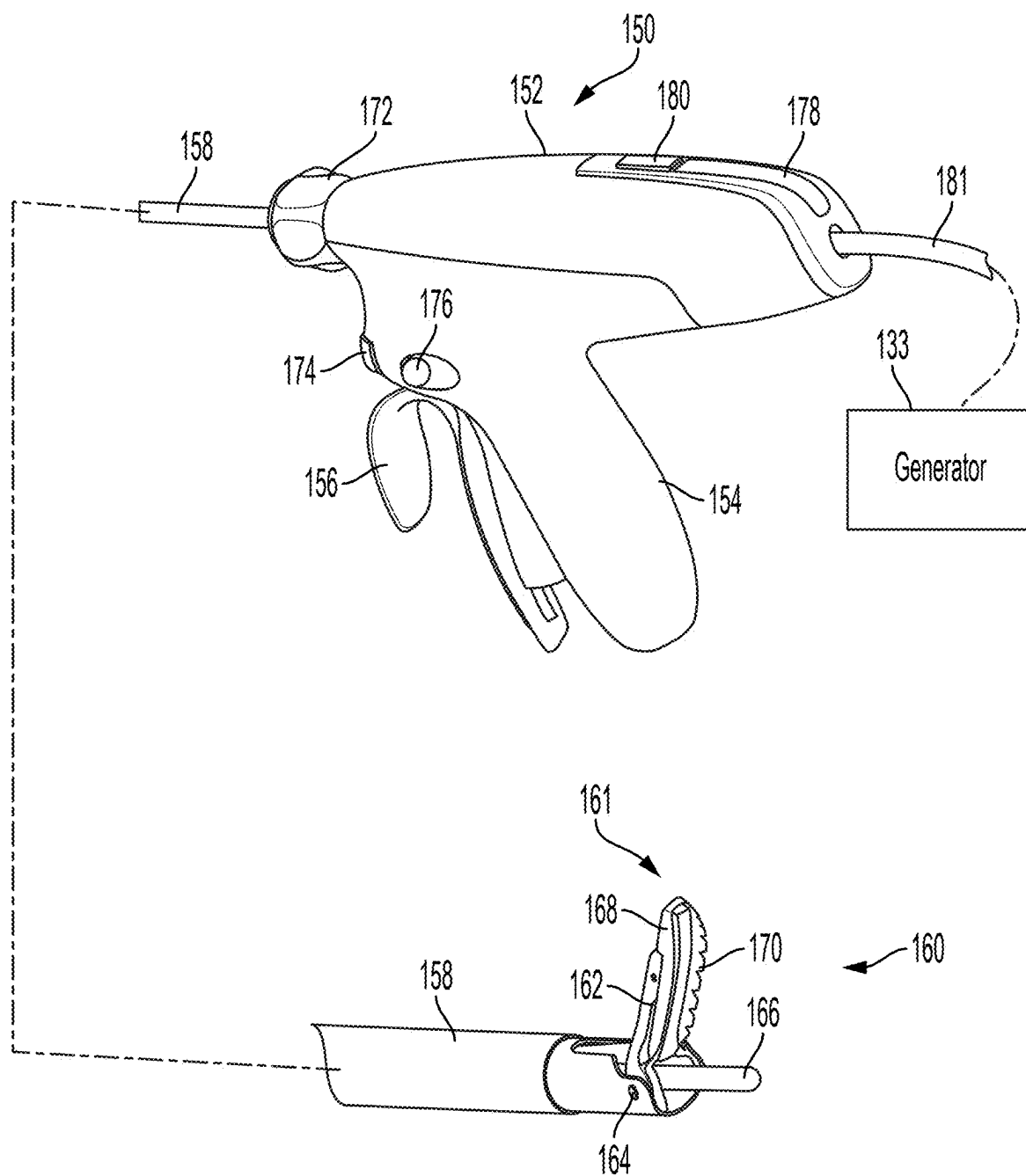
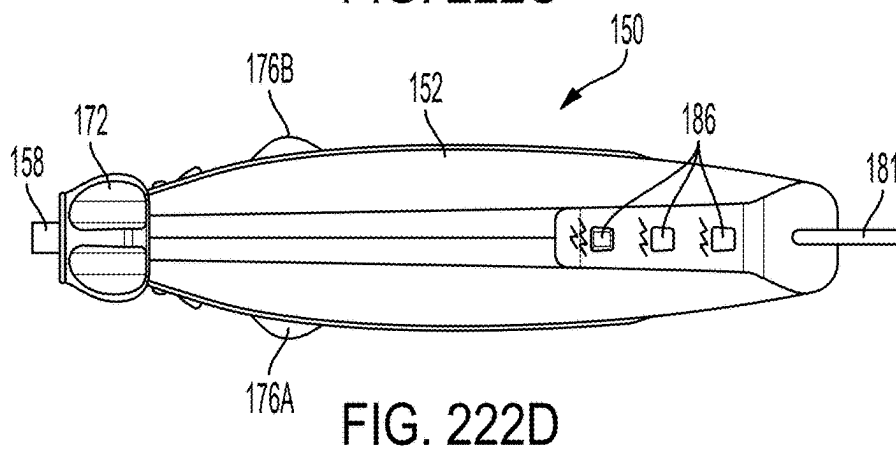
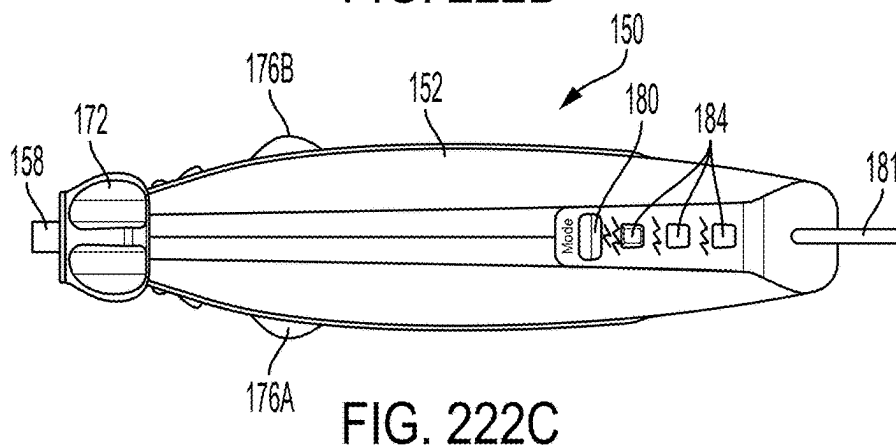
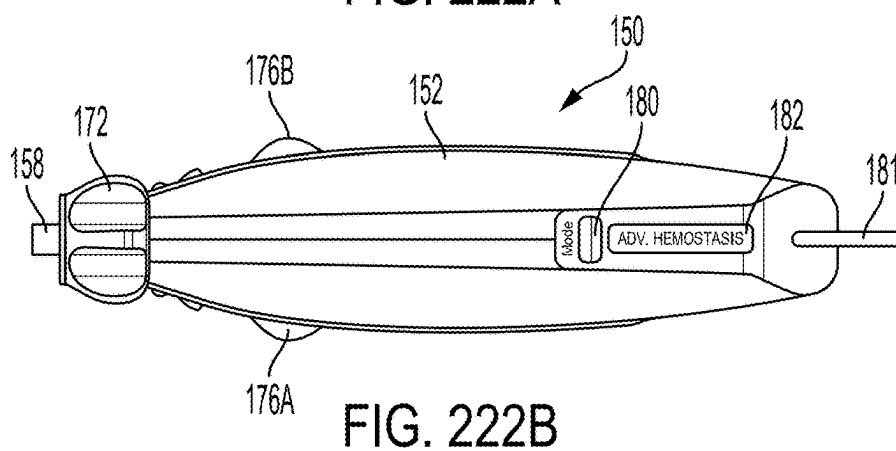
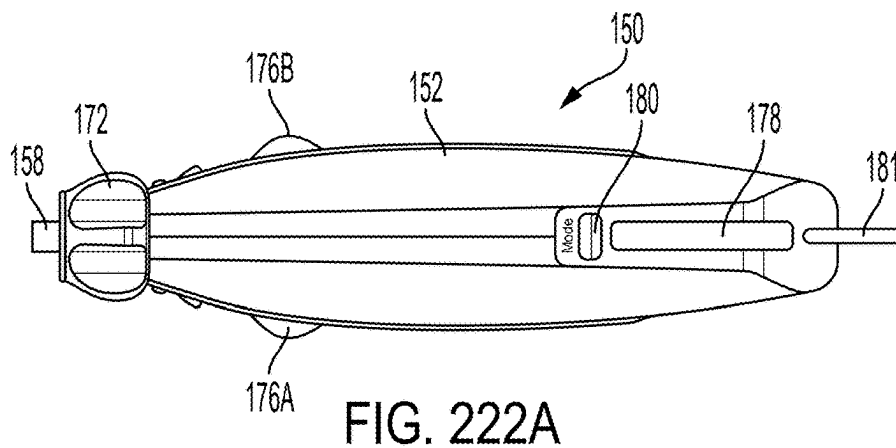


FIG. 221



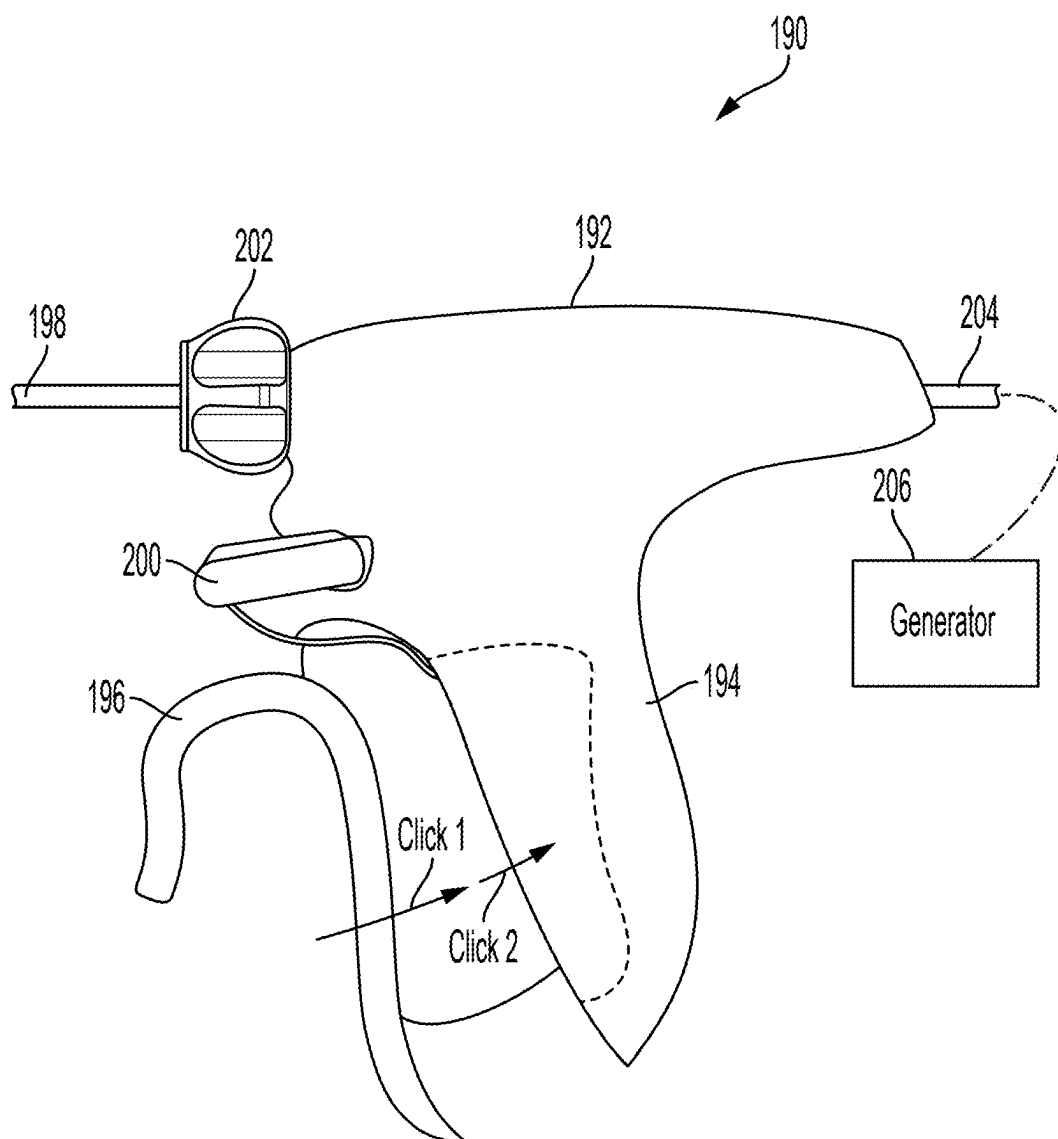


FIG. 223

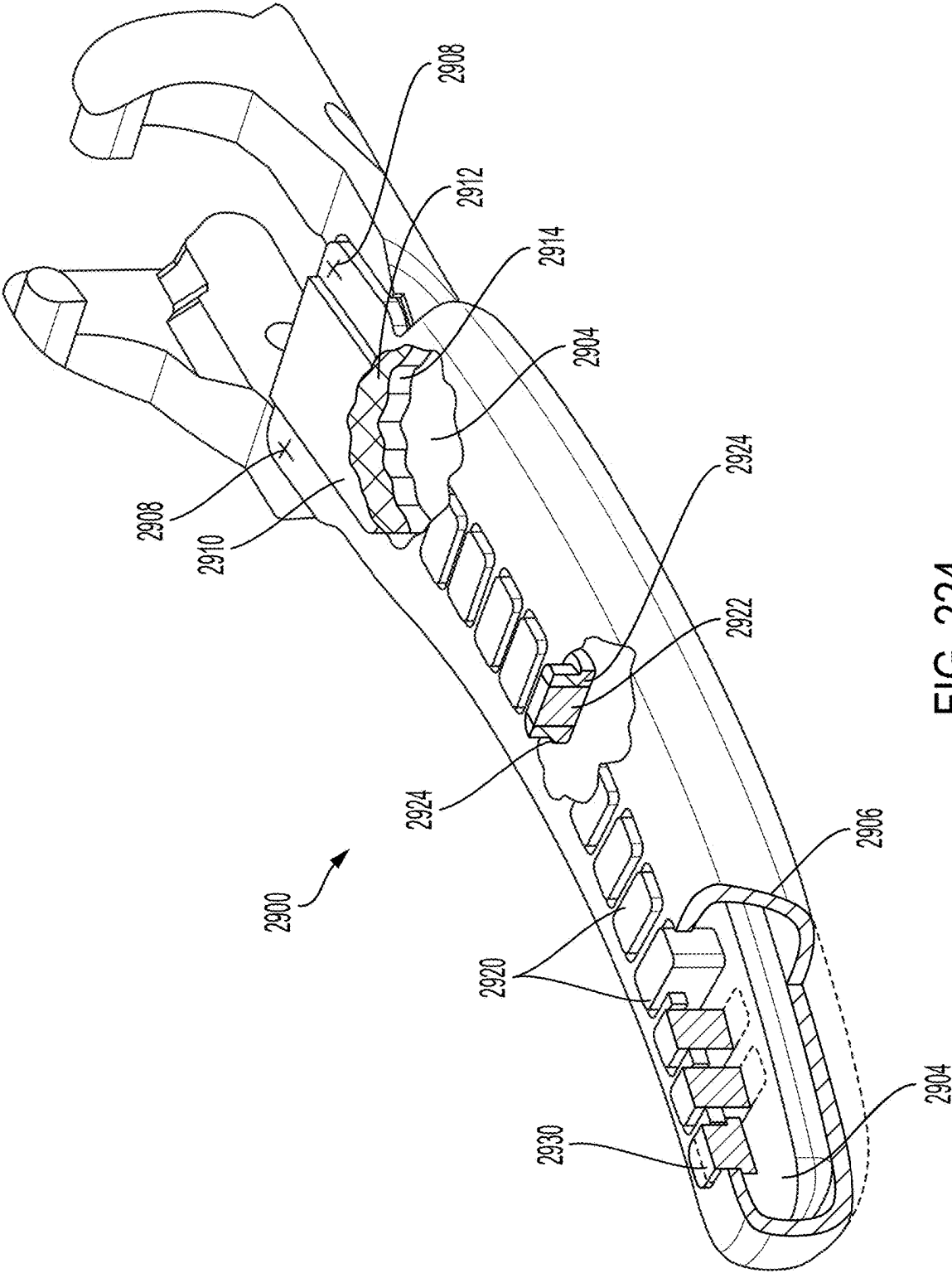


FIG. 224

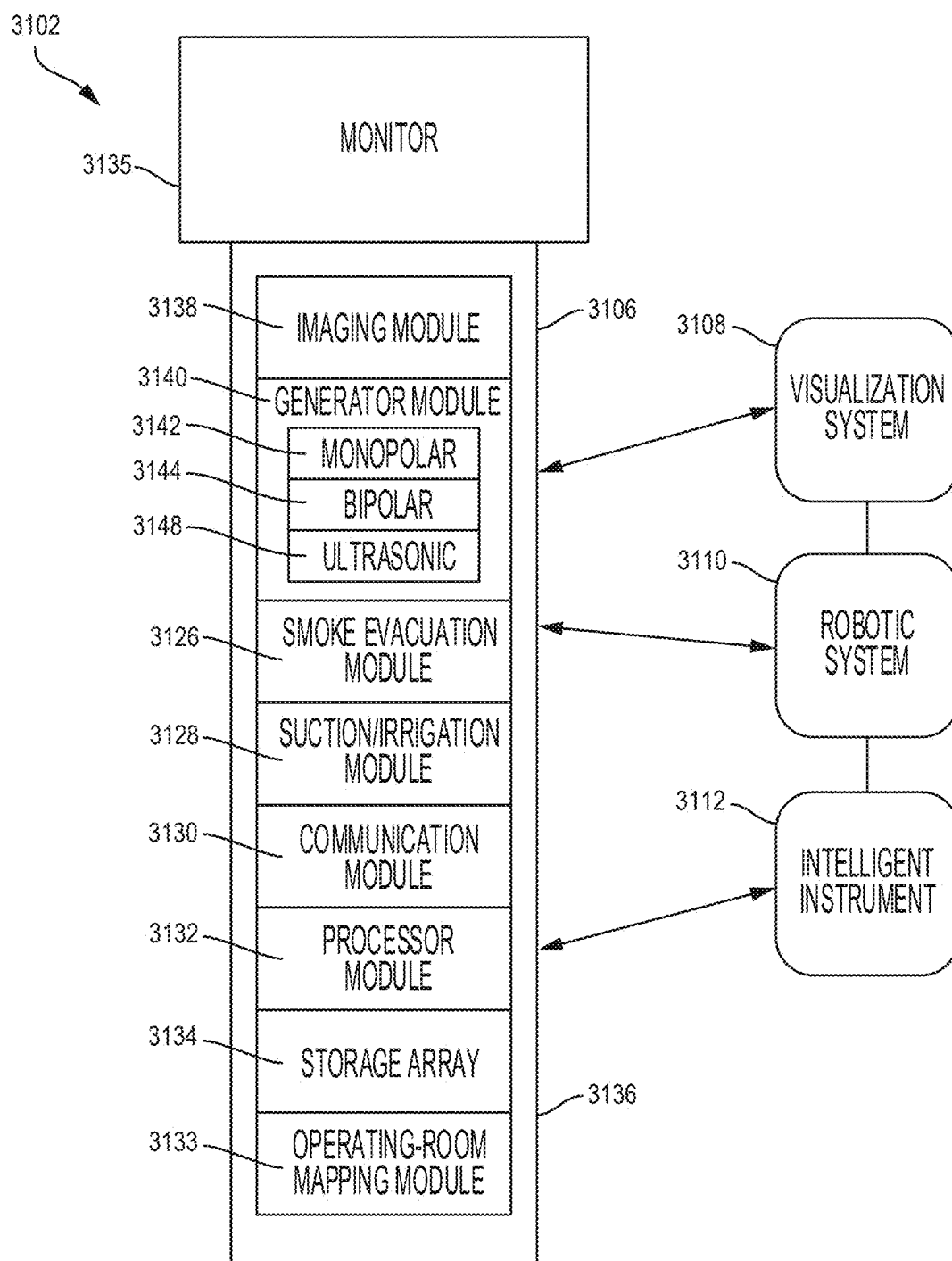


FIG. 225

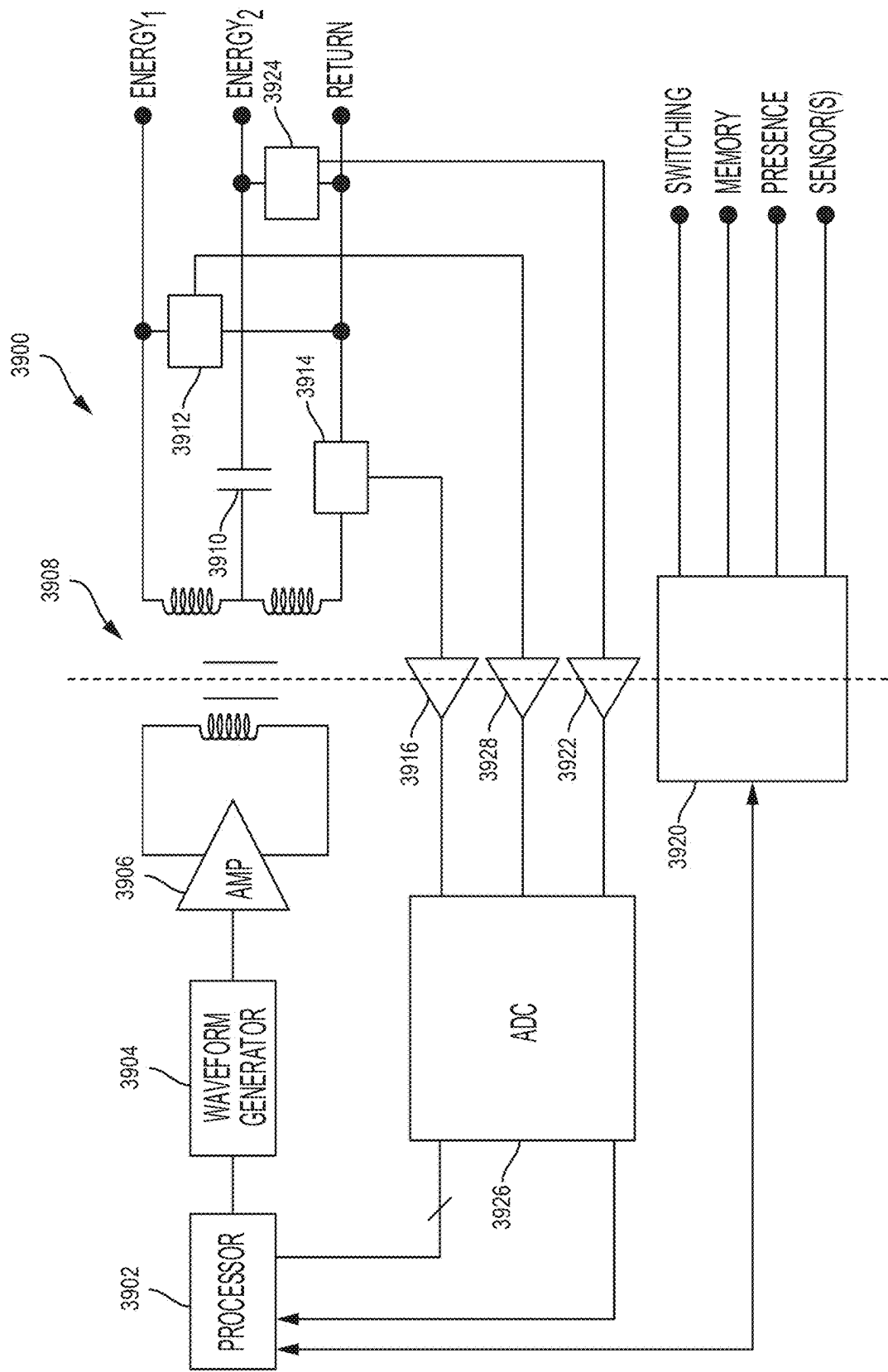


FIG. 226

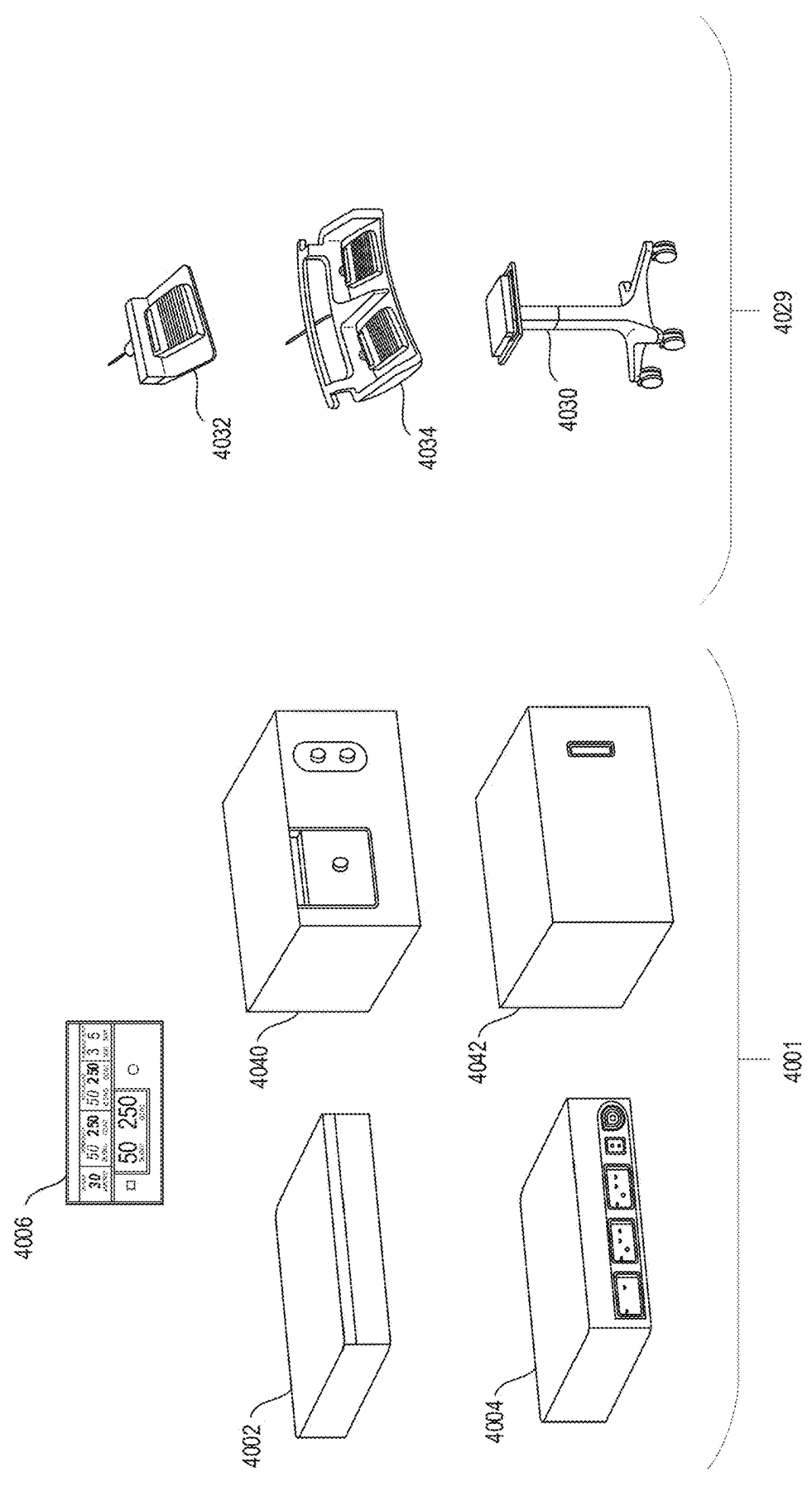


FIG. 227

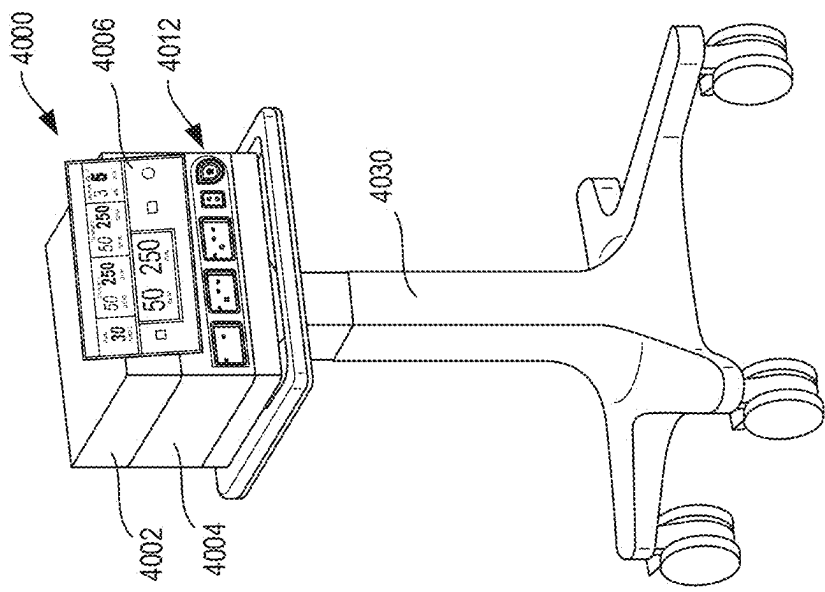


FIG. 228B

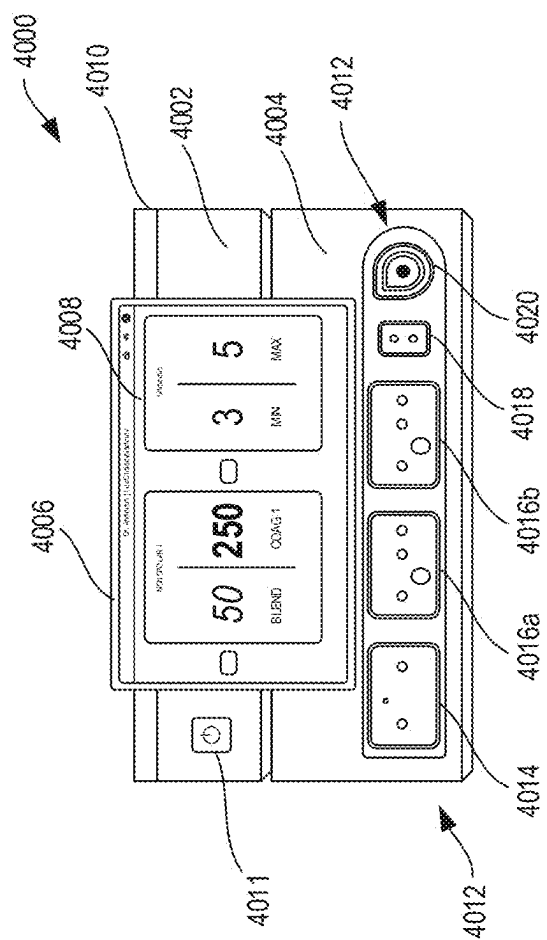
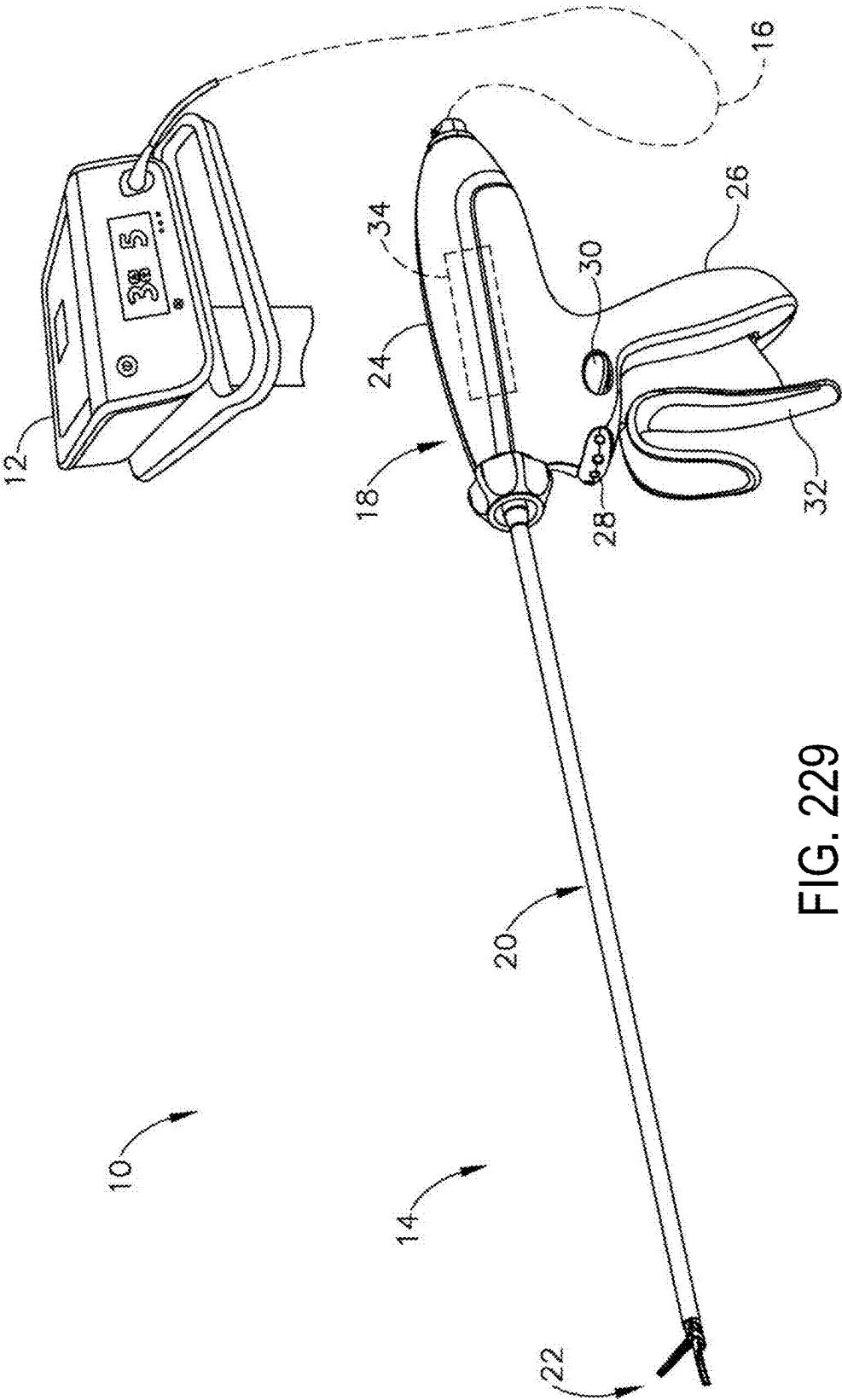


FIG. 228A



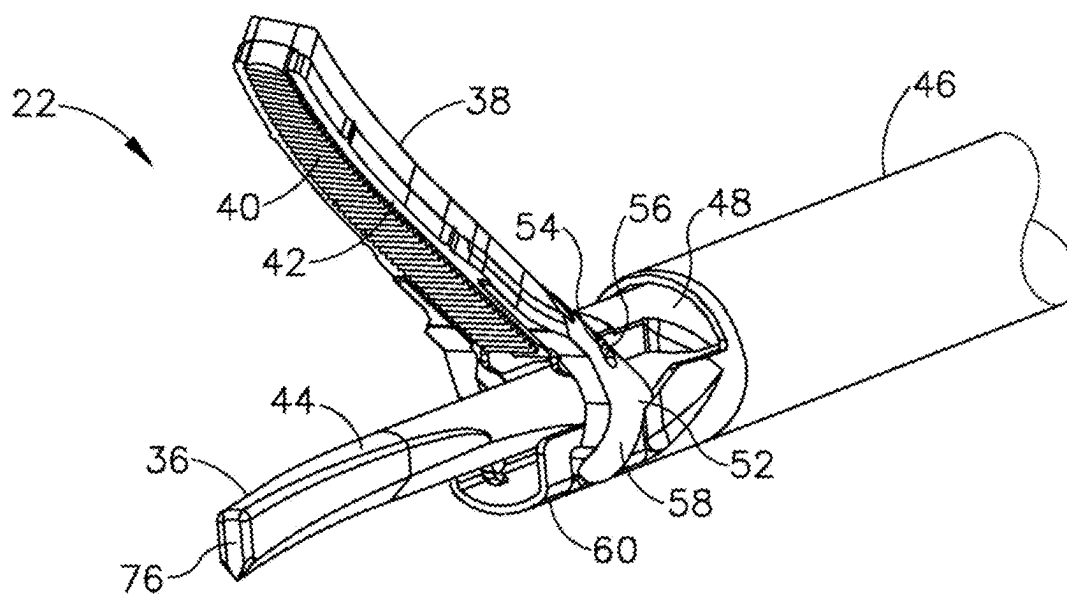


FIG. 230

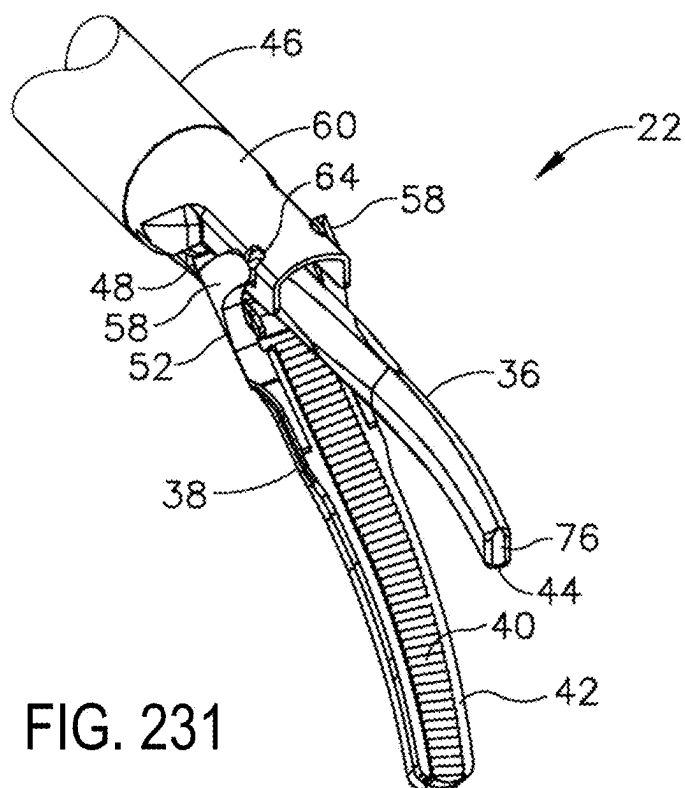


FIG. 231

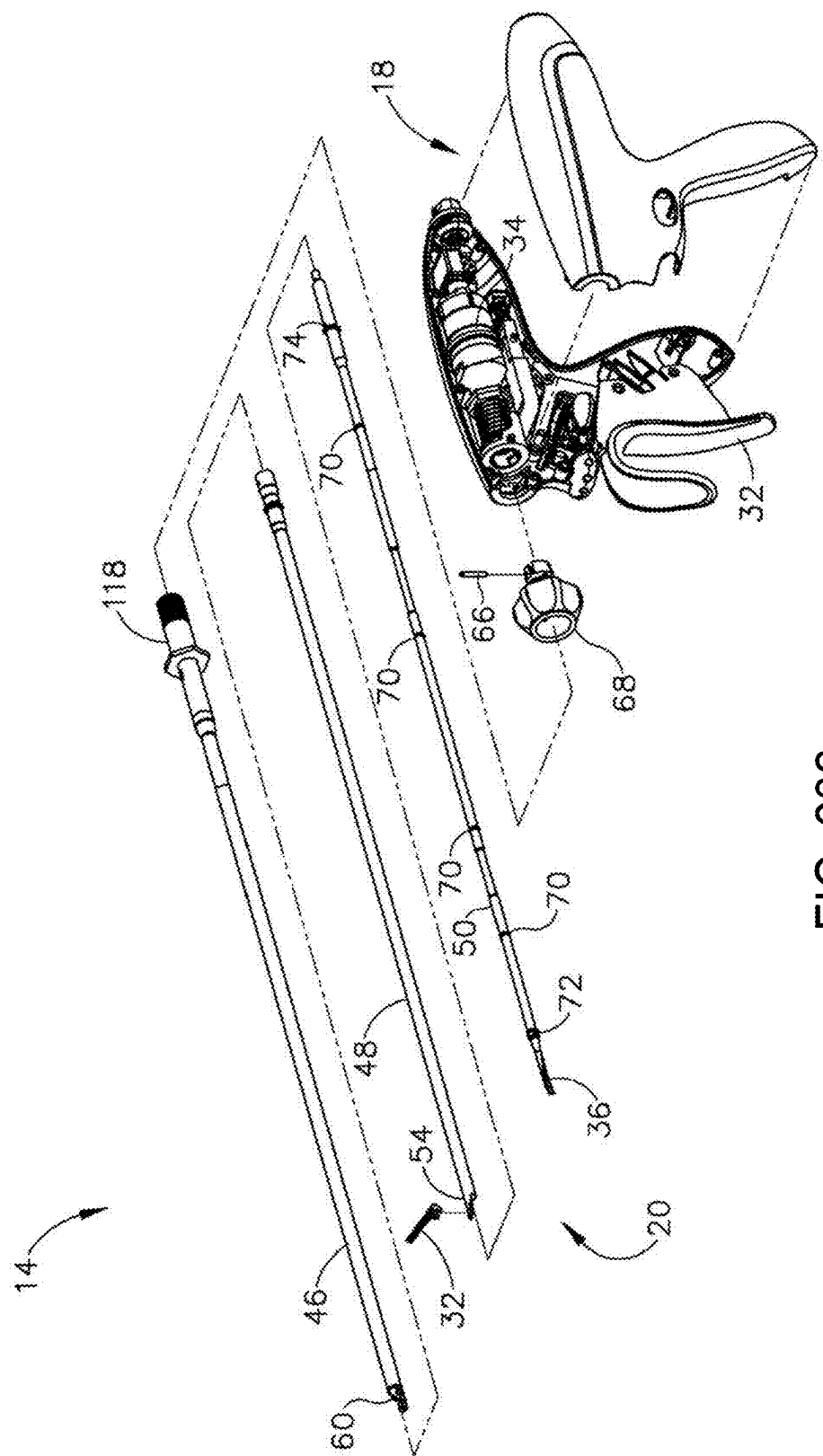


FIG. 232

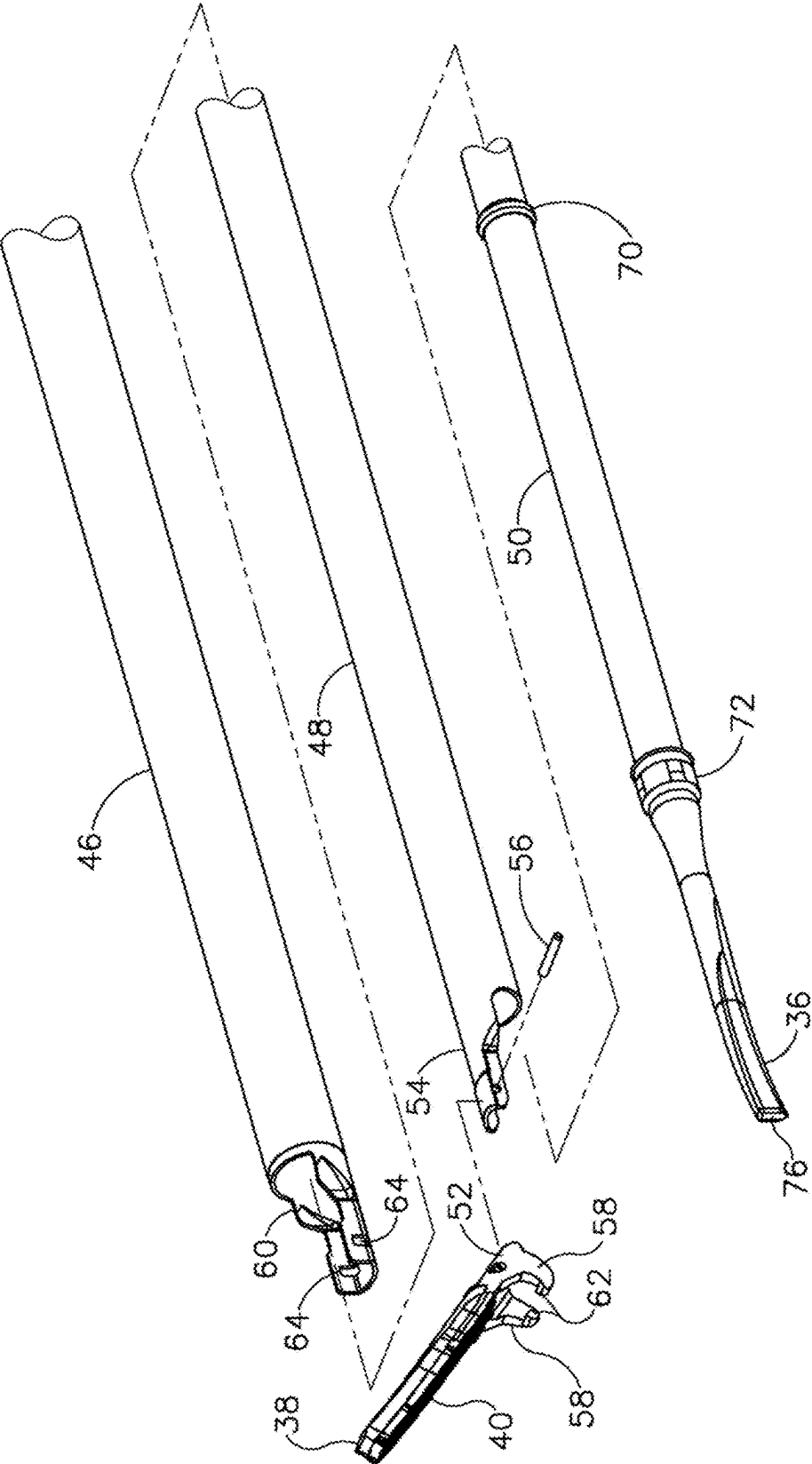


FIG. 233

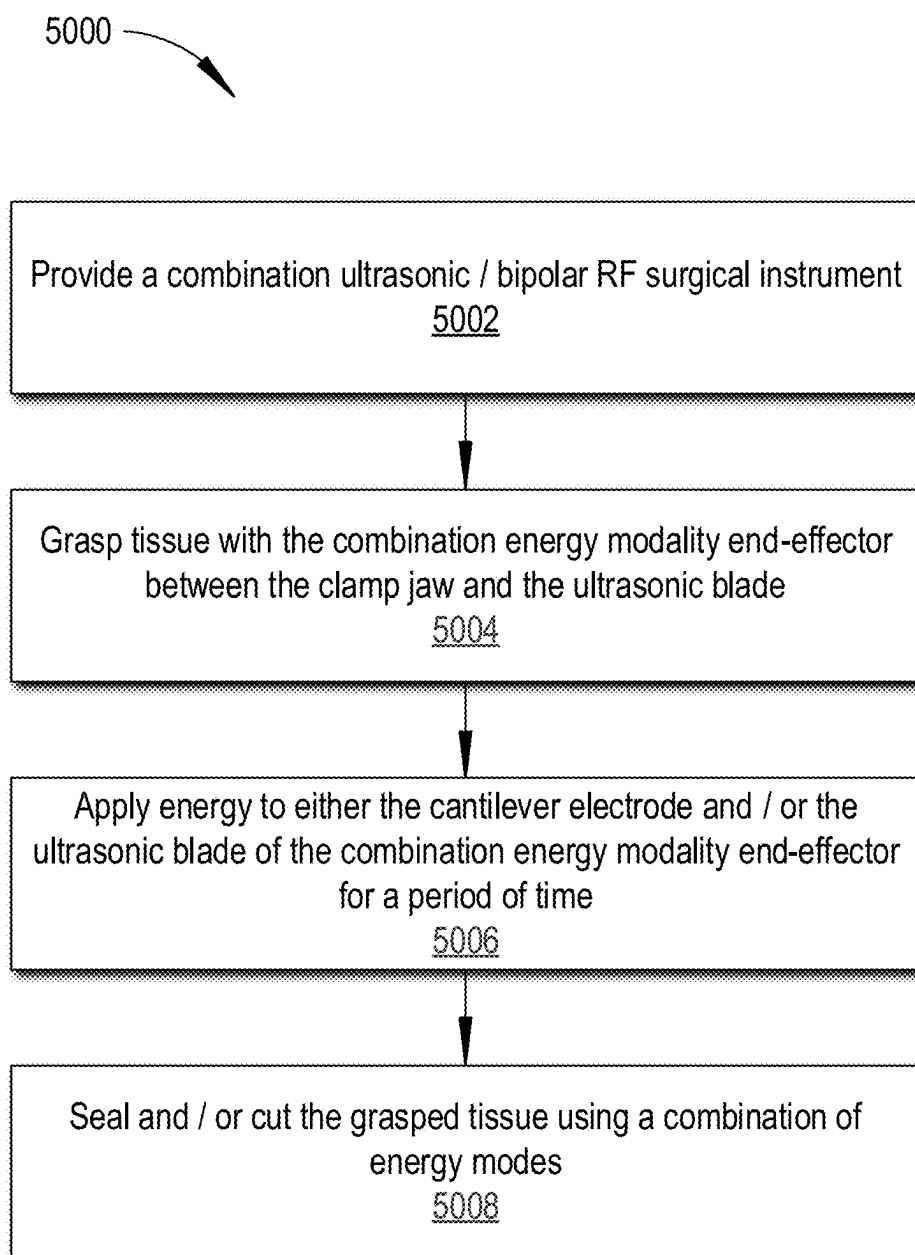


FIG. 234

**METHOD OF OPERATING A COMBINATION
ULTRASONIC / BIPOLAR RF SURGICAL
DEVICE WITH A COMBINATION ENERGY
MODALITY END-EFFECTOR**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0001] The present application is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 16/887,493, entitled METHOD OF OPERATING A COMBINATION ULTRASONIC/BIPOLAR RF SURGICAL DEVICE WITH A COMBINATION ENERGY MODALITY END-EFFECTOR, filed May 29, 2020, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. No. 62/955,292, titled COMBINATION ENERGY MODALITY END-EFFECTOR, filed Dec. 30, 2019, the disclosure of which is herein incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure generally relates to end-effectors adapted and configured to operate with multiple energy modalities to enable tissue sealing and cutting employing simultaneously, independently, or sequentially applied energy modalities. More particularly, the present disclosure relates to end-effectors adapted and configured to operate with surgical instruments that employ combined ultrasonic and electrosurgical systems, such as monopolar or bipolar radio frequency (RF), to enable tissue sealing and cutting employing simultaneously, independently, or sequentially applied ultrasonic and electrosurgical energy modalities. The energy modalities may be applied based on tissue parameters or other algorithms. The end-effectors may be adapted and configured to couple to hand held or robotic surgical systems.

BACKGROUND

[0003] Ultrasonic surgical instruments employing ultrasonic energy modalities are finding increasingly widespread applications in surgical procedures by virtue of the unique performance characteristics of such instruments. Depending upon specific instrument configurations and operational parameters, ultrasonic surgical instruments can provide substantially simultaneous cutting of tissue and hemostasis by coagulation, desirably minimizing patient trauma. The cutting action is typically realized by an end-effector, ultrasonic blade, or ultrasonic blade tip, at the distal end of the instrument, which transmits ultrasonic energy to tissue brought into contact with the end-effector. An ultrasonic end-effector may comprise an ultrasonic blade, a clamp arm, and a pad, among other components.

[0004] Some surgical instruments utilize ultrasonic energy for both precise cutting and controlled coagulation. Ultrasonic energy cuts and coagulates by vibrating a blade in contact with tissue. Vibrating at high frequencies (e.g., 55,500 times per second), the ultrasonic blade denatures protein in the tissue to form a sticky coagulum. Pressure exerted on tissue with the blade surface collapses blood vessels and allows the coagulum to form a hemostatic seal. The precision of cutting and coagulation is controlled by the surgeon's technique and adjusting the power level, blade edge, tissue traction, and blade pressure.

[0005] Electrosurgical instruments for applying electrical energy modalities to tissue to treat, seal, cut, and/or destroy tissue also are finding increasingly widespread applications in surgical procedures. An electrosurgical instrument typically includes an instrument having a distally-mounted end-effector comprising one or more than one electrode. The end-effector can be positioned against the tissue such that electrical current is introduced into the tissue. Electrosurgical instruments can be configured for bipolar or monopolar operation. During bipolar operation, current is introduced through a first electrode (e.g., active electrode) into the tissue and returned from the tissue through a second electrode (e.g., return electrode). During monopolar operation, current is introduced into the tissue by an active electrode of the end-effector and returned through a return electrode such as a grounding pad, for example, separately coupled to the body of a patient. Heat generated by the current flowing through the tissue may form hemostatic seals within the tissue and/or between tissues and thus may be particularly useful for sealing blood vessels, for example. The end-effector of an electrosurgical instrument also may include a cutting member that is movable relative to the tissue and the electrodes to transect the tissue. Electrosurgical end-effectors may be adapted and configured to couple to hand held instruments as well as robotic instruments.

[0006] Electrical energy applied by an electrosurgical instrument can be transmitted to the instrument by a generator in communication with the hand piece. The electrical energy may be in the form of radio frequency ("RF") energy. RF energy is a form of electrical energy that may be in the frequency range of 200 kilohertz (kHz) to 1 megahertz (MHz). In application, an electrosurgical instrument can transmit low frequency RF energy through tissue, which causes ionic agitation, or friction, in effect resistive heating, thereby increasing the temperature of the tissue. Because a sharp boundary is created between the affected tissue and the surrounding tissue, surgeons can operate with a high level of precision and control, without sacrificing un-targeted adjacent tissue. The low operating temperatures of RF energy is useful for removing, shrinking, or sculpting soft tissue while simultaneously sealing blood vessels. RF energy works particularly well on connective tissue, which is primarily comprised of collagen and shrinks when contacted by heat.

[0007] The RF energy may be in a frequency range described in EN 60601-2-2:2009+A11:2011, Definition 201.3.218—HIGH FREQUENCY. For example, the frequency in monopolar RF applications may be typically restricted to less than 5 MHz. However, in bipolar RF energy applications, the frequency can be almost anything. Frequencies above 200 kHz can be typically used for monopolar applications in order to avoid the unwanted stimulation of nerves and muscles that would result from the use of low frequency current. Lower frequencies may be used for bipolar applications if the risk analysis shows the possibility of neuromuscular stimulation has been mitigated to an acceptable level. Normally, frequencies above 5 MHz are not used in order to minimize the problems associated with high frequency leakage currents. Higher frequencies may, however, be used in the case of bipolar applications. It is generally recognized that 10 mA is the lower threshold of thermal effects on tissue.

[0008] Ultrasonic surgical instruments and electrosurgical instruments of the nature described herein can be configured for open surgical procedures, minimally invasive surgical

procedures, or non-invasive surgical procedures. Minimally invasive surgical procedures involve the use of a camera and instruments inserted through small incisions in order to visualize and treat conditions within joints or body cavities. Minimally invasive procedures may be performed entirely within the body or, in some circumstances, can be used together with a smaller open approach. These combined approaches, known as “arthroscopic, laparoscopic or thoracoscopic-assisted surgery,” for example. The surgical instruments described herein also can be used in non-invasive procedures such as endoscopic surgical procedures, for example. The instruments may be controlled by a surgeon using a hand held instrument or a robot.

[0009] A challenge of utilizing these surgical instruments is the inability to control and customize single or multiple energy modalities depending on the type of tissue being treated. It would be desirable to provide end-effectors that overcome some of the deficiencies of current surgical instruments and improve the quality of tissue treatment, sealing, or cutting or combinations thereof. The combination energy modality end-effectors described herein overcome those deficiencies and improve the quality of tissue treatment, sealing, or cutting or combinations thereof.

SUMMARY

[0010] In one aspect, an apparatus is provided for dissecting and coagulating tissue. The apparatus comprises a surgical instrument comprising an end-effector adapted and configured to deliver a plurality of energy modalities to tissue at a distal end thereof. The energy modalities may be applied simultaneously, independently, or sequentially. A generator is electrically coupled to the surgical instrument and is configured to supply a plurality of energy modalities to the end-effector. In one aspect, the generator is configured to supply electrosurgical energy (e.g., monopolar or bipolar radio frequency (RF) energy) and ultrasonic energy to the end-effector to allow the end-effector to interact with the tissue. The energy modalities may be supplied to the end-effector by a single generator or multiple generators.

[0011] In various aspects, the present disclosure provides a surgical instrument configured to deliver at least two energy types (e.g., ultrasonic, monopolar RF, bipolar RF, microwave, or irreversible electroporation [IRE]) to tissue. The surgical instrument includes a first activation button for activating energy, a second button for selecting an energy mode for the activation button. The second button is connected to a circuit that uses at least one input parameter to define the energy mode. The input parameter can be modified remotely through connection to a generator or through a software update.

[0012] In one aspect, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device. The combination ultrasonic/bipolar RF energy surgical device comprises an end-effector. The end-effector comprises a clamp arm and an ultrasonic blade. The clamp arm comprises a movable clamp jaw, a compliant polymeric pad, and at least one bipolar RF electrode. The at least one electrode is coupled to a positive pole of an RF generator and the ultrasonic blade is coupled to the negative pole of the RF generator. The ultrasonic blade is acoustically coupled to an ultrasonic transducer stack that is driven by an ultrasonic generator. In one aspect, the at least one electrode acts as a deflectable support with respect to an opposing ultrasonic blade. The at least one electrode crosses over the ultrasonic

blade and is configured to be deflectable with respect to the clamp arm having features to change the mechanical properties of the tissue compression under the at least one electrode. The at least one electrode includes a feature to prevent inadvertent contact between the electrode and the ultrasonic blade.

[0013] In another aspect, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device. The combination ultrasonic/bipolar RF energy surgical device comprises an end-effector. The end-effector comprises a clamp arm and an ultrasonic blade. The clamp arm comprises a movable clamp jaw, a compliant polymeric pad, and at least one bipolar RF electrode. The at least one electrode is coupled to a positive pole of an RF generator and the ultrasonic blade is coupled to the negative pole of the RF generator. The ultrasonic blade is acoustically coupled to an ultrasonic transducer stack that is driven by an ultrasonic generator. In one aspect, the movable clamp jaw comprises at least one non-biased deflectable electrode to minimize contact between the ultrasonic blade and the RF electrode. The ultrasonic blade pad contains a feature for securing the electrode to the pad. As the pad height wears or is cut through, the height of the electrode with respect to the clamp jaw is progressively adjusted. Once the clamp jaw is moved away from the ultrasonic blade, the electrode remains in its new position.

[0014] In another aspect, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device. The combination ultrasonic/bipolar RF energy surgical device comprises an end-effector. The end-effector comprises a clamp arm and an ultrasonic blade. The clamp arm comprises a movable clamp jaw, a compliant polymeric pad, and at least one bipolar RF electrode. The at least one electrode is coupled to a positive pole of an RF generator and the ultrasonic blade is coupled to the negative pole of the RF generator. The ultrasonic blade is acoustically coupled to an ultrasonic transducer stack that is driven by an ultrasonic generator. In one aspect, the at least one bipolar RF electrode is deflectable and has a higher distal bias than proximal bias. The bipolar RF electrode is deflectable with respect to the clamp jaw. The end-effector is configured to change the mechanical properties of the tissue compression proximal to distal end to create a more uniform or differing pattern of pressure than due to the clamping alone.

[0015] In another aspect, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device. The combination ultrasonic/bipolar RF energy surgical device comprises an end-effector. The end-effector comprises a clamp arm and an ultrasonic blade. The clamp arm comprises a movable clamp jaw, a compliant polymeric pad, and at least one bipolar RF electrode. The at least one electrode is coupled to a positive pole of an RF generator and the ultrasonic blade is coupled to the negative pole of the RF generator. The ultrasonic blade is acoustically coupled to an ultrasonic transducer stack that is driven by an ultrasonic generator. In one aspect, the bipolar RF electrode is deflectable and the end-effector provides variable compression/bias along the length of the deflectable electrode. The end-effector is configured to change the mechanical properties of the tissue compression under the electrodes based on clamp jaw closure or clamping amount.

[0016] In another aspect, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device. The combination ultrasonic/bipolar RF energy surgical

device comprises an end-effector. The end-effector comprises a clamp arm and an ultrasonic blade. The clamp arm comprises a movable clamp jaw, a compliant polymeric pad, and at least one bipolar RF electrode. The at least one electrode is coupled to a positive pole of an RF generator and the ultrasonic blade is coupled to the negative pole of the RF generator. The ultrasonic blade is acoustically coupled to an ultrasonic transducer stack that is driven by an ultrasonic generator. The one aspect, the pad includes asymmetric segments to provide support for the ultrasonic blade support and the electrode is movable. The asymmetric segmented pad is configured for cooperative engagement with the movable bipolar RF electrode. The segmented ultrasonic support pad extends at least partially through the bipolar RF electrode. At least one pad element is significantly taller than a second pad element. The first pad element extends entirely through the bipolar RF electrode and the second pad element extends partially through the bipolar RF electrode. The first pad element and the second pad element are made of dissimilar materials.

[0017] In another aspect, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device. The combination ultrasonic/bipolar RF energy surgical device comprises an end-effector. The end-effector comprises a clamp arm and an ultrasonic blade. The clamp arm comprises a movable clamp jaw, a compliant polymeric pad, and at least one bipolar RF electrode. The at least one electrode is coupled to a positive pole of an RF generator and the ultrasonic blade is coupled to the negative pole of the RF generator. The ultrasonic blade is acoustically coupled to an ultrasonic transducer stack that is driven by an ultrasonic generator. In one aspect, variations in the physical parameters of the electrode in combination with a deflectable electrode are employed to change the energy density delivered to the tissue and the tissue interactions. The physical aspects of the electrode vary along its length in order to change the contact area and/or the energy density of the electrode to tissue as the electrode also deflects.

[0018] In another aspect, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device. The combination ultrasonic/bipolar RF energy surgical device comprises an end-effector. The end-effector comprises a clamp arm and an ultrasonic blade. The clamp arm comprises a movable clamp jaw, a compliant polymeric pad, and at least one bipolar RF electrode. The at least one electrode is coupled to a positive pole of an RF generator and the ultrasonic blade is coupled to the negative pole of the RF generator. The ultrasonic blade is acoustically coupled to an ultrasonic transducer stack that is driven by an ultrasonic generator. In one aspect, an ultrasonic transducer control algorithm is provided to reduce the power delivered by the ultrasonic or RF generator when a short circuit of contact between the ultrasonic blade and the electrode is detected to prevent damage to the ultrasonic blade. The ultrasonic blade control algorithm monitors for electrical shorting or ultrasonic blade to electrode contact. This detection is used to adjust the power/amplitude level of the ultrasonic transducer when the electrical threshold minimum is exceeded and adjusts the transducer power/amplitude threshold to a level below the minimum threshold that would cause damage to the ultrasonic blade, ultrasonic generator, bipolar RF electrode, or bipolar RF generator. The monitored electrical parameter could be tissue impedance (Z) or electrical continuity. The power adjustment could be to shut off the

ultrasonic generator, bipolar RF generator, of the surgical device or it could be a proportionate response to either the electrical parameter, pressure, or time or any combination of these parameters.

[0019] In another aspect, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device. The combination ultrasonic/bipolar RF energy surgical device comprises an end-effector. The end-effector comprises a clamp arm and an ultrasonic blade. The clamp arm comprises a movable clamp jaw, a compliant polymeric pad, and at least one bipolar RF electrode. The at least one electrode is coupled to a positive pole of an RF generator and the ultrasonic blade is coupled to the negative pole of the RF generator. The ultrasonic blade is acoustically coupled to an ultrasonic transducer stack that is driven by an ultrasonic generator. In one aspect, the clamp jaw features or aspects are provided in the clamp ram to minimize tissue sticking and improve tissue control. The clamp arm tissue path or clamp area includes features configured to adjust the tissue path relative to the clamp arm/ultrasonic blade to create a predefined location of contact to reduce tissue sticking and charring.

[0020] In another aspect, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device. The combination ultrasonic/bipolar RF energy surgical device comprises an end-effector. The end-effector comprises a clamp arm and an ultrasonic blade. The clamp arm comprises a movable clamp jaw, a compliant polymeric pad, and at least one bipolar RF electrode. The at least one electrode is coupled to a positive pole of an RF generator and the ultrasonic blade is coupled to the negative pole of the RF generator. The ultrasonic blade is acoustically coupled to an ultrasonic transducer stack that is driven by an ultrasonic generator. In one aspect, a partially conductive clamp arm pad is provided to enable electrode wear through and minimize electrical shorting between the ultrasonic blade and the bipolar RF electrode. The clamp arm pad includes electrically conductive and non-conductive portions allowing it to act as one of the bipolar RF electrodes while also acting as the wearable support structure for the ultrasonic blade. The electrically conductive portions of the clamp ram pad are positioned around the perimeter of the pad and not positioned directly below the ultrasonic blade contact area. The electrically conductive portion is configured to degrade or wear to prevent any contact with the ultrasonic blade from interrupting the electrical conductivity of the remaining electrically conductive pad.

[0021] In addition to the foregoing, various other method and/or system and/or program product aspects are set forth and described in the teachings such as text (e.g., claims and/or detailed description) and/or drawings of the present disclosure.

[0022] The foregoing is a summary and thus may contain simplifications, generalizations, inclusions, and/or omissions of detail; consequently, those skilled in the art will appreciate that the summary is illustrative only and is NOT intended to be in any way limiting. Other aspects, features, and advantages of the devices and/or processes and/or other subject matter described herein will become apparent in the teachings set forth herein.

[0023] In one or more various aspects, related systems include but are not limited to circuitry and/or programming for effecting herein-referenced method aspects; the circuitry and/or programming can be virtually any combination of

hardware, software, and/or firmware configured to affect the herein-referenced method aspects depending upon the design choices of the system designer. In addition to the foregoing, various other method and/or system aspects are set forth and described in the teachings such as text (e.g., claims and/or detailed description) and/or drawings of the present disclosure.

[0024] Further, it is understood that any one or more of the following-described forms, expressions of forms, examples, can be combined with any one or more of the other following-described forms, expressions of forms, and examples.

[0025] The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

FIGURES

[0026] The novel features of the described forms are set forth with particularity in the appended claims. The described forms, however, both as to organization and methods of operation, may be best understood by reference to the following description, taken in conjunction with the accompanying drawings in which:

[0027] FIG. 1 is a perspective view of a clamp arm portion of an end-effector for use with a combined ultrasonic/RF device, according to at least one aspect of the present disclosure.

[0028] FIG. 2 is an exploded view of the clamp arm shown in FIG. 1, according to at least one aspect of the present disclosure.

[0029] FIGS. 3 and 4 are perspective views of the frame, according to at least one aspect of the present disclosure.

[0030] FIG. 5 is a perspective view of the electrode, according to at least one aspect of the present disclosure.

[0031] FIG. 6 is a perspective view of the clamp arm pad, according to at least one aspect of the present disclosure.

[0032] FIG. 7 is a perspective top view of the large gap pad, according to at least one aspect of the present disclosure.

[0033] FIG. 8 is a perspective top view of the small gap pad, according to at least one aspect of the present disclosure.

[0034] FIG. 9 is a perspective bottom view of the small gap pad shown in FIG. 8.

[0035] FIGS. 10A-10B, there is illustrated an electrode having a proximal end fixedly attached to a proximal end of a clamp arm and having a distal end opposite to the fixed end that is free to move or deflect, where FIG. 10A illustrates a clamp arm and FIG. 10B illustrates an electrode, according to at least one aspect of the present disclosure.

[0036] FIGS. 11A-11D illustrate an electrode having a proximal end fixedly attached to a proximal end of a clamp arm and having a distal end opposite to the fixed end that is free to move or deflect, according to at least one aspect of the present disclosure, where:

[0037] FIG. 11A is an end-effector comprising a clamp arm, a peripheral electrode defining an inner aperture, and a connection pad isolated from a conductive portion of the peripheral electrode to attach the peripheral electrode to a proximal end of the clamp arm, according to at least one aspect of the present disclosure;

[0038] FIG. 11B is a top view of the peripheral electrode and a side partial side view of the end-effector, according to at least one aspect of the present disclosure;

[0039] FIG. 11C is a side view of an end-effector comprising a clamp arm and an ultrasonic blade and a top view of a peripheral electrode configured to attach to a proximal end of the clamp arm, according to at least one aspect of the present disclosure; and

[0040] FIG. 11D is a side view of an end-effector comprising a clamp arm and an ultrasonic blade and a top view of a peripheral electrode configured to attach to a proximal end of the clamp arm, according to at least one aspect of the present disclosure.

[0041] FIGS. 12A-12D illustrate a clamp arm comprising an electrode having a proximal end fixedly attached to a proximal end of the clamp arm and having a distal end opposite to the fixed end that is free to move or deflect, according to at least one aspect of the present disclosure, where:

[0042] FIG. 12A illustrates a clamp arm comprises a jaw, a clamp arm pad, and an electrode, according to at least one aspect of the present disclosure;

[0043] FIG. 12B is a section view of the clamp arm viewed from the perspective of the distal end of the clamp arm, according to at least one aspect of the present disclosure;

[0044] FIG. 12C is a section view of the clamp arm pad with a backing, according to at least one aspect of the present disclosure; and

[0045] FIG. 12D is a clamp arm comprising a clamp jaw, a clamp arm pad, and a deflectable cantilever electrode, according to at least one aspect of the present disclosure.

[0046] FIGS. 13A-13B illustrate a clamp arm comprising an electrode and a clamp arm pad, according to at least one aspect of the present disclosure, where:

[0047] FIG. 13A shows the electrode in an unloaded undeflected state; and

[0048] FIG. 13B shows the electrode in a loaded deflected state.

[0049] FIGS. 14A-14C illustrate an electrode configuration with an end-of-life indicator, which could be sensed by a generator, to prompt device replacement, according to various aspects of the present disclosure, where:

[0050] FIG. 14A, there is illustrated an end-effector comprising a clamp arm and an ultrasonic blade, according to at least one aspect of the present disclosure; and

[0051] FIG. 14B shows the end-of-life indicator tail at the start of a procedure and FIG. 14C shows the end-of-life indicator tail at the end of the procedure; and

[0052] FIG. 14C shows the end-of-life indicator tail at the end of the procedure.

[0053] FIGS. 15A-15B illustrate an electrode configuration with an end-of-life indicator, which could be sensed by a generator, to prompt device replacement, according to various aspects of the present disclosure, where:

[0054] FIG. 15A illustrates an end-effector comprising a clamp arm and an ultrasonic blade, according to at least one aspect of the present disclosure; and

[0055] FIG. 15B illustrates the end-effector of FIG. 15A after the clamp arm pad is worn close to failure.

[0056] FIGS. 16A-16B illustrate configurations wherein the electrode can move vertically along its entire length, according to various aspects of the present disclosure, where:

[0057] FIG. 16A shows a clamp arm portion of an end-effector, according to at least one aspect of the present disclosure; and

[0058] FIG. 16B shows the wear resistant clamp arm pad is mounted to an end of the lubricious clamp arm pad to help reduce wear of the lubricious clamp arm pad.

[0059] FIG. 17 illustrates a configuration wherein the electrode can move vertically along its entire length, according to various aspects of the present disclosure.

[0060] FIGS. 18A-18D illustrate configurations wherein the electrode can move vertically along its entire length, according to various aspects of the present disclosure, where:

[0061] FIG. 18A is a section view of an end-effector comprising a clamp arm, according to at least one aspect of the present disclosure;

[0062] FIG. 18B is a section view of the end-effector showing the location of the wear resistant dowel pins in the clamp arm;

[0063] FIG. 18C is a section view of a wear resistant dowel pin, according to at least one aspect of the present disclosure; and

[0064] FIG. 18D is a section view of a wear resistant dowel pin, according to at least one aspect of the present disclosure.

[0065] FIG. 19A illustrates a portion of an end-effector comprising a clamp arm comprising an electrode with a “center pivot” configuration, according to at least one aspect of the present disclosure.

[0066] FIG. 19B illustrates a portion of an end-effector comprising a clamp arm comprising an electrode with a “center pivot” configuration, according to at least one aspect of the present disclosure.

[0067] FIG. 20 illustrates an end-effector comprising a clamp arm and an ultrasonic blade, according to at least one aspect of the present disclosure.

[0068] FIG. 21 illustrates a spring element formed into an electrode, according to at least one aspect of the present disclosure

[0069] FIGS. 22A-22C illustrates a clamp arm comprising a wave spring element supporting an electrode, according to at least one aspect of the present disclosure, where:

[0070] FIG. 22A. illustrates one aspect of the clamp arm where the wave spring element is formed integral with the resilient clamp arm pad to deflect the electrode in direction “d”;

[0071] FIG. 22B illustrates one aspect of a clamp arm where the wave spring element is positioned between the resilient clamp arm pad and the electrode to deflect the electrode in direction “d”; and

[0072] FIG. 22C illustrates one aspect of a clamp arm where the wave spring element also is formed integral with the resilient clamp arm pad to deflect the electrode in direction “d”.

[0073] FIGS. 23A-23B illustrate electrode attachment methods, according to various aspects of the present disclosure, where

[0074] FIG. 23A illustrates the electrode is disposed through protrusions in the clamp arm pad and attached to the clamp arm by a bracket; and

[0075] FIG. 23B illustrates the electrode is attached to the clamp arm pad by a cylindrical spring.

[0076] FIGS. 24A-24C illustrate electrode spring element configurations, according to various aspects of the present disclosure, where:

[0077] FIG. 24A illustrates an end-effector comprising a clamp arm and an ultrasonic blade, according to at least one aspect of the present disclosure;

[0078] FIG. 24B illustrates a clamp arm comprising a diaphragm electrode, according to at least one aspect of the present disclosure; and

[0079] FIG. 24C illustrates a clamp arm comprising a one piece spring electrode, according to at least one aspect of the present disclosure.

[0080] FIG. 25 illustrates an electrode comprising a coined bump for use with integral wear pads (not shown), according to at least one aspect of the present disclosure.

[0081] FIG. 26 illustrates springs tested under heat and tissue fluid conditions, according to at least one aspect of the present disclosure.

[0082] FIG. 27 illustrates an extended length electrode to prevent binding of the electrode as it deflects up and down, the distance “d1” of the electrode may be increased, according to at least one aspect of the present disclosure.

[0083] FIGS. 28A-28D illustrate an end-effector configured with one or more than one resistor embedded (integrated) in a clamp arm pad to sense the height “h” of the clamp arm pad, according to at least one aspect of the present disclosure, where:

[0084] FIG. 28A illustrates a new clamp arm pad and does not show any wear and the resistors and the clamp arm pad are whole and indicate a full resistance value at full height h_1 with no tissue between the clamp arm and the ultrasonic blade, according to at least one aspect of the present disclosure;

[0085] FIG. 28B illustrates the end-effector with no tissue between the clamp arm and the ultrasonic blade and with the clamp arm pad and resistors worn to a height of h_2 , which is less than h_1 due to heat generated by friction between the clamp arm and the ultrasonic blade and the resistors;

[0086] FIG. 28C illustrates the condition where the electrode is deflected to a height of h_3 , which is less than h_1 , due to thick tissue located between the clamp arm and the ultrasonic blade; and

[0087] FIG. 28D illustrates the condition where the electrode is deflected to a height of h_3 , which is less than h_1 , due to thick tissue located between the clamp arm and the ultrasonic blade.

[0088] FIG. 29 is a graphical illustration of one example of a resistance level of resistor in clamp arm pad versus pad height “h”, according to at least one aspect of the present disclosure.

[0089] FIGS. 30A-30B illustrate the resistance level (R1-R5) as a function of the height “h” of the resistor and analogously the height “h” of the clamp arm pad because the resistor is embedded in the clamp arm pad and both elements will wear at the same rate, according to at least one aspect of the present disclosure, where:

[0090] FIG. 30A shows the end-effector with a new clamp arm pad and resistor mounted in the clamp arm; and

[0091] FIG. 30B illustrates the state of the end-effector when the clamp arm pad is slightly worn to indicate a resistance level.

[0092] FIG. 31 is a graphical illustration of clamp arm pad height “h”, as measured by an integrated resistor, shown

along the vertical axis, as a function of time “t”, shown along the horizontal axis, according to at least one aspect of the present disclosure.

[0093] FIG. 32 illustrates one configuration wherein the electrode can pivot near the mid-point of the electrode, according to at least one aspect of the present disclosure, where FIG. 32 is an exploded view of a clamp arm comprising a center pivoting electrode, according to at least one aspect of the present disclosure.

[0094] FIGS. 33A-33B illustrate configurations wherein the electrode can pivot near the mid-point of the electrode, according to at least one aspect of the present disclosure, where:

[0095] FIG. 33A illustrates new pads such that at full clamp the center pivoting electrode is a predetermined angle (e.g., parallel) relative to the clamp arm jaw), according to at least one aspect of the present disclosure; and

[0096] FIG. 33B shows the change in angle θ between the center pivoting electrode relative to the clamp arm jaw 1526 in order to maintain a consistent gap between the ultrasonic blade, according to at least one aspect of the present disclosure.

[0097] FIG. 34 is a perspective section view of a clamp arm comprising a longitudinally aligned slot from a proximal end through which an electrode plate may be advanced without the clamp arm pad material, according to at least one aspect of the present disclosure.

[0098] FIG. 35A is a section view of the clamp arm shown in FIG. 34 prior to assembly, according to at least one aspect of the present disclosure.

[0099] FIG. 35B shows the clamp arm pad/electrode plate assembly slid proximally 1556 into the longitudinally aligned slot defined by the clamp jaw, according to at least one aspect of the present disclosure.

[0100] FIG. 35C shows the tab at the proximal end of the clamp arm pad locks into the proximal recess defined by the proximal end of the clamp jaw interlocking the clamp arm pad/electrode plate assembly 1560 with the clamp jaw, according to at least one aspect of the present disclosure.

[0101] FIG. 36 illustrates one aspect of a continuous spring support disposed under the electrode rather than discrete spring attachment at one end, according to at least one aspect of the present disclosure.

[0102] FIG. 37 is a perspective view of an end-effector comprising a laterally deflectable electrode, forming one pole of the electrosurgical circuit, and an ultrasonic blade, forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure.

[0103] FIG. 38 shows the laterally deflectable electrode in an undeflected state (shown in solid line) and in a deflected state (shown in broken line), according to at least one aspect of the present disclosure.

[0104] FIG. 39 is a section view of an end-effector comprising an electrode as a structural clamping member, forming one pole of the electrosurgical circuit, and an ultrasonic blade, forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure.

[0105] FIG. 40 is a section view of an end-effector comprising an electrode as a structural clamping member, forming one pole of the electrosurgical circuit, and an ultrasonic blade, forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure.

[0106] FIG. 41 is a section of an end-effector comprising an electrode as a structural clamping member, forming one

pole of the electrosurgical circuit, and an ultrasonic blade, forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure.

[0107] FIG. 42 shows the electrode in its laterally deflected state (shown in broken line), according to at least one aspect of the present disclosure.

[0108] FIG. 43 illustrates a clamp arm comprising a laterally deflectable electrode, forming one pole of the electrosurgical circuit, according to at least one aspect of the present disclosure.

[0109] FIG. 44 illustrates a clamp arm comprising a laterally deflectable electrode, forming one pole of the electrosurgical circuit, according to at least one aspect of the present disclosure.

[0110] FIGS. 45-46 illustrate an end-effector comprising a laterally deflectable electrode, forming one pole of the electrosurgical circuit, and an ultrasonic blade, forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure, where:

[0111] FIG. 45 with a distance d between the center of the electrode to the most lateral extension of the electrode; and

[0112] FIG. 46 shows a laterally deflected state of the electrode with a distance d' between the center of the electrode to the most lateral extension of the electrode.

[0113] FIG. 47 illustrates an end-effector comprising a laterally deflectable electrode, forming one pole of the electrosurgical circuit, and an ultrasonic blade, forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure.

[0114] FIG. 48 illustrates an end-effector comprising a laterally deflectable electrode, forming one pole of the electrosurgical circuit, and an ultrasonic blade, forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure.

[0115] FIG. 49 illustrates the lateral deflection of the laterally deflectable electrode in broken line in the direction of the arrows.

[0116] FIG. 50 is a magnified view of the laterally deflectable electrode where the lateral deflection is shown in broken line.

[0117] FIG. 51 illustrates an end-effector comprising a pair of metal clamp arms coated with a compliant or deformable dielectric material and a wire electrode, according to at least one aspect of the present disclosure.

[0118] FIG. 52 illustrates an end-effector comprising a pair of metal clamp arms coated with a compliant or deformable dielectric material and a wire electrode, according to at least one aspect of the present disclosure.

[0119] FIG. 53 illustrates an end-effector comprising a pair of metal clamp arms coated with a compliant or deformable dielectric material and a wire electrode, according to at least one aspect of the present disclosure.

[0120] FIG. 54 illustrates an end-effector comprising a pair of metal clamp arms coated with a compliant or deformable dielectric material and a wire electrode, according to at least one aspect of the present disclosure.

[0121] FIG. 55 illustrates a compliant or deformable dielectric material can be the same material (e.g., silicone) or can be a harder or softer material, according to at least one aspect of the present disclosure.

[0122] FIG. 56 illustrates the metal clamp arms implemented as bipolar electrodes by adding outer electrodes, according to at least one aspect of the present disclosure.

[0123] FIG. 57 illustrates bare metal or coated/covered metal clamp arms, according to at least one aspect of the present disclosure.

[0124] FIG. 58 illustrates bare metal or coated/covered metal clamp arms, according to at least one aspect of the present disclosure.

[0125] FIGS. 59-61 illustrate an effector comprising a shortened clamp arm for deflectable/cantilever electrode applications, according to various aspects of the present disclosure, where:

[0126] FIG. 59 is a side view of an end-effector comprising a shortened clamp arm, an ultrasonic blade, an electrode, and a clamp arm pad, according to at least one aspect of the present disclosure;

[0127] FIG. 60 is a top view of the end-effector, according to at least one aspect of the present disclosure; and

[0128] FIG. 61 illustrates a clamp arm comprising a clamp jaw, an electrode, and a clamp arm pad, according to at least one aspect of the present disclosure.

[0129] FIG. 62 illustrates an end-effector clamp arm comprising a clamp jaw, an electrode, and a clamp arm pad, according to at least one aspect of the present disclosure.

[0130] FIG. 63 illustrates an end-effector clamp arm comprising a clamp jaw, an electrode, and a clamp arm pad, according to at least one aspect of the present disclosure.

[0131] FIG. 64 illustrates an end-effector clamp arm comprising a clamp jaw, an electrode, and a clamp arm pad, according to at least one aspect of the present disclosure.

[0132] FIG. 65 illustrates bottom retainer tooth that is worn away such that the electrode can move toward the clamp jaw due to the pre-formed curve, according to at least one aspect of the present disclosure.

[0133] FIG. 66 illustrates an end-effector clamp arm comprising a clamp jaw, an electrode, and a clamp arm pad, according to at least one aspect of the present disclosure.

[0134] FIG. 67 illustrates a retainer wall with a tapered profile worn away such that there is sufficient melting/flowing away from the retainer wall with the tapered profile region to allow the electrode to move toward the clamp jaw due to the pre-formed curve, according to at least one aspect of the present disclosure.

[0135] FIG. 68 illustrates a clamp arm comprising a clamp jaw, an electrode, and a clamp arm pad, according to at least one aspect of the present disclosure.

[0136] FIG. 69 illustrates a magnified view of the distal end of the clamp arm with the clamp arm pad removed to expose the distal skirt, according to at least one aspect of the present disclosure.

[0137] FIGS. 70A-70C illustrate an end-effector comprising a clamp arm, an electrode, a clamp arm pad, and an ultrasonic blade, where the clamp arm pad comprises support elements to support the electrode, according to at least one aspect of the present disclosure, where:

[0138] FIG. 70A shows a new clamp arm pad and an electrode supported by support element (tab1, tab2, tab3);

[0139] FIG. 70B shows the clamp arm pad heating up and causes the first row of support elements (tab1) to melt and wear away causing the electrode to drop in height (h); and

[0140] FIG. 70C shows that as the clamp arm pad continues heating and wearing, the second row support elements (tab2) to melt and wear away causing the height (h) of the electrode to further drop down.

[0141] FIGS. 71-73 illustrate an end-effector comprising a clamp arm, an ultrasonic blade, a lattice cushion, a flexible

electrode disposed above the lattice cushion, and a plurality of hard spacers to set a gap between the flexible electrode and the ultrasonic blade, according to at least one aspect of the present disclosure, where:

[0142] FIG. 71 illustrates the clamp arm open and tissue of non-uniform thickness (T_{1a} , T_{2a} , T_{3a}) is disposed over the flexible electrode;

[0143] FIG. 72 the clamp arm is closed to compress the tissue; and

[0144] FIG. 73 is an exploded view of the end-effector shown in FIGS. 71-72.

[0145] FIGS. 74-75 illustrate an end-effector comprising a clamp arm, an electrode disposed on a plurality of springs, and an ultrasonic blade, according to at least one aspect of the present disclosure, where:

[0146] FIG. 74 illustrates the straight condition; and

[0147] FIG. 75 illustrates the deflected condition.

[0148] FIG. 76 illustrates a thick spring and a thin spring, according to at least one aspect of the present disclosure.

[0149] FIG. 77 is a top view of springs disposed in a clamp arm to show the distribution density of springs in four defined zones Zone 1-Zone 4, according to at least one aspect of the present disclosure.

[0150] FIG. 78 is a section view of a conductive polymer clamp arm pad, according to at least one aspect of the present disclosure.

[0151] FIG. 79 is a perspective view of a clamp arm pad configured to replace a conventional electrode, according to at least one aspect of the present disclosure.

[0152] FIG. 80 illustrates a clamp arm comprising the clamp arm pad described in FIG. 79, according to at least one aspect of the present disclosure.

[0153] FIG. 81 illustrates clamp arm pads configured as described in FIGS. 79-80, according to at least one aspect of the present disclosure.

[0154] FIG. 82 is a section view of a clamp arm comprising a composite clamp arm pad in contact with tissue, according to at least one aspect of the present disclosure.

[0155] FIG. 83 illustrates a clamp arm comprising a clamp jaw to support a carrier or stamping attached to the clamp jaw and a clamp arm pad, according to at least one aspect of the present disclosure.

[0156] FIG. 84 is a section view taken at section 84-84 in FIG. 83.

[0157] FIG. 85 is a section view taken at section 85-85 in FIG. 83.

[0158] FIG. 86 is a section view of an alternative implementation of a clamp arm comprising a clamp jaw, an electrically conductive pad, and an electrically non-conductive pad, according to at least one aspect of the present disclosure.

[0159] FIG. 87 is a section view of an alternative implementation of a clamp arm comprising a clamp jaw, a carrier or stamping welded to the clamp jaw, an electrically conductive pad, and an electrically non-conductive pad, according to at least one aspect of the present disclosure.

[0160] FIG. 88 illustrates insert molded electrodes, according to at least one aspect of the present disclosure.

[0161] FIG. 89 illustrates an end-effector comprising an ultrasonic blade, a clamp arm, and a clamp arm pad comprising an electrically conductive film, according to at least one aspect of the present disclosure.

[0162] FIG. 90 illustrates the clamp arm shown in FIG. 89.

[0163] FIG. 91 is a section view of the clamp arm taken along section 91-91 in FIG. 90.

[0164] FIG. 92 illustrates a clamp arm comprising a partially electrically conductive clamp arm pad, according to at least one aspect of the present disclosure.

[0165] FIG. 93 illustrates a clamp arm comprising a clamp jaw, a clamp arm pad, and a wrapped wire electrode, according to at least one aspect of the present disclosure.

[0166] FIG. 94 illustrates a clamp arm comprising a clamp jaw, a clamp arm pad, and a wrapped wire electrode, according to at least one aspect of the present disclosure.

[0167] FIG. 95 illustrates a clamp arm comprising a clamp jaw, a support comprising a hinge-like feature, an electrode, and a clamp arm pads, according to at least one aspect of the present disclosure.

[0168] FIG. 96 is a section view of the clamp arm shown in FIG. 95 in an unloaded condition through a clamp arm pad, according to at least one aspect of the present disclosure.

[0169] FIG. 97 is a section view of the clamp arm shown in FIG. 95 in a loaded condition under force F1 applied to the electrode to collapse the support comprising the hinge-like feature, according to at least one aspect of the present disclosure.

[0170] FIG. 98 illustrates a clamp arm portion of an end-effector, where the clamp arm comprises a clamp jaw, clamp arm pads, variable longitudinal support elements, bump extensions, and an electrode supported by the variable longitudinal support elements, according to at least one aspect of the present disclosure.

[0171] FIGS. 99A-99C illustrate the clamp arm shown in FIG. 98 under various load conditions, according to at least one aspect of the present disclosure, where:

[0172] FIG. 99A illustrates the clamp arm under a new firing condition with no load or evenly distributed load;

[0173] FIG. 99B illustrates the clamp arm under high load conditions, where there is a high force at the distal end and a low force at the proximal end; and

[0174] FIG. 99C illustrates the clamp arm under high load with worn down bump extensions, where there is a high load at the distal end, a low load at the proximal end, and a median load in the center.

[0175] FIG. 100 illustrates a general configuration of an end-effector comprising a clamp arm, an ultrasonic blade, an electrode, a compliant material fixed to the clamp arm to act as a spring between the electrode and the clamp arm, and a hard wear resistant material fixed to the proximal end of the electrode to set a gap between the electrode and the clamp arm, according to at least one aspect of the present disclosure.

[0176] FIG. 101 is a top view of the electrode showing apertures for receiving the compliant material therethrough.

[0177] FIGS. 102-103 shows a first configuration of an end-effector shown in FIGS. 100-101 comprising a leaf spring element at the distal end of the electrode, according to at least one aspect of the present disclosure.

[0178] FIG. 103 is a magnified view of the distal end of the electrode showing the leaf spring elements.

[0179] FIGS. 104-106 show a second configuration of the end-effector shown in FIGS. 100-101 comprising a leaf spring element at the distal end of the electrode, according to at least one aspect of the present disclosure; where

[0180] FIG. 105 is a section view of the tube showing the ultrasonic blade and the wire; and

[0181] FIG. 106 is a section view of the clamp arm showing the electrode and the leaf spring element.

[0182] FIG. 107 shows a third configuration of the end-effector shown in FIGS. 100-101 comprising a compressible material attached to the bottom portion of the distal end of the electrode, according to at least one aspect of the present disclosure.

[0183] FIG. 108 illustrates an end-effector comprising a clamp arm, an ultrasonic blade, an electrode, and a clamp arm pad, according to at least one aspect of the present disclosure.

[0184] FIGS. 109A-109F illustrate various examples of a combination ultrasonic/RF energy end-effectors comprising selectively coated components, according to at least one aspect of the present disclosure, where:

[0185] FIG. 109A illustrates an end-effector comprising selectively coated portions on the clamp arm and the ultrasonic blade, an uncoated clamp arm pad, and bare electrode portions on the clamp arm and the ultrasonic blade;

[0186] FIG. 109B illustrates an end-effector comprising selectively coated portions on the clamp arm, an uncoated ultrasonic blade, an uncoated clamp arm pad, and a bare electrode on the clamp arm;

[0187] FIG. 109C illustrates an end-effector comprising an uncoated clamp arm, selectively coated portions on the ultrasonic blade, an uncoated clamp arm pad, and a bare electrode portion on the ultrasonic blade;

[0188] FIG. 109D illustrates an end-effector comprising selectively coated portions on the clamp arm, an uncoated ultrasonic blade, an uncoated clamp arm pad, and a bare electrode on the clamp arm including a bare electrode portion at the tip of the clamp arm;

[0189] FIG. 109E illustrates an end-effector comprising an uncoated clamp arm, selectively coated portions on the ultrasonic blade, an uncoated clamp arm pad, and a bare electrode portion on the ultrasonic blade including a bare electrode portion at the tip of the ultrasonic blade; and

[0190] FIG. 109F illustrates an end-effector comprising selectively coated portions on the clamp arm and the uncoated ultrasonic blade, an uncoated clamp arm pad, and bare electrode portions on the clamp arm and the ultrasonic blade including bare electrode portions at the tip of the clamp arm and the tip of the ultrasonic blade.

[0191] FIGS. 110-112 illustrate a clamp arm comprising an I-beam shaped clamp arm pad configured for use with an end-effector comprising a clamp jaw, an ultrasonic blade (not shown), and an electrode, according to at least one aspect of the present disclosure, where:

[0192] FIG. 111 is a top view of the clamp arm showing the clamp arm pad as viewed from above the electrode; and

[0193] FIG. 112 is a top view of the electrode showing a longitudinal slot to slidably receive the clamp arm pad therein.

[0194] FIGS. 113A-113C illustrate a method of assembling the clamp arm shown in FIGS. 110-112, according to at least one aspect of the present disclosure, where:

[0195] FIG. 113A illustrates the electrode welded to the proximal end of the clamp jaw at weld points;

[0196] FIG. 113B illustrates the clamp arm pad slidably inserted into the longitudinal slot; and

[0197] FIG. 113C illustrates the hard wear resistant gap setting pad to set the gap is slidably inserted into the clamp jaw.

[0198] FIGS. 114-116 illustrate an end-effector with recesses provided within an electrode in locations of overlap with an ultrasonic blade to minimize impact between the ultrasonic blade and the electrode, according to at least one aspect of the present disclosure, where:

[0199] FIG. 114 is a section view of an end-effector comprising a clamp arm, an ultrasonic blade, an electrode, and a heavy polymer support pad, according to at least one aspect of the present disclosure;

[0200] FIG. 115 is a section view of the end-effector with a worn down heavy polymer support pad and tissue clamped between the clamp arm and the ultrasonic blade; and

[0201] FIG. 116 is a magnified view of the end-effector.

[0202] FIGS. 117-118 illustrate a clamp arm comprising a clamp jaw, a stationary gap setting pad in combination with movable floating gap setting pads, according to at least one aspect of the present disclosure, where:

[0203] FIG. 117 is a top view of a clamp arm showing the stationary gap setting pad and the movable floating gap setting pads; and

[0204] FIG. 118 is a side view of the clamp arm showing the clamp jaw, the stationary gap setting pad, the movable floating gap setting pads, the electrode, and a clamp arm pad.

[0205] FIG. 119 shows an alternative clamp arm comprising a clamp jaw, a stationary gap setting pad at a distal end, a stationary clamp arm pad, and a movable floating gap setting pad at a proximal end, according to at least one aspect of the present disclosure.

[0206] FIG. 120 shows an alternative clamp arm comprising a clamp jaw, a stationary gap setting pad at a distal end, a stationary clamp arm pad, and a movable floating gap setting pad at a proximal end, according to at least one aspect of the present disclosure.

[0207] FIGS. 121-122 illustrate a combination ultrasonic/bipolar RF energy surgical device that employs an electrode comprising small teeth, according to at least one aspect of the present disclosure, where:

[0208] FIG. 121 illustrates an end-effector comprising a clamp arm, an ultrasonic blade, an electrode comprising a plurality of teeth, a pliable clamp arm pad, and hard gap setting pads according to at least one aspect of the present disclosure; and

[0209] FIG. 122 illustrates an end-effector comprising a clamp arm, an ultrasonic blade, an electrode, a pliable clamp arm pad, and hard gap setting pads according to at least one aspect of the present disclosure.

[0210] FIGS. 123-130 illustrate a clamp arm comprising a reloadable electrode that incorporates a clamp arm pad comprising teeth, according to at least one aspect of the present disclosure, where:

[0211] FIG. 123 illustrates a reloadable electrode and a clamp arm pad, according to at least one aspect of the present disclosure;

[0212] FIG. 124 illustrates a clamp arm comprising the reloadable electrode and clamp arm pad located within a clamp jaw;

[0213] FIG. 125 is a side view of the reloadable electrode and clamp arm pad being slidably inserted into the clamp jaw from the front of the clamp jaw;

[0214] FIG. 126 is a top view of the reloadable electrode and clamp arm pad being slidably inserted into the clamp jaw from the front of the clamp jaw;

[0215] FIG. 127 is a section view of the electrode legs inserted in the clamp jaw pockets and the bent tab located in the slot in the clamp jaw;

[0216] FIG. 128 is a detailed view of the clamp arm pocket;

[0217] FIG. 129 illustrates the arm released by pressing the bent tab; and

[0218] FIG. 130 is a detailed view of the reloadable electrode and arm as a stamped component showing the extended piece of sheet metal and apertures to receive the teeth of the clamp arm pad, apertures to receive the hard wear resistant gap pads, and an aperture to receive the hard wear resistant gap pad.

[0219] FIGS. 131-133 illustrate an end-effector comprising an ultrasonic blade and a clamp arm comprising a spring-loaded clamp arm pad, according to at least one aspect of the present disclosure, where:

[0220] FIG. 131 is a perspective view of the end-effector;

[0221] FIG. 132 is a section view of the end-effector; and

[0222] FIG. 133 is an exploded view of the end-effector.

[0223] FIG. 134 illustrates a spring-loaded clamp arm pad, according to at least one aspect of the present disclosure.

[0224] FIG. 135 illustrates a clamp arm comprising a spring-loaded clamp arm pad, according to at least one aspect of the present disclosure.

[0225] FIG. 136 illustrates a clamp arm comprising a spring-loaded clamp arm pad, an electrode, and gap setting pads, according to at least one aspect of the present disclosure.

[0226] FIG. 137 illustrates a clamp arm comprising a spring-loaded clamp arm pad that is uniformly raised by an elastomeric spring, according to at least one aspect of the present disclosure.

[0227] FIG. 138 illustrates a clamp arm comprising a spring-loaded clamp arm pad, according to at least one aspect of the present disclosure.

[0228] FIGS. 139-155 illustrate two configurations of an end-effector comprising first and second configurations of floating clamp arms combined with an ultrasonic blade, according to at least one aspect of the present disclosure, where:

[0229] FIG. 139 illustrates two configurations of the end-effector;

[0230] FIG. 140 is an exploded view of a clamp jaw, electrode, and an ultrasonic blade components that are generic to the either one of the first or second configurations of the floating clamp arms;

[0231] FIG. 141 illustrates the clamp arm pad units and single piece elastic/hyper elastic block components of a first configuration of floating clamp arm;

[0232] FIG. 142 illustrates the clamp arm pad unit and multi-piece elastic/hyper elastic block components of a second configuration floating clamp arm;

[0233] FIG. 143 illustrates the clamp arm pad units layout protruding through apertures formed in the electrode without contacting each other, according to at least one aspect of the present disclosure;

[0234] FIG. 144 illustrates the apertures defined by the electrode sized and configured to receive either one of the clamp arm pad units;

[0235] FIG. 145 is an assembly view of the first configuration clamp arm comprising the single piece elastic/hyper elastic block;

[0236] FIG. 146 is a section view of a first configuration of an end-effector comprising a floating clamp arm and an ultrasonic blade, according to at least one aspect of the present disclosure;

[0237] FIG. 147 is a section view of a second configuration of an end-effector comprising a floating clamp arm and an ultrasonic blade, according to at least one aspect of the present disclosure;

[0238] FIG. 148 illustrates a method for assembling a first configuration end-effector comprising a floating clamp arm and an ultrasonic blade, according to at least one aspect of the present disclosure;

[0239] FIG. 149 illustrates the end-effector assembled in accordance with FIG. 148;

[0240] FIG. 150 illustrates a single piece elastic/hyper elastic block pushed to compression;

[0241] FIG. 151 shows a location where pressure between the ultrasonic blade and the individual clamp arm pad units can be balanced by the single piece elastic/hyper elastic block;

[0242] FIG. 152 illustrates a method for assembling a second configuration end-effector comprising a floating clamp arm and an ultrasonic blade, according to at least one aspect of the present disclosure;

[0243] FIG. 153 illustrates the end-effector assembled in accordance with FIG. 152;

[0244] FIG. 154 illustrates the multi-piece elastic/hyper elastic block pushed to compression by the individual clamp arm pad units in locations where the tissue volume is high; and

[0245] FIG. 155 shows a location where pressure between the ultrasonic blade and the individual clamp arm pad units can be balanced by the multi-piece elastic/hyper elastic block.

[0246] FIG. 156 illustrates a clamp arm comprising selectively deployable pad teeth, according to at least one aspect of the present disclosure.

[0247] FIG. 157 is a section view of the clamp arm shown in FIG. 156 comprising selectively deployable pad teeth located within the apertures in a retracted configuration, according to at least one aspect of the present disclosure.

[0248] FIG. 158 is a section view of the clamp arm shown in FIG. 156 comprising selectively deployable pad teeth in a deployed configuration, according to at least one aspect of the present disclosure.

[0249] FIG. 159 is a detail section view of the clamp arm shown in FIG. 156 comprising selectively deployable pad teeth in a retracted configuration, according to at least one aspect of the present disclosure.

[0250] FIG. 160A illustrates an end-effector configured to receive heat sink material block as shown in FIGS. 160B-160C, according to at least one aspect of the present disclosure.

[0251] FIG. 160B illustrates one example of the end-effector shown in FIG. 160A with a heat sink material block added or otherwise fixedly attached to a clamp arm pad, supported by the metal clamp jaw.

[0252] FIG. 160C shows the heat sink material block with a worn out clamp arm pad.

[0253] FIG. 161A illustrates an end-effector comprising a metal clamp jaw and an ultrasonic blade where at least some portions of the metal clamp jaw is made of a heat sink material, according to at least one aspect of the present disclosure.

[0254] FIG. 161B illustrates an alternate clamp arm pad.

[0255] FIG. 161C illustrates the heat sink material comprising either one or both of a material with high heat conductivity property and high specific heat and create large surface area feature to increase heat diffusion as heat builds up in the clamp arm pad.

[0256] FIG. 162 shows the components of the end effector described in FIGS. 160A-160C and 161A-161C.

[0257] FIG. 163 shows one heat sink structure comprising fins for increasing the surface area of the heat sink component, according to at least one aspect of the present disclosure.

[0258] FIG. 164 shows another heat sink structure comprising pores for increasing the surface area of the heat sink component, according to at least one aspect of the present disclosure.

[0259] FIG. 165 illustrates an end-effector comprising an ultrasonic blade, a clamp arm, and a watch-band style segmented electrode, according to at least one aspect of the present disclosure.

[0260] FIG. 166 is a magnified view of the distal and medial segments of the segmented electrode shown in FIG. 165.

[0261] FIG. 167 is a section view of an electrode comprising an electrically non-conductive clamp arm pad comprising a plurality of teeth, according to at least one aspect of the present disclosure.

[0262] FIG. 168 illustrates an end-effector comprising a clamp arm and an ultrasonic blade, according to at least one aspect of the present disclosure.

[0263] FIG. 169 is a graphical depiction of impedance amplitude along the left vertical axis versus time along the horizontal axis and blade movement modes along the right vertical axis versus time along the horizontal axis, according to at least one aspect of the present disclosure.

[0264] FIG. 170 illustrates a clamp arm comprising a clamp jaw, a clamp arm pad comprising exposed teeth, according to at least one aspect of the present disclosure.

[0265] FIG. 171 illustrates a clamp arm comprising a clamp jaw comprising raised sidewalls and a raised lip, and a clamp arm pad comprising a plurality of teeth, according to at least one aspect of the present disclosure.

[0266] FIGS. 172-174 illustrates an electrode with a perimeter wall, according to at least one aspect of the present disclosure, where:

[0267] FIG. 172 illustrates a clamp arm comprising an electrode with a perimeter wall supported by a clamp jaw and a clamp arm pad, according to at least one aspect of the present disclosure;

[0268] FIG. 173 is a section view of the clamp jaw taken along section 173-173 in FIG. 172; and

[0269] FIG. 174 is an alternative or additional feature where a sealant such as silicone or other fluid polymer may be provided in the space beneath the electrode and the above the clamp jaw to prevent tissue from going and accumulating under the electrode, according to at least one aspect of the present disclosure.

[0270] FIG. 175 illustrates a clamp arm comprising a clamp jaw, an electrode, a clamp arm pad (FIG. 176) comprising a plurality of teeth, and a deflectable ledge along the lateral sides of the electrode, according to at least one aspect of the present disclosure.

[0271] FIGS. 176 and 177 are section views of the clamp arm shown in FIG. 175 taken at section 176-176, where:

[0272] FIG. 176 shows the electrode and the skirt in an undeflected state; and

[0273] FIG. 177 shows the electrode and the skirt in a deflected state where the skirt shifts laterally in the direction indicated by the arrows.

[0274] FIG. 178 illustrates a clamp arm comprising a clamp jaw, an electrode, and sidewalls extending downwardly from the electrode past the sidewalls of the clamp jaw, and clamp arm pad (FIG. 179) comprising a plurality of teeth, according to at least one aspect of the present disclosure.

[0275] FIG. 179 is a section view of the clamp arm shown in FIG. 178 along section line 179-179.

[0276] FIG. 180 illustrates an alternative clamp arm to the clamp arm shown in FIGS. 178-179, according to at least one aspect of the present disclosure.

[0277] FIG. 181 is a top perspective view of an electrode comprising a lip extending downwardly from a top surface of the electrode, according to at least one aspect of the present disclosure.

[0278] FIG. 182 is a bottom perspective view of the electrode shown in FIG. 181 showing the downwardly extending lip.

[0279] FIG. 183 is a top perspective view of the electrode shown in FIGS. 181-182 with a clamp arm pad with teeth extending through the apertures.

[0280] FIG. 184 is a top perspective view of the electrode shown in FIGS. 181-182 with a clamp arm pad with teeth extending through the apertures.

[0281] FIG. 185 illustrates an end effector comprising a clamp arm and an ultrasonic blade, according to at least one aspect of the present disclosure.

[0282] FIG. 186 illustrates components of an end-effector comprising a clamp arm comprising a clamp jaw, an electrode comprising a plurality of apertures sized and configured to receive a plurality of individual tissue pad units, a single piece elastic/hyper elastic block, small wear resistant gap pads, and a large wear resistant gap pads., according to at least one aspect of the present disclosure.

[0283] FIG. 187 illustrates an assembled version of the end-effector comprising the components illustrated in FIG. 186.

[0284] FIG. 188 illustrates various views of the individual tissue pad unit, according to at least one aspect of the present disclosure.

[0285] FIG. 189 illustrates the electrode, according to at least one aspect of the present disclosure.

[0286] FIG. 190 illustrates a section view of the clamp arm showing the elastic/hyper elastic block, according to at least one aspect of the present disclosure.

[0287] FIG. 191 is a perspective view of the elastic/hyper elastic block.

[0288] FIG. 192 is perspective views of a sub-assembly comprising the clamp jaw, electrode, and the individual tissue pad units, according to at least one aspect to the present disclosure.

[0289] FIG. 193 is a perspective view of the clamp arm assembly comprising the elastic/hyper elastic block of FIG. 191 assembled to the sub-assembly, according to at least one aspect of the present disclosure.

[0290] FIG. 194 is perspective views of a sub-assembly comprising the clamp jaw, electrode, and the individual tissue pad units, according to at least one aspect to the present disclosure.

[0291] FIG. 195 illustrates a first assembly step, according to at least one aspect of the present disclosure, where the individual tissue pad units are inserted into the electrode.

[0292] FIGS. 196-198 illustrate a second assembly step, according to at least one aspect of the present disclosure, where the individual tissue pad units are locked into the electrode.

[0293] FIG. 199 illustrates a third assembly step, according to at least one aspect of the present disclosure, where the single piece elastic/hyper elastic block is bound to the clamp jaw.

[0294] FIGS. 200-205 illustrate a fourth assembly step, according to at least one aspect of the present disclosure, where the first sub-assembly produced in the second assembly step described in FIGS. 196-198 is assembled with the second sub-assembly produced in the third assembly as described in step of FIG. 199 to produce the clamp arm.

[0295] FIGS. 206-207 illustrate a fifth assembly step, according to at least one aspect of the present disclosure, where two small wear resistant gap pads are press fit into the slots defined by the electrode and the large wear resistant gap pad is slid into the slot defined by the electrode.

[0296] FIG. 208 illustrates a clamp arm comprising fill material between the electrode and the clamp jaw, according to at least one aspect of the present disclosure.

[0297] FIGS. 209-211 illustrate various apparatuses to address these desired improvements, according to at least one aspect of the present disclosure, where:

[0298] FIG. 209 that shows a clamp arm that is easier to manufacture by eliminating a separate clamp arm electrode and building up a surface of the clamp arm and eliminating the separate electrode blade;

[0299] FIG. 210 shows a clamp arm to deter tissue accumulation between the electrode and the clamp jaw by adding a skirt to the perimeter of the electrode or alternatively to the perimeter of the clamp jaw; and

[0300] FIG. 211 that shows a section view of an end-effector comprising an ultrasonic blade and a clamp arm configured to improve distal tissue grasping.

[0301] FIG. 212 illustrates a clamp arm comprising a clamp jaw, an electrode, a clamp arm pad comprising a plurality of teeth, and a curtain located on either side of the electrode to minimize pinch point of tissue, and a bull nose/clamp arm bumper to minimize delamination of the electrode by making contact with tissue first during blunt dissection, according to at least one aspect of the present disclosure.

[0302] FIGS. 213-214 illustrate an alternative clamp arm comprising a clamp jaw, an electrode, a clamp arm pad comprising a plurality of teeth, and a curtain located on either side of the clamp arm pad, according to at least one aspect of the present disclosure, where:

[0303] FIG. 213 is an exploded view of the clamp arm; and

[0304] FIG. 214 is a section view of the clamp arm.

[0305] FIG. 215 is a side view of a clamp arm, according to at least one aspect of the present disclosure.

[0306] FIG. 216 is a side view of a clamp arm with the cantilevered electrode in a deflected state to pull tissue off

the teeth of the clamp arm pad, according to at least one aspect of the present disclosure

[0307] FIG. 217 is a bottom perspective view of the clamp arm, according to at least one aspect of the present disclosure.

[0308] FIG. 218 is a bottom perspective view of the clamp arm with the cantilevered electrode in a deflected state to pull tissue off the teeth of the clamp arm pad, according to at least one aspect of the present disclosure.

[0309] FIG. 219 illustrates a surgical device comprising a mode selection button switch on the device, according to at least one aspect of the present disclosure.

[0310] FIGS. 220A-220C illustrate three options for selecting the various operating modes of the surgical device, according to at least one aspect of the present disclosure, where:

[0311] FIG. 220A shows a first mode selection option where the button switch can be pressed forward or backward to cycle the surgical instrument through the various modes;

[0312] FIG. 220B shows a second mode selection option where the button switch is pressed up or down to cycle the surgical instrument through the various modes; and

[0313] FIG. 220C shows a third mode selection option where the button switch is pressed forward, backward, up, or down to cycle the surgical instrument through the various modes.

[0314] FIG. 221 illustrates a surgical device comprising a mode selection button switch on the back of the device, according to at least one aspect of the present disclosure.

[0315] FIG. 222A shows a first mode selection option where as the mode button switch is pressed to toggle through various modes, colored light indicates the selected mode on the user interface.

[0316] FIG. 222B shows a second mode selection option where as the mode button switch is pressed to toggle through various modes a screen indicates the selected mode (e.g., LCD, e-ink).

[0317] FIG. 222C shows a third mode selection option where as the mode button switch is pressed to toggle through various modes, labelled lights indicate the selected mode.

[0318] FIG. 222D shows a fourth mode selection option where as a labeled button switch is pressed to select a mode, when a labeled button switch is selected, it is illuminated to indicate mode selected.

[0319] FIG. 223 illustrates a surgical device comprising a trigger activation mechanism, according to at least one aspect of the present disclosure.

[0320] FIG. 224 illustrates an alternative clamp arm comprising a metal clamp jaw, an electrode, a plurality of clamp arm pads, and gap pads, according to at least one aspect of the present disclosure.

[0321] FIG. 225 is a surgical system comprising a surgical hub paired with a visualization system, a robotic system, and an intelligent instrument, in accordance with at least one aspect of the present disclosure.

[0322] FIG. 226 illustrates an example of a generator, in accordance with at least one aspect of the present disclosure.

[0323] FIG. 227 is a diagram of various modules and other components that are combinable to customize modular energy systems, in accordance with at least one aspect of the present disclosure.

[0324] FIG. 228A is a first illustrative modular energy system configuration including a header module and a display screen that renders a graphical user interface (GUI)

for relaying information regarding modules connected to the header module, in accordance with at least one aspect of the present disclosure.

[0325] FIG. 228B is the modular energy system shown in FIG. 228A mounted to a cart, in accordance with at least one aspect of the present disclosure.

[0326] FIG. 229 depicts a perspective view of an exemplary surgical system having a generator and a surgical instrument operable to treat tissue with ultrasonic energy and bipolar RF energy, in accordance with at least one aspect of the present disclosure.

[0327] FIG. 230 depicts a top perspective view of an end effector of the surgical instrument of FIG. 229, having a clamp arm that provides a first electrode and an ultrasonic blade that provides a second electrode, in accordance with at least one aspect of the present disclosure.

[0328] FIG. 231 depicts a bottom perspective view of the end effector of FIG. 230, in accordance with at least one aspect of the present disclosure.

[0329] FIG. 232 depicts a partially exploded perspective view of the surgical instrument of FIG. 229, in accordance with at least one aspect of the present disclosure.

[0330] FIG. 233 depicts an enlarged exploded perspective view of a distal portion of the shaft assembly and the end effector of the surgical instrument of FIG. 229, in accordance with at least one aspect of the present disclosure.

[0331] FIG. 234 illustrates a method of operating method of operating a combination ultrasonic/bipolar RF surgical device with a combination energy modality end-effector, according to at least one aspect of the present disclosure.

DESCRIPTION

[0332] Applicant of the present application owns the following U.S. Provisional patent applications, filed on Dec. 30, 2019, the disclosure of each of which is herein incorporated by reference in its respective entirety:

[0333] U.S. Provisional Patent Application Ser. No. 62/955,294, entitled USER INTERFACE FOR SURGICAL INSTRUMENT WITH COMBINATION ENERGY MODALITY END-EFFECTOR;

[0334] U.S. Provisional Patent Application Ser. No. 62/955,299, entitled ELECTROSURGICAL INSTRUMENTS FOR COMBINATION ENERGY DELIVERY; and.

[0335] U.S. Provisional Patent Application Ser. No. 62/955,306, entitled SURGICAL INSTRUMENTS.

[0336] Applicant of the present application owns the following U.S. patent applications that were filed on even date herewith, and which are each herein incorporated by reference in their respective entireties:

[0337] Attorney Docket No. END9232USNP1/190715-1, entitled USER INTERFACE FOR SURGICAL INSTRUMENT WITH COMBINATION ENERGY MODALITY END-EFFECTOR;

[0338] Attorney Docket No. END9233USNP2/190716-2, entitled DEFLECTABLE SUPPORT OF RF ENERGY ELECTRODE WITH RESPECT TO OPPOSING ULTRASONIC BLADE;

[0339] Attorney Docket No. END9233USNP3/190716-3, entitled NON-BIASED DEFLECTABLE ELECTRODE TO MINIMIZE CONTACT BETWEEN ULTRASONIC BLADE AND ELECTRODE;

- [0340] Attorney Docket No. END9233USNP4/190716-4, entitled DEFLECTABLE ELECTRODE WITH HIGHER DISTAL BIAS RELATIVE TO PROXIMAL BIAS;
- [0341] Attorney Docket No. END9233USNP5/190716-5, entitled DEFLECTABLE ELECTRODE WITH VARIABLE COMPRESSION BIAS ALONG THE LENGTH OF THE DEFLECTABLE ELECTRODE;
- [0342] Attorney Docket No. END9233USNP6/190716-6, entitled ASYMMETRIC SEGMENTED ULTRASONIC SUPPORT PAD FOR COOPERATIVE ENGAGEMENT WITH A MOVABLE RF ELECTRODE;
- [0343] Attorney Docket No. END9233USNP7/190716-7, entitled VARIATION IN ELECTRODE PARAMETERS AND DEFLECTABLE ELECTRODE TO MODIFY ENERGY DENSITY AND TISSUE INTERACTION;
- [0344] Attorney Docket No. END9233USNP8/190716-8, entitled TECHNIQUES FOR DETECTING ULTRASONIC BLADE TO ELECTRODE CONTACT AND REDUCING POWER TO ULTRASONIC BLADE;
- [0345] Attorney Docket No. END9233USNP9/190716-9, entitled CLAMP ARM JAW TO MINIMIZE TISSUE STICKING AND IMPROVE TISSUE CONTROL; and
- [0346] Attorney Docket No. END9233USNP10/190716-10, entitled PARTIALLY CONDUCTIVE CLAMP ARM PAD TO ENABLE ELECTRODE WEAR THROUGH AND MINIMIZE SHORT CIRCUITING.
- [0347] Applicant of the present application owns the following U.S. patent applications that were filed on May 28, 2020, and which are each herein incorporated by reference in their respective entireties:
- [0348] U.S. patent application Ser. No. 16/885,813, entitled METHOD FOR AN ELECTROSURGICAL PROCEDURE;
- [0349] U.S. patent application Ser. No. 16/885,820, entitled ARTICULATABLE SURGICAL INSTRUMENT;
- [0350] U.S. patent application Ser. No. 16/885,823, entitled SURGICAL INSTRUMENT WITH JAW ALIGNMENT FEATURES;
- [0351] U.S. patent application Ser. No. 16/885,826, entitled SURGICAL INSTRUMENT WITH ROTATABLE AND ARTICULATABLE SURGICAL END EFFECTOR;
- [0352] U.S. patent application Ser. No. 16/885,838, entitled ELECTROSURGICAL INSTRUMENT WITH ASYNCHRONOUS ENERGIZING ELECTRODES;
- [0353] U.S. patent application Ser. No. 16/885,851, entitled ELECTROSURGICAL INSTRUMENT WITH ELECTRODES BIASING SUPPORT;
- [0354] U.S. patent application Ser. No. 16/885,860, entitled ELECTROSURGICAL INSTRUMENT WITH FLEXIBLE WIRING ASSEMBLIES;
- [0355] U.S. patent application Ser. No. 16/885,866, entitled ELECTROSURGICAL INSTRUMENT WITH VARIABLE CONTROL MECHANISMS;
- [0356] U.S. patent application Ser. No. 16/885,870, entitled ELECTROSURGICAL SYSTEMS WITH INTEGRATED AND EXTERNAL POWER SOURCES;
- [0357] U.S. patent application Ser. No. 16/885,873, entitled ELECTROSURGICAL INSTRUMENTS WITH ELECTRODES HAVING ENERGY FOCUSING FEATURES;
- [0358] U.S. patent application Ser. No. 16/885,879, entitled ELECTROSURGICAL INSTRUMENTS WITH ELECTRODES HAVING VARIABLE ENERGY DENSITIES;
- [0359] U.S. patent application Ser. No. 16/885,881, entitled ELECTROSURGICAL INSTRUMENT WITH MONOPOLAR AND BIPOLAR ENERGY CAPABILITIES;
- [0360] U.S. patent application Ser. No. 16/885,888, entitled ELECTROSURGICAL END EFFECTORS WITH THERMALLY INSULATIVE AND THERMALLY CONDUCTIVE PORTIONS;
- [0361] U.S. patent application Ser. No. 16/885,893, entitled ELECTROSURGICAL INSTRUMENT WITH ELECTRODES OPERABLE IN BIPOLAR AND MONOPOLAR MODES;
- [0362] U.S. patent application Ser. No. 16/885,900, entitled ELECTROSURGICAL INSTRUMENT FOR DELIVERING BLENDED ENERGY MODALITIES TO TISSUE;
- [0363] U.S. patent application Ser. No. 16/885,917, entitled CONTROL PROGRAM ADAPTATION BASED ON DEVICE STATUS AND USER INPUT;
- [0364] U.S. patent application Ser. No. 16/885,923, entitled CONTROL PROGRAM FOR MODULAR COMBINATION ENERGY DEVICE; and
- [0365] U.S. patent application Ser. No. 16/885,931, entitled SURGICAL SYSTEM COMMUNICATION PATHWAYS.
- [0366] Before explaining various forms of surgical instruments in detail, it should be noted that the illustrative forms are not limited in application or use to the details of construction and arrangement of parts illustrated in the accompanying drawings and description. The illustrative forms may be implemented or incorporated in other forms, variations and modifications, and may be practiced or carried out in various ways. Further, unless otherwise indicated, the terms and expressions utilized herein have been chosen for the purpose of describing the illustrative forms for the convenience of the reader and are not for the purpose of limitation thereof.
- [0367] Further, it is understood that any one or more of the following-described forms, expressions of forms, examples, can be combined with any one or more of the other following-described forms, expressions of forms, and examples.
- [0368] Various forms are directed to improved ultrasonic and/or electrosurgical (RF) instruments configured for effecting tissue treating, dissecting, cutting, and/or coagulation during surgical procedures. In one form, a combined ultrasonic and electrosurgical instrument may be configured for use in open surgical procedures, but has applications in other types of surgery, such as minimally invasive laparoscopic, orthoscopic, or thoracoscopic procedures, for example, non-invasive endoscopic procedures, either in hand held or and robotic-assisted procedures. Versatility is achieved by selective application of multiple energy modalities.

ties simultaneously, independently, sequentially, or combinations thereof. For example, versatility may be achieved by selective use of ultrasonic and electrosurgical energy (e.g., monopolar or bipolar RF energy) either simultaneously, independently, sequentially, or combinations thereof.

[0369] In one aspect, the present disclosure provides an ultrasonic surgical clamp apparatus comprising an ultrasonic blade and a deflectable RF electrode such that the ultrasonic blade and deflectable RF electrode cooperate to effect sealing, cutting, and clamping of tissue by cooperation of a clamping mechanism of the apparatus comprising the RF electrode with an associated ultrasonic blade. The clamping mechanism includes a pivotal clamp arm which cooperates with the ultrasonic blade for gripping tissue therebetween. The clamp arm is preferably provided with a clamp tissue pad (also known as “clamp arm pad”) having a plurality of axially spaced gripping teeth, segments, elements, or individual units which cooperate with the ultrasonic blade of the end-effector to achieve the desired sealing and cutting effects on tissue, while facilitating grasping and gripping of tissue during surgical procedures.

[0370] In one aspect, the end-effectors described herein comprise an electrode. In other aspects, the end-effectors described herein comprise alternatives to the electrode to provide a compliant coupling of RF energy to tissue, accommodate pad wear/thinning, minimize generation of excess heat (low coefficient of friction, pressure), minimize generation of sparks, minimize interruptions due to electrical shorting, or combinations thereof. The electrode is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0371] In other aspects, the end-effectors described herein comprise a clamp arm mechanism configured to apply high pressure between a pad and an ultrasonic blade to grasp and seal tissue, maximize probability that the clamp arm electrode contacts tissue in limiting or difficult scenarios, such as, for example, thin tissue, tissue under lateral tension, tissue tenting/vertical tension especially tenting tissue away from clamp arm.

[0372] In other aspects, the end-effectors described herein are configured to balance match of surface area/current densities between electrodes, balance and minimize thermal conduction from tissue interface, such as, for example, impacts lesion formation and symmetry, cycle time, residual thermal energy.

[0373] In other aspects, the end-effectors described herein are configured to minimize sticking, tissue adherence (minimize anchor points) and may comprise small polyimide pads.

[0374] In various aspects, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device. The combination ultrasonic/bipolar RF energy surgical device comprises an end-effector. The end-effector comprises a clamp arm and an ultrasonic blade. The clamp arm comprises a movable clamp jaw, a compliant polymeric pad, and at least one bipolar RF electrode. The at least one electrode is coupled to a positive pole of an RF generator and the ultrasonic blade is coupled to the negative pole of the RF generator. The ultrasonic blade is acoustically coupled to an ultrasonic transducer stack that is driven by an ultrasonic generator. In various aspects, the end-effector comprises electrode biasing mechanisms.

[0375] In one general aspect, the present disclosure is directed to a method for using a surgical device comprising a combination of ultrasonic and advanced bipolar RF energy with a movable RF electrode on at least one jaw of an end-effector. The movable RF electrode having a variable biasing force from a proximal end to a distal end of the movable RF electrode. The movable RF electrode being segmented into discrete portions than can be put in electrical communication or isolated from each other. The movable RF electrode being made of a conductive or partially conductive material. It will be appreciated that any of the end effectors described in this disclosure may be configured with an electrode biasing mechanism.

[0376] In one aspect, the present disclosure provides a limiting electrode biasing mechanism to prevent ultrasonic blade to electrode damage. Generally, in various aspects, the present disclosure provides an end-effector for use with a ultrasonic/RF combination device, where the end-effector comprises an electrode. In one aspect, the combination ultrasonic/bipolar RF energy surgical device comprises an electrode biasing mechanism. In one aspect, the limiting electrode biasing mechanism is configured to prevent or minimize ultrasonic blade to electrode damage. The electrode is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0377] In various aspects, the present disclosure provides an electrode cantilever beam fixated at only one end comprising a biasing threshold mechanism. In one aspect, the deflectable cantilever electrode is configured for combination ultrasonic/bipolar RF energy surgical devices.

[0378] In one aspect, the combination ultrasonic/RF energy surgical device comprises an ultrasonic blade, a clamp arm, and at least one electrode which crosses over the ultrasonic blade. In one aspect, the electrode is configured to be deflectable with respect to the clamp arm and includes features to change the mechanical properties of the tissue under compression between the electrode and the ultrasonic blade. In another aspect, the electrode includes a feature to prevent inadvertent contact between the electrode and the ultrasonic blade to prevent or minimize ultrasonic blade to electrode damage.

[0379] In various aspects, the electrode comprises a metallic spring element attached at a proximal end of the clamp jaw of the end-effector. The metallic spring element defines openings for receives therethrough one or more clamp arm pads (also known as “tissue pads” or “clamp tissue pads”) and comprises integrated minimum gap elements. This configuration of the electrode provides a method of preventing tissue from accumulating around the biasing mechanism that can impact the performance of the electrode. This configuration also minimizes the binding between the wear pads and the biasing spring, increases the strength of the electrode to clamp arm connection, minimizes inadvertent release of the clamp arm pads by attaching the polyimide pads to the electrode, and balance matches the surface area/current densities between electrodes. The electrode is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode is deflectable and may be referred to as a cantilever beam electrode or deflectable electrode.

[0380] FIGS. 1-9 illustrate one aspect of an end-effector comprising a deflectable/cantilever electrode configured for

use with a combination ultrasonic/bipolar RF energy device, according to at least one aspect of the present disclosure. FIG. 1 is a perspective view of a clamp arm 1000 portion of an end-effector for use with a combined ultrasonic/RF device, according to at least one aspect of the present disclosure. For conciseness and clarity of disclosure, the ultrasonic blade, which functions as the other clamp arm of the end-effector is not shown. The end-effector is configured such that the ultrasonic blade is one pole of the bipolar RF circuit and the clamp arm 1000 is the opposite pole. A consistent RF electrode gap is maintained between the clamp arm 1000 and the ultrasonic blade to prevent the ultrasonic blade from contacting the electrode resulting in blade breakage or a short circuit. Tissue under treatment is clamped and compressed between the clamp arm 1000 and the ultrasonic blade.

[0381] The clamp arm 1000 includes a frame 1002, an electrode 1004, at least one small electrically nonconductive gap pad 1006, at least one large electrically nonconductive gap pad 1008, at least one electrically nonconductive clamp arm pad 1010. In one aspect, the small and large gap pads 1006, 1008 are configured to set a gap between the electrode 1004 and the ultrasonic blade. The clamp arm pad 1010 is configured to grasp tissue between the clamp arm 1000 and the ultrasonic blade to assist with sealing and cutting of the tissue. In other aspects, the small and large nonconductive gap pads may be swapped. In other aspects, the nonconductive gap pads are simply sized differently regardless of the relative size difference between the nonconductive gap pads.

[0382] Pivotal movement of the clamp arm 1000 with respect to the end-effector is effected by the provision of at least one, and preferably a pair of, lever portions 1012 of the clamp arm 1000 frame 1002 at a proximal end 1014 thereof. The lever portions 1012 are positioned on respective opposite sides of an ultrasonic waveguide and end-effector, and are in operative engagement with a drive portion of a reciprocable actuating member. Reciprocable movement of the actuating member, relative to an outer tubular sheath and the ultrasonic waveguide, thereby effects pivotal movement of the clamp arm 1000 relative to the end-effector about pivot points 1016. The lever portions 1012 can be respectively positioned in a pair of openings defined by the drive portion, or otherwise suitably mechanically coupled therewith, whereby reciprocable movement of the actuating member acts through the drive portion and lever portions 1012 to pivot the clamp arm 1000.

[0383] FIG. 2 is an exploded view of the clamp arm 1000 shown in FIG. 1, according to at least one aspect of the present disclosure. In various aspects, the electrode 1004 is made of a metallic spring material attached at a proximal end 1014 of the frame 1002 of the clamp arm 1000 such that the electrode 1004 can deflect. The metallic spring electrode 1004 defines openings 1018 for receiving therethrough elements of the clamp arm pad 1010 and defines additional openings 1020, 1021 for receiving the gap pads 1006, 1008 to set a minimum gap between the electrode 1004 and the ultrasonic blade. At least one of the gap pads 1006 is disposed on a distal end 1022 of the electrode 1004. The gap pads 1006, 1008 are thus integrated with the electrode 1004. In this configuration, the electrode 1004 prevents tissue from accumulating around the biasing mechanism, e.g., cantilevered spring, that can impact the performance of the electrode 1004. This configuration also minimizes the binding between the wearable clamp arm pads 1010 and the biasing

spring electrode 1004, increases the strength of the electrode 1004 to the clamp arm connection, minimizes inadvertent release of the clamp arm pads 1018 by attaching the gap pads 1006, 1008 to the electrode 1004, and balance matches the surface area/current densities between electrodes. The electrode 1004 is attached to the frame 1002 by two protrusions 1024. The electrode protrusions 1024 are attached to the proximal end 1014 of the frame 1002 as shown in FIGS. 3 and 4.

[0384] FIGS. 3 and 4 are perspective views of the frame 1002, according to at least one aspect of the present disclosure. These views illustrate the connection surfaces 1026 on the proximal end 1014 of the frame 1002 for attaching the proximal end of the electrode 1004 to the frame 1002. In one aspect, the electrode protrusions 1024 are welded to the connection surfaces 1026 of the frame 1002 such that the electrode 1004 behaves in a deflectable manner.

[0385] FIG. 5 is a perspective view of the electrode 1004, according to at least one aspect of the present disclosure. This view illustrates the bias in the electrode 1004 made of spring material as indicated by the curvature of the electrode 1004 along a longitudinal length. The openings 1018, 1020, 1021 for receiving the gap pads 1006, 1008 and the clamp arm pads 1010. In one aspect, the electrode 1004 has a thickness “d” of 0.010” and may be selected within a range of thicknesses of 0.005” to 0.015”, for example. With reference also to FIGS. 8 and 9, the openings 1020 are sized and configured to receive a protrusion 1036 defined on a bottom portion of the gap pads 1006.

[0386] FIG. 6 is a perspective view of the clamp arm pad 1010, according to at least one aspect of the present disclosure. The clamp arm pad 1010 comprises a plurality of clamp arm elements 1032 protruding from a backbone 1030. Throughout this disclosure, the clamp arm elements 1032 also are referred to as “teeth.” In one aspect, the clamp arm pad 1010 defines apertures 1028 in a position where the gap pads 1006 are located on the electrode 1004. With reference also to FIGS. 8 and 9, the apertures 1028 defined by the clamp arm pad 1010 are sized and configured to receive the protrusion 1036 defined on a bottom portion of the gap pads 1006. In one aspect, the clamp arm pad 1010 material is softer than the gap pad 1006, 1008 material. In one aspect, the clamp arm pad 1010 is made of a non-stick lubricious material such as polytetrafluoroethylene (PTFE) or similar synthetic fluoropolymers of tetrafluoroethylene. PTFE is a hydrophobic, non-wetting, high density and resistant to high temperatures, and versatile material and non-stick properties. In contrast, the gap pads 1006, 1008 are made of a polyimide material, and in one aspect, is made of a durable high-performance polyimide-based plastic known under the tradename VESPEL and manufactured by DuPont or other suitable polyimide, polyimide polymer alloy, or PET (Polyethylene Terephthalate), PEEK (Polyether Ether Ketone), PEKK (Poly Ether Ketone Ketone) polymer alloy, for example. Unless otherwise noted hereinbelow, the clamp arm pads and gap pads described hereinbelow are made of the materials described in this paragraph.

[0387] FIG. 7 is a perspective top view of the large gap pad 1008, according to at least one aspect of the present disclosure. The large gap pad 1008 comprises a protrusion 1034 sized and configured to fit within the opening 1021 at the proximal end 1014 of the electrode 1004. FIG. 8 is a perspective top view of the small gap pad 1006, according to at least one aspect of the present disclosure. FIG. 9 is a

perspective bottom view of the small gap pad **1006** shown in FIG. **8**. As shown in FIGS. **8** and **9**, the small gap pads **1006** include a protrusion **1036** at the bottom portion sized and configured to be received within the openings **1020** defined by the electrode **1004** and the apertures **1028** defined by the clamp arm pad **1010**. The small and large gap pads **1006**, **1008** are made of a polyimide material, and in one aspect, is made of a durable high-performance polyimide-based plastic known under the tradename VESPEL and manufactured by DuPont. The durability of the polyimide material ensures that the electrode gap remains relatively constant under normal wear and tear.

[0388] In one aspect, the present disclosure also provides additional end-effector configurations for combination ultrasonic and bipolar RF energy devices. This portion of the disclosure provides end-effector configurations for use in combination ultrasonic and bipolar RF energy devices. In these configurations, the end-effector maintains a consistent gap between the RF electrode gap and the ultrasonic blade, which functions as one pole of the bipolar RF circuit, and the clamp arm, which functions as the opposite pole of the bipolar RF circuit. In conventional end-effector configurations, the electrode gap is set by a soft PTFE clamp arm pad which may be subject to wear during surgery. When the clamp arm pad wears through, the ultrasonic blade can contact the electrode resulting in blade breakage or an electrical short circuit, both of which are undesirable.

[0389] To overcome these and other limitations, various aspects of the present disclosure incorporate a deflectable RF electrode in combination with a clamp arm pad comprising a non-stick lubricious compliant (e.g., PTFE) pad fixed to the clamp arm. The RF electrode contains wear-resistant, electrically nonconductive pads which contact the blade to set the blade-to-electrode gap. The compliant clamp arm pad extends through openings defined by the electrode and reacts to the clamping force from the ultrasonic blade. As the compliant clamp arm pad wears, the electrode deflects to maintain a constant gap between the blade and the electrode. Such configuration provides a consistent gap between the electrode and the ultrasonic blade throughout the life of the device, prevents shorting and ultrasonic blade breakage, which can occur when the ultrasonic blade touches the electrode, and enables the electrode material to be positioned directly on the side that is opposite the ultrasonic blade to improve sealing performance. The electrode is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode may be referred to as a cantilever beam electrode or deflectable electrode.

[0390] FIGS. **10A-13B** show configurations wherein a deflectable/cantilever electrode is fixed to a clamp arm at a proximal end and can move vertically distal of the attachment point, according to various aspects of the present disclosure. In each of the configurations shown and described in connection with FIGS. **10A-13B**, the deflectable/cantilever electrode may be attached to the proximal end of the clamp arm by welding, brazing, soldering, or other suitable mounting technique. Furthermore, the clamp arm pad in each of the configurations illustrated and described herein in connection with FIGS. **10A-13B** may be made of a non-stick lubricious compliant material such as PTFE or similar synthetic fluoropolymers of tetrafluoroethylene. PTFE is a hydrophobic, non-wetting, high density and resistant to high temperatures, and versatile material and

non-stick properties. Moreover, the gap pads or plugs in each of the configurations illustrated and described herein in connection with FIGS. **10A-13B** may be made of a polyimide material, and in one aspect, is made of a durable high-performance polyimide-based plastic known under the tradename VESPEL manufactured by DuPont, or other suitable polyimide, polyimide polymer alloy, or PET, PEEK, PEKK polymer alloy, for example.

[0391] Turning now to FIGS. **10A-10B**, an electrode **1102** having a proximal end **1104** fixedly attached to a proximal end **1104** of a clamp arm **1100** and having a distal end **1106** opposite to the fixed end that is free to move or deflect is illustrated. FIG. **10A** illustrates a clamp arm **1100** and FIG. **10B** illustrates a electrode **1102**, according to at least one aspect of the present disclosure. The electrode **1102** is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode **1102** is one pole of an RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit. The electrode **1102** is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode **1102** may be referred to as a cantilever beam electrode or deflectable electrode.

[0392] Turning first to FIG. **10A**, the clamp arm **1100** comprises an electrode **1102** attached at a proximal end **1104** of the clamp arm **1100**. In one aspect, the electrode **1102** is attached to the proximal end **1104** of the clamp arm **1100** by welding or any other suitable means such as soldering, for example. This enables the electrode **1102** to move vertically at a distal end **1112** of the clamp arm **1100**. The clamp arm **1100** comprise a clamp arm pad **1108** and a resilient plug **1110** located at the distal end **1106** of the clamp arm **1100**.

[0393] Turning now to FIG. **10B**, the electrode **1102** defines a plurality of apertures **1112** to receive a plurality of protruding elements **1114** (e.g., teeth) of the clamp arm pad **1108** therethrough. The electrode **1102** comprises a connection portion **1116** to attach the electrode **1102** to the proximal end **1104** of the clamp arm **1100**. The electrode **1102** comprises a resilient plug **1110** at the distal end **1106** of the electrode **1102**. In one aspect, the resilient plug **1110** is made of silicone or other suitable material. The electrode **1102** is made an electrically conductive metal suitable to transmit RF energy to the tissue. The electrode **1102** is one pole of the bipolar RF circuit and the ultrasonic blade is another pole of the bipolar RF circuit.

[0394] FIGS. **11A-11D** illustrate an electrode having a proximal end fixedly attached to a proximal end of a clamp arm and having a distal end opposite to the fixed end that is free to move or deflect, according to at least one aspect of the present disclosure. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of an RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0395] Turning first to FIG. **11A**, there is shown an end-effector **1132** comprising a clamp arm **1120**, a peripheral electrode **1122** defining an inner aperture **1125**, and a connection pad **1124** isolated from a conductive portion of the peripheral electrode **1122** to attach the peripheral electrode **1122** to a proximal end of the clamp arm **1120**, according to at least one aspect of the present disclosure. This view also shows a connection pad **1124** at a proximal end **1128** that is used to attach the peripheral electrode **1122**

to the proximal end 1128 of the clamp arm 1120 (FIG. 11B). Also shown in this view is a portion of a clamp arm pad 1126 as seen through the aperture 1125. The peripheral electrode 1122 is fixed to the clamp arm 1120 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1122 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0396] FIG. 11B is a top view of the peripheral electrode 1122 and a partial side view of the end-effector 1132, according to at least one aspect of the present disclosure. The peripheral electrode 1122 defines a perimeter and herein may be referred to as a peripheral electrode. Also shown in this view, is an ultrasonic blade 1134 and a gap “x” defined at a distal end 1130 of the end-effector 1132 and the peripheral electrode 1122. The gap “x” is set by the thickness of the connection pad 1124. The end-effector 1132 comprises a clamp arm 1120 that is configured to pivotally open and close to grasp and compress tissue between the clamp arm 1120 and the ultrasonic blade 1134. The peripheral electrode 1122 is fixedly attached to the proximal end 1128 of the clamp arm 1120 and is free to move or deflect at the distal end 1130 of the clamp arm 1120.

[0397] FIG. 11C is a side view of an end-effector 1154 comprising a clamp arm 1156 and an ultrasonic blade 1159 and a top view of a peripheral electrode 1140 configured to attach to a proximal end 1150 of the clamp arm 1156, according to at least one aspect of the present disclosure. The peripheral electrode 1140 comprises an electrically conductive portion 1142 and an electrically non-conductive portion 1144 disposed about an outer perimeter of the electrically conductive portion 1142. A space is defined between an outer edge of the electrically conductive portion of the peripheral electrode 1140 and an outer edge of the electrically non-conductive portion of the peripheral electrode 1140. The peripheral electrode 1140 defines an aperture 1152 sized and configured to receive a clamp arm pad 1158 therethrough. The peripheral electrode 1140 is fixed to the clamp arm 1156 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the peripheral electrode 1140 may be referred to as a cantilever beam electrode or deflectable electrode.

[0398] The peripheral electrode 1140 also comprises a connection pad 1146 configured to attach the peripheral electrode 1140 to the proximal end 1150 of the clamp arm 1156. The connection pad 1146 extends from the peripheral electrode. The electrically non-conductive portion 1144 of the peripheral electrode 1140 extends to the distal end 1148 to define a space between the proximal edge of the electrically non-conductive portion 1144 and the distal edge of the electrically conductive portion 1142. The clamp arm 1156 is configured to pivotally open and close about a pivot point 1157 to grasp and compress tissue between the clamp arm 1156 and the ultrasonic blade 1159. The peripheral electrode 1140 is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the peripheral electrode 1140 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0399] The clamp arm 1156 comprises a clamp arm pad 1158 attached thereto. As the clamp arm 1156 is pivotally closed towards the ultrasonic blade 1159, the clamp arm pad 1158 is received through the aperture 1152 defined by the peripheral electrode 1140. The electrode 1140 is fixedly

attached to the proximal end 1150 of the clamp arm 1156 and is free to move or deflect at the distal end 1148 of the clamp arm 1156.

[0400] FIG. 11D is a side view of an end-effector 1174 comprising a clamp arm 1176 and an ultrasonic blade 1179 and a top view of a peripheral electrode 1160 configured to attach to a proximal end 1170 of the clamp arm 1176, according to at least one aspect of the present disclosure. The peripheral electrode 1160 comprises an electrically conductive portion 1162 and an electrically non-conductive portion 1164 disposed about an outer perimeter of the electrically conductive portion 1162 except at a distal end 1168 of the peripheral electrode 1160 where the electrically conductive portion 1162 extends the entire distance to the edge 1169 of the electrically non-conductive portion 1164. The peripheral electrode 1160 defines an aperture 1172 sized and configured to receive a clamp arm pad 1178 therethrough. The peripheral electrode 1160 also comprises a connection pad 1166 configured to attach the peripheral electrode 1160 to the proximal end 1168 of the clamp arm 1176. The conductive portion 1162 of the electrode 1160 extends to the distal end 1168 where the electrically conductive and non-conductive portions 1162, 1164 overlap to the edge 1169 such that there is no space defined between the proximal edge of the non-conductive portion 1164 and the distal edge of the electrically conductive portion 1162.

[0401] The end-effector 1174 comprises a clamp arm 1176 that is configured to pivotally open and close about a pivot point 1177 to grasp and compress tissue between the clamp arm 1176 and the ultrasonic blade 1179. The clamp arm 1176 comprises a clamp arm pad 1178 attached thereto. As the clamp arm 1176 is pivotally closed towards the ultrasonic blade 1179, the clamp arm pad 1178 is received through the aperture 1172 defined by the peripheral electrode 1160. The electrode 1160 is fixedly attached to the proximal end 1170 of the clamp arm 1176 and is free to move or deflect at the distal end 1168 of the clamp arm 1176.

[0402] FIGS. 12A-12C illustrate a clamp arm 1180 comprising an electrode 1188 having a proximal end 1194 fixedly attached to a proximal end 1194 of the clamp arm 1180 and having a distal end 1196 opposite to the fixed end that is free to move or deflect, according to at least one aspect of the present disclosure. The electrode 1188 is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit. The electrode 1188 is fixed to the clamp arm 1180 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1188 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0403] Turning first to FIG. 12A, the clamp arm 1180 comprises a jaw 1182, a clamp arm pad 1184, and an electrode 1188, according to at least one aspect of the present disclosure. A proximal end 1194 of the electrode 1188 is fixedly attached to a proximal end 1194 of the jaw 1182 to form a connection 1190. The distal end 1196 of the electrode 1188 is free to move or deflect. A plug 1186 is located at a distal end 1196 of the clamp arm 1180 to set a gap between the electrode 1188 and the ultrasonic blade (not shown). The jaw 1182 is pivotally opened and closed to grasp and compress tissue between the clamp arm pad 1184 and the ultrasonic blade (not shown).

[0404] FIG. 12B is a section view of the clamp arm 1180 viewed from the perspective of the distal end 1196 of the clamp arm 1180, according to at least one aspect of the present disclosure. This view shows the plug 1186 with no backing defining a gap 1194 between the top 1196 of the plug 1186 and the bottom 1198 of the clamp arm pad 1184. The plug 1186 protrudes below the electrode 1188 and the clamp arm pad 1184 to set the gap between the electrode 1188 and the ultrasonic blade (not shown).

[0405] FIG. 12C is a section view of the clamp arm pad 1184 with a backing 1197, according to at least one aspect of the present disclosure. The clamp arm pad 1184 protrudes through the electrode 1188.

[0406] FIG. 12D is a clamp arm 1200 comprising a clamp jaw 1202, a clamp arm pad 1204, and a deflectable cantilever electrode 1208, according to at least one aspect of the present disclosure. A proximal end 1214 of the deflectable cantilever electrode 1208 is fixedly attached to a proximal end 1204 of the clamp jaw 1202 to form a connection 1210. The distal end 1216 of the deflectable cantilever electrode 1208 is free to move or deflect. A plug 1206 is located at a distal end 1216 of the clamp arm 1200 to set a gap between the deflectable cantilever electrode 1208 and the ultrasonic blade (not shown). The jaw 1202 is pivotally opened and closed to grasp and compress tissue between the clamp arm pad 1204 and the ultrasonic blade (not shown). The deflectable cantilever electrode 1208 is fixed to the clamp jaw 1202 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the deflectable cantilever electrode 1208 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0407] FIGS. 13A-13B illustrate a clamp arm 1220 comprising an electrode 1228 and a clamp arm pad 1225, according to at least one aspect of the present disclosure. The electrode 1228 is fixed to the clamp arm 1220 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1228 may be referred to as a cantilever beam electrode or as a deflectable electrode. The clamp arm 1220 also includes wear resistant pads 1230 configured to set a gap between the electrode 1208 and the an ultrasonic blade (not shown). FIG. 13A shows the electrode 1228 in an unloaded undeflected state and FIG. 13B shows the electrode 1228 in a loaded deflected state. A proximal end 1222 of the electrode 1228 is fixedly attached to the proximal end 1224 of the clamp arm 1220. The distal end 1226 of the electrode 1228, opposite to the fixed end, is free to move or deflect when it is under load. The electrode 1228 is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode 1228 is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit. The length of the electrode 1228 is selected to accommodate burn through of the clamp arm pad 1225.

[0408] FIGS. 14A-15B illustrate an electrode configuration with an end-of-life indicator, which could be sensed by a generator, to prompt device replacement, according to various aspects of the present disclosure. Turning first to FIG. 14A, there is illustrated an end-effector 1240 comprising a clamp arm 1242 and an ultrasonic blade 1244, according to at least one aspect of the present disclosure. The clamp arm 1242 comprises an electrode 1246 configured with an end-of-life indicator which may be sensed by a generator to prompt device replacement. A proximal end 1254 of the

electrode 1246 is fixedly attached at an anchor point 1250 on a proximal 1254 end of the clamp arm 1242. The clamp arm 1242 is rotatable about a pivot point 1252 to enable the clamp arm 1242 to pivotally open and close to grasp and compress tissue between the electrode 1246 and the ultrasonic blade 1244. The electrode 1246 is fixed to the clamp arm 1242 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1242 may be referred to as a cantilever beam electrode or as a deflectable electrode. The electrode 1246 is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode 1246 is one pole of the bipolar RF circuit and the ultrasonic blade 1244 is the opposite pole of the bipolar RF circuit.

[0409] An end-of-life indicator comprising a tail 1258 made of electrically conductive electrode material is added proximal 1254 to the electrode 1246 anchor point 1250. The tail 1258 will deflect towards the ultrasonic blade 1244 as the clamp arm pads and wear resistant pads 1248 show use wear and tear during a procedure. When the tail 1258 contacts the ultrasonic blade 1244 an electrical short circuit can be detected in the device to trigger the end of life of the device and notify the surgeon to replace and dispose of the device. The electrode 1246 is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1246 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0410] FIG. 14B shows the end-of-life indicator tail 1258 at the start of a procedure and FIG. 14C shows the end-of-life indicator tail 1258 at the end of the procedure. In FIG. 14B, the end-of-life indicator tail 1258 is not in contact with the ultrasonic blade 1244 indicating that the device is in good condition. In FIG. 14C, the end-of-life indicator tail 1258 is in contact with the ultrasonic blade 1244 causing an electrical short circuit at point 1260 to indicate that the device is at the end-of-life and needs to be replaced.

[0411] FIG. 15A illustrates an end-effector 1270 comprising a clamp arm 1272 and an ultrasonic blade 1274, according to at least one aspect of the present disclosure. The clamp arm 1272 comprises an electrode 1276 configured with an end-of-life indicator 1278 which may be sensed by a generator to prompt device replacement. The end-of-life indicator 1278 comprises a tail 1288 made of electrically conductive electrode material added proximal 1282 to the electrode 1276 anchor point 1284. A new wearable clamp arm pad 1280 is fixedly attached to the clamp arm 1272 and protrudes through apertures provided in the cantilever deflectable electrode 1276 (as shown and described, for example in FIGS. 1-9). A proximal end 1282 of the electrode 1276 is fixedly attached at an anchor point 1284 on a proximal 1282 end of the clamp arm 1272. The clamp arm 1272 is rotatable about a pivot point 1284 to enable the clamp arm 1272 to pivotally open and close to grasp and compress tissue between the electrode 1276 and the ultrasonic blade 1274. The electrode 1276 is fixed to the clamp arm 1272 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1276 may be referred to as a cantilever beam electrode or as a deflectable electrode. The electrode 1276 is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load,

where the electrode **1276** is one pole of the bipolar RF circuit and the ultrasonic blade **1274** is the opposite pole of the bipolar RF circuit.

[0412] FIG. 15B illustrates the end-effector **1270** of FIG. 15A after the clamp arm pad is worn close to failure. As shown, the end-of-life indicator **1278** tail **1288** will deflect towards the ultrasonic blade **1274** as the clamp arm pads **1280** wear during a procedure and creates an electrical contact **1284** between the tail **1288** and the ultrasonic blade **1274**. As shown, components **1290**, **1292**, **1294**, **1296**, **1298** of the clamp arm pad **1280** are noticeably worn relative to the clamp arm pad **1280** shown in FIG. 15B. The worn clamp arm pads **1290-1298** cause the tail **1288** to contact **1284** the ultrasonic blade **1244** causing an electrical short circuit that can be detected either by a generator or the device to trigger a device end of life notification to the surgeon to replace and dispose of the device. A warning in that regard may be provided to a video display to alert the surgeon or may be provided to a control circuit to trigger other actions.

[0413] FIGS. 16A-16B, 17, and 18A-18D illustrate configurations wherein the electrode can move vertically along its entire length, according to various aspects of the present disclosure. Turning first to FIG. 16A, which shows a clamp arm **1300** portion of an end-effector, according to at least one aspect of the present disclosure. The clamp arm **1300** comprises a floating electrode **1308** mounted to a spring **1310** such that the floating electrode **1308** can deflect in direction “d” along its entire length. The clamp arm **1300** comprises a lubricious clamp arm pad **1302** and a wear resistant clamp arm pads **1304**, **1306** mounted into the lubricious clamp arm pad **1302**. The wear resistant clamp arm pads **1304** are used to set a gap between the floating electrode **1306** and the ultrasonic blade (not shown). For conciseness and clarity of disclosure, the ultrasonic blade portion of the end-effector is not shown. As disclosed herein, the lubricious clamp arm pad **1302** is made of a non-stick lubricious material such as PTFE or similar synthetic fluoropolymers of tetrafluoroethylene, for example, and the wear resistant clamp arm pads **1304**, **1306** are made of polyimide materials such as VESPEL, PET, PEEK, or PEKK for example. The electrode **1306** is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode **1306** is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0414] Turning now to FIG. 16B, the wear resistant clamp arm pad **1304** is mounted to an end of the lubricious clamp arm pad **1302** to help reduce wear of the lubricious clamp arm pad **1302**.

[0415] FIG. 17 illustrates a portion of an end-effector **1320** comprising a clamp arm **1322**, according to at least one aspect of the present disclosure. The clamp arm **1322** comprises a floating electrode **1324** mounted to a resilient clamp arm pad **1326** such that the floating electrode **1324** can deflect in direction “d” along its entire length. The clamp arm **1322** comprises a plurality of wear resistant clamp arm pads **1328** mounted to the resilient clamp arm pad **1326**. The wear resistant clamp arm pads **1328** are used to set a gap between the floating electrode **1324** and the ultrasonic blade (not shown). For conciseness and clarity of disclosure, the ultrasonic blade portion of the end-effector is not shown. The resilient clamp arm pad **1326** acts as a spring that the floating electrode **1324** can deflect into the resilient clamp

arm pad **1326** and move in direction “d”. The wear resistant clamp arm pads **1328** set the tissue gap between the floating electrode **1324** and the ultrasonic blade (not shown). As disclosed herein, the lubricious clamp arm pad **1326** is made of a non-stick lubricious material such as PTFE or similar synthetic fluoropolymers of tetrafluoroethylene, for example, and the wear resistant clamp arm pad **1328** are made of polyimide materials such as VESPEL, PET, PEEK, or PEKK for example. The electrode **1324** is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode **1324** is one pole of the bipolar RF circuit and the ultrasonic **1334** blade is the opposite pole of the bipolar RF circuit.

[0416] FIG. 18A is a section view of an end-effector **1330** comprising a clamp arm **1332**, according to at least one aspect of the present disclosure. The end-effector comprises a clamp arm **1332** and an ultrasonic blade **1334**. The clamp arm comprises an electrode **1336**, a resilient clamp arm pad **1338**, and a plurality of wear resistant dowel pins **1340**. The wear resistant dowel pins **1340** set the tissue gap “G” between the ultrasonic blade **1334** and the electrode **1336**.

[0417] FIG. 18B is a section view of the end-effector **1330** showing the location of the wear resistant dowel pins **1340** in the clamp arm **1332**. This view also shows the relative position of the resilient clamp arm pad **1336** relative to the wear resistant dowel pins **1340** and the electrode **1336**.

[0418] FIG. 18C is a section view of a wear resistant dowel pin **1340'**, according to at least one aspect of the present disclosure. The dowel pin **1340'** includes barb features **1342** to prevent the wear resistant dowel pin **1342'** from backing out after it is installed in the clamp arm **1332**.

[0419] FIG. 18D is a section view of a wear resistant dowel pin **1340"**, according to at least one aspect of the present disclosure. The dowel pin **1340"** includes a pierce feature **1344** to prevent the wear resistant dowel pin **1342"** from backing out after it is installed in the clamp arm **1332**.

[0420] FIG. 19A illustrates a portion of an end-effector **1350** comprising a clamp arm **1352** comprising an electrode **1354** with a “center pivot” configuration, according to at least one aspect of the present disclosure. The electrode **1354** can rotate about a pivot point **1356** near the mid-point of the electrode **1354**. The electrode **1354** is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0421] FIG. 19B illustrates a portion of an end-effector **1360** comprising a clamp arm **1362** comprising an electrode **1364** with a “center pivot” configuration, according to at least one aspect of the present disclosure. The electrode **1364** can rotate about a pivot point **1366** near the mid-point of the electrode **1364**.

[0422] FIGS. 20-27 show various design elements for the spring, electrode, and wear pads which could be incorporated into any of the configurations shown in FIGS. 10A-19B. The electrodes is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0423] FIG. 20 illustrates an end-effector **1370** comprising a clamp arm **1372** and an ultrasonic blade **1374**, according

to at least one aspect of the present disclosure. The clamp arm 1372 comprises a spring element 1380 formed into an electrode 1376. Wear pads 1378 are attached to the electrode 1376 on a side facing the ultrasonic blade 1374. In another aspect, the electrode 1376 formed into a spring 1380 may further comprise strips of metal to form a leaf spring configuration.

[0424] FIG. 21 illustrates a spring element 1382 formed into an electrode 1384, according to at least one aspect of the present disclosure. The spring element 1382 includes tabs 1386 bent at the electrode 1384. A wear pad 1388 is attached to the spring element 1382.

[0425] FIGS. 22A-22C illustrates a clamp arm 1390 comprising a wave spring element 1392 supporting an electrode 1394, according to at least one aspect of the present disclosure. The clamp arm 1390 further comprises a resilient wear clamp arm pad 1396. The wave spring element 1392 is positioned such that the electrode 1394 can deflect in direction “d”. In another aspect, wear resistant clamp arm pads (not shown) may be added to the clamp arm 1390.

[0426] FIG. 22A illustrates one aspect of the clamp arm 1390 where the wave spring element 1392 is formed integral with the resilient clamp arm pad 1396 to deflect the electrode 1392 in direction “d”. FIG. 22B illustrates one aspect of a clamp arm 1398 where the wave spring element 1392 is positioned between the resilient clamp arm pad 1396 and the electrode 1394 to deflect the electrode 1392 in direction “d”. FIG. 22C illustrates one aspect of a clamp arm 1399 where the wave spring element 1392 also is formed integral with the resilient clamp arm pad 1396 to deflect the electrode 1392 in direction “d”.

[0427] FIGS. 23A-23B illustrate electrode 1400 attachment methods, according to various aspects of the present disclosure. In FIG. 23A the electrode 1400 is disposed through protrusions 1402 in the clamp arm pad 1404 and attached to the clamp arm by a bracket 1406. In FIG. 23B the electrode 1400 is attached to the clamp arm pad 1402 by a cylindrical spring 1408.

[0428] FIGS. 24A-24C illustrate electrode spring element configurations, according to various aspects of the present disclosure. FIG. 24A illustrates an end-effector 1410 comprising a clamp arm 1412 and an ultrasonic blade 1414, according to at least one aspect of the present disclosure. The clamp arm 1412 comprises an electrode 1416 overmolded with an elastomer 1418, thus removing the need for a spring. The electrode 1416 is attached to the clamp arm 1412 by electrically conductive connections 1418. Wear resistant pads 1420 are disposed on the surface of the electrode 1416. The configuration shown in FIG. 24A provide uniform compression along the length of the electrode 1416. Electrical contact is made through the sides. The spring rate increase as it compresses against the elastomer 1418.

[0429] In other aspects, overmolded rings or pads may be provided on the ultrasonic blade 1414 in place of the wear resistant pads 1420 disposed on the surface of the electrode 1416. In another aspect, the electrode 1416 may be configured to sink into material that displaced due to a heated ultrasonic blade 1414.

[0430] FIG. 24B illustrates a clamp arm 1430 comprising a diaphragm electrode 1432, according to at least one aspect of the present disclosure. For conciseness and clarity of disclosure, the ultrasonic blade 1414 (FIG. 25A) is not shown in this view. The diaphragm electrode 1432 is attached to the clamp arm 1430 by rigid electrically con-

ductive connections 1434. The clamp arm pads 1436 are disposed on the diaphragm electrode 1432. The diaphragm electrode 1432 can be formed of a thinner metallic conductor material to flex or deflect the electrode 1432. In one aspect, the diaphragm electrode 1432 can be made in the form of a ring. The rigid connections 1434 to the clamp arm 1430 ensure electrical path. The configuration shown in FIG. 24B reduce chances of the electrode 1432 binding.

[0431] FIG. 24C illustrates a clamp arm 1440 comprising a one piece spring electrode 1442, according to at least one aspect of the present disclosure. For conciseness and clarity of disclosure, the ultrasonic blade 1414 (FIG. 25A) is not shown in this view. The one piece spring electrode 1442 includes ends 1444 folded into springs similar in shape to formed staples. The one piece spring electrode 1442 is low cost and reduces transitions in the electrical path. As shown in FIG. 24C, the curved ends 1444 of the one piece spring electrode 1442 are in electrical communication with the clamp arm 1440 at connections 1446. Clamp arm pads 1448 are disposed on the one piece spring electrode 1442.

[0432] FIG. 25 illustrates an electrode 1450 comprising a coined bump 1452 for use with integral wear pads (not shown), according to at least one aspect of the present disclosure. The coined bump 1452 is coated with an electrically insulative material 1454 to prevent shorting against the ultrasonic blade (not shown). The coined bump 1452 allows for larger springs 1456 to deflect the electrode 1450.

[0433] Following is a description of construction techniques for electrodes, pads, and springs as described herein. In one aspect, a backing may be located under the electrode. A fine blank of 300 SST may be employed with secondary machining to meet tolerances for guiding the electrode up and down and injection molding. In another aspect, the clamp arm pads may be made of silicone or plastic. The silicone or plastic clamp arm pad material may be injection molded onto the electrode. A through hole may be formed through the electrode to anchor the silicone or plastic material.

[0434] Springs 1460 as shown in FIG. 26 are tested under heat and tissue fluid conditions, according to at least one aspect of the present disclosure. The length of the spring 1460 may be increased. In one aspect, the spring 1460 may be a high temperature stainless steel coil spring. In another aspect, a PTFE clamp arm pad can be broken into more than one piece to allow room for the length of the spring 1460 to increase. The PTFE clamp arm pad can be machined. In one aspect, the clamp arm pad may be modified to remove pad material underneath the spring.

[0435] With reference to FIG. 27, to prevent binding of the electrode 1464 as it deflects up and down, the distance “d1” of the electrode may be increased, according to at least one aspect of the present disclosure.

[0436] FIGS. 28A-31 illustrate an end-effector 1470 configured with one or more than one resistor 1480 embedded (integrated) in a clamp arm pad 1478 to sense the height “h” of the clamp arm pad 1478, according to at least one aspect of the present disclosure. The end-effector 1470 comprises a clamp arm 1472, and an ultrasonic blade 1474, an electrode 1476, a clamp arm pad 1478, and one or more than one resistor 1480 embedded in the clamp arm pad 1478. The resistors 1480 melt or wear along with the clamp arm pad 1478 such that the resistance value of the resistors 1480 is analogous to the height of the clamp arm pad 1478. Accordingly, the value of the resistor 1480 may be employed to

infer the height of the clamp arm pad 1478 in real-time, the melt rate of the clamp arm pad 1478, or provide a warning signal or recommendation, or combinations thereof. The user or control circuit may employ the warning signal or recommendation to determine next steps. The warning signals are conveyed to a control circuit coupled to one or more than one the resistor 1480 via electrical conductors 1482. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0437] In FIG. 28A, the clamp arm pad 1478 is new and does not show any wear and the resistors 1480 and the clamp arm pad 1478 are whole and indicate a full resistance value at full height h_1 with no tissue between the clamp arm 1472 and the ultrasonic blade 1474. FIG. 28B illustrates the end-effector 1470 with no tissue between the clamp arm 1472 and the ultrasonic blade 1474 and with the clamp arm pad 1478 and resistors 1480 worn to a height of h_2 , which is less than h_1 due to heat 1484 generated by friction between the clamp arm 1472 and the ultrasonic blade 1474 and the resistors 1480. As the resistors 1480 are worn, the resistance level changes, e.g., may increase or decrease depending on the actual implementation, and is communicated to a control circuit via the electrical conductors 1482 coupled to the resistors 1480. One example of a resistance level R of the resistor 1480 in clamp arm pad 1478 versus pad height “h” is illustrated in FIG. 29, as described hereinbelow. As shown in FIG. 28B, the warning signal is provided when the melt rate of the clamp arm pad 1478 is too rapid and exceeds a predetermined threshold due in part because there is no tissue between the clamp arm 1472 and the ultrasonic blade 1474.

[0438] FIGS. 28C-28D illustrate the condition where the electrode 1476 is deflected to a height of h_3 , which is less than h_1 , due to thick tissue 1490 located between the clamp arm 1472 and the ultrasonic blade 1474. The thick tissue 1490 pushes down on the electrode 1476 with a force 1488. In this configuration, the clamp arm pad 1478 is still at full height h_1 or above the warning threshold and the resistance level also is at full value or above

[0439] FIG. 29 is a graphical illustration 1500 of one example of a resistance level R of the resistor 1480 in clamp arm pad 1478 versus pad height “h”, according to at least one aspect of the present disclosure. As described in connection with FIGS. 28A-28D, the measure of resistance level (R1-R5), shown along the vertical axis, will vary in accordance with pad height “h” from low to high, as shown along the horizontal axis. As shown in FIG. 29, a resistance level between R1 and R2 indicates a warning that the clamp arm pad 1472 is 50% worn, a resistance level between R2 and R3 indicates a warning signal to replace the clamp arm pad 1472 soon, and a resistance level between R3 and R4 indicates a warning signal to shut off energy to the ultrasonic blade 1474 or the device at large. As shown in FIG. 29, the resistance level R of the resistor 1480 increases as the height “h” of the clamp arm pad 1478 decreases. As shown, the curve is exponential. In various other aspects, the electrical circuit may be configured such that the curve is linear and/or the resistance level R of the resistor 1480 decreases as the height “h” of the clamp arm pad 1478 decreases, without limitation.

[0440] FIGS. 30A-30B illustrate the resistance level (R1-R5) as a function of the height “h” of the resistor 1480 and analogously the height “h” of the clamp arm pad 1478 because the resistor 1480 is embedded in the clamp arm pad 1478 and both elements will wear at the same rate. FIG. 30A shows the end-effector 1470 with a new clamp arm pad 1478 and resistor 1480 mounted in the clamp arm 1472. As shown, the ultrasonic blade 1474 is in contact with wear resistant pads 1508 which set the minimum gap between the electrode 1476 and the ultrasonic blade 1474. The clamp arm pad 148 is at full height “h” or at least above a minimum warning threshold and is in contact with the ultrasonic blade 1474. In this configuration, the resistor 1480 also is at a resistance level below the minimum warning threshold R1.

[0441] FIG. 30B illustrates the state of the end-effector 1470 when the clamp arm pad 1478 is slightly worn to indicate a resistance level R2. At the resistance level R2, the warning signal is approaching an indication to replace the clamp arm pad soon as shown in the graph 1500 in FIG. 29.

[0442] FIG. 31 is a graphical illustration 1510 of clamp arm pad 1478 height “h”, as measured by an integrated resistor 1480, shown along the vertical axis, as a function of time “t”, shown along the horizontal axis, according to at least one aspect of the present disclosure. During a first period t_1 , the warning signal indicates a good condition due to the clamp arm pad 1478 height “h” as measured by the integrated resistor 1480. The clamp arm pad 1478 may be high due to either a new pad or a pad with a very low melt rate due to tissue being located between the clamp arm 1472 and the ultrasonic blade 1474. During a second period t_2 , the warning signal indicates a bad condition because the jaw is empty with no tissue located between the clamp arm 1472 and the ultrasonic blade 1474. A bad warning signal is issued to unclamp the clamp arm 1472. This condition is detected based on the rate of change of the resistance of the integrated resistor 1480 over time as indicated by slopes S_1 , S_2 . During a third period t_3 , when tissue is reintroduced between the clamp arm 1472 and the ultrasonic blade 1474, the warning signal once again indicates a good condition based on the measured resistance value of the integrated resistor 1480 and the slope S_3 . Once again, during a fourth period t_4 , the warning signal indicates a bad condition because the jaw is empty with no tissue located between the clamp arm 1472 and the ultrasonic blade 1474. The issued warning signal indicates to either unclamp the clamp arm 1472 or temporarily shut-off energy to the ultrasonic blade 1474 or the device. This bad condition is detected based on the rate of change of resistance 1478 over time and is indicated by S_4 . In period t_5 , the warning signal once again indicates a good condition based on the measured value of the integrated resistor 1480.

[0443] FIGS. 32-33B illustrate configurations wherein the electrode can pivot near the mid-point of the electrode, according to at least one aspect of the present disclosure. FIG. 32 is an exploded view of a clamp arm 1520 comprising a center pivoting electrode 1522, according to at least one aspect of the present disclosure. The center pivoting electrode 1522 is fixed to a non-conductive rocker support 1524 at snap fit joints labeled A, B, C, D. The non-conductive rocker support 1524 is pivotally attached to a clamp arm jaw 1526 at a rocker pivot point D'. The center pivoting electrode 1522 comprises a plurality of electrically non-conductive pads 1528 to set a gap between the center pivoting electrode 1522 and the ultrasonic blade 1532 shown

in FIGS. 33A-33B. The center pivoting electrode 1522 is coupled to an RF generator via electrical conductor 1534. The electrode 1522 is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode 1522 is one pole of the bipolar RF circuit and the ultrasonic blade 1532 is the opposite pole of the bipolar RF circuit.

[0444] FIGS. 33A-33B illustrate an end-effector 1530 comprising the clamp arm 1520 shown in FIG. 32 and an ultrasonic blade 1532, according to at least one aspect of the present disclosure. In FIG. 33A, the pads 1526 are new such that at full clamp the center pivoting electrode 1522 is a predetermined angle (e.g., parallel) relative to the clamp arm jaw 1526. As the pads 1528 wear down, as shown in FIG. 33B, and in order to maintain a consistent gap between the ultrasonic blade 1532, the angle θ between the center pivoting electrode 1522 relative to the clamp arm jaw 1526 will change.

[0445] In one aspect, the present disclosure provides interlocking pad features to allow a clamp arm pad to fixate an electrode to a clamp arm support.

[0446] FIGS. 34-35C illustrate one aspect of interlocking clamp arm pads and electrode plates that act as both the restraining system and the ultrasonic support. In one aspect, the clamp arm may comprise a longitudinally aligned slot from a proximal end through which an electrode plate may be advanced without the clamp arm pad material. Once fully inserted, the clamp arm pad material pillars may be inserted from a distal end deflecting the electrode plate until they snap into position. Once interlocked together, the backside of the clamp arm pad material component prevents further proximal or distal longitudinal motion of the electrode and the front of the clamp arm pad material extending through the electrode openings form an ultrasonic blade clamp arm support. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0447] FIG. 34 is a perspective section view of a clamp arm 1540 comprising a longitudinally aligned slot 1542 from a proximal end 1556 through which an electrode plate 1544 may be advanced without the clamp arm pad 1546 material, according to at least one aspect of the present disclosure. The electrode plate 1544 defines apertures 1548 to receive protrusion elements 1550 of the clamp arm pad 1546. The clamp arm pad 1546 is disposed above a clamp jaw 1552 of the clamp arm 1540.

[0448] FIG. 35A is a section view of the clamp arm 1540 shown in FIG. 34 prior to assembly. Initially, the clamp arm pad 1546 is interlocked into the underside of the electrode plate 1544 by inserting the protruding elements 1550 of the clamp arm pad 1546 into the apertures 1548 defined by the electrode plate 1544 in the direction indicated by the arrows to form a clamp arm pad/electrode plate assembly 1560. A tab 1554 provided at the proximal end 1556 of the clamp arm pad 1546 is received or snapped into a proximal recess 1558 provided at the proximal end of the 1556 of the clamp arm 1540.

[0449] In FIG. 35B the clamp arm pad/electrode plate assembly 1560 is slid proximally 1556 into the longitudinally aligned slot 1542 defined by the clamp jaw 1552. In FIG. 35C the tab 1554 at the proximal end 1556 of the clamp arm pad 1546 locks into the proximal recess 1558 defined by

the proximal end 1556 of the clamp jaw 1552 interlocking the clamp arm pad/electrode plate assembly 1560 with the clamp jaw 1552.

[0450] In one aspect, the present disclosure provides a continuous spring support member for a deflectable combo device electrode. FIG. 36 illustrates one aspect of a continuous spring support disposed under the electrode rather than a discrete spring attached at one end of the electrode. In one aspect, an electrode may be attached to a clamp arm support via a wave spring 1570 mechanism along the entire underside of the electrode, as shown in FIG. 36. The wave spring 1570 could be a perimeter concentric oval around the electrode or could be comprised of multiple nested deflectable structures. The wave spring 1570 can be made of plastic (non-conductive) or can be electrically insulated from the electrode by a polymer, which could also be used for attachment.

[0451] In another aspect, the present disclosure provides a laterally deflectable electrode to create blade seating location. FIGS. 37-50 illustrate a laterally deflectable/parting electrode, according to various aspects of the present disclosure. In various aspects, the present disclosure provides a deflectable pad that spreads laterally when pressure applied by an ultrasonic blade exceeds a predefined level causing the electrode to separate and prevent direct contact between the electrode and the ultrasonic blade. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0452] The electrode can be supported to deflect both vertically and laterally by fixating the proximal end and creating an elastomer connection between a right and left section of the electrode which are in electrical connection with each other. As the clamp arm pad deflects away from the ultrasonic blade due to interaction between the polyamide pad and the ultrasonic blade a feature on the upper surface of the underlying clamp arm support has a distal wedge feature that separates the right and left sides at the distal end as the pad is deflected more vertically preventing contact between the blade and the electrode while allowing a zero gap state.

[0453] Thermal expansion properties of the clamp arm pad may be utilized as it is heated (10%-20%) to induce the lateral expansion. Laterally expanding electrodes also can be produced by utilizing the thermal expansion properties of the clamp arm pad due to both heat and compression by the ultrasonic blade. This can in-turn induce lateral spacing increase between the electrodes driving them out of the path of the ultrasonic blade.

[0454] FIG. 37 is a perspective view of an end-effector 1580 comprising a laterally deflectable electrode 1582, forming one pole of the electrosurgical circuit, and an ultrasonic blade 1586, forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure. The end-effector 1580 comprises a clamp arm 1584 coupled to the laterally deflectable electrode 1582 by any suitable fastener 1583 or fastening technique such as welding, laser welding, brazing, soldering, pressing, among other fastening techniques. The end-effector 1580 further comprises a clamp arm pad 1588. In FIG. 37 the laterally deflectable electrode 1582 is shown in the undeflected state.

[0455] FIG. 38 shows the laterally deflectable electrode 1582 in an undeflected state (shown in solid line) and in a deflected state 1582' (shown in broken line), according to at least one aspect of the present disclosure. The laterally deflectable electrode 1582 may be deflected by material clamped between the clamp arm 1584 and the ultrasonic blade 1586 or by the ultrasonic blade 1586 alone.

[0456] FIG. 39 is a section view of an end-effector 1600 comprising an electrode 1602 as a structural clamping member, forming one pole of the electrosurgical circuit, and an ultrasonic blade 1606, forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure. The electrode 1602 is inward facing about a clamp arm pad 1604. The electrode 1602 also comprises a vertical stiffening form 1607.

[0457] FIG. 40 is a section view of an end-effector 1608 comprising an electrode 1610 as a structural clamping member, forming one pole of the electrosurgical circuit, and an ultrasonic blade 1612, forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure. The electrode 1610 is inward facing about a clamp arm pad 1611. The electrode 1610 also comprises a vertical stiffening form 1613.

[0458] FIG. 41 is a section of an end-effector 1614 comprising an electrode 1616 as a structural clamping member, forming one pole of the electrosurgical circuit, and an ultrasonic blade 1622, forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure. The electrode 1616 is coupled to a clamp arm 1618 and is inward facing below the clamp arm 1618 to couple to a clamp arm pad 1620. The clamp arm 1618 also comprises a vertical stiffening form 1624 to secure the electrode 1616 to the clamp arm 1618. In FIG. 41 the electrode 1616 is shown in its undeflected state.

[0459] In FIG. 42 the electrode 1616 is shown in its laterally deflected state 1616' (shown in broken line), according to at least one aspect of the present disclosure. Thermal expansion of the clamp arm pad 1620 forces the electrode 1616 to deflect or expand laterally and away from the ultrasonic blade 1622 and the clamp arm pad 1620. The lateral expansion of the electrode 1616 widens or extends considerably to the expanded clamp arm pad 1620' (shown in broken line) when the clamp arm pad material (e.g., PTFE) transitions to a gel-like state at or near 330° C. Deformation is due to compressive clamp load.

[0460] FIGS. 43-44 illustrate examples of clamp arms comprising laterally expanding electrodes where the original state of the clamp arm is shown in solid line and the expanded state is shown in broken line. FIG. 43 illustrates a clamp arm 1630 comprising a laterally deflectable electrode 1632, forming one pole of the electrosurgical circuit, according to at least one aspect of the present disclosure. The clamp arm 1630 is configured for use with an end-effector comprising an ultrasonic blade forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure. The clamp arm 1630 comprises a clamp jaw 1636 coupled to the laterally deflectable electrode 1632 by any suitable fastening technique such as welding, laser welding, brazing, soldering, pressing, among other fastening techniques. The clamp arm 1630 further comprises a clamp arm pad 1638. The undeflected state is shown in solid line and the laterally deflected state of the components is shown in broken line.

[0461] FIG. 44 illustrates a clamp arm 1640 comprising a laterally deflectable electrode 1642, forming one pole of the electrosurgical circuit, according to at least one aspect of the present disclosure. The clamp arm 1640 is configured for use with an end-effector comprising an ultrasonic blade forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure. The clamp arm 1640 comprises a clamp jaw 1646 coupled to the laterally deflectable electrode 1642 by any suitable fastening technique such as welding, laser welding, press pins, among other techniques. The end-effector 1640 further comprises a clamp arm pad 1648. The undeflected state is shown in solid line and the laterally deflected state of the components is shown in broken line.

[0462] FIGS. 45-46 illustrate an end-effector 1650 comprising a laterally deflectable electrode 1652, forming one pole of the electrosurgical circuit, and an ultrasonic blade 1654, forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure. The end-effector 1650 comprises a clamp arm 1656 coupled to the laterally deflectable electrode 1652 by any suitable fastening technique such as welding, laser welding, brazing, soldering, pressing, among other fastening techniques. The end-effector 1650 further comprises a clamp arm pad 1658. The undeflected state of the electrode 1652 is shown in FIG. 45 with a distance d between the center of the electrode 1652 to the most lateral extension of the electrode 1652. The laterally deflected state of the electrode 1652 is shown in FIG. 46 with a distance d' between the center of the electrode 1652 to the most lateral extension of the electrode 1652. The lateral deflection of the electrode 1652 is due primarily to the thermal expansion and mechanical deformation, which is largely lateral. The thermal expansion and mechanical deformation forces the electrode 1652 away from the ultrasonic blade 1654, where $d' > d$. This is particularly true when the clamp arm pad 1658 material (e.g., PTFE) reaches a gel-like state at about 330° C. The clamp arm 1654 also comprises a vertical stiffening form 1655 to secure the electrode 1652 to the clamp arm 1654.

[0463] FIGS. 47-50 illustrate an end-effector 1660 comprising a laterally deflectable electrode 1662, forming one pole of the electrosurgical circuit, and an ultrasonic blade 1664, forming the other pole of the electrosurgical circuit, according to at least one aspect of the present disclosure. The end-effector 1660 comprises a clamp arm 1666 coupled to the laterally deflectable electrode 1662 by any suitable fastener 1663 or fastening technique such as welding, laser welding, brazing, soldering, pressing, among other fastening techniques. The end-effector 1660 further comprises a clamp arm pad 1668. The laterally deflectable electrode 1662 includes a notch 1670 defined on a distal end 1672. FIG. 49 illustrates the lateral deflection of the laterally deflectable electrode 1662 in broken line in the direction of the arrows 1669. FIG. 50 is a magnified view of the laterally deflectable electrode 1662 where the lateral deflection is shown in broken line.

[0464] FIGS. 51-58 illustrate an end-effector 1671 comprising a pair of metal clamp arms 1673a, 1673b coated with a compliant or deformable dielectric material 1675, and a wire electrode 1674, according to at least one aspect of the present disclosure. The compliant or deformable dielectric material 1675, such as silicone rubber, shrink tubing, or similar material is disposed over each of the metal clamp arms 1673a, 1673b. The metal clamp arms 1673a, 1673b

may be made of stainless steel, titanium, or similar metals, or alloys thereof. The wire electrode **1674** is attached to one of the metal clamp arms **1673a**. The wire electrode **1674** may be made of a metal such as stainless steel, titanium, or similar metals, or alloys thereof, with either a bare metal finish or coated with a thin dielectric material such as PTFE or coated and skived to reveal a very thin bare metal strip.

[0465] As shown in FIG. **55**, the compliant or deformable dielectric material **1675** can be the same material (e.g., silicone) or can be a harder or softer material, according to at least one aspect of the present disclosure. For example, a rigid PTFE tubing on one side or the other matched against a silicone rubber on the opposite side.

[0466] As shown in FIG. **56**, the metal clamp arms **1673a**, **1673b** can be implemented as bipolar electrodes by adding outer electrodes **1677a**, **1677b**, according to at least one aspect of the present disclosure.

[0467] As shown in FIGS. **57-58**, the metal clamp arms **1673a**, **1673b** can be bare metal or coated/covered metal such as, for example, a stainless steel structural member with a PTFE shrink tubing **1679** (thin walled) on an interior surface thereof, according to at least one aspect of the present disclosure.

[0468] Additional background disclosure may be found in U.S. Patent Application Publication Nos. 2016/0206367, 2017/0164997, 2015/0148834, which are herein incorporated by reference in their entirety.

[0469] In one aspect, the present disclosure provides asymmetric cooperation of the clamp arm/electrode/pad to effect the ultrasonic blade-RF electrode interaction. In one aspect, the present disclosure provides a shortened clamp arm. FIGS. **59-61** illustrate an effector comprising a shortened clamp arm for deflectable/cantilever electrode applications, according to various aspects of the present disclosure. In one aspect, the end-effector is configured for asymmetric cooperation of the clamp arm, electrode, and clamp arm pad to effect the ultrasonic blade/RF electrode interaction. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0470] In one aspect, a distal end of the clamp arm is shortened and a length of the clamp arm pad is kept the same length such that a distal end of the clamp arm pad extends beyond the distal end of the clamp arm. This would allow the electrode to hyper-extend to minimize potential for electrically shorting the distal end of the clamp arm. It also may have the benefit of extending the life of the clamp arm pad because of the additional exposed clamp arm pad material to be worn through. This configuration also can eliminate the use of the distal and middle gap setting clamp arm pads, previously referred to herein, for example, as wear resistant clamp arm pads for setting and maintaining a gap between the electrode and the ultrasonic blade.

[0471] FIG. **59** is a side view of an end-effector **1680** comprising a shortened clamp arm **1682**, an ultrasonic blade **1684**, an electrode **1686**, and a clamp arm pad **1688**, according to at least one aspect of the present disclosure. FIG. **60** is a top view of the end-effector **1680**. As shown in FIGS. **59-60**, the ultrasonic blade **1684** and the electrode **1686** are substantially the same length. The clamp arm **1682** is shortened to allow the electrode **1686** to overextend to

prevent an electrical short circuit. In one aspect, a gap setting pad **1690** is provided at a proximal end **1692** of the end-effector **1680**.

[0472] FIG. **61** illustrates a clamp arm **1700** comprising a clamp jaw **1702**, an electrode **1704**, and a clamp arm pad **1706**, according to at least one aspect of the present disclosure. Free up space distally on clamp arm. The clamp arm **1700** is configured for use with an end-effector comprising an ultrasonic blade as disclosed in other sections herein. This configuration frees up space distally **1708** on the clamp jaw **1702**. The clamp arm pad **1706** (e.g., PTFE) is fully supported underneath, but space is freed in the t-slot region and on the side walls to allow for more clamp arm pad **1706** burn through and further deflection of the electrode **1704** away from the ultrasonic blade (not shown).

[0473] In one aspect, the present disclosure provides an end-effector that employs the thermal behavior of the pad to deflect the electrode. In one aspect, the length of the clamp arm pad may be the same length as the ultrasonic blade and as the clamp arm pad expands or changes shape due to pressure or heat, the thermal expansion properties of the clamp arm pad material (e.g., PTFE) can be used to deflect the electrode out of the path of the ultrasonic blade.

[0474] In one aspect, a non-biased electrode and pad are provided. The non-biased but deflectable pad varies in position with respect to the clamp arm as the pad wears. The non-biased electrode is configured to minimize contact between the ultrasonic blade and the RF electrode. The clamp arm pad comprises a feature for securing the electrode to the clamp arm pad. In one aspect, as the height of the clamp arm pad wears or is cut through, the height of the electrode with respect to the clamp arm is progressively adjusted. In another aspect, once the clamp arm is moved away from the ultrasonic blade the electrode remains in its new position. The electrode is fixed to the clamp arm at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0475] Configurations of end-effectors comprising a deflectable/cantilever electrode described hereinabove with respect to FIGS. **1-61** may be combined with a biased electrode as described hereinbelow with respect to FIGS. **62-67**.

[0476] In one aspect, the present disclosure provides an end-effector for a combination ultrasonic/bipolar RF energy surgical device that employs pressure or clamp jaw compression to adjust the height of the electrode as the clamp arm pad wears. In one aspect, the clamp arm pad follows the clamp arm biased electrode with wearable stops. In one aspect, the clamp arm pad contains a feature for securing the electrode to the pad. As the pad height wears or is cut through, the electrode height with respect to the clamp arm is progressively adjusted. Once the clamp arm is moved away from the ultrasonic blade, the electrode stays in its new position.

[0477] Achieving sufficient clamp arm pad life on a combination ultrasonic/bipolar RF energy surgical device requires maintaining a sufficiently small yet non-zero clamp arm pad-to-electrode gap throughout the life of the instrument to provide desirable ultrasonic and bipolar RF tissue effects. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole

of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0478] The existing (seed) electrode is a flat electrode, which is practically horizontal or parallel to the clamp arm in the free state (no load). The electrode is fixed to the clamp arm at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode may be referred to as a cantilever beam electrode or as a deflectable/cantilever electrode. When clamped on tissue, the tissue loads the electrode, causing it to deflect toward the clamp arm.

[0479] In one aspect, the electrode “follows” the pad as it wears. In this aspect, the electrode is biased toward the clamp arm in the free state (whether by being a formed/curved electrode, or by attaching/welding the electrode non-parallel to the clamp arm) using any suitable fastening technique such as welding, laser welding, brazing, soldering, pressing, among other fastening techniques. Wearable stop features (on the pad or elsewhere) keep the electrode away from the clamp arm, until said stop features are worn away during use. Once worn away, the electrode is able to approach the clamp arm. These features could be tooth or ratchet shaped, a vertical taper, or other.

[0480] In one aspect, the present disclosure provides a deflectable/cantilever electrode, wherein in a free state, the electrode is biased toward clamp arm and may attached at an angle and made of a preformed curve using any suitable fastening technique such as welding, laser welding, brazing, soldering, pressing, among other fastening techniques.

[0481] In one aspect, the present disclosure provides an end-effector with a deflectable/cantilever electrode comprising wearable stop features to prevent the electrode from reaching or contacting the clamp arm. As the stop features wear, the electrode moves toward the clamp arm until it reaches the next stop. In one aspect, the stop features wear simultaneously with the clamp arm pad to maintain the appropriate gap between the clamp arm pad and the electrode. The features may be entirely separate from the clamp arm pad. The features can be configured to withstand clamping loads, but wear away due to heat (melting/flowing) or abrasion. Possible examples include teeth on one or more clamp arm pads (PTFE, polyimide, or other) and tapered profile on one or more clamp arm pads (PTFE, polyimide, or other).

[0482] FIG. 62 illustrates an end-effector clamp arm 1710 comprising a clamp jaw 1712, an electrode 1714, and a clamp arm pad 1716, according to at least one aspect of the present disclosure. The clamp arm 1710 is configured for use with an end-effector comprising an ultrasonic blade (not shown) as described throughout this disclosure. The clamp arm 1710 also comprises a wear resistant gap pad 1717 to set a gap between the electrode 1714 and the ultrasonic blade. As shown, in the free state, the electrode 1714 is biased in a level or horizontal 1718 orientation. The electrode 1714 is fixed to the clamp jaw 1712 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1714 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0483] FIG. 63 illustrates an end-effector clamp arm 1720 comprising a clamp jaw 1722, an electrode 1724, and a clamp arm pad 1726, according to at least one aspect of the present disclosure. The clamp arm 1720 is configured for use with an end-effector comprising an ultrasonic blade (not shown) as described throughout this disclosure. The clamp

arm 1720 also comprises a wear resistant gap pad 1727 to set a gap between the electrode 1724 and the ultrasonic blade. As shown, in the free state, the electrode 1724 is configured pre-formed, bent, or is otherwise biased toward the clamp jaw 1722 along line 1728 away from the horizontal 1718 orientation. The electrode 1724 is fixed to the clamp arm 1720 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1724 may be referred to as a cantilever beam electrode or as a deflectable electrode. To prevent the biased electrode 1724 from bending toward the clamp jaw 1722 under the biasing force, the clamp arm 1720 further comprises a retainer to prevent the biased electrode 1724 from bending toward the clamp jaw 1722 and maintaining the biased electrode 1724 in a substantially flat configuration (e.g., parallel, level, or horizontal) relative to the ultrasonic blade. Examples of retainers such as a retainer tooth 1738 and a retainer wall 1760 with a tapered profile are described below in FIGS. 64-67.

[0484] FIG. 64 illustrates an end-effector clamp arm 1730 comprising a clamp jaw 1732, an electrode 1734, and a clamp arm pad 1736, according to at least one aspect of the present disclosure. The clamp arm 1730 is configured for use with an end-effector comprising an ultrasonic blade (not shown) as described throughout this disclosure. The clamp arm 1730 also comprises a wear resistant gap pad 1737 to set a gap between the electrode 1744 and the ultrasonic blade. In the free state, the electrode 1734 is configured pre-formed curved, bent, or otherwise biased toward the clamp jaw 1732. However, a retainer tooth 1738, or similar feature, is provided on the clamp arm pad 1736 to prevent the electrode 1734 from springing in toward the clamp jaw 1732. In FIG. 65, when the bottom retainer tooth 1738 is worn away, the electrode 1734 can move toward the clamp jaw 1732 due to the pre-formed curve, according to at least one aspect of the present disclosure. The electrode 1734 is fixed to the clamp arm 1730 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1734 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0485] FIG. 66 illustrates an end-effector clamp arm 1750 comprising a clamp jaw 1752, an electrode 1754, and a clamp arm pad 1756, according to at least one aspect of the present disclosure. The clamp arm 1750 is configured for use with an end-effector comprising an ultrasonic blade (not shown) as described throughout this disclosure. The clamp arm 1750 also comprises a wear resistant gap pad 1757 to set a gap between the electrode 1754 and the ultrasonic blade. In the free state, the electrode 1754 is configured pre-formed with a curve, bent, or otherwise biased toward 1758 the clamp jaw 1752. However, a retainer wall 1760 having a tapered profile, or similar feature, is provided on the clamp arm pad 1756 to prevent the electrode 1754 from springing in toward the clamp jaw 1752.

[0486] In FIG. 67, when the tapered profile retainer wall 1760 is worn away, there is sufficient melting/flowing away from the tapered profile retainer wall 1760 region to allow the electrode 1754 to move toward the clamp jaw 1752 due to the pre-formed curve, according to at least one aspect of the present disclosure. The electrode 1754 is fixed to the clamp jaw 1752 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1754 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0487] FIG. 68 illustrates a clamp arm 1770 comprising a clamp jaw 1772, an electrode 1774, and a clamp arm pad 1776, according to at least one aspect of the present disclosure. The clamp arm 1770 is configured for use with an end-effector comprising an ultrasonic blade (not shown) as described throughout this disclosure. A partial skirt 1778 is provided around a distal end 1780 of the clamp arm 1770 to surround the distal clamp arm pad 1776. As the clamp arm pad 1776 (e.g., PTFE) heats and deforms, the clamp arm pad 1776 is contained within the distal skirt 1778 to prevent the clamp arm pad 1776 from expanding laterally. The electrode 1774 has a higher bias at the distal end 1780 than the proximal end 1782.

[0488] FIG. 69 illustrates a magnified view of the distal end 1780 of the clamp arm 1770 with the clamp arm pad 1776 removed to expose the distal skirt 1778, according to at least one aspect of the present disclosure. The electrode 1774 is fixed to the clamp arm 1770 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1774 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0489] In one aspect, electrode height-setting members are provided to engage with the electrode 1774 at different times relative to the clamp arm pad 1776 as it wears. In one aspect, height-setting members embedded in the clamp arm pad 1776 wear at different rates (but less wear relative to the PTFE clamp arm pads 1776). In another aspect, the height-setting members have different heights such that they cause a purposeful “bottoming out” of the electrode 1774 as the clamp arm pad 1776 wears.

[0490] In one aspect, the present disclosure provides an end-effector for a combination ultrasonic/bipolar RF energy surgical device that employs thermal expansion or thermal changes in the pad to move the support feature securing the electrode to the pad as the pad wears. In various aspects, as the clamp arm pad heats up during use the support component or feature holding the electrode in its current location melts/moves, which in-turn moves the support component or feature for the electrode holding feature effectively adjusting the height/position of the electrode with respect to the clamp arm. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0491] In one aspect, the present disclosure provides an end-effector for a combination ultrasonic/bipolar RF energy surgical device where the shape of the clamp arm pad or support location changes based on thermal adjustment of the clamp arm pad due to pad wear. FIGS. 70A-70C illustrate an end-effector 1790 comprising a clamp arm 1792, an electrode 1794, a clamp arm pad 1796, and an ultrasonic blade 1798, where the clamp arm pad 1796 comprises support elements 1800 to support the electrode 1794, according to at least one aspect of the present disclosure. Thermal changes in the clamp arm pad 1796 cause the electrode 1794 support elements 1800 to change, effectively adjusting the height (h) of the electrode 1794 relative to the clamp arm 1792 and the gap (g) between the ultrasonic blade 1798 and the electrode 1794 remains in a constant range. The support elements 1800 shown here as tabs (tab1, tab2, tab3), but other elements may be employed. The electrode 1794 is fixed to the clamp jaw 1792 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the

electrode 1794 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0492] FIG. 70A shows a new clamp arm pad 1796 and an electrode 1794 supported by support element 1800 (tab1, tab2, tab3). FIG. 70B shows the clamp arm pad 1796 heating up and causes the first row of support elements 1800 (tab1) to melt and wear away causing the electrode 1794 to drop in height (h). FIG. 70C shows that as the clamp arm pad 1796 continues heating and wearing, the second row support elements 1800 (tab2) to melt and wear away causing the height (h) of the electrode 1794 to further drop down.

[0493] Configurations of end-effectors comprising a deflectable/cantilever electrode described hereinabove with respect to FIGS. 1-61 may be combined with a flexible electrode disposed above a lattice cushion and a plurality of hard spacers to set a gap between the flexible electrode and the ultrasonic blade as described hereinbelow with respect to FIGS. 71-77.

[0494] Configurations of a biased electrode as described hereinabove with respect to FIGS. 62-70C may be combined with a flexible electrode disposed above a lattice cushion and a plurality of hard spacers to set a gap between the flexible electrode and the ultrasonic blade as described hereinbelow with respect to FIGS. 71-77.

[0495] Configurations of end-effectors comprising a deflectable/cantilever electrode described hereinabove with respect to FIGS. 1-61 in combination with a biased electrode as described hereinabove with respect to FIGS. 62-70C may be combined with a flexible electrode disposed above a lattice cushion and a plurality of hard spacers to set a gap between the flexible electrode and the ultrasonic blade as described hereinbelow with respect to FIGS. 71-77.

[0496] In one aspect, the present disclosure provides an end-effector for a combination ultrasonic/bipolar RF energy surgical device that employs a constant pressure distribution biasing mechanism. In one aspect, the end-effector includes an elastic compressible support for mounting and insulating a deflectable electrode. In one aspect, a hollow honeycomb or chambered elastomer support attachment cushion can be employed to allow all or part of the electrode attached to it to deflect but be biased towards the ultrasonic blade. This configuration could provide the added benefit of thermally insulating the electrode from the rest of the metallic clamp jaw. This would also provide an elastomer “curtain” around the electrode to minimize tissue accumulation behind the electrode. In one aspect, a non-strut deflectable geometry for the elastomer cells will enable the deflection force to be held constant over a predefined range of deflections. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0497] The above configuration prevents lateral skew of the electrode under compression to prevent shorting. Further, the deflectable electrode is affixed to the elastomer and the elastomer is affixed to the metallic clamp arm. The solid height of the spring is limited from driving allowable compression while maintaining as much metallic clamp arm as possible. Thermal conduction from tissue interface is balanced and minimizes impacts lesion formation and symmetry, cycle time, and residual thermal energy.

[0498] FIGS. 71-73 illustrate an end-effector 1810 comprising a clamp arm 1812, an ultrasonic blade 1814, a lattice

cushion **1816**, a flexible electrode **1818** disposed above the lattice cushion **1816**, and a plurality of hard spacers **1820** to set a gap between the flexible electrode **1818** and the ultrasonic blade **1814**, according to at least one aspect of the present disclosure. FIG. **73** is an exploded view of the end-effector **1810** shown in FIGS. **71-72**. A clamp arm pad **1822** is disposed inside a slot **1825** formed within the lattice cushion **1816**. The lattice cushion **1816** acts as a spring-like element. The hard spacers **1820** are used to set a gap between the flexible electrode **1818** and the ultrasonic blade **1814**.

[0499] In FIG. **71** the clamp arm **1812** is open and tissue **1824** of non-uniform thickness (T_{1a} , T_{2a} , T_{3a}) is disposed over the flexible electrode **1818**. In FIG. **72** the clamp arm **1812** is closed to compress the tissue **1824**. The lattice cushion **1816** on the clamp arm **1812** results in consistent tissue **1824** (T_{1b} , T_{2b} , T_{3b}) compression across variable thickness tissue **1824** (T_{1a} , T_{2a} , T_{3a}), such that:

$$\frac{T_{1a}}{T_{1b}} = \frac{T_{2a}}{T_{2b}} = \frac{T_{3a}}{T_{3b}}$$

[0500] Additional background disclosure may be found in EP3378427, WO2019/006068, which are herein incorporated by reference in their entirety.

[0501] In one aspect, the present disclosure provides an end-effector for a combination ultrasonic/bipolar RF energy surgical device with means for insuring distal tip contact with bias using a zero gap bipolar RF energy system. In various aspects, the present disclosure provides a deflectable electrode for a combination ultrasonic/bipolar RF energy surgical device with a higher distal bias than proximal bias. In one aspect, the present disclosure provides a combination energy device comprising a bipolar electrode that is deflectable with respect to the clamp arm. The combination energy device comprises features to change the mechanical properties of the tissue compression proximal to distal to create a more uniform or differing pattern of pressure than due to the clamping forces alone. In one aspect, the present disclosure provides a non-linear distal distributing mechanism and in another aspect the present disclosure provides electrical non-linear distribution of energy density. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0502] Configurations of end-effectors comprising a deflectable/cantilever electrode described hereinabove with respect to FIGS. **1-61** may be combined with a conductive polymer clamp arm pad as described hereinbelow with respect to FIGS. **78-218**.

[0503] Configurations of a biased electrode as described hereinabove with respect to FIGS. **62-70C** may be combined with a conductive polymer clamp arm pad as described hereinbelow with respect to FIGS. **78-218**.

[0504] Configurations of a flexible electrode disposed above a lattice cushion and a plurality of hard spacers to set a gap between the flexible electrode and the ultrasonic blade as described hereinabove with respect to FIGS. **71-77** may be combined with a conductive polymer clamp arm pad as described hereinbelow with respect to FIGS. **78-218**.

[0505] Configurations of a biased electrode as described hereinabove with respect to FIGS. **62-70C** may be combined with a flexible electrode disposed above a lattice cushion and a plurality of hard spacers to set a gap between the flexible electrode and the ultrasonic blade as described hereinabove with respect to FIGS. **71-77** may be combined with a conductive polymer clamp arm pad as described hereinbelow with respect to FIGS. **78-218**.

[0506] Configurations of a biased electrode as described hereinabove with respect to FIGS. **62-70C** may be combined with a flexible electrode disposed above a lattice cushion and a plurality of hard spacers to set a gap between the flexible electrode and the ultrasonic blade as described hereinabove with respect to FIGS. **71-77** may be combined with a conductive polymer clamp arm pad as described hereinbelow with respect to FIGS. **78-218**.

[0507] Configurations of end-effectors comprising a deflectable/cantilever electrode described hereinabove with respect to FIGS. **1-61** in combination with a biased electrode as described hereinabove with respect to FIGS. **62-70C** may be combined with a conductive polymer clamp arm pad as described hereinbelow with respect to FIGS. **78-218**.

[0508] Configurations of end-effectors comprising a deflectable/cantilever electrode described hereinabove with respect to FIGS. **1-61** in combination with a biased electrode as described hereinabove with respect to FIGS. **62-70C** may be combined with a flexible electrode disposed above a lattice cushion and a plurality of hard spacers to set a gap between the flexible electrode and the ultrasonic blade as described hereinabove with respect to FIGS. **71-77** may be combined with a conductive polymer clamp arm pad as described hereinbelow with respect to FIGS. **78-218**.

[0509] In one aspect, the present disclosure provides an end-effector for a combination ultrasonic/bipolar RF energy surgical device with a non-linear distal distributing mechanism. In one aspect, a variable spring bias element is provided along the length of a deflectable/cantilever electrode. In one aspect, the present disclosure provides a combination ultrasonic/RF energy surgical device having a deflectable bipolar RF electrode with respect to a clamp arm. The RF electrode having features to change the mechanical properties of the tissue compression from a proximal end to a distal end of the RF electrode. The RF electrode features create a more uniform or differing pattern of pressure along the length of the RF electrode rather than due to the clamping force alone.

[0510] In one aspect, the present disclosure provides an end-effector comprising a compressible attachment having a spring constant at the distal end that is different than a spring constant at the proximal end. The wall thickness of the cells or the number of interconnections may vary (increasing) longitudinally along the length of the clamp arm. The spring constant increases as the amount of material increases. The compressible attachment enables the distal tip spring constant to be higher than the proximal portion creating a tip loading condition. The compressible attachment may be created with 3D printing of the deformable body by creating different internal geometries moving distally along the attachment matrix. The compressible attachment may be injection molded or extruded with the wall thicknesses at one end being different than the thickness at the other end. In one aspect, the deflectable cantilever beam metal elec-

trode is hybridized with an elastomer backer located only at the distal end to produce the same effect with a linear metal spring.

[0511] FIGS. 74-75 illustrate an end-effector 1830 comprising a clamp arm 1832, an electrode 1834 disposed on a plurality of springs 1836, and an ultrasonic blade 1838, according to at least one aspect of the present disclosure. Hard spacers are not shown, but would be present to set a gap between the electrode 1834 and the ultrasonic blade 1838. The spring forces of springs S1 in Zone 1, S2 in Zone 2, S3 in Zone 3, S4 in Zone 4 are variable such that $S4 > S3 > S2 > S1$. The variable spring bias along the length of the electrode 1834 creates a tip-loading condition. The deflection of the ultrasonic blade 1838 increases in the distal direction. FIG. 74 illustrates the straight condition and FIG. 75 illustrates the deflected condition. Still with reference to FIG. 75, during low clamp loads, the ultrasonic blade 1838 remains straight 1840. Over clamping 1842 the clamp arm 1832 causes the deflection 1844 of the electrode 1834 caused by the spring 1836 loads causing deflection 1845 of the ultrasonic blade 1838. The electrode 1834 is fixed to the clamp arm 1832 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1834 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0512] FIG. 76 illustrates a thick spring 1846 and a thin spring 1848, according to at least one aspect of the present disclosure. Springs of variable thickness may be provided to produce an increase in spring constant in the distal direction. FIG. 77 is a top view of the springs 1836 disposed in the clamp arm 1832 to show the distribution density of springs 1836 in the four defined zones Zone 1-Zone 4, according to at least one aspect of the present disclosure.

[0513] Additional background disclosure may be found in U.S. Pat. No. 7,264,618, which is herein incorporated by reference in its entirety.

[0514] In one aspect, the present disclosure provides an end-effector for a combination ultrasonic/bipolar RF energy surgical device with an electrical non-linear distribution of energy density.

[0515] Additional background disclosure may be found in U.S. Pat. No. 9,867,650, which is herein incorporated by reference in its entirety.

[0516] The ultrasonic-surgical-shears may include an ultrasonic surgical blade and a clamp arm operable to open and close toward the blade and having a transversely and resiliently flexible distal tip. By “resiliently flexible distal tip” is meant that the distal tip resiliently flexes while the clamp arm is clamped closed such as when the ultrasonic-surgical-shears is used to transect and seal a blood vessel, disposed between the clamping surface and the ultrasonic surgical blade 34, whose walls have been coapted by a clamping force applied via the clamp arm. Additional background disclosure may be found in U.S. Pat. No. 8,444,663, which is incorporated herein by reference in its entirety.

[0517] In various aspects, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device comprising an ultrasonic pad with partially or fully electrically conductive portions such that the pad behaves as both the blade support/wear pad and the bipolar RF electrode. In one aspect, the present disclosure provides a partially conductive clamp arm pad to enable electrode wear and minimize short circuiting in a combination bipolar RF and ultrasonic energy device where the clamp arm pad has

conductive and non-conductive portions allowing it to act as one of the RF electrodes while also acting as a wearable support structure for the ultrasonic blade. In another aspect, the present disclosure provides conductive portions around the perimeter of the clamp arm pad and not positioned directly on the side that is opposite the ultrasonic blade contact area. In another aspect, a portion of the conductive clamp arm pad is degradable or wearable preventing contact from the ultrasonic blade from interrupting the conductivity of the remaining portions of the conductive clamp arm pad.

[0518] In one aspect, the present disclosure provides an end-effector for a combination ultrasonic/bipolar RF energy surgical device comprising a conductive polymer ultrasonic clamp arm pad. In one aspect, the end-effector comprises a clamp arm pad doped with tin oxide. FIG. 78 is a section view of a conductive polymer clamp arm pad 2440, according to at least one aspect of the present disclosure. The conductive polymer clamp arm pad 2440 comprises tin oxide 2442 (SnO_2) embedded in a polymer material 2444, such as Teflon (PTFE), to make the clamp arm pad 2440 electrically conductive. The doping may be achieved using a cold spray process. Once doped, the conductive polymer clamp arm pad 2440 can achieve traditional ultrasonic tissue clamp arm pad functions such as, for example, contacting the ultrasonic blade, absorbing heat from the ultrasonic blade, and assisting in tissue grasping and clamping. The tin oxide doped clamp arm pad 2440 functions as one of the two electrodes or poles of the bipolar RF circuit to deliver RF energy to tissue grasped between the ultrasonic blade and the clamp arm pad 2440. The tin oxide doped clamp arm pad 2440 is biocompatible, electrically conductive, thermally conductive, enables a large portion of the clamp arm pad 2440 to be used to improve wear resistance of the clamp arm pad 2440, and is white in color. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0519] In one aspect, the present disclosure provides a conductive polymer ultrasonic clamp arm pad as an electrode replacement. To improve the life of the ultrasonic clamp arm pad and improve the RF tissue effects, the present disclosure provides an electrode that is improved, easier to make, and less costly to make. In one aspect, the present disclosure provides a clamp arm pad comprising hard polyimide polymer layers and electrically conductive layers to allow the clamp arm pad to achieve traditional functions as well as carry bipolar electricity to eliminate the need for a separate electrode in the clamp arm of a combined energy end-effector. In this manner, the clamp jaw can be manufactured in a manner similar to the ultrasonic-only clamp jaw with the new clamp arm pad material swapped for the traditional ultrasonic-only clamp arm pad. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0520] Benefits include improved ultrasonic performance, including clamp arm pad wear, similar to current ultrasonic-only instruments because there are no electrode gaps between elements “squares” of polymer. The cost of the improved clamp jaw will be similar to current ultrasonic-

only clamp jaws because of the need for a separate electrode component is eliminated and provides multiple small polymer square elements. In addition, the manufacturing steps needed to make the clamp jaw are the same as the manufacturing steps required for making current ultrasonic-only clamp jaws. Manufacturing the improved clamp jaw requires only the substitution of the clamp arm pad and does not require the production of an additional electrode component to add to the clamp jaw and eliminates assembly steps.

[0521] FIG. 79 is a perspective view of a clamp arm pad 2450 configured to replace a conventional electrode, according to at least one aspect of the present disclosure. The clamp arm pad 2450 comprises electrically non-conductive layers 2452 and electrically conductive layers 2454 in a sandwich-like configuration. This configuration eliminates the need for a spring loaded electrode plate. The electrically non-conductive layers 2452 can be made of polymer, polyimide, Teflon (PTFE) and similar electrically non-conductive materials. The conductive layers 2454 may be made of thin electrically conductive polymer, metal foil, or carbon loaded material. The clamp arm pad 2450 may be manufactured such that the majority of the material contacting the ultrasonic blade are the electrically non-conductive layers 2452. In one aspect, 75% of the material contacting the ultrasonic blade is electrically non-conductive material such as PTFE. In another aspect, 85% of the material contacting the ultrasonic blade is electrically non-conductive material such as PTFE. In another aspect, 95% of the material contacting the ultrasonic blade is electrically non-conductive material such as PTFE. Additionally, as the clamp arm pad 2450 wears, the electrically conductive layers 2452 will still have available surface area to conduct RF electricity through the tissue and return electrode (e.g., ultrasonic blade).

[0522] FIG. 80 illustrates a clamp arm 2460 comprising the clamp arm pad 2450 described in FIG. 79, according to at least one aspect of the present disclosure. In the illustrated clamp arm 2460, the non-conductive layers 2452 have a large surface area compared to the conductive layers 2454, which appear as thin layers or foils.

[0523] FIG. 81 illustrates clamp arm pads configured as described in FIGS. 79-80, according to at least one aspect of the present disclosure. The first clamp arm pad 2470 is new and comprises teeth 2472 formed integrally therewith. The second clamp arm pad 2476 is new and without teeth. The third clamp arm pad 2478 worn and may be representative of either the first clamp arm pad 2470 or the second clamp arm pad 2476.

[0524] In one aspect, the present disclosure provides a composite clamp arm pad for a combination ultrasonic/bipolar RF energy surgical device. FIG. 82 is a section view of a clamp arm 2480 comprising a composite clamp arm pad 2482 in contact with tissue 2484, according to at least one aspect of the present disclosure. The end-effector 2480 comprises an upper clamp jaw 2486 and an adhesive 2488 to fixedly attach the composite clamp arm pad 2482 to the upper clamp jaw 2486. The composite clamp arm pad 2482 comprises thin electrically non-conductive layers 2490 (e.g., PTFE) and thin electrically conductive layers 2492 (e.g., thin stainless steel foils). The electrically conductive layers 2492 form the electrode portion of the composite clamp arm pad 2482. The electrically conductive layers 2492 (e.g., thin stainless steel foils) deform as the electrically non-conductive layers 2490 (e.g., PTFE) wear-away. The thickness of the electrically conductive layers 2492 enables the electrode

portion of the composite clamp arm pad 2482 to deform as the electrically non-conductive layers 2490 wear-away. Advantageously, the electrically conductive layers 2492 conduct some of the heat away from the electrically non-conductive layers 2490 to keep the composite clamp arm pad 2482 cooler. As described above, the composite clamp arm pad 2482 is fixed to the upper clamp jaw 2486 by an adhesive 2488. The adhesive 2488 may be filled with carbon to make it electrically conductive and connect the electrode portions of the composite clamp arm pad 2482 to the upper clamp jaw 2486. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0525] In one aspect, the clamp arm pad comprises cooperative conductive and insulative portions. In one aspect, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device where the clamp arm pad has conductive and non-conductive portions allowing it to act as one of the RF electrodes while also acting as the wearable support structure for the ultrasonic blade. In another aspect, the conductive portions of the clamp arm pad are disposed around the perimeter of the pad and are not positioned directly on the side that is opposite the ultrasonic blade contact area. In another aspect, the conductive portion of the clamp arm pad is degradable or wearable to prevent contact with the ultrasonic blade from interrupting the conductivity of the remaining conductive portions of the clamp arm pad.

[0526] In one aspect, the present disclosure provides a clamp arm pad for use with combination ultrasonic/bipolar RF energy devices where portions of the clamp arm pad include electrically conductive material and other portions include electrically non-conductive material. The electrode is adapted and configured for use with a combination ultrasonic/RF energy device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0527] In various aspects, the clamp arm pad may be manufactured using a variety of techniques. One technique comprises a two shot process of molding conductive and non-conductive materials in the same compression mold. This process effectively creates a single clamp arm pad with portions that can act as a bipolar RF electrode and others that will act as electrical insulators. Another technique comprises a super sonic cold spray embedding of metallic elements into a polymeric (e.g., Teflon, PTFE) pad or matrix. Another technique comprises 3D printing of multiple materials (e.g., Teflon, PTFE, and doped conductive polymer), printing/transfer printing conductive or functional inks onto clamp arm pad. Another technique comprises metals and conductive materials (e.g., graphite/carbon) may be applied to the clamp arm pad using chemical vapor deposition, physical vapor deposition, sputter deposition, vacuum deposition, vacuum metalizing, or thermal spray. Another technique comprises conductive/loading clamp arm pad electrodes provide continuity through the pad with micro randomly oriented and positioned particles or macro oriented structures (e.g., fabric, woven, long constrained fibers). Another technique comprises making the surface of the clamp arm pad conductive, providing wear-through electrodes, 3D printing, thermal spraying, cold spraying, coatings/paints/epoxies, sheet/foil/wire/film wrapping or laminating, vacuum metal-

izing, printing/transferring, among other techniques. In another technique, polymer electrodes filled with conductive material.

[0528] In one aspect, the end-effector clamp arm comprises a fixed polymer electrode. FIG. 83 illustrates a clamp arm 2500 comprising a clamp jaw 2502 to support a carrier 2504 or stamping attached to the clamp jaw 2502 and a clamp arm pad 2506, according to at least one aspect of the present disclosure. The clamp arm pad 2506 comprises an electrically conductive pad 2508 and an electrically non-conductive pad 2510. The electrically conductive pad 2508 is made of an electrically conductive polymer and acts as one of the electrodes of the bipolar RF circuit. The clamp jaw 2502 and the carrier 2504 may be made of stainless steel and attached using any suitable fastening technique such as welding, laser welding, brazing, soldering, pressing, among other fastening techniques, for example. The electrically conductive pad 2508 may comprise a polymer such as, for example, silicone, fluorosilicone, PTFE, and similar materials. The electrically conductive pad 2508 is overmolded onto the carrier 2504 using PTFE, silicone, fluorosilicone filled with silver particles, silver over aluminum, silver over copper, copper, nickel, graphite, carbon (amorphous, chopped fiber), gold, platinum, stainless steel, iron, or zinc, or combinations thereof.

[0529] FIG. 84 is a section view taken at section 84-84 in FIG. 83 and FIG. 85 is a section view taken at section 85-85 in FIG. 83. The sections views 84-84 and 85-85 show the clamp arm 2500 comprising the clamp jaw 2502, the support carrier 2504, the electrically conductive pad 2508, and the electrically non-conductive pad 2510.

[0530] FIG. 86 is a section view of an alternative implementation of a clamp arm 2520 comprising a clamp jaw 2522, an electrically conductive pad 2524, and an electrically non-conductive pad 2526, according to at least one aspect of the present disclosure. The electrically conductive pad 2524 is made of an electrically conductive polymer and acts as one of the electrodes in the bipolar RF circuit.

[0531] FIG. 87 is a section view of an alternative implementation of a clamp arm 2530 comprising a clamp jaw 2532, a carrier 2534 or stamping welded to the clamp jaw 2532, an electrically conductive pad 2536, and an electrically non-conductive pad 2538, according to at least one aspect of the present disclosure. The electrically conductive pad 2536 is made of an electrically conductive polymer and acts as one of the electrodes in the bipolar RF circuit. The electrically conductive pad 2536 is overmolded over the carrier 2534 or stamping.

[0532] In one aspect, the end-effector clamp arm comprises a film over metal insert molded electrode assembly. In one aspect, a film may be provided over a metal (e.g., stainless steel) insert molded electrode assembly. A film over metal such as stainless steel can be insert molded to form an electrode assembly. The film on the insert molded electrode may be etched to form micro-holes, slots, honeycomb, among other patterns, to enable conduction of RF energy as well as to cut the periphery of the component. The film may be formed onto or bond onto a stainless steel electrode using IML/FIM (In-Mold Labeling/Film Insert Molding) processes described hereinbelow. The charged film electrode may be placed into a polymer injection mold tool to mold a polymer to the back of the electrode and film. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflect-

able under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0533] FIG. 88 illustrates insert molded electrodes 2540, according to at least one aspect of the present disclosure. The insert molded electrode 2540 comprises an electrically conductive element 2546, a molded polymer pad 2548, and a film 2542 coating. Features 2550 such as micro-holes, slots, honeycomb, or similar features, are formed in the film 2542 to allow the passage of RF energy. Retention features 2552 also are formed on the film 2542. The side walls 2558 of the film 2542 extend below the bottom of the polymer pad 2548 and may be folded around the bottom of the polymer pad 2548 and over molded with retention posts. The retention features 2552 are molded into the holes 2554 defined by the film 2542. Although the two insert molded electrodes 2540 are shown with a gap between them, in actuality, the two insert molded electrodes 2540 are fit line-to-line 2556 via mold pressure.

[0534] The conductive element 2546 may be made of an electrically conductive metal such as stainless steel or similar conductive material. The conductive element 2546 can be about 0.010" thick and may be selected within a range of thicknesses of 0.005" to 0.015" and can be formed by tamping or machining. The film 2544 can be about 0.001" to 0.002" thick and may be made of polyimide, polyester, or similar materials. Alternatively to mechanical retention, such as posts, the film 2544 can be directly bonded to the conductive element 2546. One example includes DuPont Pyralux HXC Kapton film with epoxy adhesive backing having a thickness of 0.002".

[0535] Advantageously, the non-stick surface prevents tissue from sticking to the insert molded electrode 2540. The non-stick surface eliminates short circuiting of opposing electrodes by setting a gap within the range of 0.002" to 0.004" along the entire length of the insert molded electrode 2540. The non-stick surface minimizes lateral spread of RF energy due to coverage of side walls 2558 of the insert molded electrode 2540. Also, the insert molded electrode 2540 exhibits structural soundness and provides an easier more robust electrical connection than a multi-layer flexible circuit.

[0536] In one aspect, the end-effector comprises a conductive clamp arm and pad constructs for combination ultrasonic/bipolar RF energy surgical devices. In one aspect, the present disclosure provides a clamp arm assembly comprising a conductive or selectively conductive thin film, foil, or laminate that is applied to, around or on the clamp arm assembly to serve as a durable "pole" in a combination ultrasonic/bipolar RF energy surgical device. Further, an algorithm, software, or logic is provided to manage conditions of electrical short circuiting. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0537] FIG. 89 illustrates an end-effector 2560 comprising an ultrasonic blade 2562, a clamp arm 2564, and a clamp arm pad 2566 comprising an electrically conductive film 2568, according to at least one aspect of the present disclosure.

[0538] FIG. 90 illustrates the clamp arm 2564 shown in FIG. 89. The clamp arm 2564 comprising a clamp jaw 2570

to support the clamp arm pad **2566**. A thin electrically conductive film **2568** is disposed over the clamp arm pad **2566** to form an electrode of one of the poles of the bipolar RF circuit.

[0539] FIG. **91** is a section view of the clamp arm **2564** taken along section **91-91** in FIG. **90**. The clamp jaw **2570** can be made of metal such as stainless steel. The clamp arm pad **2566** can be made of an electrically non-conductive complaint material such as PTFE, silicone, high temperature polymer, or similar materials. The electrically conductive film **2568** or foil can be made of an electrically conductive material such as titanium, silver, gold, aluminum, zinc, and any alloys thereof including stainless steel.

[0540] FIG. **92** illustrates a clamp arm **2580** comprising a partially electrically conductive clamp arm pad **2582**, according to at least one aspect of the resent disclosure. An electrically conductive foil **2584** covers a portion of an electrically non-conductive pad **2586**. The electrically non-conductive pad **2588** at the proximal end **2590** sets a gap between the clamp arm pad **2582** and the ultrasonic blade.

[0541] Elements of the electrically conductive film **2568**, foil, or laminate may include, for example, a single layer of thin conductive material such as metals (titanium, silver, gold, zinc, aluminum, magnesium, iron, etc. and their alloys or stainless steels), plated metals (nickel and then gold over copper, for example) or polymers filled heavily with conductive materials such as metal powder, or filings. Preferably, it is a biocompatible metal foil such as titanium, silver, gold, zinc, or stainless steel selected from a thickness within the range of 0.001" to 0.008" (0.025 mm-0.20 mm).

[0542] The film **2568**, foil, or laminate may include a thin polymer coating, film or layer covering the thin conductive material described above. This coating, film or layer is highly resistive, that is, it is not an effective conductor of bipolar RF energy to adjacent tissue. The coating may be perforated to allow for energy delivery from the electrode to tissue.

[0543] The conductive material may be perforated or contain holes or windows through the full thickness of the conductive material to minimize the thermal capacitance of this layer (testing has shown that long and/or thick foils result in longer transection times due to thermal energy being removed from the treatment sight. These perforations, holes or windows also may allow for retention of the foil to other parts or layers. These perforations, holes or windows may be patterned across the entire foil sheet or may be localized at the treatment site or away from the treatment site such as, for example, on the sides of the clamp arm only.

[0544] If present, the thin polymer coating, film or layer may be perforated or contain full thickness holes or windows such that the conductive film, foil or laminate is in direct communication with tissue for delivery of bipolar radiofrequency energy to the tissue. For coatings, these holes or windows may be formed by selective coating or coating removal.

[0545] Ideally, the conductive film **2568**, foil, or laminate is in direct contact with the clamp arm structure that is typically fabricated from stainless steel. The resulting conductive path then allows for simplicity of construction in that the path is formed by necessary structural component, namely a support tube or actuator that connects directly to the clamp arm and then the conductive film, foil or laminate.

[0546] In one aspect, the conductive film **2568**, foil, or laminate is backed by a relatively soft, high temperature,

low wear polymer or elastomer pad made from materials such as PTFE, silicone, polyimide, high temperature thermoplastics, among other materials. The compliance of this relatively soft pad allows for a wide range of component tolerances to obtain a zero or near zero gap between the jaw and the ultrasonic blade along its full tissue effecting length when the jaw is fully closed, thus allowing tissue to be sealed and cut along this length. The compliance also eliminates or greatly dampens any audible vibration of the conductive layer that may occur when the ultrasonic blade is closed against the conductive layer.

[0547] The conductive film **2568**, foil, or laminate may include a rigid to semi-rigid polymer on its backside/back surface (that is the surface away from the tissue and toward the clamp arm). This part is made from injection moldable polymers or polymer alloys and adhered to the film, foil or laminate by way of Film Insert Molding (FIM) or In-Mold Labeling (IML).

[0548] In testing, thin stainless steel, copper, or aluminum foils are quiet in operation (no "screeching" or emitting of obtuse squeals). The thin stainless steel, copper, or aluminum foils provide a robust surface against which the ultrasonic blade can act. Robust enough that materials such as silicone rubber that would otherwise tear and serve as a poor pad material are usable and do not easily tear or split.

[0549] The proximal portion of the jaw clamping surface may not include the conductive film, foil or laminate because this area of the jaw contacts the blade first and will be more likely result in shunting of power/shorting in this area.

[0550] In one aspect, the present disclosure provides a short circuit mitigation algorithm for activating an output including bipolar RF energy.

[0551] A short alert is not given to the user if it occurs after the energy delivered for the activation exceeds a threshold amount (thereby indicating that the tissue thinned but has likely received an adequate dose of bipolar RF energy for the sealing, coagulation of tissue), or an activation time threshold has been exceeded (again, thereby indicating that the tissue has thinned but has likely received an adequate dose), or both energy and activation time thresholds have been exceeded.

[0552] A process of making a film over stainless steel insert molded electrode assembly comprises etching the film and forming apertures (micro-holes, slots, or honeycomb) for passing RF energy; cutting periphery of the electrode component; forming a film onto/bond onto stainless steel electrode if needed; placing the charged film and electrode into a polymer injection mold tool; molding the polymer to the back of the electrode and film.

[0553] FIG. **93** illustrates a clamp arm **2600** comprising a clamp jaw **2602**, a clamp arm pad **2604**, and a wrapped wire electrode **2606**, according to at least aspect of the present disclosure. The clamp jaw **2602** is made of stainless steel and the clamp arm pad **2604** is made of a polymer (e.g., PTFE). The wrapped wire electrode **2606** is made an electrical conductor.

[0554] FIG. **94** illustrates a clamp arm **2610** comprising a clamp jaw **2612**, a clamp arm pad **2614**, and a wrapped wire electrode **2616**, according to at least aspect of the present disclosure. The clamp jaw **2612** is made of stainless steel and the clamp arm pad **2614** is made of a polymer (e.g., PTFE). The wrapped wire electrode **2616** is made an electrical conductor.

[0555] Additional disclosure material may be found in U.S. Pat. No. 9,764,164; U.S. Patent Application Publication No. 2017/0164997; U.S. Patent Application Publication No. 2017/0056059; and U.S. Pat. No. 7,442,193, each of which is herein incorporated by reference in their entirety.

[0556] In one aspect, the clamp arm pad can be comprised of a typically non-conductive material, such as PTFE, for example, which can be impregnated with electrically conductive particles, such as medical grade stainless steel, for example, such that the pad is sufficiently conductive to permit current to flow between the ultrasonic blade and the clamp arm.

[0557] In some variations, the clamp arm pad itself is conductive. By way of example only, the clamp arm pad may be formed of a molded, carbon filled PTFE.

[0558] In various aspects, the present disclosure provides an end-effector for a combination ultrasonic/bipolar RF energy surgical device comprising an electrode support mechanism. In one aspect, the electrode support mechanism is configured to prevent delamination of electrode from the clamp arm. In one aspect, the electrode is attached to the clasp with a continuous adhesive to prevent incidental of a flexible electrode from the clamp arm. In one aspect, the present disclosure provides flexible electrodes adhered to a mechanical support member as described herein. The support member may be a metallic plastic hybrid with an integrated hinge to minimize the likelihood of the structural mechanical support member separating from the clamp arm. The delamination could be between the structural mechanical support member and an electrode member. The electrode member is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0559] FIG. 95 illustrates a clamp arm 1850 comprising a clamp jaw 1852, a support comprising a hinge-like feature 1854, an electrode 1856, and clamp arm pads 1858, according to at least one aspect of the present disclosure. The clamp arm 1850 is configured for use with an end-effector comprising an ultrasonic blade (not shown) as described throughout this disclosure. The hinge-like feature 1854 is disposed around the perimeter of the electrode 1856 to prevent delamination. The electrode 1856 is fixed to the clamp jaw 1852 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1856 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0560] FIG. 96 is a section view of the clamp arm 1850 shown in FIG. 95 in an unloaded condition through a clamp arm pad 1858, according to at least one aspect of the present disclosure. FIG. 97 is a section view of the clamp arm 1850 shown in FIG. 95 in a loaded condition under force F_1 applied to the electrode 1856 to collapse the support comprising the hinge-like feature 1854, according to at least one aspect of the present disclosure.

[0561] In one aspect, the present disclosure provides a support and electrode attachment member. In one aspect, the electrode is the support structure and is attached at a predefined location that needs to be stress protected. In another aspect, the mechanical support could be a metal substrate for easy attachment to the surrounding jaw using

any suitable fastening technique such as welding, laser welding, brazing, soldering, pressing, among other fastening techniques.

[0562] In one aspect, the pressure properties of the end-effector can be changed based on support. In one aspect, the present disclosure provides a variable compression/bias along the length of an electrode for a combination ultrasonic/bipolar RF energy device. The combination energy device comprises a bipolar RF electrode that is deflectable with respect to the clamp arm having features to change the mechanical properties of the tissue compression under the electrodes based on jaw closure or clamping amount. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0563] Additional background disclosure may be found in U.S. Pat. No. 6,533,784, which is herein incorporated by reference in its entirety.

[0564] In one aspect, the variable longitudinal supports also act as interactive supports. The integrated interactive support features are configured to effect the compressibility or spring constant of a compliant support structure. The support structure could also be a plastic metallic composite or overmolded part to give the integrated interactive support features that would limit the maximum deflection of the support structure via plastic bump extensions that would interact with the underlying jaw along its length. The integrated interactive support features also may be spring-like features that enable the adjustment of the spring constant of the electrode either along its length or proportionally to the amount of deflection already induced.

[0565] FIG. 98 illustrates a clamp arm 1860 portion of an end-effector, where the clamp arm 1860 comprises a clamp jaw 1862, clamp arm pads 1864, variable longitudinal support elements 1866, bump extensions 1867, and an electrode 1868 supported by the variable longitudinal support elements 1866, according to at least one aspect of the present disclosure. The clamp arm 1860 is configured for use with an end-effector comprising an ultrasonic blade (not shown) as described throughout this disclosure. The variable longitudinal support elements 1866 is configured to limit maximum deflection of the electrode 1868. The variable longitudinal support elements 1866 comprises a plurality of springs with variable force (F_1, F_2, F_3, F_4). The bump extensions 1867 limit the maximum deflection of the variable longitudinal support elements 1866 with variable longitudinal spring forces (S_1, S_2, S_3, S_4) where:

$$S_4 > S_3 > S_2 > S_1 > F_4 > F_3 > F_2 > F_1$$

[0566] FIGS. 99A-99C illustrate the clamp arm 1860 shown in FIG. 98 under various load conditions, according to at least one aspect of the present disclosure. FIG. 99A illustrates the clamp arm 1860 under a new firing condition either with no load or evenly distributed load 1869. In FIG. 99B the clamp arm 1860 is illustrated under high load conditions, where there is a high force 1870 at the distal end and a low force 1874 at the proximal end. As shown, the distal bump extension 1867 prevents maximum deflection of the electrode 1868. In FIG. 99C the clamp arm 1860 is shown under high load with worn down bump extensions 1867, where there is a high load 1870 at the distal end, a low

load **1874** at the proximal end, and a median load **1872** in the center. The bump extensions **1867** protect the electrode **1868** from maximum deflection.

[0567] In one aspect, the present disclosure provides an end-effector comprising integrated proximal/distal springs. A spring mechanism may be added to both the distal and proximal ends of an electrode. The spring mechanism has multiple configurations including integrally stamped leaf springs, separate springs, or flexible materials acting as springs. These springs could be configured and tuned for a desired pressure profile. Proximal loading may be reduced to improve the life of the clamp arm pad or increase loading distally to provide better tip performance. All of the following configurations include a metal electrode located on top of the clamp arm, a wear resistant material on the metal electrode to set the gap between the metal electrode and the ultrasonic blade, and a more compliant material on the clamp arm. The metal electrode also is connected in some way to one pole of a bipolar RF generator and opposite the clamp arm is a titanium ultrasonic blade that is connected to the other pole of the bipolar RF generator. The titanium ultrasonic blade can vibrate when driven by an ultrasonic transducer.

[0568] FIG. **100** illustrates a general configuration of an end-effector **1880**, according to at least one aspect of the present disclosure. FIG. **101** is a top view of the electrode **1886** showing apertures **1887** for receiving the compliant material **1888** therethrough. With reference now to FIGS. **100-101**, the end-effector **1880** comprises a clamp arm **1882**, an ultrasonic blade **1884**, an electrode **1886**, a compliant material **1888**, and a hard wear resistant material **1890**. The compliant material **1888** is fixed to the clamp arm **1882** to act as a spring between the electrode **1886** and the clamp arm **1882**. The hard wear resistant material **1890** is fixed to the proximal end of the electrode **1886** to set a gap between the electrode **1886** and the clamp arm **1882**.

[0569] FIGS. **102-103** shows a first configuration of the end-effector **1880** shown in FIGS. **100-101**, according to at least one aspect of the present disclosure. The end-effector **1880** comprises a leaf spring element **1892** at the distal end **1894** of the electrode **1886**. The proximal end **1896** of the electrode **1886** is attached to the clamp arm **1882** at point **1889**. FIG. **103** is a magnified view of the distal end **1894** of the electrode **1886** showing the leaf spring elements **1892**. The leaf spring elements **1892** are symmetrically disposed on the distal end of the electrode **1886**.

[0570] FIGS. **104-106** show a second configuration of the end-effector **1880** shown in FIGS. **100-101**, according to at least one aspect of the present disclosure. The end-effector **1880** comprises a leaf spring element **1896** at the distal end **1894** of the electrode **1886**. The electrode **1886** is attached to the RF generator via a wire **1898** provided through a tube **1900**. FIG. **105** is a section view of the tube **1900** showing the ultrasonic blade **1884** and the wire **1898**. FIG. **106** is a section view of the clamp arm **1882** showing the electrode **1886** and the leaf spring element **1896**.

[0571] FIG. **107** shows a third configuration of the end-effector **1880** shown in FIGS. **100-101**, according to at least one aspect of the present disclosure. The end-effector comprises a compressible material **1902** attached to the bottom portion of the distal end **1894** of the electrode **1886**. The distal end **1894** of the clamp arm **1882** defines a pocket **1904** to receive the compressible material.

[0572] Additional background disclosure may be found in WO 2017/198672, which is herein incorporated by reference in its entirety.

[0573] In various aspects, the present disclosure provides an end-effector for a combination ultrasonic/bipolar RF energy surgical device configured to lower tissue gap for RF welding while minimizing short circuiting the electrode to the ultrasonic blade. In one aspect, components of the end-effector comprise a DLC or PTFE (Teflon) coating to prevent short circuiting the ultrasonic blade to the electrode while enabling a zero gap. In one aspect, an end-effector comprising non-conductive supports defines a minimum gap between an electrode and ultrasonic blade to prevent the opposed electrode contact with the ultrasonic blade in the return path defined by the ultrasonic blade and waveguide. In one aspect, a first polymer is positioned for interaction with the ultrasonic blade and creates appropriate pressure while minimizing damage to the ultrasonic blade, and a second polymer spacer is provided to avoid damage by the ultrasonic blade and to minimize zero gap impacts with the flexible metal electrode. In another aspect, the proximal end of the flexible metal electrode has a feature that interacts with the jaw support designed to induce a deflection of the flexible metal electrode once the jaw is closed beyond a predetermined level.

[0574] In various aspects, the present disclosure provides an end-effector for a combination ultrasonic/bipolar RF energy surgical device configured to lower tissue gap for RF welding while minimizing short circuiting the electrode to the ultrasonic blade. In one aspect, the electrodes may be separated into right and left electrodes with a distal non-conductive zone aligned with the blade projection. In one aspect, the ultrasonic blade is selectively insulated. In a combination ultrasonic/RF energy surgical instrument, there exists the risk that the positive electrode of the bipolar RF circuit will short circuit with the negative electrode of the bipolar RF circuit. Generally, the clamp arm functions as the positive (hot) electrode and the ultrasonic blade functions as the negative (cold) electrode of the bipolar RF circuit. In use, it is desirable to prevent or minimize the likelihood of the positive electrode electrically contacting the ultrasonic blade and thus creating a short circuit condition. Accordingly, it is desirable to selectively coat particular areas of the ultrasonic blade to prevent or minimize short circuiting the clamp arm to the ultrasonic blade as described hereinbelow. Further, it may be desirable to selectively coat other components of the end-effector to electrically isolate portions of the end-effector from adjacent tissue. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0575] FIG. **108** illustrates an end-effector **1910** comprising a clamp arm **1912**, an ultrasonic blade **1914**, an electrode **1916**, and a clamp arm pad **1918**, according to at least one aspect of the present disclosure. The ultrasonic blade **1914** comprises an electrically insulative material **1920** on select areas of ultrasonic blade **1914** to prevent electrical shorting in the event of the ultrasonic blade **1914** contacting the upper jaw electrode **1916**. In one aspect, the electrically insulative material **1920** may be applied by vapor deposition or other coating techniques. The electrically insulative material **1920** may be deposited by masking or machining off areas of the

ultrasonic blade **1914** that are required to be electrically conductive. The electrically insulative material **1920** deposited on the ultrasonic blade **1914** may be a fluoropolymer material known under the tradename XYLAN, PTFE, nanocomposite coatings such as diamond-like carbon (DLC) coatings, or ceramic coatings, for example.

[0576] In one aspect, the present disclosure provides an end-effector comprising an offset electrode with a deflectable portion. In one aspect, the offset electrode may be provided in combination with a deflectable portion.

[0577] In one aspect, the present disclosure provides an end-effector that includes selectively coated components. The selective coating of end-effector components to electrically isolate portions of the end-effector from adjacent tissue. Potential advantages of selectively coating end-effector components to electrically isolate portions of the end-effector from adjacent tissue include reduce the likelihood of inadvertent RF injury to tissue, focus RF energy to desired tissue effects for shorter activation times and reduced thermal spread, and provide active electrode tips to allow for precise spot coagulation and touch up.

[0578] The electrically insulative coatings may be selectively applied in thick, thin, or in between layers depending on a desirable outcome. It may be advantageous to provide thin, electrically insulative and thermally dissipative coatings such as fluoropolymer coatings known under the tradename XYLAN, PTFE, nanocomposite coatings such as DLC coatings, or ceramic coatings, example. The coatings may be applied to masked components or selectively removed by buffing, grinding, machining, laser, or similar technique to expose the underlying electrically conductive electrode surface.

[0579] FIGS. 109A-109F illustrate various examples of combination ultrasonic/RF energy end-effectors comprising selectively coated components, according to at least one aspect of the present disclosure. In various aspects, the end-effectors include coated components, uncoated components, and bare electrode components as described hereinbelow.

[0580] In FIG. 109A, an end-effector **1930** comprises selectively coated portions **1932** on the clamp arm **1934** and the ultrasonic blade **1936**, an uncoated clamp arm pad **1938**, and bare electrode **1939** portions on the clamp arm **1934** and the ultrasonic blade **1936**.

[0581] In FIG. 109B, an end-effector **1940** comprises selectively coated portions **1942** on the clamp arm **1934**, an uncoated ultrasonic blade **1936**, an uncoated clamp arm pad **1938**, and a bare electrode **1944** on the clamp arm **1934**.

[0582] In FIG. 109C, an end-effector **1950** comprises an uncoated clamp arm **1934**, selectively coated portions **1952** on the ultrasonic blade **1936**, an uncoated clamp arm pad **1938**, and a bare electrode **1939** portion on the ultrasonic blade **1936**.

[0583] In FIG. 109D, an end-effector **1960** comprises selectively coated portions **1962** on the clamp arm **1934**, an uncoated ultrasonic blade **1936**, an uncoated clamp arm pad **1938**, and a bare electrode **1939** on the clamp arm **1934** including a bare electrode portion at the tip **1964** of the clamp arm **1934**.

[0584] In FIG. 109E, an end-effector **1970** comprises an uncoated clamp arm **1934**, selectively coated portions **1972** on the ultrasonic blade **1936**, an uncoated clamp arm pad

1938, and a bare electrode **1939** portion on the ultrasonic blade **1936** including a bare electrode portion at the tip **1974** of the ultrasonic blade **1936**.

[0585] In FIG. 109F, an end-effector **1980** comprises selectively coated portions **1982** on the clamp arm **1934** and the uncoated ultrasonic blade **1936**, an uncoated clamp arm pad **1938**, and bare electrode portions **1939** on the clamp arm **1934** and the ultrasonic blade **1936** including bare electrode portions at the tip **1984** of the clamp arm **1934** and the tip **1986** of the ultrasonic blade **1936**.

[0586] In various aspects, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device comprising an ultrasonic pad configuration. The end-effector comprises a compression element to limit heat generation in the end-effector or any of its components. In one aspect, the present disclosure provides an asymmetric segmented ultrasonic support clamp arm pad for cooperative engagement with a movable RF electrode configured for use with a combination ultrasonic/RF energy device. In another aspect, the asymmetric segmented ultrasonic support clamp arm pad extends at least partially through the RF electrode. In another aspect, at least one element of the clamp arm pad is significantly taller than a second element of the clamp arm pad. In another aspect, a first clamp arm pad extends entirely through and a second clamp arm pad extends only partially through the electrode. In another aspect, the first element of the clamp arm pad and the second element of the clamp arm pad are made of dissimilar materials. Additional background material can be found in U.S. Patent Application Publication No. 2017/0164997, which is incorporated herein by reference in its entirety.

[0587] In one aspect, the end-effector comprises a deflectable electrode configuration. In one aspect the present disclosure provides a segmented ultrasonic support clamp arm pad which extends at least partially through an RF electrode. In another aspect, at least one pad element is significantly taller than a second element. In another aspect, the first pad element extends entirely through the electrode and the second pad element extends partially through the electrode. In another aspect, the first pad element and the second pad element are made of dissimilar materials.

[0588] In one aspect, the present disclosure provides an electrode comprising destructive and non-destructive portions. The description of FIGS. 1-9 is herein incorporated by reference. The clamp arm pad **1010** shown in connection with FIGS. 1-9 provides larger resilient (e.g., PTFE) pad areas while still allowing the electrode **1004** to deflect. The electrode is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode may be referred to as a cantilever beam electrode or as a deflectable electrode. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0589] In one aspect, the end-effector comprises a clamp arm pad in an I-beam configuration. The I-beam shaped clamp arm pad (e.g., Teflon or PTFE) is provided such that the bottom of the clamp arm pad slides into the clamp arm, and the top of the clamp arm pad slides above the electrode. The top portion of the clamp arm pad may be shaped in a modified cobblestone fashion to increase electrode surface area on the side that is opposite the ultrasonic blade. This

configuration simultaneously provides a pad at the midline and flanges on the clamp arm pad over the electrode to prevent the clamp arm pad from delaminating from the clamp arm. The proximal hard (e.g., polyimide) gap setting pad slides in from the back (proximal end) to provide gap setting structure and a “plug” to hold the clamp arm pad in place.

[0590] FIGS. 110-112 illustrate a clamp arm 1990 comprising an I-beam shaped clamp arm pad 1992 configured for use with an end-effector comprising a clamp jaw 1994, an ultrasonic blade (not shown), and an electrode 1996, according to at least one aspect of the present disclosure. The clamp arm pad 1992 has an I-beam configuration comprising top and bottom lateral portions 1998, 2000 separated by a mesial portion 2002 to define a cross-sectional shape of an “I”. The electrode 1996 is disposed between the top and bottom lateral portions 1998, 2000 of the I-beam shaped clamp arm pad 1992. The electrode 1996 is fixed to the clamp jaw 1994 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 1996 may be referred to as a cantilever beam electrode or as a deflectable electrode. The clamp jaw 1994 defines a longitudinal slot 2010 to slidably receive the bottom lateral portions 1998, 2000 and a portion of the mesial portion 2002 of the I-beam shaped clamp arm pad 1992 therethrough.

[0591] FIG. 111 is a top view of the clamp arm 1990 showing the clamp arm pad 1992 as viewed from above the electrode 1996. FIG. 112 is a top view of the electrode 1996 showing a longitudinal slot 2010 to slidably receive the clamp arm pad 1992 therein. With reference to FIGS. 111-112, the top portions 1998 of the I-beam segments of the clamp arm pad 1992, referred to as the flanges, are positioned above the electrode 1996 to prevent the clamp arm pad 1992 from delaminating from the clamp jaw 1994. Also shown in this view is a hard wear resistant gap setting pad 2004 to set a gap between the electrode 1996 and the ultrasonic blade (not shown). The deflectable electrode 1996 is fixedly attached to the clamp jaw 1994 by welding the proximal end 2006 of the electrode 1996 to the clamp jaw 1994 by welding at weld points 2008. Other attachment techniques may be employed such as laser welding, brazing, soldering, pressing, among other fastening techniques.

[0592] FIGS. 113A-113C illustrate a method of assembling the clamp arm 1990 shown in FIGS. 110-112, according to at least one aspect of the present disclosure. In FIG. 113A the electrode 1996 is welded to the proximal end 2006 of the clamp jaw 1994 at weld points 2008. In FIG. 113B the clamp arm pad 1992 is slidably inserted into the longitudinal slot 2010. In FIG. 113C the hard wear resistant gap setting pad 2004 to set the gap is slidably inserted into the clamp jaw 1994.

[0593] The clamp arm 1990 described in connection with FIGS. 110-113C provides an electrode 1996 that deflects when clamped on tissue. Tissue accumulation points between the clamp arm pad 1992 segments is eliminated and the electrode 1996 delamination is minimized. The clamp arm 1990 also eliminates small gap setting pads and easy to manufacture.

[0594] In one aspect, the end-effector comprises interactive electrode recesses and ultrasonic blade support features. FIGS. 114-116 illustrate an end-effector 2020 comprising an electrode 2026 defining recesses 2030 in overlapping locations with an ultrasonic blade 2024 to minimize impact

between the ultrasonic blade 2024 and the electrode 2026, according to at least one aspect of the present disclosure. FIG. 114 is a section view of an end-effector 2020 comprising a clamp arm 2022, an ultrasonic blade 2024, an electrode 2026, and a heavy polymer support pad 2028 (e.g., polyamide pad), according to at least one aspect of the present disclosure. FIG. 115 is a section view of the end-effector 2020 with a worn down heavy polymer support pad 2028 and tissue 2032 clamped between the clamp arm 2022 and the ultrasonic blade 2024. In the clamped state, the electrode 2026 is substantially flush with the ultrasonic blade 2024. FIG. 116 is a magnified view of the end-effector 2020. In the fully clamped state, the electrode 2026 is substantially flush with the ultrasonic blade 2024. As the ultrasonic blade 2024 rests in the recess 2030 defined by the heavy polymer support pad 2028 a plane 2034 defined by the top of the electrode 2026 is substantially flush or even with a plane 2036 defined by the bottom of the ultrasonic blade 2024 as it rests within the recess 2030 defined by the heavier polymer support pad 2028.

[0595] With reference now to FIGS. 114-116, along the centerline (CL) of the electrode 2026, the electrode 2026 defines a recess 2029 and the heavy polymer support pad 2028 defines an integrated recess 2030 that corresponds to the location of the ultrasonic blade 2024 to minimize interaction on the electrode 2026 and the ultrasonic blade 2024. The location of these recesses 2030 may coincide with the location of the heavier polymer support pad 2028 that is not subjected to damage by contact with the ultrasonic blade 2024. In this way the electrode 2026 may be substantially flush with the ultrasonic blade 2024 when clamped in the fully clamped state and yet not in metallic contact with the ultrasonic blade 2024. An additional hole defined within the recessed area could be used to heat stake or fix the heavy polymer support pad 2028 to the electrode recess 2030. The electrode is adapted and configured for use with a combination ultrasonic/RF energy device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0596] In one aspect, the end-effector comprises a polyimide front pad. FIGS. 117-118 illustrates a clamp arm 2040 comprising a clamp jaw 2042, a stationary gap setting pad 2044 in combination with movable floating gap setting pads 2046, according to at least one aspect of the present disclosure. This configuration provides a composite arrangement to both create ultrasonic blade pressure while also setting a minimum gap between an electrode 2048 and the ultrasonic blade. The electrode 2048 is adapted and configured for use with a combination ultrasonic/RF energy device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0597] FIG. 117 is a top view of a clamp arm 2040 showing the stationary gap setting pad 2044 and the movable floating gap setting pads 2046. FIG. 118 is a side view of the clamp arm 2040 showing the clamp jaw 2042, the stationary gap setting pad 2044, the movable floating gap setting pads 2046, the electrode 2048, and a clamp arm pad 2050. The most distal 2052 stationary pad 2044 is attached to the clamp jaw 2042 in a similar fashion as the clamp arm pad 2050. The most distal 2052 stationary gap setting pad 2044 will improve tip pressure/grasping and decreases the

likelihood of short circuiting the electrode **2048** to the ultrasonic blade due to burn through of the stationary gap setting pad **2044**.

[0598] FIGS. **119-120** show an alternative clamp arm **2060** comprising a clamp jaw **2074**, a stationary gap setting pad **2064** at a distal end **2062**, a stationary clamp arm pad **2066**, and a movable floating gap setting pad **2070** at a proximal end **2068**, according to at least one aspect of the present disclosure. The clamp arm **2060** also comprises an electrode **2072**.

[0599] In one aspect, the end-effector comprises a deflectable/cantilever electrode with teeth. Tissue retention in an ultrasonic device can be difficult due to the smooth nature of the ultrasonic blade. Additional gripping features can be added to increase the tissue gripping force. The electrode element of the device is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0600] FIGS. **121-122** illustrate a combination ultrasonic/bipolar RF energy surgical device that employs an electrode comprising small teeth, according to at least one aspect of the present disclosure. The electrode teeth can add to the tissue retention. Teeth that pass through the electrode can be added to the clamp jaw. This configuration is relatively easy to implement and provides better tissue retention in an ultrasonic device. In one aspect, the present disclosure provides a surgical device comprising an ultrasonic blade and an RF electrode for cutting and sealing tissue where the RF electrode is implemented as a flexible member and where the flexible electrode comprises a plurality of teeth for facilitating the tissue grasping function. In another aspect, the present disclosure provides a surgical device comprising an ultrasonic blade and an RF electrode for cutting and sealing tissue where the RF electrode is implemented as a flexible member and where the rigid movable jaw comprises a plurality of teeth protruding through the flexible electrode for facilitating the tissue grasping function.

[0601] FIG. **121** illustrates an end-effector **2080** comprising a clamp arm **2082**, an ultrasonic blade **2084**, an electrode **2086** comprising a plurality of teeth **2088**, a pliable clamp arm pad **2090**, and hard gap setting pads **2092**, **2093** according to at least one aspect of the present disclosure. The clamp arm **2082** comprises a clamp jaw **2094** that is pivotally movable about a pivot point **2096**. The electrode **2086** is fixed to the clamp jaw **2094** at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode **2086** may be referred to as a cantilever beam electrode or as a deflectable electrode. The electrode **2086** is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode **2086** is one pole of the bipolar RF circuit and the ultrasonic blade **2084** is the opposite pole of the bipolar RF circuit.

[0602] FIG. **122** illustrates an end-effector **2100** comprising a clamp arm **2102**, an ultrasonic blade **2104**, an electrode **2106**, a pliable clamp arm pad **2110**, and hard gap setting pads **2112**, **2113** according to at least one aspect of the present disclosure. The clamp arm **2102** comprises a clamp jaw **2114** comprising teeth **2108** that extend through the electrode **2106**. The clamp jaw **2114** is pivotally movable about a pivot point. The electrode **2106** is fixed to the clamp jaw **2114** at the proximal end and is free to deflect at the

distal end. Accordingly, throughout this disclosure the electrode **2106** may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0603] In one aspect, the end-effector comprises an ultrasonic reloadable electrode subassembly. FIGS. **123-130** illustrate a clamp arm **2120** comprising a reloadable electrode **2122** that incorporates a clamp arm pad **2126** comprising teeth **2123**, according to at least one aspect of the present disclosure. The reloadable electrode **2122** provides an advantageous solution to challenges typically encountered with electrodes such as wear, tissue accumulation, and manufacturing difficulties. The clamp arm **2120** includes a pivot point **2121** that enables the clamp arm **2120** to pivotally open and close. The electrode **2122** is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode **2122** is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0604] FIG. **123** illustrates a reloadable electrode **2122** and a clamp arm pad **2126**, according to at least one aspect of the present disclosure. FIG. **124** illustrates a clamp arm **2120** comprising the reloadable electrode **2122** and clamp arm pad **2126** located within a clamp jaw **2127**. Hard wear resistant gap pads **2125**, **2129** are disposed on or adjacent to the reloadable electrode **2122** to set a gap between the reloadable electrode **2122** and an ultrasonic blade (not shown). The reloadable electrode **2122** includes an arm **2124** formed of an extended piece of sheet metal bent into a spring that can be reloaded into the clamp arm **2120** of the end-effector. The clamp arm pad **2126** (Teflon, PTFE) can be trapped in a spring form **2128** of the arm **2124** and retained until loaded into the clamp arm **2120**. The sides **2130** of the reloadable electrode **2122** are curved inwards to prevent tissue from accumulating between the clamp arm **2120** and the arm **2124**. The reloadable electrode **2122** is fixed to the clamp jaw **2127** at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the reloadable electrode **2122** may be referred to as a cantilever beam electrode or as a deflectable electrode. FIG. **125** is a side view and FIG. **126** is a top view of the reloadable electrode **2122** and clamp arm pad **2126** in the process of being slidably inserted into the clamp jaw **2127** from the front (distal end) of the clamp jaw **2127**. The arm **2124** is assembled from the front of the clamp arm **2120** and is retained by a bent tab **2132** that locates a slot **2134** in the clamp jaw **2127**. The tab **2132** can be depressed to replace the reloadable electrode **2122** and the clamp jaw **2127** during the procedure. The reloadable electrode **2122** comprises retention legs **2136** that interface with pockets **2138** in the clamp arm **2120**. The retention legs **2136** locate and keep the clamp arm **2120** from releasing during operation. A slot **2135** defined by the clamp jaw **2127** receives the reloadable electrode **2122**. The assembly process costs less than welding and allows for reclamation of components if defects are found. Alternative aspects include the direction in which the reloadable electrode **2122** and clamp arm pad **2126** are loaded or reloaded into the clamp jaw **2127**, the shape and number of pads of the clamp arm pad **2126**, the configuration of the proximal pad, and the configuration of the electrode retention legs **2136**.

[0605] Although the reloadable electrode **2122** is depicted as being loaded from the front (distal end) of the clamp jaw **2127**, alternatively, the reloadable electrode **2122** could be

loaded from the back (proximal end), sides, or top of the clamp jaw **2127**. The shape of the clamp arm pad **2126** and the number of pads can vary to accommodate a particular clinical task. In this aspect, the proximal clamp arm pad **2126** is located on the clamp arm **2120**. In a replaceable aspect, the proximal clamp arm pad **2126** could be incorporated onto the reload side. The number and size of the electrode retention legs **2136** can vary. Another retention feature besides the electrode retention legs **2136** can be used.

[0606] FIG. **127** is a section view of the electrode legs **2136** inserted in the clamp jaw **2127** pockets **2138** and the bent tab **2132** located in the slot **2134** defined by the clamp jaw **2127** and FIG. **128** is a detailed view of the clamp arm pocket **2138**. FIG. **129**, shows how the arm **2124** can be released by pressing down the bent tab **2132**. FIG. **130** is a detailed view of the reloadable electrode **2122** and arm **2124** formed as a stamped component showing the extended piece of sheet metal and apertures **2142** sized and configured to receive the teeth **2123** protruding from the clamp arm pad **2126**, apertures **2144** sized and configured to receive the hard wear resistant gap pads **2125**, and an aperture **2146** sized and configured to receive the hard wear resistant gap pad **2129**.

[0607] The reloadable electrode **2122** provides several advantageous and benefits including preventing short circuiting of the reloadable electrode **2122** to the ultrasonic blade **2104**, overcoming the application of insufficient tissue pressure due to wear of the polymeric (e.g., Teflon PTFE) clamp arm pad **2126**, improving tip grasping, reducing tissue snagging and tissue sticking, reducing tissue accumulation between the replaceable electrode **2124** and the clamp arm **2120**, preventing the replaceable electrode **2124** from delaminating from the clamp arm **2120**, improved manufacturability, and/or improved reliability/mission life.

[0608] In one aspect, the end-effector comprises a replenishing pad configuration. The present disclosure provides an end-effector configured for use with combination ultrasonic/bipolar (RF) electrosurgical instruments. The end-effector maintains a consistent gap between the bipolar RF electrode and the ultrasonic blade in combination ultrasonic/Bipolar RF energy devices where the ultrasonic blade is one pole of the bipolar RF circuit and the clamp arm is the opposite pole of the bipolar RF circuit. In conventional end-effector configurations, the gap between the bipolar RF electrode and the ultrasonic blade is set by a soft polymeric (e.g., PTFE) clamp arm pad, which is subject to wear during a surgical procedure. When the polymeric pad wears through, there's the peril that the ultrasonic blade can contact the electrode resulting in breakage of the ultrasonic or an electrode to ultrasonic blade short circuit. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0609] In one aspect, the present disclosure provides an end-effector that incorporates an RF electrode fixed to the clamp arm. The electrode comprises wear-resistant, non-conductive pads, which contact the ultrasonic blade to set the blade-to-electrode gap. These wear-resistant gap setting pads also react to the clamp force applied to the ultrasonic blade. The electrode defines a series of holes through which a compliant polymeric (TEFLON/PTFE) clamp arm pad protrudes and contacts the ultrasonic blade when in the

clamped position. The compliant polymeric clamp arm pad is supported by a spring element positioned between the clamp arm pad and the clamp arm. The spring element maintains consistent contact between the clamp arm pad and the ultrasonic blade. As the compliant polymeric clamp arm pad wears, additional material is pushed through the electrode by the spring to maintain clamp arm pad to ultrasonic blade contact throughout the life of the combination ultrasonic/bipolar RF energy device. The spring may be a compression spring, leaf spring, or an elastomer, among other types of springs.

[0610] FIGS. **131-133** illustrate an end-effector **2150** comprising an ultrasonic blade **2152** and a clamp arm **2154** comprising a spring-loaded clamp arm pad **2156**, according to at least one aspect of the present disclosure. FIG. **131** is a perspective view of the end-effector, FIG. **132** is a section view of the end-effector **2150**, and FIG. **133** is an exploded view of the end-effector **2150**. The clamp arm **2154** further comprises a clamp jaw **2158** defining an aperture **2160** sized and configured to receive therein an electrode **2162** and a clamp arm pad teeth **2155** assembly. The electrode **2162** further defines apertures **2164** sized and configured to receive the clamp arm pad teeth **2155** therethrough. A plurality of springs **2168** are located in corresponding bosses **2170**. A backing plate **2172** is locked to the clamp jaw **2158** and compresses the springs **2168** to apply a load to the clamp arm pad **2155**. The load applied to the clamp arm pad **2155** causes the spring-loaded clamp arm pad **2156** to remain in contact with the ultrasonic blade **2152** as the spring-loaded clamp arm pad **2156** wears during use. The electrode **2162** is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode **2162** is one pole of the bipolar RF circuit and the ultrasonic blade **2152** is the opposite pole of the bipolar RF circuit.

[0611] FIG. **134** illustrates a spring-loaded clamp arm pad **2180**, according to at least one aspect of the present disclosure. The clamp arm pad **2180** comprises a clamp jaw **2182**, and electrode **2184**, a compliant clamp arm pad **2186** (e.g., PTFE), and a hard gap setting pad **2188** (e.g., polyimide). A spring force is applied to the clamp arm pad **2186** to move the clamp arm pad **2186** relative to the electrode **2184**, while the electrode **2184** remains fixed.

[0612] FIG. **135** illustrates a clamp arm **2190** comprising a spring-loaded clamp arm pad **2192**, according to at least one aspect of the present disclosure. The clamp arm **2190** further comprises a backing plate **2194** and a plurality of springs **2196** located between a clamp jaw **2198** and a backing plate **2200**. The base of the clamp arm pad **2192** is fixed to the backing plate **2194** and projections **2202** (e.g., teeth) of the clamp arm pad **2192** extend through apertures **2204** defined by an electrode **2206**. In one aspect, the clamp arm **2190** further comprises a plurality of gap setting pads **2208** made of a hard wear resistant material to set a gap between the electrode **2206** and an ultrasonic blade (not shown). In one aspect, the spring-loaded clamp arm pad **2192** is made of a lubricous, non-stick, compliant material such as PTFE and the gap setting pads **2208** are made of a harder wear resistant material such as polyimide.

[0613] FIG. **136** illustrates a clamp arm **2210** comprising a spring-loaded clamp arm pad **2212**, an electrode **2214**, and gap setting pads **2216**, according to at least one aspect of the present disclosure. Cuts **2218** in the clamp arm pad **2212** define projections **2215** that act as spring elements. The base

of the clamp arm pad **2212** is fixed to the back of the electrode **2214** and projections **2211** (e.g., teeth) of the clamp arm pad **2212** extend through apertures **2213** defined by the electrode **2214**. In one aspect, the spring-loaded clamp arm pad **2212** is made of a lubricous, non-stick, compliant material such as PTFE and the gap setting pads **2216** are made of a harder wear resistant material such as polyimide.

[0614] FIG. 137 illustrates a clamp arm **2220** comprising a spring-loaded clamp arm pad **2222** that is uniformly raised by an elastomeric spring **2224**, according to at least one aspect of the present disclosure. The elastomeric spring **2224** may be formed of rubber, foam, or other resilient material. The elastomeric spring **2224** is positioned over a hard plastic gap setting pad **2226**. In one aspect, the spring-loaded clamp arm pad **2222** is made of a lubricous, non-stick, compliant material such as PTFE and the gap setting pads **2226** are made of a harder wear resistant material such as polyimide.

[0615] FIG. 138 illustrates a clamp arm **2230** comprising a spring-loaded clamp arm pad **2232**, according to at least one aspect of the present disclosure. A spring **2234** applies a load to the clamp arm pad **2232** such that clamp arm pad **2232** maintains intimate contact with the ultrasonic blade during a surgical procedure.

[0616] In one aspect, the end-effector comprises a floating clamp arm pad. FIGS. 139-155 illustrate two configurations of an end-effector **2240** comprising first and second configurations of floating clamp arms **2242**, **2244** combined with an ultrasonic blade **2246**, according to at least one aspect of the present disclosure. The end-effector **2240** configurations comprise clamp arm pad units **2248**, **2250** supported by an elastic/hyper elastic block **2252**, **2254** and an electrode **2255** for use in combination with the ultrasonic blade **2246** to reduce wear of the clamp arm pad units **2248**, **2250** and increase cutting speed where tissue volume is high. The elastic/hyper elastic block **2252**, **2254** may be formed as a single piece or multiple individual pieces **2250**. The individual clamp arm pad units **2248**, **2250** may be biased against the elastic/hyper elastic block **2254** to accommodate a high pressure area between the floating clamp arm **2244** and the ultrasonic blade **2246**. The electrode **2255** is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode **2255** is one pole of the bipolar RF circuit and the ultrasonic blade **2246** is the opposite pole of the bipolar RF circuit.

[0617] In one aspect, the floating clamp arms **2242**, **2244** include multiple individual clamp arm pad units **2248**, **2250** where each of the individual clamp arm pad units **2248**, **2250** seats against a single piece block **2252** or multi-piece block **2254** attached to the floating clamp arms **2242**, **2244**. Each of the individual clamp arm pad units **2248**, **2250** comprise collars **2264** to constrain movement of the individual clamp arm pad units **2248**, **2250** along electrode **2255** channels and between the clamp jaw **2260** and the electrode **2255**. The single piece block **2252** or multi-piece block **2254** is made of an elastic or hyper elastic material to provide spring-like characteristics for balancing pressure between the sides of the floating clamp arms **2242**, **2244** and the side of the ultrasonic blade **2246**. The material may be porous, rubber, or similar electrically insulative materials. The compressible blocks **2252**, **2254** enable the individual clamp arm pad units **2248**, **2250** to float. The blocks **2252**, **2254** are pushed to compression by the individual clamp arm pad

units **2248**, **2250** where tissue volume is high, thus causing higher pressure on tissue to increase cutting speed in that area, and increasing hemostasis. The compressed block **2252**, **2254** will continue to push the individual clamp arm pad units **2248**, **2250** back against the tissue until the tissue is fully cut.

[0618] FIG. 140 is an exploded view of a clamp jaw **2260**, electrode **2255**, and ultrasonic blade **2246** components that are generic to the either one of the first or second configurations of the floating clamp arms **2242**, **2244**.

[0619] FIG. 141 illustrates the clamp arm pad units **2248** and single piece elastic/hyper elastic block **2252** components of a first configuration of floating clamp arm **2242**. The single piece elastic/hyper elastic block **2252** supports the clamp arm pad units **2248** and may be made of porous, rubber, or similar electrically insulative materials.

[0620] FIG. 142 illustrates the clamp arm pad unit **2250** and multi-piece elastic/hyper elastic block **2254** components of a second configuration floating clamp arm **2244**. Each of the multi-piece elastic/hyper elastic blocks **2254** supports each of the clamp arm pad units **2250** and may be made of porous, rubber, or similar electrically insulative materials. Each of the multi-piece elastic/hyper elastic blocks **2254** is bound to the each individual clamp arm pad units **2250**. Each of these configurations will be described in more detail hereinbelow.

[0621] FIG. 143 illustrates the clamp arm pad units **2250** layout protruding through apertures **2262** defined by the electrode **2255** without contacting each other, according to at least one aspect of the present disclosure.

[0622] FIG. 144 illustrates the apertures **2262** defined by the electrode **2255** sized and configured to receive either one of the clamp arm pad units **2248**, **2250**. The apertures **2262** define channels **2266** to guide movement of the individual clamp arm pad units **2248**, **2250**. The collars **2264** on the individual clamp arm pad units **2248**, **2250** constrict movement of the individual clamp arm pad units **2248**, **2250** between the floating clamp arm **2242**, **2244** and the electrode **2255** and prevents the individual clamp arm pad units **2248**, **2250** from escaping out of the electrode **2255**.

[0623] FIG. 145 is an assembly view of the first configuration clamp arm **2248** comprising the single piece elastic/hyper elastic block **2252**. FIG. 146 is a section view of a first configuration of an end-effector **2270** comprising a floating clamp arm **2242** and an ultrasonic blade **2246**, according to at least one aspect of the present disclosure. The floating clamp arm **2242** comprises a clamp jaw **2260**, a plurality of individual clamp arm pad units **2248** located through apertures **2262** defined by the electrode **2255** and supported by the single piece elastic/hyper elastic block **2252**. The plurality of individual clamp arm pad units **2248** engages tissue **2268** located between the ultrasonic blade **2246** and the electrode **2255**.

[0624] FIG. 147 is a section view of a second configuration of an end-effector **2280** comprising a floating clamp arm **2244** and an ultrasonic blade **2246**, according to at least one aspect of the present disclosure. The floating clamp arm **2244** comprises a clamp jaw **2261**, a plurality of individual clamp arm pad units **2250** located through apertures **2262** defined by the electrode **2255** and supported by the multi-piece elastic/hyper elastic block **2254**. The plurality of individual clamp arm pad units **2250** engage tissue **2268** located between the ultrasonic blade **2246** and the electrode **2255**.

[0625] FIG. 148 illustrates a method 2290 of assembling a first configuration end-effector 2292 comprising a floating clamp arm 2242 and an ultrasonic blade 2246, according to at least one aspect of the present disclosure. A first subassembly 2296 comprises the individual clamp arm pad units 2248 assembled to the electrode 2255. A second subassembly 2298 comprises the single piece elastic/hyper elastic block 2252 assembled to the clamp jaw 2260 by welding or screws. A third subassembly 2300 comprises the first subassembly 2296 assembled to the second subassembly 2298 to produce a fully assembled floating clamp arm 2242. The end-effector 2292 is completed by adding the ultrasonic blade 2246 to the third-subassembly 2300.

[0626] FIG. 149 illustrates the end-effector 2292 assembled in accordance with FIG. 148. As shown in FIG. 150 the single piece elastic/hyper elastic block 2252 is compressed by the individual clamp arm pad units 2248 in locations where the volume of tissue 2268 is high, thus causing higher pressure on the tissue 2268 to increase cutting speed in that area, possibly increasing hemostasis. The compressed single piece elastic/hyper elastic block 2252 will continue to push on the individual clamp arm pad units 2248 against the tissue 2268 until the tissue 2268 is fully cut and the individual clamp arm pad units 2248 will restore to their original position.

[0627] FIG. 151 shows a location 2249 where pressure between the ultrasonic blade 2246 and the individual clamp arm pad units 2248 can be balanced by the single piece elastic/hyper elastic block 2252 when the ultrasonic blade 2246 contacts the individual clamp arm pad units 2248 directly, thus reducing wear of the individual clamp arm pad units 2248.

[0628] FIG. 152 illustrates a method 2310 for assembling a second configuration end-effector 2312 comprising a floating clamp arm 2244 and an ultrasonic blade 2246, according to at least one aspect of the present disclosure. A first subassembly 2316 comprises the individual clamp arm pad units 2250 and multi-piece elastic/hyper elastic block 2254 assembled to the electrode 2255. A second subassembly 2318 comprises the first subassembly 2316 assembled to the clamp jaw 2260 by any suitable fastening technique such as welding, laser welding, brazing, soldering, pressing, among other fastening techniques such as screws. The end-effector 2312 is completed by adding the ultrasonic blade 2246 to the second-subassembly 2318.

[0629] FIG. 153 illustrates the end-effector 2312 assembled in accordance with FIG. 152. As shown in FIG. 154 the multi-piece elastic/hyper elastic block 2254 is pushed to compression by the individual clamp arm pad units 2250 in locations where the tissue 2268 volume is high, thus causing higher pressure on the tissue 2268 to increase cutting speed in that area, possibly increasing hemostasis. The compressed multi-piece elastic/hyper elastic block 2254 will continue to push on the individual clamp arm pad units 2250 against the tissue 2268 until the tissue 2268 is fully cut and the individual clamp arm pad units 2250 will restore to their original position.

[0630] FIG. 155 shows a location 2251 where pressure between the ultrasonic blade 2246 and the individual clamp arm pad units 2250 can be balanced by the multi-piece elastic/hyper elastic block 2254 when the ultrasonic blade 2246 contacts the individual clamp arm pad units 2250 directly, thus reducing wear of the individual clamp arm pad units 2250.

[0631] In one aspect, the end-effector comprises selectively deployable teeth. In combination ultrasonic/bipolar RF energy devices it may be desirable to have selectively deployable teeth to facilitate different energy modalities. This enables multi-functionality without making significant configuration trade-offs to. In a retracted configuration, the ultrasonic pad teeth provide increased surface area contact between tissue and electrodes and would be ideal for applying a single RF energy modality for sealing tissue. In a fully deployed configuration, the ultrasonic pad teeth provide low risk of metal-to-metal contact between the ultrasonic blade and the electrode and would be ideal for tissue manipulation and applying a single ultrasonic energy modality. In a partially deployed configuration, the ultrasonic pad teeth are well suited for application of both ultrasonic and RF energy modalities simultaneously. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0632] FIG. 156 illustrates a clamp arm 2320 comprising selectively deployable pad teeth 2330, according to at least one aspect of the present disclosure. The clamp arm 2320 comprises a clamp jaw 2322, an electrode 2324, a plurality of apertures 2326 defined by the electrode 2324, and a driver tab 2328. The plurality of apertures 2326 defined by the electrode 2324 include deployable pad teeth 2330. The driver tab 2328 is used to deploy or retract the deployable pad teeth 2330 such that the deployable pad teeth 2330 are retracted when the driver tab 2328 is retracted and the deployable pad teeth 2330 are deployed when the driver tab 2328 is advanced. In another aspect, this configuration may be reversed such that the deployable pad teeth 2330 are deployed when the driver tab 2328 is retracted and the deployable pad teeth 2330 are retracted when the driver tab 2328 is advanced. In one aspect, the pad tooth plugs 2330 are made of lubricious compliant polymeric materials such as Teflon/PTFE.

[0633] Similar to a staple cartridge, the clamp arm 2320 subassembly is formed by nesting the driver tab 2328 into a channel of the clamp jaw 2322. Selectively deployable pad teeth 2330 are placed on a driver 2332 (FIGS. 157-158) and are in the subassembly through interference with the electrode 2324. To complete the subassembly, the electrode 2324 can be fixedly attached to the clamp jaw 2322 using any suitable fastening technique such as welding, laser welding, brazing, soldering, pressing, among other fastening techniques. Translating the driver 2332 within the clamp jaw 2322 causes the selectively deployable pad teeth 2330 to raise or lower relative to the surface of the electrode 2324.

[0634] FIG. 157 is a section view of the clamp arm 2320 shown in FIG. 156 comprising selectively deployable pad teeth 2330 located within the apertures 2326 in a retracted configuration, according to at least one aspect of the present disclosure. When the driver 2332 is retracted proximally 2334, as shown, the selectively deployable pad teeth 2330 retract below the electrode 2324.

[0635] FIG. 158 is a section view of the clamp arm 2320 shown in FIG. 156 comprising selectively deployable pad teeth 2330 in a deployed configuration, according to at least one aspect of the present disclosure. When the driver 2332 is advanced distally 2336, as shown, the selectively deployable pad teeth 2330 deploy above the electrode 2324.

[0636] FIG. 159 is a detail section view of the clamp arm 2320 shown in FIG. 156 comprising selectively deployable pad teeth 2330 in a retracted configuration, according to at least one aspect of the present disclosure. Each of the selectively deployable pad teeth 2330 have a shoulder 2338 that is greater than the apertures 2326 defined by the electrode 2324 to retain the selectively deployable pad teeth 2330 within the clamp jaw 2322. The selectively deployable pad teeth 2330 geometry 2340 is sized and configured to deploy through the apertures 2326. Interference geometry 2342 can be round (as shown), angled, or a combination thereof to optimize a desired force profile for deployment of the selectively deployable pad teeth 2330. Additional handle mechanisms (push/pull rod, additional tube drive) already known in the art may be employed to actuate the driver 2332.

[0637] In one aspect, the end-effector comprises a clamp arm pad with a heat sink cooling mechanism. In one aspect, the clamp arm pad comprises a heat sink block to protect the clamp arm pad of an ultrasonic device by reducing heat accumulation on the clamp arm pad. Reducing heat accumulation on the clamp arm pad may be achieved in a variety of techniques. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0638] FIG. 160A illustrates an end-effector 2350 configured to receive heat sink material block 2356 as shown in FIGS. 160B-160C, according to at least one aspect of the present disclosure. The end-effector 2350 comprises a clamp jaw 2352 and an ultrasonic blade 2354. FIG. 160B illustrates one example of the end-effector 2350 shown in FIG. 160A with a heat sink material block 2356 added or otherwise fixedly attached to a clamp arm pad 2358, supported by the metal clamp jaw 2352. FIG. 160C shows the heat sink material block 2356 with a worn out clamp arm pad 2359. The heat sink material block 2356 reduces heat accumulation and will protect the clamp arm pad 2359 from excessive heating due to contact with ultrasonic blade 2354.

[0639] FIG. 161A illustrates an end-effector 2360 comprising a metal clamp jaw 2362 and an ultrasonic blade 2364 where at least some portions of the metal clamp jaw is made of a heat sink material, according to at least one aspect of the present disclosure. In one aspect, the entire metal clamp jaw 2362 is made of heat sink material and is closely fixed to a clamp arm pad 2366, shown in FIG. 161B, and an alternate clamp arm pad 2368, shown in FIG. 161C. The heat sink material may comprise either one or both of a material with high heat conductivity property and high specific heat and create large surface area feature to increase heat diffusion as heat builds up in the clamp arm pad 2366, 2368 due to contact with the ultrasonic blade 2364.

[0640] In connection with FIGS. 160A-160C and 161A-161C, the heat sink material should have high heat conductivity performance such as, for example, aluminum, copper, and alloys thereof, or other similar materials. In addition, the heat sink material creates a big surface area on the heat sink component to accelerate heat diffusion or dissipation. The most common heat sink materials are aluminum alloys. Aluminum alloy 1050 has one of the higher thermal conductivity values at 229 W/m·K but is mechanically soft.

Aluminum alloys 6060 (low stress), 6061, 6063 can be used, with thermal conductivity values of 166 and 201 W/m·K, respectively.

[0641] FIG. 162 shows the components of the end effector 2350, 2360 described in FIGS. 160A-160C and 161A-161C.

[0642] FIG. 163 shows one heat sink structure 2367 comprising fins 2369 for increasing the surface area of the heat sink component, according to at least one aspect of the present disclosure.

[0643] FIG. 164 shows another heat sink structure 2370 comprising pores 2372 for increasing the surface area of the heat sink component, according to at least one aspect of the present disclosure.

[0644] In various aspects, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical instrument comprising an end-effector with a clamp arm actuation or pivot mechanism. The clamp arm actuation or pivot mechanism changes the closure gap, angle, or levelness based on the loading experienced during clamping.

[0645] In various aspects, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical instrument comprising an end-effector with an electrode configuration. In one aspect, the end-effector comprises a variable electrode with a deflectable portion. The electrode physical parameters in combination with an electrode may be varied to change the energy density and tissue interactions. An electrode is provided for use with an ultrasonic blade and RF energy electrosurgical system where the physical aspects of the electrode vary along its length in order to change the contact area and/or the energy density of the electrode to tissue as the electrode also deflects. The electrode is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode may be referred to as a cantilever beam electrode or as a deflectable electrode. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0646] In one aspect, the end-effector comprises longitudinal variations in electrode size. The electrode can vary in width from the proximal end to the distal end proportionate to the clamp arm width change. The size of the electrode can be varied to vary the energy density and contact area. In one aspect, the width of the electrically active portion of the electrode could change proximal to distal to create constant energy density limiting compression or amplifying the concentrating effect. The electrode is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode may be referred to as a cantilever beam electrode or as a deflectable electrode. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0647] In one aspect, the present disclosure provides an end-effector comprising local adjustment of the electrode to provide compliant tissue coupling. In one aspect, a segmented electrode configuration comprises independently deflectable portions. Each segment of the segmented electrode is capable of deflecting independently. In one aspect,

the segmented electrode comprises independently deflectable electrodes for use in a combination ultrasonic/bipolar RF energy device. Each segment of the segmented electrode may have a separate spring rate along the length of the clamp arm jaw. This configuration may provide variable spring/compression rates. In other aspects, this configuration may enable only a portion of the electrode to be deflectable. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0648] In one aspect, the end-effector comprises deflectable electrodes broken into multiple elements. In one aspect, the deflectable electrodes are configured as watch band-style electrode elements. The physical aspects of the deflectable electrode are configured to vary along its length in order to change the contact area and/or the energy density of the deflectable electrode to tissue as the deflectable electrode deflects.

[0649] In one aspect, the deflectable electrode may comprise a plurality of segments. In one aspect, the deflectable electrode may comprise three elements although three or more elements may be employed depending on the desired pressure profile. These segments may be joined together such that they pivot around each other like an articulation mechanism. Each electrode segment may comprise a spring type mechanism to control the resistance to compression. The spring mechanism may comprise multiple possible configurations including being integrated with stamped leaf springs, separate springs, or flexible materials acting as springs.

[0650] In one aspect, springs could be designed and tuned to apply the exact desired pressure profile. The proximal loading can be reduced to improve pad life or increase loading distally to provide better performance at the tip of the end effector. All configurations may comprise a fundamental metal electrode located above the clamp arm and a wear resistant material on the electrode and a more compliant material on the clamp arm.

[0651] In one aspect, the electrode also may be connected to the electrical return path (cold) of a bipolar circuit. Opposite the clamp arm is a titanium ultrasonic blade that acts as the electrical source path (hot) of the bipolar RF circuit. The ultrasonic blade is configured to oscillate mechanically to generate heat by creating friction between tissue and the ultrasonic blade. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0652] FIG. 165 illustrates an end-effector **2380** comprising an ultrasonic blade **2382**, a clamp arm **2384**, and a watch-band style segmented electrode **2386**, according to at least one aspect of the present disclosure. The end-effector **2380** further comprises spring elements **2388** configured to bias the segmented electrode **2386**. In the illustrated aspect, the segmented electrode **2386** comprises three segments **2386a**, **2386b**, **2386c** linked by pins **2387a**, **2387b**. Three springs **2388a**, **2388b**, **2388c** are positioned between each of the electrode segments **2386a**, **2386b**, **2386c** and the clamp jaw **2385** to apply distal, medial, and proximal bias to the

electrode segments **2386a**, **2386b**, **2386c**, respectively. In other aspects, the segmented electrode **2386** may comprise additional or fewer elements.

[0653] FIG. 166 is a magnified view of the distal and medial segments **2386a**, **2386b** of the segmented electrode **2386** shown in FIG. 165. The distal segment **2386a** comprises a rounded distal end **2390**, an opening **2392** to receive a pin **2387a**, and a grooved surface **2396** at a proximal end **2398**. The medial segment **2386b** comprises a cylindrical insert **2400** that is received in the proximal grooved surface **2396** of the distal segment **2386a** and the pin **2387a** rotatably fixes the distal and medial segments **2386a**, **2386b**. Accordingly, the medial segment **2386b** is placed into the grooved surface **2396** of the distal segment **2386a** and the pin **2387a** is placed through the opening **2392** to hold distal segment **2386a** and the medial segment **2386b** together and allowing the distal segment **2386a** and the medial segment **2386b** to pivot against each other by way of the grooved surface **2396** and the cylindrical insert **2400**.

[0654] Additional background disclosure may be found in European Patent No. EP3420980, which is herein incorporated by reference in its entirety.

[0655] In one aspect, the present disclosure provides an end-effector comprising an electrode configured to minimize tissue sticking to the electrode. In one aspect, the electrode comprises a cooperative coating to minimize adhesion and focus energy. In one aspect, the present disclosure provides an apparatus and several techniques to prevent tissue charring due to electrical insulative properties and to improve easy clean off of accumulated material due to low frictional properties. The material could be a high melt temperature material like Teflon (PTFE) with a predefined opening or could be a DLC (diamond like coating) with high resistance and dielectric breakdown properties. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0656] In various aspects, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device configured to prevent energy activation when the clamping pressure is above a predefined threshold. In one aspect, a generator algorithm adaptation is provided to minimize damage to the ultrasonic blade, the electrode, the RF generator, or the ultrasonic generator when the electrode contacts or short circuit to the ultrasonic blade. The ultrasonic power control is based on the detection of a short circuit condition by the RF electrode. In one aspect, an ultrasonic transducer algorithm is provided to reduce power when the ultrasonic blade to electrode contact is detected to prevent damage to the ultrasonic blade. The ultrasonic blade control algorithm is configured for use with a combination ultrasonic/bipolar RF energy device. The algorithm monitors short circuits or more generally monitors contact between the ultrasonic blade and the clamp arm electrode. Detection of short circuits and contact between the ultrasonic blade and the clamp arm electrode may be employed in a logic to adjust the power/amplitude level of the ultrasonic transducer when a minimum electrical threshold is exceeded and adjusts the power threshold of the ultrasonic transducer to a level below the minimum electrical threshold which would cause damage to the ultrasonic blade, the clamp arm electrode, or the RF bipolar generator, for example. In another

aspect, one of the electrical parameters that may be monitored includes tissue impedance or continuity. In another aspect, the power/amplitude level adjustment of the ultrasonic transducer may include shutting off the system or it may include a proportionate response to either the electrical parameter, pressure, or time, or any combination of these parameters, for example.

[0657] In one aspect, the present disclosure provides a logic circuit or algorithm to control the combination ultrasonic/bipolar RF energy surgical device. In one aspect, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device configured to execute a set of machine executable instructions defining an ultrasonic blade control algorithm to monitor for short circuits between the ultrasonic blade and the electrode or contact between the ultrasonic blade and the electrode. Detection of short circuits and contact information is used to adjust the power/amplitude level of the drive signal for the ultrasonic transducer when a predefined electrical threshold minimum is exceeded. The algorithm then adjusts the transducer power/amplitude threshold to a suitable level below that which would cause damage to the ultrasonic blade, the electrode, the bipolar RF generator, the ultrasonic generator, among other component failures. In another aspect, the monitored electrical parameter may be tissue impedance or continuity. In another aspect, the power/amplitude adjustment could be to shut off the system or it could be a proportionate response to either the electrical parameter, pressure, or time or any combination of these parameters.

[0658] In combination ultrasonic/bipolar RF energy devices there exists a risk of unintended contact between the ultrasonic blade and the electrode causing a short circuit and may lead to damage to the ultrasonic blade, the electrode, the bipolar RF generator, the ultrasonic generator, among other components of the system. The following disclosure provides a logic technique for minimizing such damage due to any unintended contact between the ultrasonic blade and the electrode. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0659] FIG. 167 is a section view of an electrode 2410 comprising an electrically non-conductive clamp arm pad 2412 comprising a plurality of teeth 2414, according to at least one aspect of the present disclosure. FIG. 168 illustrates an end-effector 2420 comprising a clamp arm 2422 and an ultrasonic blade 2424, according to at least one aspect of the present disclosure. The clamp arm 2412 comprises an electrically conductive electrode 2410 and an electrically non-conductive clamp arm pad 2412.

[0660] With reference now to FIGS. 167-168, as described herein, the electrode 2410 is made of an electrically conductive material and functions as one pole of the bipolar RF circuit and an ultrasonic blade 2424 functions as the other pole of the bipolar RF circuit. The electrically non-conductive clamp arm pad 2412 is compliant and is made of Teflon (PTFE) or similar polymers and is used to contact tissue and provide support for the ultrasonic blade 2424. In one aspect, the electrode 2410 is the positive active (hot) pole of the bipolar RF circuit and the ultrasonic blade 2424 is the negative return (cold) pole of the bipolar RF circuit. The electrode 2410 comprises an additional distal conductive element 2418 at the distal end 2416 of the electrode 2410

such that when the distal conductive element 2414 briefly contacts the ultrasonic blade 2424 an algorithm is executed to reduce the power to the ultrasonic blade 2424. The power is reduced when the algorithm detects contact between the conductive element 2414 and the ultrasonic blade 2424 to minimize damage to the electrode 2410, ultrasonic blade 2424, bipolar RF generator, or the ultrasonic generator, among other components. In one aspect, contact between the conductive element 2414 or electrode 2410 and the ultrasonic blade 2424 may be detected by measuring the impedance over time.

[0661] FIG. 169 is a graphical depiction 2430 of impedance amplitude along the left vertical axis versus time along the horizontal axis and blade movement modes along the right vertical axis versus time along the horizontal axis, according to at least one aspect of the present disclosure. Accordingly, the bipolar RF generator or the ultrasonic generator may comprise control circuits that detect the impedance and alter the amplitude and/or frequency of the ultrasonic blade 2424 when the ultrasonic blade 2424 contacts the distal conductive element 2416 or other portions of the electrode 2410 on the clamp arm 2422. When the ultrasonic blade 2424 contacts the distal conductive element 2416 or other portions of the electrode 2410, the impedance 2432 through the device will drop to a low value and can be interpreted as a short circuit or contact. To prevent damage to the electrode 2410 or the ultrasonic blade 2424 the movement of the ultrasonic blade 2424 would be altered to minimize electrode 2410 wear/damage. When the impedance 2432 is above a threshold value 2434, the ultrasonic blade 2424 operates in mode "A". When the impedance 2432 is below the threshold 2434 the ultrasonic blade 2424 movement is optimized for cutting only and operates in mode "B" to reduce the time that the ultrasonic blade 2424 is active while in contact with the distal conductive element 2414 or the electrode 2410.

[0662] Additional background disclosure may be found in U.S. Pat. Nos. 8,253,303; 6,454,781; 9,017,326; and U.S. Patent Application Publication No. US2010/022568, each of which is herein incorporated by reference in its entirety.

[0663] In one aspect, the present disclosure provides an apparatus for driving an end-effector coupled to an ultrasonic drive system of a surgical instrument combined with any of the deflectable and/or cantilevered electrodes as described herein. The method comprises an ultrasonic generator configured to couple to an ultrasonic drive system of an ultrasonic instrument, the ultrasonic drive system comprising an ultrasonic transducer coupled to a waveguide and an end-effector coupled to the waveguide. The generator is configured to generate a first ultrasonic drive signal, actuate the ultrasonic transducer with the first ultrasonic drive signal for a first period, generate a second ultrasonic drive signal by the generator, and actuate the ultrasonic transducer with the second ultrasonic drive signal for a second period, subsequent to the first period. The first drive signal is different from the second drive signal over the respective first and second periods. The first and second drive signals define a step function waveform over the first and second periods.

[0664] In another aspect, the apparatus further comprises a measurement module to monitor a measurable characteristic of the ultrasonic drive system. The ultrasonic generator is configured to generate any one of the first and second drive signals based on a measured characteristic.

[0665] In another aspect, the measurement module further comprises a radio frequency (RF) generator coupled to the end-effector. The radio frequency generator is configured to generate therapeutic monopolar or bipolar RF energy or sub-therapeutic monopolar or bipolar RF energy. A clamp arm assembly is coupled to a distal end of the ultrasonic drive system. The clamp arm assembly comprises an electrically conductive portion forming a return (negative or cold) electrode coupled to the RF generator. The clamp arm assembly is operatively coupled to the end-effector to grasp tissue therebetween.

[0666] In another aspect, the present disclosure provides a temperature or impedance sensing device for monitoring the temperature or impedance of tissue grasped in the end-effector of a therapeutic ultrasonic cutting and coagulating instrument while the tissue is being heated by the friction generated by the ultrasonic vibrations of the ultrasonic blade portion of the end-effector. One or more sensors are located at the end-effector, preferably on a clamping member. The signals generated by the sensors provide feedback to control circuitry and a device for setting the function of the instrument either in cutting mode or coagulating mode.

[0667] In another aspect, a surgical instrument for ultrasonic surgical system is disclosed for dissecting, cutting and/or coagulating tissue during, e.g., an endoscopic procedure, has conductors placed in electrical communication with power source. Algorithms controlling ultrasonic drive in response to monitored electrical characteristic

[0668] In another aspect, the present disclosure provides an apparatus for monitoring tissue impedance Z_t . The tissue impedance Z_t may be monitored by an impedance module in accordance with the following process. A measurable RF current i_1 is conveyed through a first energizing conductor a to the ultrasonic blade, through the tissue, and back to the impedance module through a conductive jacket and a second conductor b. As the tissue is desiccated and cut by the friction generated by ultrasonic action of the blade acting against one or more clamp arm pads, the impedance Z of the tissue increases and thus the current in the return path, i.e., the second conductor b, decreases. The impedance module measures the tissue impedance Z_t and conveys a representative signal to the analog-to-digital converter (ADC) whose digital output is provided to a processor. The processor calculates the tissue impedance Z_t based on these measured values of v_{rf} and i_{rf} . The processor steps the frequency by any suitable increment or decrement in response to changes in tissue impedance Z_t . The processor controls the drive signals and can make any necessary adjustments in amplitude and frequency in response to the tissue impedance Z_t . In another aspect, the processor can cut off the drive signal when the tissue impedance Z_t reaches a predetermined threshold value.

[0669] In various aspects, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device configured to control tissue effects, charring, and/or tissue sticking to the clamp arm, ultrasonic blade, or electrode. In one aspect, a clamp arm jaw is configured with features or aspects to minimize tissue sticking to end-effector components and improve tissue control. In one aspect, the tissue path or clamp area control features are provided to adjust the tissue path relative to the clamp arm/ultrasonic blade to create a predefined location of contact reducing sticking of tissue to the clamp arm or ultrasonic blade and charring of tissue due to excess heat or prolonged application of heat.

[0670] In one aspect, the end-effector comprises a side guard to prevent tissue accumulation on end-effector components. In one aspect, the end-effector comprises tissue path or clamp arm area control features to adjust the tissue path relative to the clamp arm/blade to create a predefined location of contact. The control features are configured to reduce sticking and charring of tissue to the end-effector elements such as the clamp arm, clamp jaw, clamp arm pad, or ultrasonic blade.

[0671] In one aspect, the present disclosure provides a clamp arm with raised sidewalls or guards to surround the electrode to prevent exposure and prevent tissue from entering the area inside the sidewalls. In one implementation, the sidewalls of the clamp arm may be extruded out and around the clamp arm pad to an extent, but not enough to cause the tissue to begin to fold. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0672] FIG. 170 illustrates a clamp arm 2620 comprising a clamp jaw 2622, a clamp arm pad 2624 comprising exposed teeth 2626, according to at least one aspect of the present disclosure. The exposed teeth 2626 prevent tissue from entering the area inside the exposed teeth 2626.

[0673] FIG. 171 illustrates a clamp arm 2630 comprising a clamp jaw 2632 comprising raised sidewalls 2634 and a raised lip 2636, and a clamp arm pad 2638 comprising a plurality of teeth 2639, according to at least one aspect of the present disclosure. The raised sidewalls 2634 and raised lip 2636 prevent tissue from accumulating inside 2637 the sidewalls 2634 and the raised lip 2636.

[0674] In one aspect, the present disclosure provides an end-effector comprising an electrode with a perimeter wall and silicone seal located on the clamp arm. FIGS. 172-174 illustrates an electrode comprising a perimeter wall. FIG. 172 illustrates a clamp arm 2640 comprising an electrode 2642 having a perimeter wall 2644 supported by a clamp jaw 2646 and a clamp arm pad 2647, according to at least one aspect of the present disclosure. FIG. 173 is a section view of the clamp jaw 2640 taken along section 173-173 in FIG. 172. Since it would be desirable to prevent or minimize tissue penetration and accumulation/sticking between the electrode 2642 and the clamp jaw 2644, in one aspect, the perimeter wall 2644 around the electrode 2642 is sized and configured to prevent tissue from going and accumulating under the electrode 2642 in the space 2643 defined between the electrode 2642, the clamp jaw 2646, and the clamp arm pad 2647. The electrode 2642 is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode 2642 is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0675] FIG. 174 is an alternative or additional feature of the electrode comprising a perimeter wall shown in FIGS. 172-173. In the aspect illustrated in FIG. 174, a sealant 2648 such as silicone or other fluid polymer may be provided in the space 2645 beneath the electrode 2642 and above the clamp jaw 2646 to prevent tissue from going and accumulating under the electrode 2642, according to at least one aspect of the present disclosure.

[0676] In one aspect, the present disclosure provides an end-effector comprising a skirt for deflectable/cantilevered

electrode. It is common for tissue to stick to RF electrodes and fluid to accumulate between the electrode and the clamp jaw after heating in RF based energy surgical devices. Thus, it would be desirable to mitigate tissue/fluid accumulation between an electrode and a clamp arm or prevent tissue pinching. In one aspect, FIGS. 175-177 of the present disclosure illustrate an electrode to prevent or minimize tissue and fluid accumulation between the electrode and the clamp arm thereby limiting the deflecting function of the electrode or inhibiting the return of the electrode to the neutral position when unclamped. The electrode is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode may be referred to as a cantilever beam electrode or as a deflectable electrode. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0677] Overlapping configuration of skirt for adding incremental stiffness to the deflectable electrode. When deflected, the skirt displaces outwardly to clean/expel tissue around the clamp arm.

[0678] FIG. 175 illustrates a clamp arm 2650 comprising a clamp jaw 2652, an electrode 2654, a clamp arm pad 2656 (FIG. 176) comprising a plurality of teeth 2657, and a deflectable ledge 2658 along the lateral sides of the electrode 2654, according to at least one aspect of the present disclosure. FIGS. 176 and 177 are section views of the clamp arm 2650 shown in FIG. 175 taken at section 176-176. FIG. 176 shows the electrode 2654 and the skirt 2658 in an undeflected state and FIG. 177 shows the electrode 2654 and the skirt 2658 in a deflected state where the skirt 2658 shifts laterally in the direction indicated by the arrows 2659.

[0679] With reference now to FIGS. 175-176, the clamp arm 2650 is configured for use with an end-effector comprising an ultrasonic blade for use in a combined ultrasonic/bipolar RF energy device. A proximal end 2660 of the electrode 2654 is attached to the clamp jaw 2652 by any suitable fastener 2653 or fastening technique such as welding, laser welding, brazing, soldering, pressing, among other fastening techniques, for example. A distal end 2662 of the electrode 2654 is free to move as in a cantilever. As previously described, the clamp arm pad 2656 is made of a compliant material such as Teflon (PTFE) although similar polymers could be substituted therefor. The electrode 2654 is fixed to the clamp jaw 2652 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 2654 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0680] The deflectable ledge comprises an overmolded elastomeric skirt 2658 on the lower perimeter of the electrode 2654 disposed between the electrode 2654 and the clamp jaw 2652 and overlapping the electrode 2654. The overmolded skirt 2658 prevents tissue and fluid from accumulating or getting pinched between the electrode 2654 and the clamp jaw 2652 when they are in the deflected state. Additionally, the overlapping configuration of the overmolded skirt 2658 to the clamp jaw 2652 may stiffen the electrode 2654 if required for a particular implementation. When the electrode 2654 is deflected toward the clamp jaw 2652, the overmolded skirt 2658 displaces outwardly to

clean and expel tissue from the periphery of the clamp arm 2658 electrode 2654 or the clamp jaw 2652.

[0681] In one aspect, the present disclosure provides an end-effector comprising an electrode curtain. It is common for tissue to stick between the clamp jaw and the RF electrodes after heating in RF based energy surgical devices. Thus, as previously discussed, it would be desirable to reduce tissue build up between the electrode and the clamp arm in a combination ultrasonic/bipolar RF energy surgical device. Accordingly, as shown in FIGS. 178-180 of the present disclosure, various clamp arms are configured to reduce tissue build up and sticking points. In one aspect, the clamp arms comprise an electrode, which extends downwardly along the sides of the clamp arm akin to a curtain to reduce gaps where tissue can enter and build up during RF sealing. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0682] FIG. 178 illustrates a clamp arm 2670 comprising a clamp jaw 2672, an electrode 2674, and sidewalls 2676 extending downwardly from the electrode 2674 past the sidewalls 2678 of the clamp jaw 2672, and clamp arm pad 2677 (FIG. 179) comprising a plurality of teeth 2679, according to at least one aspect of the present disclosure. FIG. 179 is a section view of the clamp arm 2670 shown in FIG. 178 along section line 179-179. With reference to FIGS. 178-179, the clamp arm 2670 is configured for use with an end-effector comprising an ultrasonic blade for use in a combination ultrasonic/bipolar RF energy surgical device. The sidewalls 2676 of the electrode 2674 cover the lateral sides of the clamp jaw 2672 and extend around the distal end 2684 of the clamp jaw 2672 to close off the gap and prevent tissue build up in the space 2680 defined between the electrode 2674 and the clamp jaw 2672. The arrows 2682 indicate the deflection path of the electrode 2674. As previously described, the clamp arm pad 2686 is made of a compliant material such as PTFE or similar polymer material. The electrode 2674 is attached to the proximal end 2688 of the clamp jaw 2672 by any suitable fastener 2673 or fastening technique such as welding, laser welding, brazing, soldering, pressing, among other fastening techniques, for example. The electrode 2674 is fixed to the clamp jaw 2672 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 2674 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0683] FIG. 180 illustrates an alternative clamp arm 2690 to the clamp arm 2670 shown in FIGS. 178-179, according to at least one aspect of the present disclosure. The clamp arm 2690 comprises a clamp jaw 2692, an electrode 2694 with sidewalls 2696 extending downwardly from the electrode 2694 past the sidewalls 2698 of the clamp jaw 2692, and a clamp arm pad 2697. The sidewalls 2698 prevent tissue build up in the space 2691 defined between the electrode 2694 and the clamp jaw 2692. The clamp arm 2690 further comprises a deflection stop surface 2700 away from a tissue interface. The electrode 2694 is fixed to the clamp jaw 2692 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 2694 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0684] In one aspect, the present disclosure provides an end-effector comprising a deflectable/cantilevered electrode comprising a shield lip to prevent tissue accumulation in the end-effector or components thereof. It is common for tissue to stick between the clamp jaw and the RF electrodes after heating in RF based energy surgical devices. Thus, as previously discussed, it would be desirable to prevent or minimize tissue accumulation between an electrode and a clamp jaw. FIG. 181 is a top perspective view of an electrode 2710 comprising a lip 2712 extending downwardly from a top surface 2714 of the electrode 2710, according to at least one aspect of the present disclosure. FIG. 182 is a bottom perspective view of the electrode 2710 shown in FIG. 181 showing the downwardly extending lip 2712. FIGS. 183-184 is a top perspective view of the electrode 2710 shown in FIGS. 181-182 with a clamp arm pad 2718 with teeth 2719 extending through the apertures 2716.

[0685] With reference now to FIGS. 181-184, the electrode 2710 defines apertures 2716 to receive elements of a clamp arm pad therethrough. The electrode 2710 is configured for use with an end-effector comprising an ultrasonic blade for use in an ultrasonic/bipolar RF combination energy device. The electrode 2710 is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 2710 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0686] Accordingly, FIGS. 181-184 shield lip 2712 for the electrode 2710 to prevent tissue accumulation in the space defined between the electrode 2710 and the clamp jaw. The lip 2712 surrounds the clamp arm and the electrode 2710 to prevent tissue from entering the space between the electrode 2710 and the clamp jaw. An insulating coating could be added to the lip 2712 to prevent any electrode activity from coming from the lip 2712. The electrode 2710 is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode may be referred to as a cantilever beam electrode or as a deflectable electrode. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0687] In one aspect, the present disclosure provides an end-effector comprising a deflectable/cantilevered electrode with compressed block and individual clamp arm pad units. In one aspect, the deflectable/cantilevered electrode with compressed block and individual clamp arm pad units is configured to reduce trade off decisions between ultrasonic devices and bipolar RF devices with precise dissection and multiple reliable sealing functionalities. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0688] In one aspect, the end-effector utilizes individual clamp arm pads with collar features to prevent the electrode delaminating when the jaw is open, and utilizes a pre-compressed block to prevent tissue fragments from going into side crevices between the electrode and the clamp jaw to lower the risk of getting stuck.

[0689] In one aspect, the end-effector architecture includes an end-effector clamp arm side that has multiple individual clamp arm pad units. Each of the individual clamp arm pad units has a collar feature to constrain the electrode from delaminating when the jaw is open, increase the pad volume against to ultrasonic blade tip, and a foot/tip features to get aligned and constrained with a slot feature on the clamp arm.

[0690] In another aspect, the end-effector comprises an electrode having a pad slot feature in specific direction to allow assembly of the electrode and help hold the clamp arm pad in position.

[0691] In yet another aspect, the end-effector comprises a compressed block between the electrode and the clamp jaw to prevent tissue fragments going into side crevices and lower the risk of getting stuck, assist holding parts together, and ensure no gap between the pad collar and the electrode before clamping tissue.

[0692] FIG. 185 illustrates an end effector 2720 comprising a clamp arm 2722 and an ultrasonic blade 2724, according to at least one aspect of the present disclosure. With reference also to FIGS. 186-187, the clamp arm 2722 comprises a clamp jaw 2726, an electrode 2728 defining a plurality of apertures 2730 sized and configured to receive a plurality of individual tissue pad units 2732, a single piece elastic/hyper elastic block 2734, small wear resistant gap pads 2736, and a large wear resistant gap pads 2738. The single piece elastic/hyper elastic block 2734 is made of elastic or hyper elastic material to provide spring-like characteristics such as, for example, porosint, rubber or sponge, among others. The electrode 2728 functions as one pole of the bipolar RF circuit and the ultrasonic blade 2724 functions as the opposite pole of the bipolar RF circuit. The electrode 2728 is fixed to the clamp jaw 2726 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 2728 may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0693] FIG. 188 illustrates various views of the individual tissue pad unit 2732, according to at least one aspect of the present disclosure. The tissue pad unit 2732 comprises a pad collar 2742, a pad neck 2746, a pad foot 2748, and a pad tip 2750. When tissue is clamped, tissue will push the electrode 2728 to a deflected state, at the same time the electrode 2728 will push the slightly compressed elastic block 2734 to a more compressed state. When the tissue is fully cut, the more compressed block 2734 will push on the electrode 2728 back against the tissue pad collar 2740, the block 2734 will remain in a slightly compressed state to maintain zero gap between the tissue pad units 2732 and the electrode 2728. When the clamp jaw 2726 is open, the tissue pad collar 2740 will prevent the electrode 2728 from delaminating.

[0694] FIG. 189 illustrates the electrode 2728, according to at least one aspect of the present disclosure. The electrode 2728 defines a plurality of apertures 2730. The plurality of apertures 2730 function as pad slots on the electrode 2728 having a similar shape as the pad foot 2748 shown in FIG. 188. The electrode 2728 also defines two slots 2752 to receive the small wear resistant gap pads 2736 and another slot 2754 to receive the large wear resistant gap pads 2738.

[0695] FIG. 190 illustrates a section view of the clamp arm 2722 showing the elastic/hyper elastic block 2734, according to at least one aspect of the present disclosure. In one aspect, the elastic/hyper elastic block 2734 functions as a

pre-compressed block 2734 to push the electrode 2728 against the tissue pad collar 2740 and to push the clamp arm 2722 against the pad foot 2748 at contact area (shown at arrows 2756) to hold the parts together and ensure there is no gap between the pad collar 2742 and the electrode 2728 before clamping tissue.

[0696] FIG. 191 is a perspective view of the elastic/hyper elastic block 2734. FIGS. 192 and 194 are perspective views of a sub-assembly 2760 comprising the clamp jaw 2726, electrode 2728, and the individual tissue pad units 2732, according to at least one aspect to the present disclosure. This view illustrates side crevices 2758 formed between the electrode 2728 and the clamp jaw 2726. FIG. 193 is a perspective view of the clamp arm 2722 assembly comprising the elastic/hyper elastic block 2734 of FIG. 191 assembled to the sub-assembly 2760, according to at least one aspect of the present disclosure. With reference now to FIGS. 191-194, the pre-compressed block 2734 prevents tissue fragments from going into the side crevices 2758 between the electrode 2728 and the clamp jaw 2726 to lower risk of getting stuck.

[0697] FIG. 195 illustrates a first assembly step, according to at least one aspect of the present disclosure. In this step the individual tissue pad units 2732 are inserted into the electrode 2728. The tissue pad foot 2748 is aligned with the electrode pad slot 2730 and insert the tissue pad unit 2732 into the electrode pad slot 2730 until the tissue pad collar 2742 sits against the electrode 2728.

[0698] FIGS. 196-198 illustrate a second assembly step, according to at least one aspect of the present disclosure. In this step, the individual tissue pad units 2732 are locked into the electrode 2728. The tissue pad unit 2732 is rotated counterclockwise for 35° (see also FIG. 189), at this 'lock' position the pad foot 2748 and the pad collar 2742 feature pair prevent the tissue pad unit 2732 from falling off from electrode 2728. Also at this position, the pad tips 2750 are roughly aligned to assemble into the clamp arm 2722. This creates a first sub-assembly 2762.

[0699] FIG. 199 illustrates a third assembly step, according to at least one aspect of the present disclosure. In this step the single piece elastic/hyper elastic block 2734 is bound to the clamp jaw 2726. This creates a second sub-assembly 2764.

[0700] FIGS. 200-205 illustrate a fourth assembly step, according to at least one aspect of the present disclosure. In this step the first sub-assembly 2762 produced in the second assembly step described in FIGS. 196-198 is assembled with the second sub-assembly 2764 produced in the third assembly as described in step of FIG. 199 to produce the clamp arm 2722. The pad tips 2750 are roughly aligned with the clamp arm slot 2766. The first sub-assembly 2762 is then inserted into the second sub-assembly 2764. The clamp arm slot 2766 feature will function to assist aligning the pad tips 2750 and constrain the tissue pad unit 2732 rotation D.O.F. along the pad shaft direction shown by the arrow 2768. Then 3 displacement D.O.F. and 3 rotation D.O.F. of the tissue pad units 2732 are fully constrained by the electrode 2728 and the clamp arm 2722. The electrode 2728 is bound to the clamp arm 2722 at bound points 2770. After assembly, the single piece elastic/hyper elastic block 2734 will be pre-compressed. The clamp arm feature 2772 prevents the tissue pad units 2732 from coming out in the direction shown by the arrow 2774.

[0701] FIGS. 206-207 illustrate a fifth assembly step, according to at least one aspect of the present disclosure. In this step the two small wear resistant gap pads 2736 are press fit into the slots 2752 defined by the electrode 2728 and the large wear resistant gap pad 2738 is slid into the slot 2754 defined by the electrode 2728.

[0702] In one aspect, the present disclosure provides an end-effector comprising a filled gap between the electrode and the clamp arm. It is common for tissue to stick between the clam jaw and the RF electrodes after heating in RF based energy surgical devices. Thus, as previously discussed, it would be desirable to prevent or minimize tissue sticking in the clamp arm in between the clamp jaw and the electrode. Accordingly, FIG. 208 illustrates a clamp arm 2780 comprising fill material 2782 between the electrode 2784 and the clamp jaw 2786, according to at least one aspect of the present disclosure. The fill material 2782 may be woven nylon or a 3D printed structure. The fill material 2782 prevents tissue from getting into the space 2788 between the electrode 2784 and the clamp jaw 2786. The 3D printed structure may be printed to be a specific shape in order to maximize compression. The electrode 2784 is fixed to the clamp jaw 2786 at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode 2784 may be referred to as a cantilever beam electrode or as a deflectable electrode. The electrode 2784 is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode 2784 is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0703] In one aspect, the present disclosure provides an end-effector comprising a deflectable/cantilever electrode. It would be desirable to improve the ease of manufacturing of the electrode, reduce tissue accumulation between the electrode and the clamp jaw. It is further desirable to prevent or minimize tissue from sticking to the electrode. FIGS. 209-211 illustrate various apparatuses to address these desired improvements, according to at least one aspect of the present disclosure. The electrode is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode may be referred to as a cantilever beam electrode or as a deflectable electrode. The electrode is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0704] In the aspect shown in FIG. 209, a clamp arm 2790 that is easier to manufacture by eliminating a separate clamp arm electrode and building up a surface 2792 of the clamp arm and eliminating the separate electrode blade. The clamp arm now becomes the electrode. Also overmolding the polyimide (Vespel) clamp arm pads 2794 onto the clamp jaw 2796.

[0705] In the aspect shown in FIG. 210, a clamp arm 2800 to deter tissue accumulation between the electrode 2802 and the clamp jaw 2804 by adding a skirt 2806 to the perimeter of the electrode 2802 or alternatively to the perimeter of the clamp jaw 2804. This still allows the electrode 2802 to deflect but still deter tissue accumulation. The electrode 2802 is fixed to the clamp jaw 2804 at the proximal end and is free to deflect at the distal end. Accordingly, throughout

this disclosure the electrode **2802** may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0706] In the aspect shown in FIG. **211**, a section view of an end-effector **2810** comprising an ultrasonic blade **2812** and a clamp arm **2814** configured to improve distal tissue grasping. The clamp arm **2814** defines a curvature **2816** at the distal end of the polyimide gap arm pad and/or the distal end of the teeth of the clamp arm pad (PTFE).

[0707] In one aspect, the present disclosure provides an end-effector configured to minimize tissue sticking via electrode geometry and a clamp arm bumper/bull element. It is common for tissue to stick to RF electrodes after heating in RF based energy surgical devices. Thus, as previously discussed, it would be desirable to minimize or prevent tissue sticking to the RF electrode. This may be achieved by a new electrode geometry and adding a clamp arm bumper/bull nose to a clamp jaw to minimize electrode delamination. FIG. **212** illustrates a clamp arm **2820** comprising a clamp jaw **2822**, an electrode **2824**, a clamp arm pad **2826** comprising a plurality of teeth **2836**, and a curtain **2828** located on either side of the electrode **2824** to minimize pinch point of tissue, and a bull nose/clamp arm bumper **2830** to minimize delamination of the electrode **2824** by making contact with tissue first during blunt dissection, according to at least one aspect of the present disclosure. The clamp arm **2820** further comprises a hard gap setting pad **2832** at a proximal end **2834** of the clamp arm **2820**. The clamp arm pad **2826** is made of a complaint polymer such as Teflon (PTFE) and the hard gap setting pad **2832** is made of hard polyimide material. The electrode **2824** is fixed to the clamp jaw **2822** at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode **2824** may be referred to as a cantilever beam electrode or as a deflectable electrode. The electrode **2824** is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode **2824** is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0708] FIGS. **213-214** illustrate an alternative clamp arm **2840** comprising a clamp jaw **2842**, an electrode **2844**, a clamp arm pad **2846** comprising a plurality of teeth **2854**, and a curtain **2848** located on either side of the clamp arm pad **2846**, according to at least one aspect of the present disclosure. FIG. **213** illustrates an assembled view of the clamp arm **2840**, FIG. **213** is an exploded view of the clamp arm **2840**, and FIG. **214** is a section view of the clamp arm **2840** taken along section **214-214**. The curtain **2848** prevent tissue pinching and improves manufacturability of the clamp arm **2840**. The proximal end **2850** of the clamp arm **2840** comprises a hard gap setting pad **2852**. The electrode **2844** comprises a plurality of apertures **2854** to receive the plurality of the clamp arm pad **2846** teeth. The clamp arm pad **2846** is made of a complaint polymer such as Teflon (PTFE) and the hard gap setting pad **2852** is made of hard polyimide material. The electrode **2844** is fixed to the clamp jaw **2842** at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode **2844** may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0709] In one aspect, the present disclosure provides an end-effector comprising a deflectable electrode comprising anti-tissue sticking features. It is common for tissue to stick to RF electrodes after heating in RF based energy surgical

devices. The tissue typically separates and flows into spaces around the electrode and then re-solidifies to form anchor points to the larger tissue body. The anchored tissue needs to be either prevented or detached to ensure no tissue sticking. FIGS. **215-218** illustrate a clamp arm **2860** comprising a clamp jaw **2862**, a cantilevered electrode **2864**, a clamp arm pad **2866** comprising teeth **2868** received through apertures **2870** defined by the cantilevered electrode **2864**, and hard gap setting pads **2872**, **2874** according to at least one aspect of the present disclosure. The electrode **2864** is adapted and configured for use with a combination ultrasonic/bipolar RF energy surgical device and is deflectable under load, where the electrode **2864** is one pole of the bipolar RF circuit and the ultrasonic blade is the opposite pole of the bipolar RF circuit.

[0710] FIG. **215** is a side view and FIG. **217** is a bottom perspective view of the clamp arm **2860**, according to at least one aspect of the present disclosure. Both views show the cantilevered electrode **2864** in an undeflected state. FIG. **216** is a side view and FIG. **218** is a bottom perspective view of the clamp arm **2860** with the cantilevered electrode **2864** in a deflected state to pull tissue off the teeth **2868** of the clamp arm pad **2866**, according to at least one aspect of the present disclosure.

[0711] With reference now to FIGS. **215-217**, in one method, the cantilevered electrode **2864** moves past the end of the pad teeth **2868** to pull the tissue out of the anchor spots before the tissue cools. The cantilevered electrode **2864** can move independently from the clamp jaw **2862** and the clamp arm pad **2866** with a spring force. The apertures **2870** in the cantilevered electrode **2864** for the clamp arm pad **2866** can move off the clamp arm pad **2866** to ensure no anchor are present spots after sealing.

[0712] In various aspects, the present disclosure provides combination ultrasonic/bipolar RF energy surgical devices and systems. Various forms are directed to user interfaces for surgical instruments with ultrasonic and/or electrosurgical (RF) end-effectors configured for effecting tissue treating, dissecting, cutting, and/or coagulation during surgical procedures. In one form, a user interface is provided for a combined ultrasonic and electrosurgical instrument that may be configured for use in open surgical procedures, but has applications in other types of surgery, such as minimally invasive laparoscopic procedures, for example, non-invasive endoscopic procedures, either in hand held or and robotic-assisted procedures. Versatility is achieved by selective application of multiple energy modalities simultaneously, independently, sequentially, or combinations thereof. For example, versatility may be achieved by selective use of ultrasonic and electrosurgical energy (e.g., monopolar or bipolar RF energy) either simultaneously, independently, sequentially, or combinations thereof.

[0713] In one aspect, the present disclosure provides a user interface for an apparatus comprising an ultrasonic blade and clamp arm with a deflectable RF electrode such that the ultrasonic blade and deflectable RF electrode cooperate to effect sealing, cutting, and clamping of tissue by cooperation of a clamping mechanism of the apparatus comprising the RF electrode with an associated ultrasonic blade. The clamping mechanism includes a pivotal clamp arm which cooperates with the ultrasonic blade for gripping tissue therebetween. The clamp arm is preferably provided with a clamp tissue pad (also known as "clamp arm pad") having a plurality of axially spaced gripping teeth, segments,

elements, or individual units which cooperate with the ultrasonic blade of the end-effector to achieve the desired sealing and cutting effects on tissue, while facilitating grasping and gripping of tissue during surgical procedures.

[0714] In one aspect, the end-effectors described herein comprise an electrode. In other aspects, the end-effectors described herein comprise alternatives to the electrode to provide a compliant coupling of RF energy to tissue, accommodate pad wear/thinning, minimize generation of excess heat (low coefficient of friction, pressure), minimize generation of sparks, minimize interruptions due to electrical shorting, or combinations thereof. The electrode is fixed to the clamp jaw at the proximal end and is free to deflect at the distal end. Accordingly, throughout this disclosure the electrode may be referred to as a cantilever beam electrode or as a deflectable electrode.

[0715] In other aspects, the end-effectors described herein comprise a clamp arm mechanism configured to high pressure between a pad and an ultrasonic blade to grasp and seal tissue, maximize probability that the clamp arm electrode contacts tissue in limiting or difficult scenarios, such as, for example, thin tissue, tissue under lateral tension, tissue tenting/vertical tension especially tenting tissue away from clamp arm.

[0716] In other aspects, the end-effectors described herein are configured to balance match of surface area/current densities between electrodes, balance and minimize thermal conduction from tissue interface, such as, for example, impacts lesion formation and symmetry, cycle time, residual thermal energy. In other aspects, the end-effectors described herein are configured to minimize sticking, tissue adherence (minimize anchor points) and may comprise small polyimide pads.

[0717] In various aspects, the present disclosure provides a surgical device configured to deliver at least two energy types (e.g., ultrasonic, monopolar RF, bipolar RF, microwave, or irreversible electroporation [IRE]) to tissue. The surgical device includes a first activation button switch for activating energy, a second button switch for selecting an energy mode for the activation button switch. The second button switch is connected to a circuit that uses at least one input parameter to define the energy mode. The input parameter can be modified remotely through connection to a generator or through a software update.

[0718] In one aspect, at least one of the energy modes is a simultaneous blend of RF and ultrasonic energy, and the input parameter represents a duty cycle of the RF and ultrasonic energy.

[0719] In one aspect, the second button switch is configurable to select from a list of predefined modes and the number of modes in the list is defined by a second input parameter defined by a user.

[0720] In one aspect, the input parameter is either duty cycle, voltage, frequency, pulse width, or current.

[0721] In one aspect, the device also includes a visual indicator of the selected energy mode within the portion of device in the surgical field

[0722] In one aspect, the second button switch is a separate control from the end effector closure trigger.

[0723] In one aspect, the second button switch is configured to be activated second stage of the closure trigger. The first stage of the closure trigger in the closing direction is to actuate the end effector.

[0724] In one aspect, at least one of the energy modes is selected from ultrasonic, RF bipolar, RF monopolar, microwave, or IRE.

[0725] In one aspect, at least one of the energy modes is selected from ultrasonic, RF bipolar, RF monopolar, microwave, or IRE and is configured to be applied in a predefined duty cycle or pulsed algorithm.

[0726] In one aspect, at least one of the energy modes is selected from a sequential application of two or more of the following types of energy: ultrasonic, RF bipolar, RF monopolar, microwave, or IRE.

[0727] In one aspect, at least one of the energy modes is a simultaneous blend of two or more of the following types of energy: ultrasonic, RF bipolar, RF monopolar, microwave, and IRE.

[0728] In one aspect, at least one of the energy modes is a simultaneous blend of two or more of the following types of energy: ultrasonic, RF bipolar, RF monopolar, microwave, and IRE followed sequentially by one or more of the aforementioned energies.

[0729] In one aspect, at least one of the energy modes is one off the following types of energy: Ultrasonic, RF bipolar, RF monopolar, microwave, and IRE followed sequentially by a simultaneous blend of two or more of the aforementioned energies.

[0730] In one aspect, at least one of the energy modes is procedure or tissue specific predefined algorithm.

[0731] In one aspect, at least one of the energy modes is compiled from learned surgical behaviors or activities.

[0732] In one aspect, the input parameter is at least one of: energy type, duty cycle, voltage, frequency, pulse width, current, impedance limit, activation time, or blend of energy.

[0733] In one aspect, the second button switch is configurable to select from a list of predefined modes and the number of modes in the list is either predefined or defined by a second input parameter defined by a user.

[0734] In one aspect, the aforementioned energy modes are made available to the user through software updates to the generator.

[0735] In one aspect, the aforementioned energy modes are made available to the user through software updates to the device.

[0736] In one aspect, the preferred selections by the user are made available to multiple generators through either networking, the cloud, or manual transfer.

[0737] In one aspect, the device also includes a visual indicator of the selected energy mode within the portion of device in the surgical field.

[0738] As used herein a button switch can be a manually, mechanically, or electrically operated electromechanical device with one or more sets of electrical contacts, which are connected to external circuits. Each set of electrical contacts can be in one of two states: either “closed” meaning the contacts are touching and electricity can flow between them, or “open”, meaning the contacts are separated and the switch is electrically non-conducting. The mechanism actuating the transition between these two states (open or closed) can be either an “alternate action” (flip the switch for continuous “on” or “off”) or “momentary” (push for “on” and release for “off”) type.

[0739] In one aspect, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device comprising on device mode selection and visual feedback. As surgical devices evolve and become more capable, the

number of specialized modes in which they can be operated increases. Adding extra button switches on a device to accommodate these new additional modes would complicate the user interface and make the device more difficult to use. Accordingly, the present disclosure provides techniques for assigning different modes to a single physical button switch, which enables a wider selection of modes without adding complexity to the housing design (e.g., adding more and more button switches). In one aspect, the housing is in the form of a handle or pistol grip.

[0740] As more specialized modes become available, there is a need to provide multiple modes to a surgeon using the surgical device without creating a complex user interface. Surgeons want to be able to control the mode selection from the sterile field rather than relying on a circulating nurse at the generator. Surgeon want real time feedback so they are confident they know which mode is selected.

[0741] FIG. 219 illustrates a surgical device 100 comprising a mode selection button switch 130 on the device 100, according to at least one aspect of the present disclosure. The surgical device 100 comprises a housing 102 defining a handle 104 in the form of a pistol grip. The housing 102 comprises a trigger 106 which when squeezed is received into the internal space defined by the handle 104. The trigger 106 is used to operate a clamp arm 111 portion of an end-effector 110. A clamp jaw 112 is pivotally movable about pivot point 114. The housing 102 is coupled to the end-effector 110 through a shaft 108, which is rotatable by a knob 122.

[0742] The end-effector 110 comprises a clamp arm 111 and an ultrasonic blade 116. The clamp arm 111 comprises a clamp jaw 112, an electrode 118, and a clamp arm pad 120. In one aspect, the clamp arm pad 120 is made of a non-stick lubricious material such as PTFE or similar synthetic fluoropolymers of tetrafluoroethylene. PTFE is a hydrophobic, non-wetting, high density and resistant to high temperatures, and versatile material and non-stick properties. The clamp arm pad 120 is electrically non-conductive. In contrast, the electrode 118 is made of an electrically conductive material to deliver electrical energy such as monopolar RF, bipolar RF, microwave, or irreversible electroporation (IRE), for example. The electrode 118 may comprises gap setting pads made of a polyimide material, and in one aspect, is made of a durable high-performance polyimide-based plastic known under the tradename VESPEL and manufactured by DuPont or other suitable polyimide, polyimide polymer alloy, or or PET (Polyethylene Terephthalate), PEEK (Polyether Ether Ketone), PEKK (Poly Ether Ketone Ketone) polymer alloy, for example. Unless otherwise noted hereinbelow, the clamp arm pads and gap pads described hereinbelow are made of the materials described in this paragraph.

[0743] The electrode 118 and the ultrasonic blade 116 are coupled to the generator 133. The generator 133 is configured to drive RF, microwave, or IRE energy to the electrode 118. The generator 133 also is configured to drive an ultrasonic transducer acoustically coupled to the ultrasonic blade 116. In certain implementations, the electrode 118 is one pole of an electrical circuit and the ultrasonic blade 116 is the opposite pole of the electrical circuit. The housing 102 includes a switch 124 to activate the ultrasonic blade 116. The circuit may be contained in the housing 102 or may reside in the generator 133. The surgical device 100 is

coupled to the generator 133 via a cable 131. The cable 131 conducts signals for the electrosurgical functions and the ultrasonic transducer.

[0744] In various aspects, the surgical device 100 is configured to deliver at least two energy types (e.g., ultrasonic, monopolar RF, bipolar RF, microwave, or irreversible electroporation [IRE]) to tissue located in the end-effector 110 between the clamp arm 111 and the ultrasonic blade 116. The housing 102 of the surgical device 100 includes a first activation button switch 126 for activating energy and a second “mode” button switch 130 for selecting an energy mode for the activation button switch. The second button switch 130 is connected to a circuit that uses at least one input parameter to define the energy mode. The input parameter can be modified remotely through connection to a generator or through a software update. The energy mode is displayed on a user interface 128.

[0745] In one aspect, the surgical instrument 100 provides mode switching through the on device directional selector “mode” button switch 130. The user can press the mode button switch 130 to toggle through different modes and the colored light on the user interface 128 indicates the selected mode.

[0746] According to various aspects of the present disclosure, different modes of operation can be assigned to the surgical device by pressing the “mode” button switch 130, where each time the mode button switch 130 is pressed, or pushed and held, the surgical device 100 toggles through the available modes, which are displayed on the user interface 128. Once a mode is selected, the generator 133 will provide the appropriate generator tone and the surgical device 100 will have a lighted indicator on the user interface 128 to indicate which mode was selected.

[0747] In the example illustrated in FIG. 219, the “mode” selection button switch 130 is placed symmetrically on both sides of the housing 102. This enables both a right and left handed surgeon to select/toggle through modes without using a second hand. In this aspect, the “mode” selection button switch 130 can toggle in many different directions, which enables the surgeon to select from a list of options and navigate more complex selections remotely from the sterile field without having to ask a circulator to make adjustments at the generator 133. The lighted indicator on the user interface 128 of the surgical device 100, in addition to generator 133 tones, gives the surgeon feedback on which mode is selected.

[0748] FIGS. 220A-220C illustrate three options for selecting the various operating modes of the surgical device 100, according to at least one aspect of the present disclosure. In addition to the colored light user interface 128 on the housing 102 of the surgical device 100, feedback for mode selection is audible and/or visible through the generator 133 interface where the generator 133 announces the selected mode verbally and/or shows a description of the selected mode on a screen of the generator 133.

[0749] FIG. 220A shows a first mode selection option 132A where the button switch 130 can be pressed forward 136 or backward 134 to cycle the surgical instrument 100 through the various modes.

[0750] FIG. 220B shows a second mode selection option 132B where the button switch 130 is pressed up 140 or down 138 to cycle the surgical instrument 100 through the various modes.

[0751] FIG. 220C shows a third mode selection option 132C where the button switch 130 is pressed forward 136, backward 134, up 149, or down 138 to cycle the surgical instrument 100 through the various modes.

[0752] FIG. 221 illustrates a surgical device 150 comprising a mode selection button switch 180 on the back of the device 150, according to at least one aspect of the present disclosure. The surgical device 150 comprises a housing 152 defining a handle 154 in the form of a pistol grip. The housing 152 comprises a trigger 156 which when squeezed is received into the internal space defined by the handle 154. The trigger 156 is used to operate a clamp arm 161 portion of an end-effector 160. A clamp jaw 162 is pivotally movable about pivot point 164. The housing 152 is coupled to the end-effector 160 through a shaft 158, which is rotatable by a knob 172.

[0753] The end-effector 160 comprises a clamp arm 161 and an ultrasonic blade 166. The clamp arm 161 comprises a clamp jaw 162, an electrode 168, and a clamp arm pad 170. In one aspect, the clamp arm pad 170 is made of a non-stick lubricious material such as PTFE or similar synthetic fluoropolymers of tetrafluoroethylene. PTFE is a hydrophobic, non-wetting, high density and resistant to high temperatures, and versatile material and non-stick properties. The clamp arm pad 170 is electrically non-conductive. In contrast, the electrode 168 is made of an electrically conductive material to deliver electrical energy such as monopolar RF, bipolar RF, microwave, or irreversible electroporation (IRE), for example. The electrode 168 may comprise gap setting pads made of a polyimide material, and in one aspect, is made of a durable high-performance polyimide-based plastic known under the tradename VESPEL and manufactured by DuPont or other suitable polyimide, polyimide polymer alloy, or or PET (Polyethylene Terephthalate), PEEK (Polyether Ether Ketone), PEKK (Poly Ether Ketone Ketone) polymer alloy, for example. Unless otherwise noted hereinbelow, the clamp arm pads and gap pads described hereinbelow are made of the materials described in this paragraph.

[0754] The electrode 168 and the ultrasonic blade 166 are coupled to the generator 133. The generator 133 is configured to drive RF, microwave, or IRE energy to the electrode 168. The generator 133 also is configured to drive an ultrasonic transducer acoustically coupled to the ultrasonic blade 166. In certain implementations, the electrode 168 is one pole of an electrical circuit and the ultrasonic blade 166 is the opposite pole of the electrical circuit. The housing 152 includes a switch 174 to activate the ultrasonic blade 166. The circuit may be contained in the housing 152 or may reside in the generator 133. The surgical device 150 is coupled to the generator 133 via a cable 181. The cable 181 conducts signals for the electrosurgical functions and the ultrasonic transducer.

[0755] In various aspects, the surgical device 100 is configured to deliver at least two energy types (e.g., ultrasonic, monopolar RF, bipolar RF, microwave, or irreversible electroporation [IRE]) to tissue located in the end-effector 110 between the clamp arm 111 and the ultrasonic blade 116. The housing 102 of the surgical device 100 includes a first activation button switch 126 for activating energy and a second “mode” button switch 130 for selecting an energy mode for the activation button switch. The second button switch 130 is connected to a circuit that uses at least one input parameter to define the energy mode. The input parameter can be modified remotely through connection to

a generator or through a software update. The energy mode is displayed on a user interface 128.

[0756] In one aspect, the surgical instrument 150 provides mode switching through the on device directional selector “mode” button switch 180. The user can press the mode button switch 180 to toggle through different modes and the colored light on the user interface 178 indicates the selected mode.

[0757] According to various aspects of the present disclosure, different modes of operation can be assigned to the surgical device by pressing the “mode” button switch 180, where each time the mode button switch 180 is pressed, or pushed and held, the surgical device 150 toggles through the available modes, which are displayed on the user interface 178. Once a mode is selected, the generator 133 will provide the appropriate generator tone and the surgical device 150 will have a lighted indicator on the user interface 178 to indicate which mode was selected.

[0758] In the example illustrated in FIG. 221, the “mode” selection button switch 180 is placed on the back of the surgical device 150. The location of the “mode” selection button switch 180 is out of the reach of the surgeon’s hand holding the surgical device 150 so a second hand is required to change modes. This is intended to prevent inadvertent activation. In order to change modes, a surgeon must use her second hand to intentionally press the mode button switch 180. The lighted indicator on the user interface 178 of the surgical device 150, in addition to generator tones gives the surgeon feedback on which mode is selected.

[0759] FIG. 222A shows a first mode selection option where as the mode button switch 180 is pressed to toggled through various modes, colored light indicates the selected mode on the user interface 178.

[0760] FIG. 222B shows a second mode selection option where as the mode button switch 180 is pressed to toggle through various modes a screen 182 indicates the selected mode (e.g., LCD, e-ink).

[0761] FIG. 222C shows a third mode selection option where as the mode button switch 180 is pressed to toggle through various modes, labelled lights 184 indicate the selected mode.

[0762] FIG. 222D shows a fourth mode selection option where as a labeled button switch 186 is pressed to select a mode, when a labeled button switch 180 is selected, it is illuminated to indicate mode selected

[0763] In one aspect, the present disclosure provides a combination ultrasonic/bipolar RF energy surgical device comprising energy activation with trigger closure. As more functionality is added to advanced energy surgical devices additional button switches or controls are added to the surgical devices. The additional button switches or controls make these advanced energy surgical devices complicated and difficult to use. Additionally, when using an advanced energy surgical device to control bleeding, difficult to use user interfaces or difficult to access capability will cost critical time and attention during a surgical procedure.

[0764] According to the present disclosure, monopolar RF energy or advanced bipolar RF energy is activated by closing the trigger by squeezing the trigger past a first closure click to a second activation click and holding closed until energy delivery is ceased by the power source in the generator. Energy also can be immediately reapplied by slightly releasing and re-squeezing the trigger as many times as desired.

[0765] FIG. 223 illustrates a surgical device 190 comprising a trigger 196 activation mechanism, according to at least one aspect of the present disclosure. The surgical device 190 comprises a housing 192 defining a handle 194 in the form of a pistol grip. The housing 192 comprises a trigger 196 which when squeezed is received into the internal space defined by the handle 194. The housing 192 is coupled to an end-effector through a shaft 198, which is rotatable by a knob 202. The surgical device 190 is coupled to a generator 206 via a cable 204. The cable 204 conducts signals for the electrosurgical functions and the ultrasonic transducer.

[0766] The trigger 196 is configured to operate a clamp arm portion of an end-effector and to trigger electrosurgical energy, thus eliminating the activation button switch 126, 176 shown in FIGS. 219 and 221. The trigger 196 closes to a first audible and tactile click to close the jaws for grasping tissue and further closes to a second audible and tactile click to activate electrosurgical energy such as monopolar or bipolar RF, Microwave, or IRE energy. The full sequence is completed by activating the front button switch which cuts using ultrasonic energy.

[0767] Procedure for operating the surgical device 190: squeeze the trigger 196 to a first audible and tactile click; verify targeted tissue in jaws; activate RF energy by further squeezing the trigger 196 to a second audible and tactile click until end tone is heard; cut by pressing ultrasonic front switch 200 until tissue divides.

[0768] Modified procedure for operating the surgical instrument 190 for additional capability: activate RF energy with the trigger 196 and hold while simultaneously activation the front button switch 200 to activate the ultrasonic transducer, which will result in simultaneous application of electrosurgical and ultrasonic energy modalities being delivered to the tissue at the same time.

[0769] In an alternative implementation, the front button switch 200 for activating ultrasonic energy may be toggled to different speeds via a mode selector on the surgical device 190 or on the power source generator 206.

[0770] The surgical instruments 100, 150, 190 and associated algorithms described above in connection with FIGS. 219-223 comprising the end-effectors described in FIGS. 1-218 may be implemented in the following surgical hub system in conjunction with the following generator and modular energy system, for example.

[0771] FIG. 224 illustrates an alternative clamp arm comprising a metal clamp jaw, an electrode, a plurality of clamp arm pads, and gap pads, according to at least one aspect of the present disclosure. FIG. 224 illustrates an alternative clamp arm 2900 comprising a metal clamp jaw 2904, an electrode 2906, a plurality of clamp arm pads 2920 extend through holes in the electrode 2906, a gap pad 2930, and a gap pad 2910, according to at least one aspect of the present disclosure. The electrode 2906 is attached to the metal jaw 2904 at weld locations 2908. The electrode 2906 wraps around the metal clamp jaw 2904 and electrode 2906 can deflect. The gap pad 2910 has a top PI layer 2912 and a bottom elastomer layer 2914 for pressure control that is attached directly to the metal clamp jaw 2904. The clamp arm pads 2920 are attached directly to the metal clamp jaw 2904 and are composite pads with a high pressure center zone 2922 made of PTFE for reduced heat and an outer zone 2924 made of PI for electrode 2906 deflection.

[0772] In one aspect, the combination ultrasonic/bipolar RF energy surgical device is configured to operate within a

surgical hub system. FIG. 225 is a surgical system 3102 comprising a surgical hub 3106 paired with a visualization system 3108, a robotic system 3110, and an intelligent instrument 3112, in accordance with at least one aspect of the present disclosure. Referring now to FIG. 225, the hub 3106 is depicted in communication with a visualization system 3108, a robotic system 3110, and a handheld intelligent surgical instrument 3112 configured in a similar manner to the surgical instruments 100, 150, 190 as described in FIGS. 219-224. The hub 3106 includes a hub display 3135, an imaging module 3138, a generator module 3140, a communication module 3130, a processor module 3132, and a storage array 3134. In certain aspects, as illustrated in FIG. 225, the hub 3106 further includes a smoke evacuation module 3126 and/or a suction/irrigation module 3128.

[0773] During a surgical procedure, energy application to tissue, for sealing and/or cutting, is generally associated with smoke evacuation, suction of excess fluid, and/or irrigation of the tissue. Fluid, power, and/or data lines from different sources are often entangled during the surgical procedure. Valuable time can be lost addressing this issue during a surgical procedure. Detangling the lines may necessitate disconnecting the lines from their respective modules, which may require resetting the modules. The hub modular enclosure 3136 offers a unified environment for managing the power, data, and fluid lines, which reduces the frequency of entanglement between such lines.

[0774] Aspects of the present disclosure present a surgical hub for use in a surgical procedure that involves energy application to tissue at a surgical site. The surgical hub includes a hub enclosure and a combo generator module slidably receivable in a docking station of the hub enclosure. The docking station includes data and power contacts. The combo generator module includes two or more of an ultrasonic energy generator component, a bipolar RF energy generator component, and a monopolar RF energy generator component that are housed in a single unit. In one aspect, the combo generator module also includes a smoke evacuation component, at least one energy delivery cable for connecting the combo generator module to a surgical instrument, at least one smoke evacuation component configured to evacuate smoke, fluid, and/or particulates generated by the application of therapeutic energy to the tissue, and a fluid line extending from the remote surgical site to the smoke evacuation component.

[0775] In one aspect, the fluid line is a first fluid line and a second fluid line extends from the remote surgical site to a suction and irrigation module slidably received in the hub enclosure. In one aspect, the hub enclosure comprises a fluid interface.

[0776] Certain surgical procedures may require the application of more than one energy type to the tissue. One energy type may be more beneficial for cutting the tissue, while another different energy type may be more beneficial for sealing the tissue. For example, a bipolar generator can be used to seal the tissue while an ultrasonic generator can be used to cut the sealed tissue. Aspects of the present disclosure present a solution where a hub modular enclosure 136 is configured to accommodate different generators, and facilitate an interactive communication therebetween. One of the advantages of the hub modular enclosure 136 is enabling the quick removal and/or replacement of various modules.

[0777] Aspects of the present disclosure present a modular surgical enclosure for use in a surgical procedure that involves energy application to tissue. The modular surgical enclosure includes a first energy-generator module, configured to generate a first energy for application to the tissue, and a first docking station comprising a first docking port that includes first data and power contacts, wherein the first energy-generator module is slidably movable into an electrical engagement with the power and data contacts and wherein the first energy-generator module is slidably movable out of the electrical engagement with the first power and data contacts,

[0778] Further to the above, the modular surgical enclosure also includes a second energy-generator module configured to generate a second energy, different than the first energy, for application to the tissue, and a second docking station comprising a second docking port that includes second data and power contacts, wherein the second energy-generator module is slidably movable into an electrical engagement with the power and data contacts, and wherein the second energy-generator module is slidably movable out of the electrical engagement with the second power and data contacts.

[0779] In addition, the modular surgical enclosure also includes a communication bus between the first docking port and the second docking port, configured to facilitate communication between the first energy-generator module and the second energy-generator module.

[0780] In one aspect, the present disclosure provides a generator configured to drive the combination ultrasonic/bipolar RF energy surgical device. FIG. 226 illustrates an example of a generator 3900, in accordance with at least one aspect of the present disclosure. As shown in FIG. 226, the generator 3900 is one form of a generator configured to couple to a surgical instrument 100, 150, 190 as described in FIGS. 219-224, and further configured to execute adaptive ultrasonic and electrosurgical control algorithms in a surgical data network comprising a modular communication hub as shown in FIG. 225. The generator 3900 is configured to deliver multiple energy modalities to a surgical instrument. The generator 3900 provides RF and ultrasonic signals for delivering energy to a surgical instrument either independently or simultaneously. The RF and ultrasonic signals may be provided alone or in combination and may be provided simultaneously. As noted above, at least one generator output can deliver multiple energy modalities (e.g., ultrasonic, bipolar or monopolar RF, irreversible and/or reversible electroporation, and/or microwave energy, among others) through a single port, and these signals can be delivered separately or simultaneously to the end effector to treat tissue.

[0781] The generator 3900 comprises a processor 3902 coupled to a waveform generator 3904. The processor 3902 and waveform generator 3904 are configured to generate a variety of signal waveforms based on information stored in a memory coupled to the processor 3902, not shown for clarity of disclosure. The digital information associated with a waveform is provided to the waveform generator 3904 which includes one or more DAC circuits to convert the digital input into an analog output. The analog output is fed to an amplifier 3906 for signal conditioning and amplification. The conditioned and amplified output of the amplifier 3906 is coupled to a power transformer 3908. The signals are coupled across the power transformer 3908 to the secondary

side, which is in the patient isolation side. A first signal of a first energy modality is provided to the surgical instrument between the terminals labeled ENERGY1 and RETURN. A second signal of a second energy modality is coupled across a capacitor 3910 and is provided to the surgical instrument between the terminals labeled ENERGY2 and RETURN. It will be appreciated that more than two energy modalities may be output and thus the subscript “n” may be used to designate that up to n ENERGYn terminals may be provided, where n is a positive integer greater than 1. It also will be appreciated that up to “n” return paths RETURNn may be provided without departing from the scope of the present disclosure.

[0782] A first voltage sensing circuit 3912 is coupled across the terminals labeled ENERGY1 and the RETURN path to measure the output voltage therebetween. A second voltage sensing circuit 3924 is coupled across the terminals labeled ENERGY2 and the RETURN path to measure the output voltage therebetween. A current sensing circuit 3914 is disposed in series with the RETURN leg of the secondary side of the power transformer 3908 as shown to measure the output current for either energy modality. If different return paths are provided for each energy modality, then a separate current sensing circuit should be provided in each return leg. The outputs of the first and second voltage sensing circuits 3912, 3924 are provided to respective isolation transformers 3916, 3922 and the output of the current sensing circuit 3914 is provided to another isolation transformer 3918. The outputs of the isolation transformers 3916, 3928, 3922 in the on the primary side of the power transformer 3908 (non-patient isolated side) are provided to a one or more ADC circuit 3926. The digitized output of the ADC circuit 3926 is provided to the processor 3902 for further processing and computation. The output voltages and output current feedback information can be employed to adjust the output voltage and current provided to the surgical instrument and to compute output impedance, among other parameters. Input/output communications between the processor 3902 and patient isolated circuits is provided through an interface circuit 3920. Sensors also may be in electrical communication with the processor 3902 by way of the interface circuit 3920.

[0783] In one aspect, the impedance may be determined by the processor 3902 by dividing the output of either the first voltage sensing circuit 3912 coupled across the terminals labeled ENERGY1/RETURN or the second voltage sensing circuit 3924 coupled across the terminals labeled ENERGY2/RETURN by the output of the current sensing circuit 3914 disposed in series with the RETURN leg of the secondary side of the power transformer 3908. The outputs of the first and second voltage sensing circuits 3912, 3924 are provided to separate isolations transformers 3916, 3922 and the output of the current sensing circuit 3914 is provided to another isolation transformer 3918. The digitized voltage and current sensing measurements from the ADC circuit 3926 are provided to the processor 3902 for computing impedance. As an example, the first energy modality ENERGY1 may be ultrasonic energy and the second energy modality ENERGY2 may be RF energy. Nevertheless, in addition to ultrasonic and bipolar or monopolar RF energy modalities, other energy modalities include irreversible and/or reversible electroporation and/or microwave energy, among others. Also, although the example illustrated in FIG. 226 shows a single return path RETURN may be provided

for two or more energy modalities, in other aspects, multiple return paths RETURNn may be provided for each energy modality ENERGYn. Thus, as described herein, the ultrasonic transducer impedance may be measured by dividing the output of the first voltage sensing circuit 3912 by the current sensing circuit 3914 and the tissue impedance may be measured by dividing the output of the second voltage sensing circuit 3924 by the current sensing circuit 3914.

[0784] As shown in FIG. 226, the generator 3900 comprising at least one output port can include a power transformer 3908 with a single output and with multiple taps to provide power in the form of one or more energy modalities, such as ultrasonic, bipolar or monopolar RF, irreversible and/or reversible electroporation, and/or microwave energy, among others, for example, to the end effector depending on the type of treatment of tissue being performed. For example, the generator 3900 can deliver energy with higher voltage and lower current to drive an ultrasonic transducer, with lower voltage and higher current to drive RF electrodes for sealing tissue, or with a coagulation waveform for spot coagulation using either monopolar or bipolar RF electro-surgical electrodes. The output waveform from the generator 3900 can be steered, switched, or filtered to provide the frequency to the end effector of the surgical instrument. The connection of an ultrasonic transducer to the generator 3900 output would be preferably located between the output labeled ENERGY1 and RETURN as shown in FIG. 225. In one example, a connection of RF bipolar electrodes to the generator 3900 output would be preferably located between the output labeled ENERGY2 and RETURN. In the case of monopolar output, the preferred connections would be active electrode (e.g., pencil or other probe) to the ENERGY2 output and a suitable return pad connected to the RETURN output.

[0785] Additional details are disclosed in U.S. Patent Application Publication No. 2017/0086914, titled TECHNIQUES FOR OPERATING GENERATOR FOR DIGITALLY GENERATING ELECTRICAL SIGNAL WAVEFORMS AND SURGICAL INSTRUMENTS, which published on Mar. 30, 2017, which is herein incorporated by reference in its entirety.

[0786] In one aspect, the present disclosure provides a modular energy system configured to drive the combination ultrasonic/bipolar RF energy surgical device. FIG. 227 is a diagram of various modules and other components that are combinable to customize modular energy systems, in accordance with at least one aspect of the present disclosure. FIG. 228A is a first illustrative modular energy system configuration including a header module and a display screen that renders a graphical user interface (GUI) for relaying information regarding modules connected to the header module, in accordance with at least one aspect of the present disclosure. FIG. 228B is the modular energy system shown in FIG. 228A mounted to a cart, in accordance with at least one aspect of the present disclosure.

[0787] With reference now to FIGS. 226-228B, ORs everywhere in the world are a tangled web of cords, devices, and people due to the amount of equipment required to perform surgical procedures. Surgical capital equipment tends to be a major contributor to this issue because most surgical capital equipment performs a single, specialized task. Due to their specialized nature and the surgeons' needs to utilize multiple different types of devices during the course of a single surgical procedure, an OR may be forced

to be stocked with two or even more pieces of surgical capital equipment, such as energy generators. Each of these pieces of surgical capital equipment must be individually plugged into a power source and may be connected to one or more other devices that are being passed between OR personnel, creating a tangle of cords that must be navigated. Another issue faced in modern ORs is that each of these specialized pieces of surgical capital equipment has its own user interface and must be independently controlled from the other pieces of equipment within the OR. This creates complexity in properly controlling multiple different devices in connection with each other and forces users to be trained on and memorize different types of user interfaces (which may further change based upon the task or surgical procedure being performed, in addition to changing between each piece of capital equipment). This cumbersome, complex process can necessitate the need for even more individuals to be present within the OR and can create danger if multiple devices are not properly controlled in tandem with each other. Therefore, consolidating surgical capital equipment technology into singular systems that are able to flexibly address surgeons' needs to reduce the footprint of surgical capital equipment within ORs would simplify the user experience, reduce the amount of clutter in ORs, and prevent difficulties and dangers associated with simultaneously controlling multiple pieces of capital equipment. Further, making such systems expandable or customizable would allow for new technology to be conveniently incorporated into existing surgical systems, obviating the need to replace entire surgical systems or for OR personnel to learn new user interfaces or equipment controls with each new technology.

[0788] A surgical hub can be configured to interchangeably receive a variety of modules, which can in turn interface with surgical devices (e.g., a surgical instrument or a smoke evacuator) or provide various other functions (e.g., communications). In one aspect, a surgical hub can be embodied as a modular energy system 4000, which is illustrated in connection with FIGS. 227-228B. The modular energy system 4000 can include a variety of different modules 4001 that are connectable together in a stacked configuration. In one aspect, the modules 4001 can be both physically and communicably coupled together when stacked or otherwise connected together into a singular assembly. Further, the modules 4001 can be interchangeably connectable together in different combinations or arrangements. In one aspect, each of the modules 4001 can include a consistent or universal array of connectors disposed along their upper and lower surfaces, thereby allowing any module 4001 to be connected to another module 4001 in any arrangement (except that, in some aspects, a particular module type, such as the header module 4002, can be configured to serve as the uppermost module within the stack, for example). In an alternative aspect, the modular energy system 4000 can include a housing that is configured to receive and retain the modules 4001, as is shown in FIG. 225. The modular energy system 4000 can also include a variety of different components or accessories that are also connectable to or otherwise associatable with the modules 4001. In another aspect, the modular energy system 4000 can be embodied as a generator module 3140, 3900 (FIGS. 225-226) of a surgical hub 3106. In yet another aspect, the modular energy system 4000 can be a distinct system from a surgical hub 3106. In such aspects, the modular energy

system **4000** can be communicably couplable to a surgical hub **3106** for transmitting and/or receiving data therebetween.

[0789] The modular energy system **4000** can be assembled from a variety of different modules **4001**, some examples of which are illustrated in FIG. 227. Each of the different types of modules **4001** can provide different functionality, thereby allowing the modular energy system **4000** to be assembled into different configurations to customize the functions and capabilities of the modular energy system **4000** by customizing the modules **4001** that are included in each modular energy system **4000**. The modules **4001** of the modular energy system **4000** can include, for example, a header module **4002** (which can include a display screen **4006**), an energy module **4004**, a technology module **4040**, and a visualization module **4042**. In the depicted aspect, the header module **4002** is configured to serve as the top or uppermost module within the modular energy system stack and can thus lack connectors along its top surface. In another aspect, the header module **4002** can be configured to be positioned at the bottom or the lowermost module within the modular energy system stack and can thus lack connectors along its bottom surface. In yet another aspect, the header module **4002** can be configured to be positioned at an intermediate position within the modular energy system stack and can thus include connectors along both its bottom and top surfaces. The header module **4002** can be configured to control the system-wide settings of each module **4001** and component connected thereto through physical controls **4011** thereon and/or a graphical user interface (GUI) **4008** rendered on the display screen **4006**. Such settings could include the activation of the modular energy system **4000**, the volume of alerts, the footswitch settings, the settings icons, the appearance or configuration of the user interface, the surgeon profile logged into the modular energy system **4000**, and/or the type of surgical procedure being performed. The header module **4002** can also be configured to provide communications, processing, and/or power for the modules **4001** that are connected to the header module **4002**. The energy module **4004**, which can also be referred to as a generator module **3140**, **3900** (FIGS. 225-226), can be configured to generate one or multiple energy modalities for driving electrosurgical and/or ultrasonic surgical instruments connected thereto, such as is described above in connection with the generator **3900** illustrated in FIG. 226. The technology module **4040** can be configured to provide additional or expanded control algorithms (e.g., electrosurgical or ultrasonic control algorithms for controlling the energy output of the energy module **4004**). The visualization module **4042** can be configured to interface with visualization devices (i.e., scopes) and accordingly provide increased visualization capabilities.

[0790] The modular energy system **4000** can further include a variety of accessories **4029** that are connectable to the modules **4001** for controlling the functions thereof or that are otherwise configured to work on conjunction with the modular energy system **4000**. The accessories **4029** can include, for example, a single-pedal footswitch **4032**, a dual-pedal footswitch **4034**, and a cart **4030** for supporting the modular energy system **4000** thereon. The footswitches **4032**, **4034** can be configured to control the activation or function of particular energy modalities output by the energy module **4004**, for example.

[0791] By utilizing modular components, the depicted modular energy system **4000** provides a surgical platform that grows with the availability of technology and is customizable to the needs of the facility and/or surgeons. Further, the modular energy system **4000** supports combo devices (e.g., dual electrosurgical and ultrasonic energy generators) and supports software-driven algorithms for customized tissue effects. Still further, the surgical system architecture reduces the capital footprint by combining multiple technologies critical for surgery into a single system.

[0792] The various modular components utilizable in connection with the modular energy system **4000** can include monopolar energy generators, bipolar energy generators, dual electrosurgical/ultrasonic energy generators, display screens, and various other modules and/or other components, some of which are also described above in connection with FIGS. 1-224.

[0793] Referring now to FIG. 228A, the header module **4002** can, in some aspects, include a display screen **4006** that renders a GUI **4008** for relaying information regarding the modules **4001** connected to the header module **4002**. In some aspects, the GUI **4008** of the display screen **4006** can provide a consolidated point of control of all of the modules **4001** making up the particular configuration of the modular energy system **4000**. In alternative aspects, the header module **4002** can lack the display screen **4006** or the display screen **4006** can be detachably connected to the housing **4010** of the header module **4002**. In such aspects, the header module **4002** can be communicably couplable to an external system that is configured to display the information generated by the modules **4001** of the modular energy system **4000**. For example, in robotic surgical applications, the modular energy system **4000** can be communicably couplable to a robotic cart or robotic control console, which is configured to display the information generated by the modular energy system **4000** to the operator of the robotic surgical system. As another example, the modular energy system **4000** can be communicably couplable to a mobile display that can be carried or secured to a surgical staff member for viewing thereby. In yet another example, the modular energy system **4000** can be communicably couplable to a surgical hub **4100** or another computer system that can include a display **4104**. In aspects utilizing a user interface that is separate from or otherwise distinct from the modular energy system **4000**, the user interface can be wirelessly connectable with the modular energy system **4000** as a whole or one or more modules **4001** thereof such that the user interface can display information from the connected modules **4001** thereon.

[0794] Referring still to FIG. 228A, the energy module **4004** can include a port assembly **4012** including a number of different ports configured to deliver different energy modalities to corresponding surgical instruments that are connectable thereto. In the particular aspect illustrated in FIGS. 227-228B, the port assembly **4012** includes a bipolar port **4014**, a first monopolar port **4016a**, a second monopolar port **4018b**, a neutral electrode port **4018** (to which a monopolar return pad is connectable), and a combination energy port **4020**. However, this particular combination of ports is simply provided for illustrative purposes and alternative combinations of ports and/or energy modalities may be possible for the port assembly **4012**.

[0795] As noted above, the modular energy system 4000 can be assembled into different configurations. Further, the different configurations of the modular energy system 4000 can also be utilizable for different surgical procedure types and/or different tasks. For example, FIGS. 228A and 228B illustrate a first illustrative configuration of the modular energy system 4000 including a header module 4002 (including a display screen 4006) and an energy module 4004 connected together. Such a configuration can be suitable for laparoscopic and open surgical p

[0796] FIGS. 229-233 illustrate an example surgical system 10 with ultrasonic and electrosurgical features including any one of the end-effectors, surgical instruments, and generators described herein. FIG. 229 depicts a surgical system 10 including a generator 12 and a surgical instrument 14. The surgical instrument 14 is operatively coupled with the generator 12 via a power cable 16. The generator 12 is operable to power the surgical instrument 14 to deliver ultrasonic energy for cutting tissue, and electrosurgical bipolar RF energy (i.e., therapeutic levels of RF energy) for sealing tissue. In one aspect, the generator 12 is configured to power the surgical instrument 14 to deliver ultrasonic energy and electrosurgical bipolar RF energy simultaneously or independently.

[0797] The surgical instrument 14 of the present example comprises a handle assembly 18, a shaft assembly 20 extending distally from the handle assembly 18, and an end effector 22 arranged at a distal end of the shaft assembly 20. The handle assembly 18 comprises a body 24 including a pistol grip 26 and energy control buttons 28, 30 configured to be manipulated by a surgeon. A trigger 32 is coupled to a lower portion of the body 24 and is pivotable toward and away from the pistol grip 26 to selectively actuate the end effector 22, as described in greater detail below. In other suitable variations of the surgical instrument 14, the handle assembly 18 may comprise a scissor grip configuration, for example. An ultrasonic transducer 34 is housed internally within and supported by the body 24. In other configurations, the ultrasonic transducer 34 may be provided externally of the body 24.

[0798] As shown in FIGS. 230 and 231, the end effector 22 includes an ultrasonic blade 36 and a clamp arm 38 configured to selectively pivot toward and away from the ultrasonic blade 36, for clamping tissue therebetween. The ultrasonic blade 36 is acoustically coupled with the ultrasonic transducer 34, which is configured to drive (i.e., vibrate) the ultrasonic blade 36 at ultrasonic frequencies for cutting and/or sealing tissue positioned in contact with the ultrasonic blade 36. The clamp arm 38 is operatively coupled with the trigger 32 such that the clamp arm 38 is configured to pivot toward the ultrasonic blade 36, to a closed position, in response to pivoting of the trigger 32 toward the pistol grip 26. Further, the clamp arm 38 is configured to pivot away from the ultrasonic blade 36, to an open position (see e.g., FIGS. 229-231), in response to pivoting of the trigger 32 away from the pistol grip 26. Various suitable ways in which the clamp arm 38 may be coupled with the trigger 32 will be apparent to those of ordinary skill in the art in view of the teachings provided herein. In some versions, one or more resilient members may be incorporated to bias the clamp arm 38 and/or the trigger 32 toward the open position.

[0799] A clamp pad 40 is secured to and extends distally along a clamping side of the clamp arm 38, facing the ultrasonic blade 36. The clamp pad 40 is configured to

engage and clamp tissue against a corresponding tissue treatment portion of the ultrasonic blade 36 when the clamp arm 38 is actuated to its closed position. At least a clamping-side of the clamp arm 38 provides a first electrode 42, referred to herein as clamp arm electrode 42. Additionally, at least a clamping-side of the ultrasonic blade 36 provides a second electrode 44, referred to herein as a blade electrode 44. The electrodes 42, 44 are configured to apply electrosurgical bipolar RF energy, provided by the generator 12, to tissue electrically coupled with the electrodes 42, 44. The clamp arm electrode 42 may serve as an active electrode while the blade electrode 44 serves as a return electrode, or vice-versa. The surgical instrument 14 may be configured to apply the electrosurgical bipolar RF energy through the electrodes 42, 44 while vibrating the ultrasonic blade 36 at an ultrasonic frequency, before vibrating the ultrasonic blade 36 at an ultrasonic frequency, and/or after vibrating the ultrasonic blade 36 at an ultrasonic frequency.

[0800] As shown in FIGS. 229-233, the shaft assembly 20 extends along a longitudinal axis and includes an outer tube 46, an inner tube 48 received within the outer tube 46, and an ultrasonic waveguide 50 supported within the inner tube 48. As seen best in FIGS. 230-233, the clamp arm 38 is coupled to distal ends of the inner and outer tubes 46, 48. In particular, the clamp arm 38 includes a pair of proximally extending clevis arms 52 that receive therebetween and pivotably couple to a distal end 54 of the inner tube 48 with a pivot pin 56 received through bores formed in the clevis arms 52 and the distal end 54 of the inner tube 48. The first and second clevis fingers 58 depend downwardly from the clevis arms 52 and pivotably couple to a distal end 60 of the outer tube 46. Specifically, each clevis finger 58 includes a protrusion 62 that is rotatably received within a corresponding opening 64 formed in a sidewall of the distal end 60 of the outer tube 46.

[0801] In the present example, the inner tube 48 is longitudinally fixed relative to the handle assembly 18, and the outer tube 46 is configured to translate relative to the inner tube 48 and the handle assembly 18, along the longitudinal axis of the shaft assembly 20. As the outer tube 46 translates distally, the clamp arm 38 pivots about the pivot pin 56 toward its open position. As the outer tube 46 translates proximally, the clamp arm 38 pivots in an opposite direction toward its closed position. A proximal end of the outer tube 46 is operatively coupled with the trigger 32, for example via a linkage assembly, such that actuation of the trigger 32 causes translation of the outer tube 46 relative to the inner tube 48, thereby opening or closing the clamp arm 38. In other suitable configurations not shown herein, the outer tube 46 may be longitudinally fixed and the inner tube 48 may be configured to translate for moving the clamp arm 38 between its open and closed positions.

[0802] The shaft assembly 20 and the end effector 22 are configured to rotate together about the longitudinal axis, relative to the handle assembly 18. A retaining pin 66, shown in FIG. 232, extends transversely through the proximal portions of the outer tube 46, the inner tube 48, and the waveguide 50 to thereby couple these components rotationally relative to one another. In the present example, a rotation knob 68 is provided at a proximal end portion of the shaft assembly 20 to facilitate rotation of the shaft assembly 20, and the end effector 22, relative to the handle assembly 18. The rotation knob 68 is secured rotationally to the shaft assembly 20 with the retaining pin 66, which extends

through a proximal collar of the rotation knob **68**. It will be appreciated that in other suitable configurations, the rotation knob **68** may be omitted or substituted with alternative rotational actuation structures.

[0803] The ultrasonic waveguide **50** is acoustically coupled at its proximal end with the ultrasonic transducer **34**, for example by a threaded connection, and at its distal end with the ultrasonic blade **36**, as shown in FIG. **233**. The ultrasonic blade **36** is shown formed integrally with the waveguide **50** such that the blade **36** extends distally, directly from the distal end of the waveguide **50**. In this manner, the waveguide **50** acoustically couples the ultrasonic transducer **34** with the ultrasonic blade **36**, and functions to communicate ultrasonic mechanical vibrations from the transducer **34** to the blade **36**. Accordingly, the ultrasonic transducer **34**, the waveguide **50**, and the ultrasonic blade **36** together define an acoustic assembly. During use, the ultrasonic blade **36** may be positioned in direct contact with tissue, with or without assistive clamping force provided by the clamp arm **38**, to impart ultrasonic vibrational energy to the tissue and thereby cut and/or seal the tissue. For example, the blade **36** may cut through tissue clamped between the clamp arm **38** and a first treatment side of the blade **36**, or the blade **36** may cut through tissue positioned in contact with an oppositely disposed second treatment side of the blade **36**, for example during a “back-cutting” movement. In some variations, the waveguide **50** may amplify the ultrasonic vibrations delivered to the blade **36**. Further, the waveguide **50** may include various features operable to control the gain of the vibrations, and/or features suitable to tune the waveguide **50** to a selected resonant frequency. Additional features of the ultrasonic blade **36** and the waveguide **50** are described in greater detail below.

[0804] The waveguide **50** is supported within the inner tube **48** by a plurality of nodal support elements **70** positioned along a length of the waveguide **50**, as shown in FIGS. **232-233**. Specifically, the nodal support elements **70** are positioned longitudinally along the waveguide **50** at locations corresponding to acoustic nodes defined by the resonant ultrasonic vibrations communicated through the waveguide **50**. The nodal support elements **70** may provide structural support to the waveguide **50**, and acoustic isolation between the waveguide **50** and the inner and outer tubes **46, 48** of the shaft assembly **20**. In variations, the nodal support elements **70** may comprise o-rings. The waveguide **50** is supported at its distal-most acoustic node by a nodal support element in the form of an overmold member **72**, shown in FIG. **233**. The waveguide **50** is secured longitudinally and rotationally within the shaft assembly **20** by the retaining pin **66**, which passes through a transverse through-bore **74** formed at a proximally arranged acoustic node of the waveguide **50**, such as the proximal-most acoustic node, for example.

[0805] In the present example, a distal tip **76** of the ultrasonic blade **36** is located at a position corresponding to an anti-node associated with the resonant ultrasonic vibrations communicated through the waveguide **50**. Such a configuration enables the acoustic assembly of the instrument **14** to be tuned to a preferred resonant frequency f_0 when the ultrasonic blade **36** is not loaded by tissue. When the ultrasonic transducer **34** is energized by the generator **12** to transmit mechanical vibrations through the waveguide **50** to the blade **36**, the distal tip **76** of the blade **36** is caused to oscillate longitudinally in the range of approximately 20 to

120 microns peak-to-peak, for example, and in some instances in the range of approximately 20 to 50 microns, at a predetermined vibratory frequency f_0 of approximately 50 kHz, for example. When the ultrasonic blade **36** is positioned in contact with tissue, the ultrasonic oscillation of the blade **36** may simultaneously sever the tissue and denature the proteins in adjacent tissue cells, thereby providing a coagulative effect with minimal thermal spread.

[0806] FIG. **234** illustrates a method **5000** of operating method of operating a combination ultrasonic/bipolar RF surgical device with a combination energy modality end-effector, according to at least one aspect of the present disclosure. With reference also to FIGS. **1-233**, the method comprises providing **5002** any one of the combination ultrasonic/bipolar RF surgical instruments **10, 100, 150, 190** described throughout this disclosure. The surgical instrument **10, 100, 150, 190** is coupled to a generator **12, 133, 206, 3140, 3900, 4000** (modular energy system), as described throughout this disclosure. The surgical instrument **10, 100, 150, 190** comprises a housing and a combination energy modality end-effector according to any one of the combination energy modality end-effector implementations described in FIGS. **1-233** throughout this disclosure. The end-effector comprises a clamp arm and an ultrasonic blade acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point, a clamp arm pad fixed to the clamp jaw, and a cantilever electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect, the cantilever electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith. The cantilever electrode is configured to electrically couple to an opposite pole of the electrical generator.

[0807] The method **5000** further comprises grasping **5004** tissue with the combination energy modality end-effector between the clamp jaw and the ultrasonic blade. The method **5000** further comprises applying **5006** energy to either the cantilever electrode and/or the ultrasonic blade of the combination energy modality end-effector for a period of time. The applied energy may be an RF energy mode applied to the cantilever electrode and/or an ultrasonic energy mode applied to the ultrasonic transducer to drive the ultrasonic blade. The method **5000** further comprises sealing and/or cutting **5008** the grasped tissue using a combination of energy modes. In one aspect, the ultrasonic energy modes may be applied to the cantilever electrode or the ultrasonic blade (e.g., and the tissue) simultaneously over the entire period. In another aspect, the ultrasonic energy modes may be applied to the cantilever electrode or the ultrasonic blade (e.g., and the tissue) may be applied separately (e.g., independently) over the entire period without overlapping the energy modes. In another aspect, the ultrasonic energy modes may be applied to the cantilever electrode or the ultrasonic blade (e.g., and the tissue) simultaneously over at least a portion of the period and then separately (e.g., independently) over at least another portion of the period. In another aspect, the ultrasonic energy modes may be applied to the cantilever electrode or the ultrasonic blade (e.g., and the tissue) may be applied separately (e.g., independently) over at least a portion of the period and then simultaneously for at least another portion of the period. In any of the above,

the energy levels of either energy mode may be varied or maintained constant throughout the period until a desired tissue seal and cut is achieved.

[0808] The end-effector used in the method 5000 be any of one the following aspects of end-effectors described throughout this disclosure. In one aspect, the end-effector may comprise an electrode comprising a resilient plug at a distal end.

[0809] In another aspect, the end-effector may comprise a peripheral electrode defines a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith.

[0810] In another aspect, the end-effector may comprise a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the peripheral electrode and another portion extending through the inner aperture beyond the surface of the peripheral electrode to contact tissue.

[0811] In another aspect, the end-effector may comprise an electrode comprising an end-of-life indicator which can be sensed by the electrical generator to prompt replacement of the electrode.

[0812] In another aspect, the end-effector may comprise a floating electrode configured to deflect along its entire length, the floating electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith.

[0813] In another aspect, the end-effector may comprise a resilient clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0814] In another aspect, the end-effector may comprise a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0815] In another aspect, the end-effector may comprise a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0816] In another aspect, the end-effector may comprise a spring disposed between the electrode and the clamp jaw.

[0817] In another aspect, the end-effector may comprise an electrode overmolded with an elastomer.

[0818] In another aspect, the end-effector may comprise a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0819] In another aspect, the end-effector may comprise one or more than one resistor embedded in the clamp arm pad to sense the height "h" of the clamp arm pad.

[0820] In another aspect, the end-effector may comprise a clamp arm pad comprising a longitudinally aligned slot configured to receive an electrode plate, wherein the interlocked electrode plate and clamp arm pad serve as both the restraining system and the ultrasonic support.

[0821] In another aspect, the end-effector may comprise a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0822] In another aspect, the end-effector may comprise a lattice cushion defining a longitudinal slot, wherein the lattice cushion acts as a spring-like element. In another aspect, the end-effector may comprise a flexible electrode disposed above the lattice cushion defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith. In another aspect, the end-effector may comprise a clamp arm pad disposed inside the longitudinal slot defined by the lattice cushion, fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0823] In another aspect, the end-effector may comprise a retainer configured to oppose the biasing force and to maintain the electrode in a substantially flat configuration relative to the ultrasonic blade.

[0824] In another aspect, the end-effector may comprise a skirt provided around a distal end of the clamp jaw to surround the distal end of the clamp arm pad.

[0825] In another aspect, the end-effector may comprise a clamp arm pad comprising multiple support elements arranged in multiple rows to support the electrode.

[0826] In another aspect, the end-effector may comprise at least one spring disposed in each of a plurality of zones, wherein a spring bias force of a first spring in a first zone is different from the spring bias force bias of a second spring in a second zone. In another aspect, the end-effector may comprise an electrode disposed along the plurality of zones and in contact with the each of the springs to apply a variable spring bias along the length of the electrode, wherein the electrode is fixed to the clamp jaw at a proximal end and free to deflect at a distal end.

[0827] In another aspect, the end-effector may comprise an electrode supported by a variable longitudinal support elements, and wherein the variable longitudinal support elements apply a variable force on the electrode from the proximal end to the distal end.

[0828] In another aspect, the end-effector may comprise a compliant material fixed to the clamp jaw to support the electrode and act as a spring between the electrode and the clamp jaw.

[0829] In another aspect, the end-effector may comprise a leaf spring element disposed at a distal end of the electrode.

[0830] In another aspect, the end-effector may comprise a compressible material attached to a distal end of the electrode.

[0831] In another aspect, the end-effector may comprise an I-beam shaped clamp arm pad, wherein the clamp arm pad defines top and bottom lateral portions separated by a mesial portion to define the shape of an "I", wherein the electrode is disposed between the top and bottom lateral portions of the I-beam shaped clamp arm pad.

[0832] In another aspect, the end-effector may comprise a clamp jaw, electrode, and clamp arm pad that defines recesses along a longitudinal length that coincides with the ultrasonic blade.

[0833] In another aspect, the end-effector may comprise movable floating gap setting pads to create a composite arrangement to both create ultrasonic blade pressure while also setting a minimum gap between the electrode and the ultrasonic blade.

[0834] In another aspect, the end-effector may comprise a watch-band style segmented electrode.

[0835] In another aspect, the end-effector may comprise an ultrasonic blade comprising a vapor deposition of electrically insulative material on select areas of the ultrasonic blade to prevent electrical shorting in the event of the ultrasonic blade contacting the upper jaw electrode. In another aspect, the end-effector may comprise a clamp arm, ultrasonic blade, or both, comprising selectively coated components.

[0836] In another aspect, the end-effector may comprise an electrically non-conductive clamp arm pad comprising a plurality of teeth.

[0837] In another aspect, the end-effector may comprise a clamp arm comprising control features to adjust tissue path relative to the clamp arm or ultrasonic blade to create a predefined location of contact.

[0838] In another aspect, the end-effector may comprise a conductive polymer clamp arm pad.

[0839] In another aspect, the end-effector may comprise a clamp arm pad comprising electrically non-conductive layers and electrically conductive layers arranged in a sandwich like configuration.

[0840] In another aspect, the end-effector may comprise a composite clamp arm pad comprising electrically non-conductive layers and electrically conductive layers, a carrier attached to the clamp jaw, and a clamp arm pad comprising an electrically conductive pad and an electrically non-conductive pad.

EXAMPLES

[0841] Examples of the method according to various aspects of the present disclosure are provided below. An aspect of the method may include any one or more than one, and any combination of, the examples described below.

[0842] Example 1—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises an electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect. The electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith, wherein the electrode is configured to couple to an opposite pole of the electrical generator. The clamp arm also comprises a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0843] Example 2—The end-effector of Example 1, further comprising at least one gap pad fixed to the electrode to set a gap between the ultrasonic blade and the electrode.

[0844] Example 3—The end-effector of Example 2, wherein the gap pad is located at a proximal end of the electrode.

[0845] Example 4—The end-effector of any one of Examples 2-3, wherein the gap pad is made of a material having a hardness that is greater than a hardness of the clamp arm pad such that the gap pad has longer wear life than the clamp arm pad.

[0846] Example 5—The end-effector of any one of Examples 2-4, wherein the clamp arm pad is made of a polymer material and the gap pad is made of a polyimide material.

[0847] Example 6—The end-effector of any one of Examples 2-5, wherein the gap pad is made of a non-compliant material and the clamp arm pad is made of a compliant material.

[0848] Example 7—The end-effector of any one of Examples 2-6, wherein the clamp arm pad is made of a polymer material and the gap pad is made of a polyimide material.

[0849] Example 8—The end-effector of any one of Examples 1-7, wherein the electrode comprises a gap pad at a proximal end of the electrode, a gap pad at a distal end of the electrode, and a gap pad located between the proximal end and the distal end.

[0850] Example 9—The end-effector of any one of Examples 1-8, wherein the clamp arm pad comprises a base portion fixedly attached to the clamp jaw and a plurality of teeth extending from the base in a direction toward the electrode.

[0851] Example 10—The end-effector of any one of Examples 1-9, wherein the electrode defines a plurality of apertures.

[0852] Example 11—The end-effector of any one of Examples 1-10, further comprising at least one gap pad fixed to the electrode to set a gap between the ultrasonic blade and the electrode. The clamp arm pad comprises a base portion fixedly attached to the clamp jaw and a plurality of teeth extending from the base in a direction toward the electrode. The electrode defines a plurality of apertures sized and configured to receive the at least one gap pad and the plurality of teeth therethrough.

[0853] Example 12—The end-effector of any one of Examples 2-11, wherein the electrode defines an aperture at the proximal defining an open end to slidably receive one of the gap pads.

[0854] Example 13—The end-effector of any one of Examples 1-12, further comprising a gap pad located at the distal end of the electrode and a gap pad located between the distal and the proximal end of the electrode, wherein the proximal gap pad is sized larger than the distal and the medial gap pads.

[0855] Example 14—The end-effector of any one of Examples 1-13, wherein the proximal end of the electrode is welded, brazed, or soldered to the proximal end of the clamp jaw.

[0856] Example 15—A surgical instrument comprising a housing and an end-effector as defined by Examples 1-14. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0857] Example 16—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises an electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect. The electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith. The electrode is configured to couple to an opposite pole of the electrical generator, and the electrode comprises a resilient plug at a distal end. The clamp arm also comprises a clamp arm pad fixed to the clamp jaw and having at least a portion

disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0858] Example 17—The end-effector of Example 16, wherein the resilient plug is made of silicone.

[0859] Example 18—The end-effector of Example 16-17, further comprising at least one gap pad fixed to the electrode to set a gap between the ultrasonic blade and the electrode.

[0860] Example 19—The end-effector of Example 18, wherein the gap pad is located at a proximal end of the electrode.

[0861] Example 20—The end-effector of any one of Examples 18-19, wherein the gap pad is made of a material having a hardness that is greater than a hardness of the clamp arm pad such that the gap pad has longer wear life than the clamp arm pad.

[0862] Example 21—The end-effector of any one of Examples 18-20, wherein the clamp arm pad is made of a polymer material and the gap pad is made of a polyimide material.

[0863] Example 22—The end-effector of any one of Examples 18-21, wherein the gap pad is made of a non-compliant material and the clamp arm pad is made of a compliant material.

[0864] Example 23—The end-effector of any one of Examples 18-22, wherein the clamp arm pad is made of a polymer material and the gap pad is made of a polyimide material.

[0865] Example 24—The end-effector of any one of Examples 16-23, wherein the electrode comprises a gap pad at a proximal end of the electrode, a gap pad at a distal end of the electrode, and a gap pad located between the proximal end and the distal end.

[0866] Example 25—The end-effector of any one of Examples 16-24, wherein the clamp arm pad comprises a base portion fixedly attached to the clamp jaw and a plurality of teeth extending from the base in a direction toward the electrode.

[0867] Example 26—The end-effector of any one of Examples 16-25, wherein the electrode defines a plurality of apertures.

[0868] Example 27—The end-effector of any one of Examples 16-26, further comprising at least one gap pad fixed to the electrode to set a gap between the ultrasonic blade and the electrode. The clamp arm pad comprises a base portion fixedly attached to the clamp jaw and a plurality of teeth extending from the base in a direction toward the electrode. The electrode defines a plurality of apertures sized and configured to receive the at least one gap pad and the plurality of teeth therethrough.

[0869] Example 28—The end-effector of any one of Examples 18-27, wherein the electrode defines an aperture at the proximal defining an open end to slidably receive one of the gap pads.

[0870] Example 29—The end-effector of any one of Examples 16-28, further comprising a gap pad located at the distal end of the electrode and a gap pad located between the distal and the proximal end of the electrode, wherein the proximal gap pad is sized larger than the distal and the medial gap pads.

[0871] Example 30—The end-effector of any one of Examples 16-29, wherein the proximal end of the electrode is welded, brazed, or soldered to the proximal end of the clamp jaw.

[0872] Example 31—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises a peripheral electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect. The electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith. The peripheral electrode is configured to couple to an opposite pole of the electrical generator, and wherein the peripheral electrode defines an inner aperture. The clamp arm also a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the peripheral electrode and another portion extending through the inner aperture beyond the surface of the peripheral electrode to contact tissue.

[0873] Example 32—The end-effector of Example 31, wherein the electrode further comprises a connection pad isolated from the peripheral electrode to attach the peripheral electrode to a proximal end of the clamp arm.

[0874] Example 33—The end-effector of Example 32, wherein the connection pad sets a gap “x” defined between the peripheral electrode and the ultrasonic blade.

[0875] Example 34—The end-effector of any one of Examples 31-33 wherein the electrode further comprises a connection pad extending from the peripheral electrode to attach the peripheral electrode to a proximal end of the clamp arm.

[0876] Example 35—The end-effector of any one of Examples 31-34, wherein the peripheral electrode comprises an electrically non-conductive portion to define a space between an outer of the electrically conductive portion of the peripheral electrode and an outer edge of the peripheral electrode.

[0877] Example 36—The end-effector of any one of Examples 31-35, wherein the peripheral electrode comprises an electrically conductive portion that extends the entire distance to the distal edge of the electrically non-conductive portion.

[0878] Example 37—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises an electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect. The electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith. The electrode is configured to couple to an opposite pole of the electrical generator, wherein the electrode comprises an end-of-life indicator which can be sensed by the electrical generator to prompt replacement of the electrode. The clamp arm also comprises a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0879] Example 38—The end-effector of Example 37, wherein the end-of-life indicator comprises a tail made of electrically conductive electrode material added to the proximal end of the electrode anchor point.

[0880] Example 39—The end-effector of Example 38, wherein the tail is configured to deflect towards the ultrasonic blade as the clamp arm pads and wear resistant pads show wear and tear use.

[0881] Example 40—The end-effector of any one of Examples 38-39, wherein the tail contact with the ultrasonic blade creates an electrical short circuit that can be detected to trigger an end notification.

[0882] Example 41—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises a floating electrode configured to deflect along its entire length. The floating electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith. The floating electrode is configured to couple to an opposite pole of the electrical generator. The clamp arm also comprises a resilient clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0883] Example 42—The end-effector of Example 41, wherein the floating electrode is mounted to a spring.

[0884] Example 43—The end-effector of any one of Examples 41-42, wherein the clamp arm comprises a plurality of wear resistant clamp arm pads mounted to the resilient clamp arm pad.

[0885] Example 44—The end-effector of any one of Examples 41-43, wherein the resilient clamp arm pad acts as a spring such that the floating electrode can deflect into the resilient clamp arm pad and move in a direction toward the clamp jaw.

[0886] Example 45—The end-effector of any one of Examples 41-44, further comprising a wear resistant clamp arm pad to set a tissue gap between the floating electrode and the ultrasonic blade.

[0887] Example 46—The end-effector of any one of Examples 41-45, further comprising a plurality of wear resistant dowel pins sized and configured to set a tissue gap between the ultrasonic blade and the floating electrode.

[0888] Example 47—The end-effector of Example 46, wherein the dowel pin comprises barb features to prevent the wear resistant dowel pin from backing out after it is installed in the clamp arm.

[0889] Example 48—The end-effector of any one of Examples 46-47, wherein the dowel pin comprises a pierce feature to prevent the wear resistant dowel pin from backing out after it is installed in the clamp arm.

[0890] Example 49—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to an opposite pole of the electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises an electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect. The electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith. The electrode is configured to couple to a pole of an electrical generator, wherein the electrode comprises a center pivot configuration to enable the electrode to rotate about the center pivot point. The clamp arm also

comprises a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0891] Example 50—The end-effector of Example claim 49, wherein the electrode is formed into a spring element.

[0892] Example 51—The end-effector of any one of Examples 49-50, wherein the spring may further comprise strips of metal to form a leaf spring configuration.

[0893] Example 52—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises an electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect. The electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith. The electrode is configured to couple to an opposite pole of the electrical generator. The clamp arm also comprises a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue. The clamp arm also comprises a spring disposed between the electrode and the clamp jaw.

[0894] Example 53—The end-effector of Example 52, wherein the electrode comprises wear pads attached to the electrode on a side facing the ultrasonic blade.

[0895] Example 54—The end-effector of any one of Examples 52-53, wherein the clamp arm further comprises a wave spring element to support the electrode.

[0896] Example 55—The end-effector of Example 54, wherein the wave spring element is formed integral with the clamp arm pad to deflect the electrode in a direction toward the clamp jaw.

[0897] Example 56—The end-effector of any one of Examples 54-55, wherein the wave spring element is positioned between the clamp arm pad and the electrode to deflect the electrode in a direction toward the clamp jaw.

[0898] Example 57—The end-effector of any one of Examples 54-56, wherein the wave spring element is formed integral with the clamp arm pad to deflect the electrode in a direction toward the clamp jaw.

[0899] Example 58—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises an electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect. The electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith. The electrode is configured to couple to an opposite pole of the electrical generator, wherein the electrode is overmolded with an elastomer. The clamp arm also comprises a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0900] Example 59—The end-effector of Example 58, wherein the electrode overmolded with the elastomer provides uniform compression along a length of the electrode.

[0901] Example 60—The end-effector of any one of Examples 58-59, wherein the electrode comprises electrical contacts an lateral portions thereof.

[0902] Example 61—The end-effector of any one of Examples 58-60, wherein the spring rate increase as the electrode overmolded with the elastomer compresses against the elastomer.

[0903] Example 62—The end-effector of any one of Examples 58-61, further comprising overmolded rings or pads on the ultrasonic blade.

[0904] Example 63—The end-effector of any one of Examples 58-62, where the electrode comprises a diaphragm attached to the clamp jaw by rigid electrically conductive connections.

[0905] Example 64—The end-effector of Example 63, wherein clamp arm pads are disposed on the diaphragm electrode.

[0906] Example 65—The end-effector of any one of Examples 63-64, wherein the diaphragm electrode can be formed of a thin metallic conductor material to flex or deflect the electrode.

[0907] Example 66—The end-effector of any one of Examples 63-65, wherein the diaphragm electrode is made in the form of a ring.

[0908] Example 67—The end-effector of any one of Examples 58-66, wherein the electrode overmolded with the elastomer comprises ends folded into springs and are in electrical contact with the clamp jaw.

[0909] Example 68—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises an electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect. The electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith, wherein the electrode is configured to couple to an opposite pole of the electrical generator. The clamp arm also comprises a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue. The clamp arm also comprises one or more than one resistor embedded in the clamp arm pad to sense the height “h” of the clamp arm pad.

[0910] Example 69—The end-effector of Example 68, wherein the one or more than one resistor is configured to melt or wear along with the clamp arm pad such that a resistance value of the one or more than one resistor is analogous to the height of the clamp arm pad.

[0911] Example 70—A surgical instrument comprising a housing and an end-effector as defined by Examples 16-69. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0912] Example 71—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises an electrode plate defining a surface configured to contact

tissue and apply electrical energy to the tissue in contact therewith. The electrode plate is configured to couple to an opposite pole of the electrical generator. The clamp arm also comprises a clamp arm pad comprising a longitudinally aligned slot configured to receive the electrode plate. The interlocked electrode plate and clamp arm pad serve as both the restraining system and the ultrasonic support.

[0913] Example 72—The end-effector of Example 71, wherein the longitudinally aligned slot extends from a proximal end through which the electrode plate can be advanced without clamp arm pad material.

[0914] Example 73—The end-effector of any one of Examples 71-72, further comprising a tab at a proximal end of the clamp arm pad.

[0915] Example 74—The end-effector of any one of Examples 71-73, further comprising a proximal recess at a proximal end of the of the clamp jaw, wherein the proximal recess is configured to receive the tab therein to snap fit and lock the clamp arm pad into the clamp jaw.

[0916] Example 75—The end-effector of any one of Examples 71-74, wherein the clamp arm pad comprises a plurality of projections sized and configured to be received through apertures defined the electrode plate.

[0917] Example 76—A surgical instrument comprising a housing and an end-effector as defined by Examples 71-75. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0918] Example 77—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises a laterally deflectable electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith. The electrode is configured to couple to an opposite pole of the electrical generator. The clamp arm also comprises a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0919] Example 78—The end-effector of Example 77, wherein the electrode comprises a proximal end fixed to a proximal end of the clamp jaw and a distal end that is free to deflect vertically and laterally by fixating the proximal end and creating an elastomer connection between a right and left section of the electrode which are in electrical connection with each other.

[0920] Example 79—The end-effector of any one of Examples 77-78, wherein the laterally deflectable electrode is configured to expand by thermal expansion of the clamp arm pad due to heat and deformation due to compressive clamp load.

[0921] Example 80—The end-effector of any one of Examples 77-79, wherein the laterally deflectable electrode comprises a notch defined on a distal end thereof.

[0922] Example 81—The end-effector of any one of Examples 77-80, wherein the laterally deflectable electrode comprises a vertical stiffening form to secure the deflectable electrode to the clamp jaw.

[0923] Example 82—The end-effector of any one of Examples 77-81, wherein the laterally deflectable electrode is a structural clamping member and is inward facing about the clamp arm pad.

[0924] Example 83—The end-effector of any one of Examples 77-82, wherein the laterally expandable electrode extends toward the expanded clamp arm pad when the clamp arm pad material transitions to a gel-like state at or near 330° C.

[0925] Example 84—A surgical instrument comprising a housing and an end-effector as defined by Examples 77-83. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0926] Example 85—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises a lattice cushion defining a longitudinal slot, wherein the lattice cushion acts as a spring-like element. The clamp arm also comprises a flexible electrode disposed above the lattice cushion defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith. The electrode is configured to couple to an opposite pole of the electrical generator. The clamp arm also comprises a clamp arm pad disposed inside the longitudinal slot defined by the lattice cushion. The clamp arm pad is fixed to the clamp jaw and has at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0927] Example 86—The end-effector of Example 85, further comprising a plurality of hard spacers disposed on the flexible electrode to set a gap between the flexible electrode and the ultrasonic blade.

[0928] Example 87—The end-effector of any one of Examples 85-86, wherein in a closed position of the clamp arm, the lattice cushion applies consistent tissue compression across variable thickness tissue T1a, T2a, T3a.

[0929] Example 88—The end-effector of Example 87, wherein the consistent tissue compression across variable thickness tissue T1a, T2a, T3a is defined by:

$$\frac{T_{1a}}{T_{1b}} = \frac{T_{2a}}{T_{2b}} = \frac{T_{3a}}{T_{3b}}$$

[0930] Example 89—A surgical instrument comprising a housing and an end-effector as defined by Examples 85-88. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0931] Example 90—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises an electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect. The electrode defining a surface configured to contact tissue and

apply electrical energy to the tissue in contact therewith. The electrode is configured to couple to an opposite pole of the electrical generator, wherein the electrode is biased in a direction toward the clamp jaw. The clamp arm also comprises a retainer configured to oppose the biasing force and to maintain the electrode in a substantially flat configuration relative to the ultrasonic blade. The clamp arm also comprises a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[0932] Example 91—The end-effector of Example 90, wherein the retainer is configured as a tooth.

[0933] Example 92—The end-effector of Example 90, wherein the retainer is configured as a wall with a tapered profile.

[0934] Example 93—The end-effector of any one of Examples 90-92, wherein in a free-state, the electrode is pre-formed in a shape biased toward the clamp jaw.

[0935] Example 94—The end-effector of any one of Examples 90-93, wherein in a free-state, the electrode is bent in a shape biased toward the clamp jaw.

[0936] Example 95—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises an electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect. The electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith. The electrode is configured to couple to an opposite pole of the electrical generator, wherein the electrode is biased in a direction toward the clamp jaw. The clamp arm also comprises a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue. The clamp arm also comprises a skirt provided around a distal end of the clamp jaw to surround the distal end of the clamp arm pad.

[0937] Example 96—The end-effector of Example 95, wherein the skirt is configured to contain the clamp arm pad as the clamp arm pad heats and deforms to prevent the clamp arm pad from expanding laterally.

[0938] Example 97—The end-effector of any one of Examples 95-96, where in the electrode is biased.

[0939] Example 98—The end-effector of any one of Examples 95-97, where in the electrode has a higher bias at the distal end compared to the proximal end.

[0940] Example 99—A surgical instrument comprising a housing and an end-effector as defined by Examples 90-98. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0941] Example 100—An end-effector comprising a clamp arm and an ultrasonic blade that is configured to couple to an ultrasonic transducer and to a pole of an electrical generator. The clamp arm comprises a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point. The clamp arm also comprises an electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free

to deflect. The electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith. The electrode is configured to couple to an opposite pole of the electrical generator. The clamp arm also comprises a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue. The clamp arm pad comprises multiple support elements arranged in multiple rows to support the electrode. The clamp arm also comprises a skirt provided around a distal end of the clamp jaw to surround the distal end of the clamp arm pad.

[0942] Example 101—The end-effector of Example 100, wherein as the clamp arm pad heats up, a row of support elements melt and wear away causing the electrode to drop in height.

[0943] Example 102—The end-effector of Example 101, wherein as the clamp arm pad continues to heat up, additional rows of support elements melt and wear away causing the height of the electrode to further drop in height.

[0944] Example 103—A surgical instrument comprising a housing and an end-effector as defined by Examples 100-102. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0945] Example 104—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw defining a plurality of zones along the clamp jaw. The clamp arm also comprises at least one spring disposed in each of the plurality of zones. A spring bias force of a first spring in a first zone is different from the spring bias force bias of a second spring in a second zone. The clamp arm also comprises an electrode disposed along the plurality of zones and in contact with each of the springs to apply a variable spring bias along the length of the electrode. The electrode is fixed to the clamp jaw at a proximal end and free to deflect at a distal end.

[0946] Example 105—The end-effector of Example 104, wherein the clamp jaw defines at least a first zone Z1 at a proximal end of the clamp arm and a second zone Z2 at a distal end of the clamp arm. The clamp arm further comprises at least one spring S1 disposed in the first zone Z1 and at least one spring S2 disposed in the second zone Z2. The spring bias force of the at least one spring S1 in the first zone Z1 and the spring bias force of the at least one spring S2 in the second zone Z2 are variable such that $S2 > S1$.

[0947] Example 106—The end-effector of any one of Examples 104-105, wherein the clamp jaw defines at least a first zone Z1 at a proximal end of the clamp arm and a second zone Z2 at a distal end of the clamp arm. The clamp arm further comprises a plurality of springs S1 disposed in the first zone Z1 and a plurality of springs S2 disposed in the second zone Z2. The spring bias force of the plurality of springs S1 in the first zone Z1 and the spring bias force of the plurality of springs S2 in the second zone Z2 are variable such that $S2 > S1$.

[0948] Example 107—The end-effector of any one of Examples 104-106, wherein variable spring bias along the length of the electrode creates a tip-loading condition.

[0949] Example 108—The end-effector of any one of Examples 104-107, wherein a deflection of the ultrasonic blade increases in the distal direction.

[0950] Example 109—The end-effector of any one of Examples 104-108, wherein under low clamp load conditions, the ultrasonic blade remains straight and under over clamping conditions the clamp arm causes the deflection of the electrode caused by the spring loads causing deflection of the ultrasonic blade.

[0951] Example 110—The end-effector of any one of Examples 104-109, further comprising a plurality of hard spacers to set a gap between the electrode and the ultrasonic blade.

[0952] Example 111—A surgical instrument comprising a housing and an end-effector as defined by Examples 104-110. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0953] Example 112—An end-effector comprising a clamp arm, and an ultrasonic blade. The clamp arm comprises a clamp jaw, a plurality of variable longitudinal support elements, and an electrode fixed to the clamp jaw at a proximal end and free to deflect at a distal end. The electrode is supported by the variable longitudinal support elements. The variable longitudinal support elements apply a variable force on the electrode from the proximal end to the distal end.

[0954] Example 113—The end-effector of Example 112, wherein the variable longitudinal support elements limit a maximum deflection of the electrode.

[0955] Example 114—The end-effector of any one of Examples 112-113, wherein the variable longitudinal support elements comprises a plurality of springs with variable force (F).

[0956] Example 115—The end-effector of any one of Examples 112-114, wherein the electrode further comprises bump extensions located on the electrode between the electrode and the clamp jaw.

[0957] Example 116—The end-effector of Example 115, wherein the bump extensions located on the electrode between the electrode and the clamp jaw limit the maximum deflection of the variable longitudinal support elements with a variable longitudinal spring force(S).

[0958] Example 117—The end-effector of Example 116, wherein the variable longitudinal spring force(S) is less than the variable force (F).

[0959] Example 118—The end-effector of any one of Examples 112-117, wherein the clamp arm further comprises clamp arm pads.

[0960] Example 119—A surgical instrument comprising a housing and an end-effector as defined by Examples 112-118. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0961] Example 120—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw, an electrode, and a compliant material fixed to the clamp jaw to support the electrode and act as a spring between the electrode and the clamp jaw.

[0962] Example 121—The end-effector of Example 120, wherein the clamp arm further comprises a hard wear resistant material fixed to the electrode to set a gap between the electrode and the ultrasonic blade.

[0963] Example 122—The end-effector of any one of Examples 120-121, wherein the electrode comprises a hard

wear resistant material fixed to a proximal end of the electrode and a hard wear resistant material fixed to a distal end of the electrode.

[0964] Example 123—The end-effector of any one of Examples 120-122, wherein the electrode defines apertures for receiving the compliant material therethrough.

[0965] Example 124—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw, an electrode, and a leaf spring element disposed at a distal end of the electrode.

[0966] Example 125—The end-effector of Example 124, wherein a proximal end of the electrode is fixedly attached to the clamp jaw.

[0967] Example 126—The end-effector of any one of Examples 124-125, further comprising a hard wear resistant pad disposed on a proximal end of the electrode to set a gap between the electrode and the clamp arm.

[0968] Example 127—The end-effector of any one of Examples 124-126, wherein the leaf spring defines symmetric elements on lateral sides of the distal end of the electrode.

[0969] Example 128—An end-effector comprising a clamp arm, and an ultrasonic blade. The clamp arm comprises a clamp jaw, an electrode, and a compressible material attached to a distal end of the electrode.

[0970] Example 129—The end-effector of Example 128, wherein the compressible material attached to the distal end of the electrode is disposed between the electrode and the clamp jaw.

[0971] Example 130—The end-effector of any one of Examples 128-129, wherein a distal end of the clamp jaw defines a pocket to receive the compressible material.

[0972] Example 131—A surgical instrument comprising a housing and an end-effector as defined by Examples 120-130. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0973] Example 132—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw, an electrode fixed to the clamp jaw at a proximal end and free to deflect at a distal end, and an I-beam shaped clamp arm pad. The clamp arm pad defines top and bottom lateral portions separated by a mesial portion to define the shape of an “I”. The electrode is disposed between the top and bottom lateral portions of the I-beam shaped clamp arm pad.

[0974] Example 133—The end-effector of Example 132, wherein the top portions of the I-beam segments positioned above the electrode prevent the clamp arm pad from delaminating.

[0975] Example 134—The end-effector of any one of Examples 132-133, further comprising a hard wear resistant gap setting pad disposed at the proximal end of the electrode to set a gap between the electrode and the ultrasonic blade.

[0976] Example 135—The end-effector of any one of Examples 132-134, wherein the electrode defines a slot open proximal end to slidably receive the mesial portion of the I-beam shaped clamp arm pad.

[0977] Example 136—The end-effector of any one of Examples 132-135, wherein the clamp jaw defines a longitudinal slot to slidably receive the bottom lateral portions and a portion of the mesial portion of the I-beam shaped clamp arm pad therethrough.

[0978] Example 137—A surgical instrument comprising a housing and an end-effector as defined by Examples 132-136. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0979] Example 138—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw, an electrode fixed to the clamp jaw at a proximal end and free to deflect at a distal end, and a clamp arm pad. The clamp jaw, the electrode, and the clamp arm pad define recesses along a longitudinal length that coincides with the ultrasonic blade.

[0980] Example 139—The end-effector of Example 138, wherein the recesses minimize the impact between the ultrasonic blade and the electrode.

[0981] Example 140—The end-effector of any one of Examples 138-139, wherein the clamp arm pad is a heavy polymer support pad.

[0982] Example 141—The end-effector of any one of Examples 138-140, wherein the electrode is almost flush with the ultrasonic blade when in the most clamped state and yet not in metallic contact with the ultrasonic blade.

[0983] Example 142—The end-effector of any one of Examples 138-141, further comprising an additional hole defined within the recessed area to heat stake or fix the heavy polymer support pad to the electrode recess.

[0984] Example 143—A surgical instrument comprising a housing and an end-effector as defined by Examples 138-142. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0985] Example 144—An end-effector comprising a clamp arm, and an ultrasonic blade. The clamp arm comprises a clamp jaw, an electrode fixed to the clamp jaw at a proximal end and free to deflect at a distal end, a stationary gap setting pad, and movable floating gap setting pads. The movable floating gap setting pads create a composite arrangement to both create ultrasonic blade pressure while also setting a minimum gap between the electrode and the ultrasonic blade.

[0986] Example 145—The end-effector of Example 144, wherein the clamp arm further comprises a clamp arm pad attached to the clamp jaw.

[0987] Example 146—The end-effector of any one of Examples 144-145, wherein a distal most stationary gap setting pad is attached to the clamp jaw.

[0988] Example 147—The end-effector of any one of Examples 144-146, wherein a distal stationary gap setting pad will improve tip pressure/grasping and decrease chances of short circuiting the electrode to the ultrasonic blade due to burn through of the stationary gap setting pad.

[0989] Example 148—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw, an electrode fixed to the clamp jaw at a proximal end and free to deflect at a distal end, a stationary gap setting pad at a distal end, a stationary clamp arm pad; and a movable floating gap setting pad at a proximal end. The movable floating gap setting pad creates a composite arrangement to both create ultrasonic blade pressure while also setting a minimum gap between the electrode and the ultrasonic blade.

[0990] Example 149—A surgical instrument comprising a housing and an end-effector as defined by Examples 144-148. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0991] Example 150—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw and a watch-band style segmented electrode.

[0992] Example 151—The end-effector of Example 150, further comprises spring elements to bias the watch-band style segmented electrode.

[0993] Example 152—The end-effector of any one of Examples 150-151, wherein the segmented electrode comprises three segments linked by pins.

[0994] Example 153—The end-effector of any one of Examples 150-152, further comprising three springs positioned between each of the electrode segments and the clamp jaw to apply distal, medial, and proximal bias to the electrode segments, respectively.

[0995] Example 154—The end-effector of any one of Examples 150-153, wherein a distal segment comprises a rounded distal end, an opening to receive a pin, and a grooved surface at a proximal end.

[0996] Example 155—The end-effector of any one of Examples 150-154, wherein the medial segment comprises a cylindrical insert that is received in the proximal grooved surface of the distal segment and the pin rotatably fixes the distal and medial segments.

[0997] Example 156—The end-effector of any one of Examples 150-155, wherein the medial segment is placed into the grooved surface of the distal segment and the pin is placed through the medial segment and the grooved segment to hold the medial segment and the grooved segment together and the grooved surface and the cylindrical insert allows the medial segment and the grooved segment to pivot against each other.

[0998] Example 157—A surgical instrument comprising a housing and an end-effector as defined by Examples 150-156. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[0999] Example 158—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw and an electrode fixed to the clamp jaw at a proximal end and free to deflect at a distal end. The ultrasonic blade comprises a vapor deposition of electrically insulative material on select areas of the ultrasonic blade to prevent electrical shorting in the event of the ultrasonic blade contacting the upper jaw electrode.

[1000] Example 159—The end-effector of Example 158, wherein the electrically insulative material is deposited by masking the ultrasonic blade.

[1001] Example 160—The end-effector of any one of Examples 158-159, wherein the electrically insulative material is machined off areas of the ultrasonic blade that needs to be electrically conductive.

[1002] Example 161—The end-effector of any one of Examples 158-160, wherein the electrically insulative material is any one of a fluoropolymer coating, a nanocomposite coating, or a ceramic coating, or combinations thereof.

[1003] Example 162—A surgical instrument comprising a housing and an end-effector as defined by Examples 158-161. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[1004] Example 163—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw and an electrode fixed to the clamp jaw at a proximal end and free to deflect at a distal end. The clamp arm, the ultrasonic blade, or both comprise selectively coated components.

[1005] Example 164—The end-effector of Example 163, wherein the clamp jaw and the ultrasonic blade comprise selectively coated portions, and wherein the clamp arm pad and the electrode are uncoated.

[1006] Example 165—The end-effector of any one of Examples 163-164, wherein the clamp jaw comprises selectively coated portions, and wherein the ultrasonic blade, clamp arm pad, and the electrode are uncoated.

[1007] Example 166—The end-effector of any one of Examples 163-165, wherein the ultrasonic blade comprises selectively coated portions, and wherein the clamp jaw, clamp arm pad, and the electrode are uncoated.

[1008] Example 167—The end-effector of any one of Examples 163-166, wherein the clamp jaw comprises selectively coated portions, and wherein the ultrasonic blade, the clamp arm pad, and the electrode are uncoated, and wherein the clamp jaw comprises a bare electrode portion at a tip of the clamp jaw.

[1009] Example 168—The end-effector of any one of Examples 163-167, wherein the ultrasonic blade comprises selectively coated portions, and wherein the clamp jaw, the clamp arm pad, and the electrode are uncoated, and wherein the ultrasonic blade comprises a bare electrode portion at a tip of the ultrasonic blade.

[1010] Example 169—The end-effector of any one of Examples 163-168, wherein the clamp jaw and the ultrasonic blade comprise selectively coated portions, and wherein the clamp arm pad and the electrode are uncoated, and wherein the clamp jaw comprises a bare electrode portion at a tip of the clamp jaw, and wherein the ultrasonic blade comprises a bare electrode portion at a tip of the ultrasonic blade.

[1011] Example 170—The end-effector of any one of Examples 163-169, wherein the electrode is an offset electrode.

[1012] Example 171—A surgical instrument comprising a housing and an end-effector as defined by Examples 163-170. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[1013] Example 172—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw, an electrode fixed to the clamp jaw at a proximal end and free to deflect at a distal end, and an electrically non-conductive clamp arm pad comprising a plurality of teeth.

[1014] Example 173—The end-effector of Example 172, wherein the electrode comprises an additional distal conductive element at the distal end of the electrode such that when the distal conductive element briefly contacts the ultrasonic blade an algorithm is executed to reduce the

power to the ultrasonic blade when conductive element to ultrasonic blade contact is detected to minimize damage to the electrode and/or the ultrasonic blade.

[1015] Example 174—The end-effector of Example 173, wherein contact of the distal conductive element with the ultrasonic blade is detected by measuring impedance over time.

[1016] Example 175—A surgical instrument comprising a housing and an end-effector as defined by Examples 172-174. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[1017] Example 176—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw and an electrode fixed to the clamp jaw at a proximal end and free to deflect at a distal end. The clamp arm also comprises control features to adjust tissue path relative to the clamp arm or ultrasonic blade to create a predefined location of contact.

[1018] Example 177—The end-effector of Example 176, wherein the control features are configured to reduce sticking and charring of tissue to the clamp arm or ultrasonic blade.

[1019] Example 178—The end-effector of any one of Examples 176-177, wherein the clamp jaw comprises raised sidewalls or guards to surround the electrode to prevent exposure and prevent tissue from entering the area inside the sidewalls.

[1020] Example 179—The end-effector of any one of Examples 178, wherein the raised sidewalls of the clamp jaw are extruded out and around the clamp arm pad.

[1021] Example 180—The end-effector of any one of Examples 176-179, wherein the clamp arm pad comprises exposed teeth to prevent tissue from entering the area inside the exposed teeth.

[1022] Example 181—The end-effector of any one of Examples 176-180, wherein the clamp jaw comprises raised sidewalls and a raised lip to prevent tissue from accumulating inside the sidewalls and the raised lip, and the clamp arm pad comprises a plurality of teeth.

[1023] Example 182—A surgical instrument comprising a housing and an end-effector as defined by Examples 176-181. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrode is electrically coupled to an opposite pole of the electrical generator.

[1024] Example 183—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw and a conductive polymer clamp arm pad.

[1025] Example 184—The end-effector of Example 183, wherein the conductive polymer clamp arm pad comprises tin oxide (SnO_2) embedded in a polymer material comprising PTFE to make the clamp arm pad electrically conductive.

[1026] Example 185—The end-effector of Example 184, wherein the tin oxide is embedded in the polymer material using a cold spray process doping process.

[1027] Example 186—The end-effector of any one of Examples 183-185, wherein the conductive polymer clamp arm pad is configured to contact the ultrasonic blade, absorb heat from the ultrasonic blade, and assist in tissue grasping and clamping.

[1028] Example 187—The end-effector of any one of Examples 184-186, wherein the tin oxide clamp arm pad is configured as one of two electrodes or poles of the bipolar RF circuit to deliver RF energy to tissue grasped between the ultrasonic blade and the conductive polymer clamp arm pad.

[1029] Example 188—The end-effector of any one of Examples 184-187, wherein the tin oxide clamp arm pad is biocompatible, electrically conductive, thermally conductive, enables a large portion of the clamp arm pad to be used to improve wear resistance of the clamp arm pad, and is white in color.

[1030] Example 189—A surgical instrument comprising a housing and an end-effector as defined by Examples 183-188. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the conductive polymer clamp arm pad is electrically coupled to an opposite pole of the electrical generator.

[1031] Example 190—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw and a clamp arm pad. The clamp arm pad comprises electrically non-conductive layers and electrically conductive layers arranged in a sandwich like configuration.

[1032] Example 191—The end-effector of Example 190, wherein the electrically non-conductive layers are made of polymer, polyimide, or PTFE, or any combination thereof.

[1033] Example 192—The end-effector of any one of Examples 190-191, wherein the electrically conductive layers are made of thin electrically conductive polymer, metal foil, or carbon loaded material, or any combination thereof.

[1034] Example 193—The end-effector of any one of Examples 190-192, wherein a majority of material contacting the ultrasonic blade is the electrically non-conductive layer material.

[1035] Example 194—The end-effector of any one of Examples 190-193, wherein at least 74% of the material contacting the ultrasonic blade is the electrically non-conductive layer material.

[1036] Example 195—The end-effector of any one of Examples 190-194, wherein at least 84% of the material contacting the ultrasonic blade is the electrically non-conductive layer material.

[1037] Example 196—The end-effector of any one of Examples 190-195, wherein at least 94% of the material contacting the ultrasonic blade is the electrically non-conductive layer material.

[1038] Example 197—The end-effector of any one of Examples 190-196, wherein a surface area of the non-conductive layers is greater than a surface area of the conductive layers.

[1039] Example 198—The end-effector of any one of Examples 190-197, wherein the clamp arm pad further comprises teeth formed integrally therewith.

[1040] Example 199—A surgical instrument comprising a housing and an end-effector as defined by Examples 190-198. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrically conductive layers of the clamp arm pad are electrically coupled to an opposite pole of the electrical generator.

[1041] Example 200—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw and a composite clamp arm pad attached

to the clamp jaw. The composite clamp arm pad comprises electrically non-conductive layers and electrically conductive layers.

[1042] Example 201—The end-effector of Example 200, wherein the electrically non-conductive layers are made of polymer and the electrically conductive layers are made of metal.

[1043] Example 202—The end-effector of any one of Examples 200-201, wherein the polymer comprises PTFE and the metal comprises stainless steel.

[1044] Example 203—The end-effector of any one of Examples 200-202, wherein a thickness of the electrically conductive layers enables the electrically conductive layers to deform as the electrically non-conductive layers wear.

[1045] Example 204—The end-effector of any one of Examples 200-203, wherein the composite clamp arm pad is attached to the clamp jaw by an adhesive.

[1046] Example 205—The end-effector of Example 204, wherein the adhesive comprises carbon to make it electrically conductive and connect the electrically conductive layers of the composite clamp arm pad to the clamp jaw.

[1047] Example 206—A surgical instrument comprising a housing and an end-effector as defined by Examples 200-205. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the electrically conductive layers of the composite the clamp arm pad are electrically coupled to an opposite pole of the electrical generator.

[1048] Example 207—An end-effector comprising a clamp arm and an ultrasonic blade. The clamp arm comprises a clamp jaw, a carrier attached to the clamp jaw, and a clamp arm pad comprising an electrically conductive pad and an electrically non-conductive pad.

[1049] Example 208—The end-effector of Example 207, wherein the electrically conductive pad is made of an electrically conductive polymer.

[1050] Example 209—The end-effector of any one of Examples 207-208, wherein the clamp jaw and the carrier are made of metal.

[1051] Example 210—The end-effector of Example 209, wherein the metal is stainless steel.

[1052] Example 211—The end-effector of any one of Examples 207-210, wherein the electrically non-conductive pad comprises a polymer.

[1053] Example 212—The end-effector of any one of Examples 207-211, wherein the electrically conductive pad is overmolded over the carrier.

[1054] Example 213—The end-effector of any one of Examples 207-212, wherein the carrier is a metal stamping.

[1055] Example 214—A surgical instrument comprising a housing and an end-effector as defined by Examples 207-213. The ultrasonic blade is acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator and the conductive pad of the clamp arm pad is electrically coupled to an opposite pole of the electrical generator.

[1056] Example 215—A method of operating method of operating a combination ultrasonic/bipolar RF surgical instrument with a combination energy modality end-effector, the combination ultrasonic/bipolar RF surgical instrument comprising: end-effector, comprising: a clamp arm; and an ultrasonic blade acoustically coupled to an ultrasonic transducer and electrically coupled to a pole of an electrical generator; wherein the clamp arm comprises: a clamp jaw

having a proximal end and a distal end, the proximal end pivotally movable about a pivot point; a clamp arm pad fixed to the clamp jaw; and a cantilever electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect, the cantilever electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith, wherein the cantilever electrode is electrically coupled to an opposite pole of the electrical generator; the method comprising: providing the combination ultrasonic/bipolar RF surgical instrument; grasping tissue with the end-effector between the clamp jaw and the ultrasonic blade; applying energy to either the cantilever electrode or the ultrasonic blade of the combination energy modality end-effector for a period of time; sealing or cutting the grasped tissue using a combination of energy modes delivered by the generator.

[1057] Example 216—The method of Example 215, wherein applying energy to either the cantilever electrode or the ultrasonic blade of the combination energy modality end-effector for a period of time comprises applying RF and ultrasonic energy modes.

[1058] Example 217—The method of Example 216, comprising applying the RF and ultrasonic energy modes to the cantilever electrode or the ultrasonic blade simultaneously over the entire period.

[1059] Example 218—The method of Example 216, comprising applying the RF and ultrasonic energy modes to the cantilever electrode or the ultrasonic blade separately over the entire period without overlapping the energy modes.

[1060] Example 219—The method of Example 216, comprising applying the RF and ultrasonic energy modes to the cantilever electrode or the ultrasonic blade simultaneously over at least a portion of the period and then separately for at least another portion of the period.

[1061] Example 220—The method of Example 216, comprising applying the RF and ultrasonic energy modes to the cantilever electrode or the ultrasonic blade separately for at least a portion of the period and then simultaneously for at least another portion of the period.

[1062] Example 221—The method of any one of any one of Examples 215-220, comprising applying the RF and ultrasonic energy modes to the cantilever electrode or the ultrasonic blade in a variable manner over at least a portion of the period until a desired tissue seal and cut is achieved.

[1063] Example 222—The method of any one of any one of Examples 215-220, comprising applying the RF and ultrasonic energy modes to the cantilever electrode or the ultrasonic blade in a constant manner over at least a portion of the period until a desired tissue seal and cut is achieved.

[1064] Example 223—The method of any one of any one of Examples 215-222, wherein the cantilever electrode comprises a resilient plug at a distal end.

[1065] Example 224—The method of any one of any one of Examples 215-222, wherein the cantilever electrode comprises a peripheral electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect and the clamp arm pad is fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the peripheral electrode and another portion extending through the inner aperture beyond the surface of the peripheral electrode to contact tissue.

[1066] Example 225—The method of any one of any one of Examples 215-222, wherein the cantilever electrode com-

prises an end-of-life indicator which can be sensed by the electrical generator to prompt replacement of the electrode.

[1067] Example 226—The method of any one of Examples 215-222, wherein the cantilever electrode comprises a floating electrode configured to deflect along its entire length.

[1068] Example 227—The method of any one of Examples 215-222, wherein the clamp arm pad is fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the electrode and another portion extending beyond the surface of the electrode to contact tissue.

[1069] Example 228—The method of any one of Examples 215-222, wherein the clamp arm comprises a spring disposed between the electrode and the clamp jaw.

[1070] Example 229—The method of any one of Examples 215-222, wherein the cantilever electrode is overmolded with an elastomer.

[1071] Example 230—The method of any one of Examples 215-222, wherein the clamp arm pad comprises one or more than one resistor embedded in the clamp arm pad to sense the height “h” of the clamp arm pad.

[1072] Example 231—The method of any one of Examples 215-222, wherein the clamp arm pad comprising a longitudinally aligned slot configured to receive an electrode plate, wherein the electrode plate and clamp arm pad are configured to both restrain and support the ultrasonic blade.

[1073] Example 232—The method of any one of Examples 215-222, the clamp arm comprises a lattice cushion defining a longitudinal slot, wherein the lattice cushion acts as a spring-like element and the cantilever electrode comprises a flexible member disposed above the lattice cushion and the clamp arm pad is disposed inside the longitudinal slot defined by the lattice cushion and is fixed to the clamp jaw.

[1074] Examples 233—The method of any one of Examples 215-222, where in the end-effector comprises a retainer configured to oppose the biasing force to maintain the cantilever electrode in a substantially flat configuration relative to the ultrasonic blade.

[1075] Example 234—The method of any one of Examples 215-222, wherein the clamp arm comprises a skirt provided around a distal end of the clamp jaw to surround the distal end of the clamp arm pad.

[1076] Example 235—The method of any one of Examples 215-222, wherein the clamp arm define a plurality of zones and at least one spring disposed in each of the plurality of zones, wherein a spring bias force of a first spring in a first zone is different from the spring bias force of a second spring in a second zone and the cantilever electrode is disposed along the plurality of zones and in contact with the each of the springs to apply a variable spring bias along the length of the electrode.

[1077] Example 236—The method of any one of Examples 215-222, wherein the cantilever electrode is supported by variable longitudinal support elements, wherein the variable longitudinal support elements apply a variable force on the electrode from the proximal end to the distal end.

[1078] Example 237—The method of any one of Examples 215-222, wherein the clamp arm comprises a compliant material fixed to the clamp jaw to support the cantilever electrode and is configured as a spring between the electrode and the clamp jaw.

[1079] Example 238—The method of any one of Examples 215-222, wherein the clamp arm comprises a leaf spring element disposed at a distal end of the cantilever electrode.

[1080] Example 239—The method of any one of Examples 215-222, wherein the clamp arm comprises a compressible material attached to a distal end of the electrode.

[1081] Example 240—The method of any one of Examples 215-222, wherein the clamp arm comprises an I-beam shaped clamp arm pad, wherein the clamp arm pad defines top and bottom lateral portions separated by a mesial portion to define the shape of an “I”, wherein the electrode is disposed between the top and bottom lateral portions of the I-beam shaped clamp arm pad.

[1082] Example 241—The method of any one of Examples 215-222, wherein the clamp jaw, the cantilever electrode, and the clamp arm pad define recesses along a longitudinal length that coincides with the ultrasonic blade.

[1083] Example 242—The method of any one of Examples 215-222, wherein the clamp arm comprise movable floating gap setting pads to create a composite arrangement to both create ultrasonic blade pressure while also setting a minimum gap between the electrode and the ultrasonic blade.

[1084] Example 243—The method of any one of Examples 215-222, wherein the cantilever electrode is configured as a segmented electrode.

[1085] Example 244—The method of any one of Examples 215-222, wherein the wherein the ultrasonic blade comprises a vapor deposition of electrically insulative material on select areas of the ultrasonic blade to prevent electrical shorting in the event of the ultrasonic blade contacting the upper jaw electrode.

[1086] Example 245—The method of any one of Examples 215-222, wherein the wherein the clamp arm, the ultrasonic blade, or both comprise selectively coated components.

[1087] Example 246—The method of any one of Examples 215-222, wherein the clamp arm comprises an electrically non-conductive clamp arm pad comprising a plurality of teeth.

[1088] Example 247—The method of any one of Examples 215-222, wherein the clamp arm comprises control features to adjust tissue path relative to the clamp arm or ultrasonic blade to create a predefined location of contact.

[1089] Example 248—The method of any one of Examples 215-222, wherein the clamp arm comprises a conductive polymer clamp arm pad.

[1090] Example 249—The method of any one of Examples 215-222, wherein the a clamp arm pad comprises electrically non-conductive layers and electrically conductive layers arranged in a sandwich like configuration.

[1091] Example 250—The method of any one of Examples 215-222, wherein the clamp arm comprises a composite clamp arm pad attached to the clamp jaw, wherein the composite clamp arm pad comprises electrically non-conductive layers and electrically conductive layers.

[1092] Example 251—The method of any one of Examples 215-222, wherein the clamp arm comprises a carrier attached to the clamp jaw and a clamp arm pad comprising an electrically conductive pad and an electrically non-conductive pad.

[1093] While several forms have been illustrated and described, it is not the intention of Applicant to restrict or limit the scope of the appended claims to such detail. Numerous modifications, variations, changes, substitutions, combinations, and equivalents to those forms may be implemented and will occur to those skilled in the art without departing from the scope of the present disclosure. Moreover, the structure of each element associated with the described forms can be alternatively described as a means for providing the function performed by the element. Also, where materials are disclosed for certain components, other materials may be used. It is therefore to be understood that the foregoing description and the appended claims are intended to cover all such modifications, combinations, and variations as falling within the scope of the disclosed forms. The appended claims are intended to cover all such modifications, variations, changes, substitutions, modifications, and equivalents.

[1094] The foregoing detailed description has set forth various forms of the devices and/or processes via the use of block diagrams, flowcharts, and/or examples. Insofar as such block diagrams, flowcharts, and/or examples contain one or more functions and/or operations, it will be understood by those within the art that each function and/or operation within such block diagrams, flowcharts, and/or examples can be implemented, individually and/or collectively, by a wide range of hardware, software, firmware, or virtually any combination thereof. Those skilled in the art will recognize that some aspects of the forms disclosed herein, in whole or in part, can be equivalently implemented in integrated circuits, as one or more computer programs running on one or more computers (e.g., as one or more programs running on one or more computer systems), as one or more programs running on one or more processors (e.g., as one or more programs running on one or more microprocessors), as firmware, or as virtually any combination thereof, and that designing the circuitry and/or writing the code for the software and/or firmware would be well within the skill of one of skill in the art in light of this disclosure. In addition, those skilled in the art will appreciate that the mechanisms of the subject matter described herein are capable of being distributed as one or more program products in a variety of forms, and that an illustrative form of the subject matter described herein applies regardless of the particular type of signal bearing medium used to actually carry out the distribution.

[1095] Instructions used to program logic to perform various disclosed aspects can be stored within a memory in the system, such as dynamic random access memory (DRAM), cache, flash memory, or other storage. Furthermore, the instructions can be distributed via a network or by way of other computer readable media. Thus a machine-readable medium may include any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computer), but is not limited to, floppy diskettes, optical disks, compact disc, read-only memory (CD-ROMs), and magneto-optical disks, read-only memory (ROMs), random access memory (RAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic or optical cards, flash memory, or a tangible, machine-readable storage used in the transmission of information over the Internet via electrical, optical, acoustical or other forms of propagated signals (e.g., carrier waves, infrared signals, digital signals,

etc.). Accordingly, the non-transitory computer-readable medium includes any type of tangible machine-readable medium suitable for storing or transmitting electronic instructions or information in a form readable by a machine (e.g., a computer).

[1096] As used in any aspect herein, the term “control circuit” may refer to, for example, hardwired circuitry, programmable circuitry (e.g., a computer processor including one or more individual instruction processing cores, processing unit, processor, microcontroller, microcontroller unit, controller, digital signal processor (DSP), programmable logic device (PLD), programmable logic array (PLA), or field programmable gate array (FPGA)), state machine circuitry, firmware that stores instructions executed by programmable circuitry, and any combination thereof. The control circuit may, collectively or individually, be embodied as circuitry that forms part of a larger system, for example, an integrated circuit (IC), an application-specific integrated circuit (ASIC), a system on-chip (SoC), desktop computers, laptop computers, tablet computers, servers, smart phones, etc. Accordingly, as used herein “control circuit” includes, but is not limited to, electrical circuitry having at least one discrete electrical circuit, electrical circuitry having at least one integrated circuit, electrical circuitry having at least one application specific integrated circuit, electrical circuitry forming a general purpose computing device configured by a computer program (e.g., a general purpose computer configured by a computer program which at least partially carries out processes and/or devices described herein, or a microprocessor configured by a computer program which at least partially carries out processes and/or devices described herein), electrical circuitry forming a memory device (e.g., forms of random access memory), and/or electrical circuitry forming a communications device (e.g., a modem, communications switch, or optical-electrical equipment). Those having skill in the art will recognize that the subject matter described herein may be implemented in an analog or digital fashion or some combination thereof.

[1097] As used in any aspect herein, the term “logic” may refer to an app, software, firmware and/or circuitry configured to perform any of the aforementioned operations. Software may be embodied as a software package, code, instructions, instruction sets and/or data recorded on non-transitory computer readable storage medium. Firmware may be embodied as code, instructions or instruction sets and/or data that are hard-coded (e.g., nonvolatile) in memory devices.

[1098] As used in any aspect herein, the terms “component,” “system,” “module” and the like can refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution.

[1099] As used in any aspect herein, an “algorithm” refers to a self-consistent sequence of steps leading to a desired result, where a “step” refers to a manipulation of physical quantities and/or logic states which may, though need not necessarily, take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It is common usage to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like. These and similar terms may be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities and/or states.

[1100] A network may include a packet switched network. The communication devices may be capable of communicating with each other using a selected packet switched network communications protocol. One example communications protocol may include an Ethernet communications protocol which may be capable permitting communication using a Transmission Control Protocol/Internet Protocol (TCP/IP). The Ethernet protocol may comply or be compatible with the Ethernet standard published by the Institute of Electrical and Electronics Engineers (IEEE) titled “IEEE 802.3 Standard”, published in December 2008 and/or later versions of this standard. Alternatively or additionally, the communication devices may be capable of communicating with each other using an X.25 communications protocol. The X.25 communications protocol may comply or be compatible with a standard promulgated by the International Telecommunication Union-Telecommunication Standardization Sector (ITU-T). Alternatively or additionally, the communication devices may be capable of communicating with each other using a frame relay communications protocol. The frame relay communications protocol may comply or be compatible with a standard promulgated by Consultative Committee for International Telegraph and Telephone (CCITT) and/or the American National Standards Institute (ANSI). Alternatively or additionally, the transceivers may be capable of communicating with each other using an Asynchronous Transfer Mode (ATM) communications protocol. The ATM communications protocol may comply or be compatible with an ATM standard published by the ATM Forum titled “ATM-MPLS Network Interworking 2.0” published August 2001, and/or later versions of this standard. Of course, different and/or after-developed connection-oriented network communication protocols are equally contemplated herein.

[1101] Unless specifically stated otherwise as apparent from the foregoing disclosure, it is appreciated that, throughout the foregoing disclosure, discussions using terms such as “processing,” “computing,” “calculating,” “determining,” “displaying,” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system’s registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

[1102] One or more components may be referred to herein as “configured to,” “configurable to,” “operable/operative to,” “adapted/adaptable,” “able to,” “conformable/conformed to,” etc. Those skilled in the art will recognize that “configured to” can generally encompass active-state components and/or inactive-state components and/or standby-state components, unless context requires otherwise.

[1103] The terms “proximal” and “distal” are used herein with reference to a clinician manipulating the handle portion of the surgical instrument. The term “proximal” refers to the portion closest to the clinician and the term “distal” refers to the portion located away from the clinician. It will be further appreciated that, for convenience and clarity, spatial terms such as “vertical,” “horizontal,” “up,” and “down” may be used herein with respect to the drawings. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and/or absolute.

[1104] Those skilled in the art will recognize that, in general, terms used herein, and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes but is not limited to,” etc.). It will be further understood by those within the art that if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to claims containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should typically be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

[1105] In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should typically be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, typically means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). In those instances where a convention analogous to “at least one of A, B, or C, etc.” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (e.g., “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, etc.). It will be further understood by those within the art that typically a disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms unless context dictates otherwise. For example, the phrase “A or B” will be typically understood to include the possibilities of “A” or “B” or “A and B.”

[1106] With respect to the appended claims, those skilled in the art will appreciate that recited operations therein may generally be performed in any order. Also, although various operational flow diagrams are presented in a sequence(s), it should be understood that the various operations may be performed in other orders than those which are illustrated, or may be performed concurrently. Examples of such alternate orderings may include overlapping, interleaved, interrupted, reordered, incremental, preparatory, supplemental, simulta-

neous, reverse, or other variant orderings, unless context dictates otherwise. Furthermore, terms like “responsive to,” “related to,” or other past-tense adjectives are generally not intended to exclude such variants, unless context dictates otherwise.

[1107] It is worthy to note that any reference to “one aspect,” “an aspect,” “an exemplification,” “one exemplification,” and the like means that a particular feature, structure, or characteristic described in connection with the aspect is included in at least one aspect. Thus, appearances of the phrases “in one aspect,” “in an aspect,” “in an exemplification,” and “in one exemplification” in various places throughout the specification are not necessarily all referring to the same aspect. Furthermore, the particular features, structures or characteristics may be combined in any suitable manner in one or more aspects.

[1108] Any patent application, patent, non-patent publication, or other disclosure material referred to in this specification and/or listed in any Application Data Sheet is incorporated by reference herein, to the extent that the incorporated materials is not inconsistent herewith. As such, and to the extent necessary, the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference. Any material, or portion thereof, that is said to be incorporated by reference herein, but which conflicts with existing definitions, statements, or other disclosure material set forth herein will only be incorporated to the extent that no conflict arises between that incorporated material and the existing disclosure material.

[1109] In summary, numerous benefits have been described which result from employing the concepts described herein. The foregoing description of the one or more forms has been presented for purposes of illustration and description. It is not intended to be exhaustive or limiting to the precise form disclosed. Modifications or variations are possible in light of the above teachings. The one or more forms were chosen and described in order to illustrate principles and practical application to thereby enable one of ordinary skill in the art to utilize the various forms and with various modifications as are suited to the particular use contemplated. It is intended that the claims submitted herewith define the overall scope.

1. An end-effector comprising:

an ultrasonic blade configured to acoustically couple to an ultrasonic transducer and to electrically couple to a pole of an electrical generator; and

a clamp arm comprising:

a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point;

a peripheral cantilever electrode having a proximal end fixed to the proximal end of the clamp jaw and a distal end that is free to deflect, the peripheral cantilever electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith, wherein the peripheral cantilever electrode is configured to electrically couple to an opposite pole of the electrical generator, and wherein the peripheral cantilever electrode defines an inner aperture; and

a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the peripheral cantilever electrode and another por-

tion extending through the inner aperture beyond the surface of the peripheral cantilever electrode to contact tissue.

2. The end-effector of claim 1, wherein the peripheral cantilever electrode further comprises a connection pad isolated from the peripheral cantilever electrode to attach the peripheral cantilever electrode to a proximal end of the clamp arm.

3. The end-effector of claim 2, wherein the connection pad sets a gap between the peripheral cantilever electrode and the ultrasonic blade.

4. The end-effector of claim 3, wherein the gap is defined by a thickness of the connection pad.

5. The end-effector of claim 1, wherein the peripheral cantilever electrode further comprises a connection pad extending from the peripheral cantilever electrode to attach the peripheral cantilever electrode to a proximal end of the clamp arm.

6. The end-effector of claim 1, wherein the peripheral cantilever electrode comprises an electrically non-conductive portion to define a space between an outer edge of the electrically conductive portion of the peripheral cantilever electrode and an outer edge of the peripheral cantilever electrode.

7. An end-effector comprising:

an ultrasonic blade configured to couple to acoustically couple to an ultrasonic transducer and to electrically couple to a pole of an electrical generator; and

a clamp arm comprising:

a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point;

a floating cantilever electrode configured to deflect along its entire length, the floating cantilever electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith, wherein the floating cantilever electrode is configured to electrically couple to an opposite pole of the electrical generator; and

a resilient clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the floating cantilever electrode and another portion extending beyond the surface of the floating cantilever electrode to contact tissue.

8. The end-effector of claim 7, wherein the floating cantilever electrode is mounted to a spring.

9. The end-effector of claim 7, wherein the clamp arm comprises a plurality of wear resistant clamp arm pads mounted to the resilient clamp arm pad.

10. The end-effector of claim 7, wherein the resilient clamp arm pad acts as a spring and wherein the floating cantilever electrode is configured to deflect into the resilient clamp arm pad and move in a direction toward the clamp jaw.

11. The end-effector of claim 7, further comprising a wear resistant clamp arm pad to set a tissue gap between the floating cantilever electrode and the ultrasonic blade.

12. The end-effector of claim 7, further comprising a plurality of wear resistant dowel pins sized and configured to set a tissue gap between the ultrasonic blade and the floating cantilever electrode.

13. The end-effector of claim 12, wherein at least one of the plurality of wear resistance dowel pins comprises one of

barb features or pierce features to prevent the wear resistant dowel pins from backing out from the clamp arm when embedded in the clamp arm.

14. An end-effector comprising:

an ultrasonic blade configured to acoustically couple to an ultrasonic transducer and to electrically couple to a pole of an electrical generator; and

a clamp arm comprising:

a clamp jaw having a proximal end and a distal end, the proximal end pivotally movable about a pivot point;

a laterally deflectable cantilever electrode defining a surface configured to contact tissue and apply electrical energy to the tissue in contact therewith, wherein the laterally deflectable cantilever electrode is configured to electrically couple to an opposite pole of the electrical generator; and

a clamp arm pad fixed to the clamp jaw and having at least a portion disposed between the clamp jaw and the laterally deflectable cantilever electrode and another portion extending beyond the surface of the laterally deflectable cantilever electrode to contact tissue.

15. The end-effector of claim **14**, wherein the clamp arm pad is configured to expand due to at least one of heat or compression.

16. The end-effector of claim **15**, wherein the laterally deflectable cantilever electrode is configured to deflect laterally due to expansion of the clamp arm pad.

17. The end-effector of claim **14**, wherein the laterally deflectable cantilever electrode comprises a notch defined on a distal end thereof.

18. The end-effector of claim **14**, wherein the laterally deflectable cantilever electrode comprises a vertical stiffening form to secure the deflectable cantilever electrode to the clamp jaw.

19. The end-effector of claim **14**, wherein the laterally deflectable cantilever electrode comprises a proximal end fixed to a proximal end of the clamp jaw and a distal end that is free to deflect vertically and laterally.

20. The end-effector of claim **14**, wherein the laterally deflectable cantilever electrode is configured to be deflected by at least one of material clamped between the clamp arm and the ultrasonic blade, the ultrasonic blade, or the clamp arm pad.

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